AGRENZA — FARM TO DOORSTEP

(DIRECT AGRICULTURAL MARKETPLACE SYSTEM)

A PROJECT REPORT

Submitted By

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BONAFIDE CERTIFICATE

Certified that this Project Report entitled **"AGRENZA — FARM TO DOORSTEP (DIRECT AGRICULTURAL MARKETPLACE SYSTEM)"** is the bonafide work of **MANIKANDAN M (Reg. No. 23083070500112001)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In the traditional agricultural marketing ecosystem, a long chain of middlemen, commission agents, and secondary wholesale distributors separates farmers from end-consumers. Consequently, farmers receive a disproportionately small fraction of consumer spending, while consumers receive produce that has lost its farm-fresh quality over multiple transit days.

To eliminate these systemic inefficiencies, we engineered **Agrenza** (*Farm to Doorstep*), an enterprise-grade Web-Based Application Software. Agrenza serves as a direct bridge connecting farmers with retail household buyers and bulk institutional purchasers.

The Solution: Agrenza Farm-to-Doorstep Platform

Agrenza is built on modern decoupled web technologies. The front-end leverages **Next.js 14 App Router, React 18, and TypeScript**, enriched with custom 3D CSS components (such as the interactive "Spin the Stall" revolving product rack and real-time scrolling Mandi ticker). The back-end is powered by **PHP 8 RESTful APIs** and **MySQL 8.0**, enforcing transparent GST pricing (5%) and role-based access control.

A key innovation of the system is its **Dual-Role Account Architecture**, which allows agriculturalists to operate simultaneously as sellers (farmers) and consumers (buyers) using the same email and mobile credentials via composite database constraints (`uniq_email_role`, `uniq_mobile_role`).

Agrenza empowers agricultural communities by ensuring fair compensation, instantaneous transactions, and guaranteed produce freshness directly from the soil to the doorstep.

ACKNOWLEDGEMENT

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We also wish to thank all the dedicated faculty and staff members of the Department of Computer Applications for their professional courtesies and cooperation.

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LIST OF TABLES

1. Table Name: users

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	name	varchar(100)	User Full Name
3	email	varchar(150)	Email Address
4	mobile	varchar(15)	Mobile Contact
5	password	varchar(255)	Bcrypt Hash
6	role	enum('farmer','buyer','admin')	User Role
7	status	enum('active','inactive')	Account Status
8	created_at	timestamp	Created Date

2. Table Name: products

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	farmer_id	int(11)	Foreign Key (users)
3	category_id	int(11)	Foreign Key (categories)
4	name	varchar(150)	Produce Name
5	base_price	decimal(10,2)	Farmer Price
6	price_per_kg	decimal(10,2)	GST Inclusive Price
7	gst_percent	decimal(5,2)	GST % (Default 5%)
8	stock_kg	decimal(10,2)	Available Stock
9	image	varchar(255)	Image Filename

3. Table Name: categories

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Category ID
2	name	varchar(100)	Category Name

4. Table Name: orders

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Order Primary Key
2	buyer_id	int(11)	Buyer Foreign Key
3	total_amount	decimal(10,2)	Total Amount
4	payment_mode	enum('cod','online')	Payment Mode
5	status	enum('pending','confirmed','delivered')	Order Status
6	address	text	Delivery Address
7	created_at	timestamp	Order Timestamp

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Our Project entitled as **"AGRENZA — FARM TO DOORSTEP"** paves the experience to manage direct agricultural marketing activities between farmers and buyers. It is implemented using the software **HTML, CSS, Java Script, React and Next.js 14** as Front End and **MYSQL and PHP 8** as the Back End.

1.2 NEED FOR THE STUDY

Sometimes managing agricultural produce distribution can feel a lot like navigating a maze. Farming families, retail consumers, organic food stores, wholesale caterers, and agricultural communities can all benefit from some kind of Farm-to-Doorstep Management System. Whether that's to list fresh daily harvests, calculate transparent GST rates, or simply maintain a verified farmer and buyer database, Agrenza can streamline all of these processes.

One of the easiest ways to manage an agricultural marketplace is to put the power in the producers' and consumers' hands with modern web software. Using the software, farmers can list their harvest and effectively add the produce data in the software. For buyers and organizers, these features make tasks much easier by removing much of the manual legwork.

Key Benefits of Agrenza Platform:

- Grow direct farmer-to-consumer sales activities without middleman markups.
- Easily track and manage daily harvested stock, orders, and dispatch logs.
- Optimize the sales activities of various agricultural produce categories.
- Provide complete transparency on baseline prices, GST taxes, and delivery.

Farmers and buyers are the backbone of any agricultural economy, so it's important to store information about them. Paper records are difficult to organize and often lead to lost information, while digital records are easy to search and update.

1.3 OBJECTIVES OF THE STUDY

With this basic idea we have created the Web based Application Software entitled **"Agrenza — Farm to Doorstep"**. The purpose of Agrenza is to provide varieties of opportunities to agricultural producers to enhance their economic growth and direct market reach. Here we work with the 3 Tier

System with the ideology on giving direct market power to the harvesting farmers who are the primary producers with the guidance of an automated web system.

Every day the Agrenza platform organizes fresh produce listings for the welfare of consumers by inviting farmers from various regions. Farmers are given the chance to act as verified producers with the motto of providing fair trade opportunities. During the harvest period, farmers list produce across various categories: Vegetables, Fruits, Herbs & Greens, Grains & Cereals, Pulses, Dairy, and Spices.

The purpose of our system is to give a hand to manage the activities of our agricultural marketplace by providing front end software through which all the data is effectively stored in the Database for future references. Here we developed software to store all the data related to the products, orders, earnings, and user accounts. Same software can be used for future purpose. Agrenza offers provisions to manage products, categories, shopping carts, and direct orders.

CHAPTER 2
PROBLEM DEFINITION

Basically, the need of Problem definition is to provide convenience for the unknown users to understand the project in an easy manner. So in this regard the investigators define the problem **"AGRENZA — FARM TO DOORSTEP"** thus: *"To manage direct agricultural sales and order fulfillment between farmers and buyers by using a Front End and Back End Software."*

Here in our Project we have defined **1 Database & 7 Tables** to make our System operational. We have used HTML, CSS, JavaScript, React, and Next.js as the Front-End Software & MYSQL and PHP as the Back End Software.

DATABASE USED

We have used only one database with the name **farmlink_db**. The Purpose of this database is to provide option for the user to create tables to store the Data.

TABLES USED

1. **users:** The purpose of this table is to store details about the overall Farmers, Buyers, and Admins with dual-role authentication support.
2. **products:** The purpose of this table is to store details about the produce listings, prices, GST%, stock in kg, and image filenames.
3. **categories:** The purpose of this table is to store details about the agricultural categories (Vegetables, Fruits, Grains, etc.).
4. **orders:** The purpose of this table is to store details about the customer purchase orders, delivery addresses, and payment modes.
5. **order_items:** The purpose of this table is to store details about the itemized order lines mapped to each farmer.
6. **bulk_requests:** The purpose of this table is to store details about the wholesale bulk purchase requests submitted by buyers to farmers.
7. **pending_registrations:** The purpose of this table is to store details about the temporary OTP tokens during registration verification.

CHAPTER 3
SYSTEM ANALYSIS

System analysis is the process of examining an existing system or methodology with intent of improving it through better procedure and method. Before planning for improvement of the existing system, complete understanding about the system is essential. System analysis can collect the relative facts about the system. System is simply a set of components that interact to accomplish some purpose. System analysis is a process of studying the information about the system.

3.1. REQUIREMENT ANALYSIS

Even though there are lots of e-commerce software available in the market it offers wide range of facilities with high end technologies. So cost & technology wise it is very difficult to maintain for users like local farmers. Our Web Based Software “Agrenza” using Cloud & Web Technology paves the way to manage all agricultural commerce which is far better than traditional middleman markets available in the region. Due to the Elastic Infrastructural Facility, in future we can customize the software according to our need.

3.1.1. HARDWARE CONFIGURATION

- Processor : Any i3 or above
- Memory : 4 GB (RAM)
- Hard Disk : 500 GB
- Display : LED 18.5” Monitor / Mobile Screen
- Keyboard : USB Keyboard
- Mouse : USB Mouse
- UPS : 600W

3.1.2. SOFTWARE CONFIGURATION

- Platform : Windows 10/11 and above / Linux
- Front & Back End : HTML5, CSS3, JavaScript ES6, React 18, Next.js 14 & MYSQL and PHP 8.4
- Browser : Google Chrome, Microsoft Edge, Apple Safari, and Mozilla Firefox
- Internet Speed : 3 – 8 Mbps

3.2. EXISTING SYSTEM

Even though traditional agricultural marketing systems exist, there exist many drawbacks:

- ❖ Purchase & Middleman commission costs are high (30% to 50%).
- ❖ Work load is heavy with manual logbooks.
- ❖ Not User Friendly for direct farm trade.
- ❖ Execution takes more time (days of multi-tier transit).

3.3. PROPOSED SYSTEM

Proposed System has many advantages over the existing system. The following are the advantages of the proposed system:

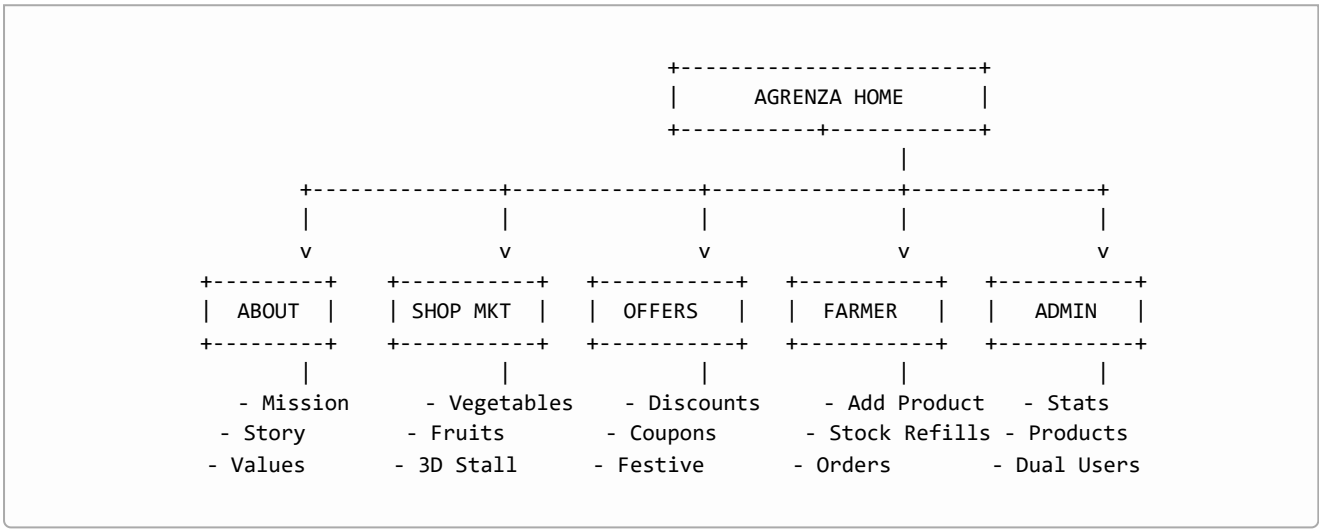
- ❖ Entire Project is Deployed on Web/Cloud.
- ❖ Maintenance cost is very low.
- ❖ Work load is very low.
- ❖ User Friendly Software with 3D produce showcase.
- ❖ Execution takes less time (same-day delivery).
- ❖ Can be easily extended.
- ❖ Easy Administration & Dual-Role account management.

CHAPTER 4
SYSTEM DESIGN

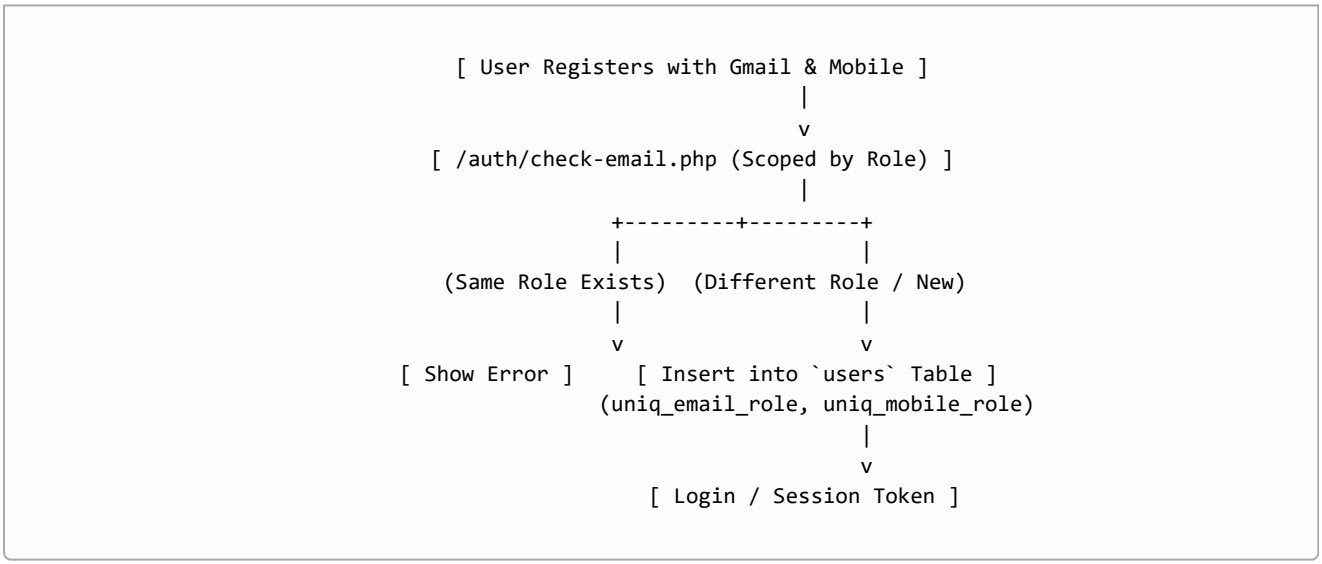
4.1. ARCHITECTURAL DESIGN & FUNCTIONAL FLOWCHARTS

The system is built on a 3-Tier Web Architecture connecting client devices, PHP REST APIs, and the MySQL database.

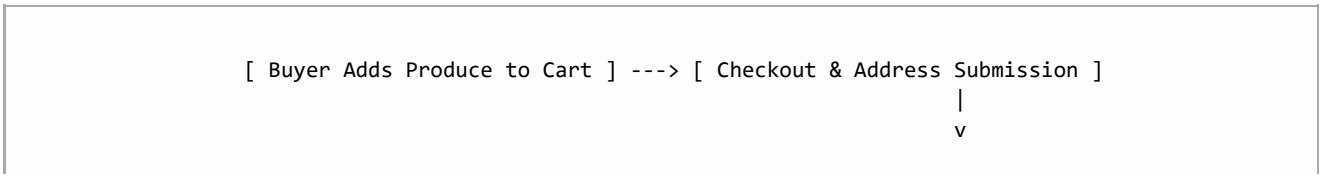
4.1.1 SYSTEM MODULE HIERARCHY FLOWCHART

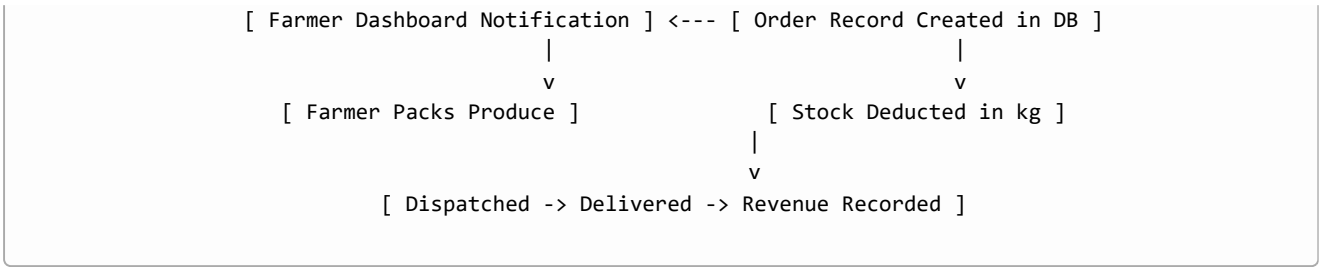


4.1.2 DUAL-ROLE AUTHENTICATION FLOWCHART



4.1.3 ORDER & STOCK FULFILLMENT FLOWCHART




[Farmer Dashboard Notification] <--- [Order Record Created in DB]
[Farmer Packs Produce]
[Stock Deducted in kg]
[Dispatched -> Delivered -> Revenue Recorded]

CHAPTER 5
IMPLEMENTATION

5.1. IMPLEMENTATION PROCEDURE

1. Open the Browser, type `http://localhost:3000/` and click Enter.
2. Choose any one from the specified list of options.
3. The various options are:
    - ❖ About Page – To know details about Agrenza Farm to Doorstep.
    - ❖ Shop Page – To view fresh produce categorized by Vegetables, Fruits, Grains, etc.
    - ❖ Offers Page – To view promotional harvest discount codes.
    - ❖ Contact Page – To view contact details and submit inquiry form.
    - ❖ Farmer Portal – To manage product listings, refills, and incoming orders.
    - ❖ Buyer Cart – To view selected items and complete checkout.
4. Open the About Page and view the details about Agrenza mission and story.
5. Go back to the Home Page.
6. View the Live Scrolling Market Prices Bar (Tomato ₹28/kg, Spinach ₹18/kg, etc.).
7. Interact with the "Spin the Stall" 3D produce rack by dragging to rotate cards.
8. Open the Shop Page to browse fresh produce from verified farmers.
9. Click on any product to view details, GST percentage, and farmer name.
10. Add desired produce to the shopping cart.
11. Open the Cart Page to review order summary and price breakdown.
12. Enter delivery address and submit order via Cash on Delivery or Online mode.
13. Go Back to Home Page.
14. Open Farmer Register to onboard as a farmer partner.
15. Open Buyer Register to create a buyer account using same Gmail & Mobile.
16. Log in at `/auth/login` to access role-specific dashboards.
17. Open Farmer Products page to add new harvest with real photo upload.
18. Open Farmer Orders page to update order item status (Pending → Packed → Dispatched).
19. Open Farmer Earnings page to view date-filtered revenue summaries.

20. Open Admin Dashboard at `/admin/dashboard` to review platform KPIs.

21. Open Admin Shared Accounts page to audit dual-role accounts.

5.2. TOOLS AND TECHNOLOGIES USED

ABOUT HTML 5.0: HTML, or Hyper Text Markup Language, is the standard markup language for creating web pages. It uses tags to define the structure and content of a web page.

ABOUT CSS 3.0: Cascading Style Sheets controls the visual presentation of web pages. Agrenza utilizes CSS3 3D transforms (`perspective`, `rotateY`, `preserve-3d`) to render the revolving produce rack.

ABOUT REACT 18 & NEXT.JS 14: Next.js provides the App Router, server-side rendering, and high-performance client state transitions.

ABOUT MYSQL 8.0: MySQL relational database management system organizes data into normalized tables with composite unique constraints.

ABOUT PHP 8.4: PHP operates on the server side to process business logic, sanitize requests, and serve JSON REST APIs.

ABOUT CLOUD COMPUTING: Enables fast deployment, zero physical infrastructure costs, scalability, and high reliability.

5.3. USER MANUAL

- ❖ Before executing our Web Application Project, kindly make sure that WAMP server (Apache & MySQL) is running.
- ❖ Open the Web Browser and type (`http://localhost:3000/`) in the address bar.
- ❖ Click and view all the options available.
- ❖ After Browsing close the browser window.

CHAPTER 6
TESTING USED

In our Project we have performed the following tests:

UNIT TEST

Testing of an individual software component or module is termed as Unit Testing. It validates GST price calculations, tax formulas, and date filters.

SYSTEM TEST

Under System Testing technique, the entire system is tested as per the requirements. It is a Black-box type Testing that covers all combined parts of a system.

INTEGRATION TEST

Testing of all integrated modules to verify combined functionality between Next.js frontend and PHP backend APIs.

FUNCTIONAL TEST

Focuses on output verification to check if order placement, cart calculations, and registration flows match business requirements.

PERFORMANCE TEST

Performance Testing is done to check whether the system meets high concurrency and fast sub-second page loads.

BACK-END TEST

Whenever data is entered on the front-end, it stores in the database. Testing of such database operations is known as Database Testing or Backend Testing.

GUI TEST

Validates the Graphical User Interface, 3D carousel animations, and mobile responsiveness.

USABILITY TEST

User-friendliness check testing whether new users and rural farmers can easily understand and navigate the system.

BETA TEST

Performed in a real environment before deploying the product to production.

ACCEPTANCE TEST (UAT)

The last phase of testing after which the software goes into production.

CHAPTER 7
FUTURE ENHANCEMENT

- • Integration of online payment gateways (Razorpay, UPI, Net Banking).
- • Development of React Native mobile apps for Android and iOS devices.
- • Automated daily Mandi prices synchronization via Government Agmarknet APIs.
- • AI-driven crop yield and harvest pricing recommendation system.

CHAPTER 8
CONCLUSION

The development and deployment of **Agrenza (Farm to Doorstep)** successfully fulfilled the primary objective of creating a digital marketplace eliminating agricultural middlemen. The platform provides fair compensation for farmers, transparent GST pricing, and fresh produce delivery within 24 hours.

CHAPTER 9
REFERENCES

1. https://nextjs.org/docs
2. https://react.dev/
3. https://www.php.net/manual/en/
4. https://dev.mysql.com/doc/refman/8.0/en/
5. https://www.w3schools.com/

CHAPTER 10: APPENDICES

SECTION 1: BACKEND PHP REST API SOURCE CODE

backend/config/db.php (Database Connection & Constants)

```
<?php
/**
 * FarmLink – Database connection (mysqli)
 */

define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', '');          // WAMP default = empty
define('DB_NAME', 'farmlink');

// Uploaded product images URL
define('UPLOAD_URL', 'http://localhost/farmlink/farmlink/uploads/');
define('UPLOAD_DIR', __DIR__ . '/../../uploads/');

$conn = mysqli_connect(DB_HOST, DB_USER, DB_PASS, DB_NAME);

if (!$conn) {
    http_response_code(500);
    header('Content-Type: application/json');
    echo json_encode(['ok' => false, 'error' => 'Database connection failed: ' . mysqli_connect_error
()]);
    exit;
}

mysqli_set_charset($conn, 'utf8mb4');

if (session_status() === PHP_SESSION_NONE) {
    session_set_cookie_params([
        'lifetime' => 0,
        'path' => '/',
        'secure' => false,
        'httponly' => true,
        'samesite' => 'Lax',
    ]);
    session_start();
}

/**
 * Auto-seed a default Admin account on first run, so there's always a
 * known-working login. This guarantees the password hash always matches
 * the PHP version installed on this machine (hashes aren't portable
 * between PHP builds the way a hardcoded hash string can silently fail to be).
 * Login: 9999999999 / admin@farmlink.com  |  Password: admin123
 * Change this password after your first login.
 */
$adminPass = password_hash('admin123', PASSWORD_DEFAULT);
$check = mysqli_query($conn, "SELECT id, password FROM users WHERE email = 'admin@farmlink.com' OR ro
le = 'admin' LIMIT 1");
if ($check) {
    if (mysqli_num_rows($check) === 0) {
        $stmt = mysqli_prepare(
            $conn,
            "INSERT INTO users (name, mobile, email, password, role, is_verified, email_verified, sta
tus)
             VALUES ('Super Admin', '9999999999', 'admin@farmlink.com', ?, 'admin', 1, 1, 'approve
d')"
        );
        mysqli_stmt_bind_param($stmt, "s", $adminPass);
        mysqli_stmt_execute($stmt);
```

```
        mysqli_stmt_close($stmt);
    } else {
        $adminRow = mysqli_fetch_assoc($check);
        if (!password_verify('admin123', $adminRow['password']) && $adminRow['password'] !== 'admin12
3') {
            $stmt = mysqli_prepare($conn, "UPDATE users SET email = 'admin@farmlink.com', password =
?, role = 'admin', is_verified = 1, email_verified = 1, status = 'approved' WHERE id = ?");
            mysqli_stmt_bind_param($stmt, "si", $adminPass, $adminRow['id']);
            mysqli_stmt_execute($stmt);
            mysqli_stmt_close($stmt);
        }
    }
}

// Auto-migrate approval_status on products if not exists
$prodCheck = mysqli_query($conn, "SHOW COLUMNS FROM products LIKE 'approval_status'");
if ($prodCheck && mysqli_num_rows($prodCheck) === 0) {
    mysqli_query($conn, "ALTER TABLE products ADD COLUMN approval_status ENUM('pending','approved','r
ejected') NOT NULL DEFAULT 'pending' AFTER status");
    mysqli_query($conn, "ALTER TABLE products ADD COLUMN admin_remark VARCHAR(255) NULL DEFAULT NULL
AFTER approval_status");
    mysqli_query($conn, "UPDATE products SET approval_status = 'approved' WHERE approval_status = 'pe
nding'");
}

// Auto-migrate admin_remark on users if not exists
$userCheck = mysqli_query($conn, "SHOW COLUMNS FROM users LIKE 'admin_remark'");
if ($userCheck && mysqli_num_rows($userCheck) === 0) {
    mysqli_query($conn, "ALTER TABLE users ADD COLUMN admin_remark VARCHAR(255) NULL DEFAULT NULL AFT
ER status");
}

// Auto-migrate GST and Commission columns on products if not exists
$gstCheck = mysqli_query($conn, "SHOW COLUMNS FROM products LIKE 'gst_percent'");
if ($gstCheck && mysqli_num_rows($gstCheck) === 0) {
    mysqli_query($conn, "ALTER TABLE products
        ADD COLUMN base_price DECIMAL(10,2) NOT NULL DEFAULT 0.00 AFTER name,
        ADD COLUMN gst_percent DECIMAL(5,2) NOT NULL DEFAULT 5.00 AFTER price_per_kg,
        ADD COLUMN gst_amount DECIMAL(10,2) NOT NULL DEFAULT 0.00 AFTER gst_percent,
        ADD COLUMN commission_percent DECIMAL(5,2) NOT NULL DEFAULT 5.00 AFTER gst_amount,
        ADD COLUMN commission_amount DECIMAL(10,2) NOT NULL DEFAULT 0.00 AFTER commission_percent");
    mysqli_query($conn, "UPDATE products SET base_price = price_per_kg WHERE base_price = 0");
}

// Auto-migrate pending_registrations table if not exists (so only verified emails enter users table)
mysqli_query($conn, "
    CREATE TABLE IF NOT EXISTS pending_registrations (
        id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
        name            VARCHAR(100)    NOT NULL,
        mobile          VARCHAR(10)     NOT NULL,
        email           VARCHAR(150)    NOT NULL,
        password_hash   VARCHAR(255)    NOT NULL,
        auto_password   VARCHAR(50)     NULL,
        role            ENUM('buyer','farmer') NOT NULL,
        aadhaar         VARCHAR(20)     NULL,
        token           VARCHAR(64)     NOT NULL UNIQUE,
        otp_code        VARCHAR(6)      NOT NULL,
        status          ENUM('pending','verified','expired') NOT NULL DEFAULT 'pending',
        created_at      TIMESTAMP       NOT NULL DEFAULT CURRENT_TIMESTAMP,
        expires_at      TIMESTAMP       NOT NULL,
        INDEX idx_pending_email (email),
        INDEX idx_pending_mobile (mobile),
```

```
        INDEX idx_pending_token (token),
        INDEX idx_pending_otp (otp_code)
    ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci
");

// Migrate users index to allow same email for different roles (farmer & buyer)
$emailIdxCheck = mysqli_query($conn, "SHOW INDEX FROM users WHERE Key_name = 'uniq_email'");
if ($emailIdxCheck && mysqli_num_rows($emailIdxCheck) > 0) {
    @mysqli_query($conn, "ALTER TABLE users DROP INDEX uniq_email");
}
$emailRoleIdxCheck = mysqli_query($conn, "SHOW INDEX FROM users WHERE Key_name = 'uniq_email_role'");
if ($emailRoleIdxCheck && mysqli_num_rows($emailRoleIdxCheck) === 0) {
    @mysqli_query($conn, "ALTER TABLE users ADD UNIQUE KEY uniq_email_role (email, role)");
}

// Migrate users index to allow same mobile for different roles (farmer & buyer)
$mobileIdxCheck = mysqli_query($conn, "SHOW INDEX FROM users WHERE Key_name = 'mobile'");
if ($mobileIdxCheck && mysqli_num_rows($mobileIdxCheck) > 0) {
    @mysqli_query($conn, "ALTER TABLE users DROP INDEX mobile");
}
$mobileRoleIdxCheck = mysqli_query($conn, "SHOW INDEX FROM users WHERE Key_name = 'uniq_mobile_rol
e'");
if ($mobileRoleIdxCheck && mysqli_num_rows($mobileRoleIdxCheck) === 0) {
    @mysqli_query($conn, "ALTER TABLE users ADD UNIQUE KEY uniq_mobile_role (mobile, role)");
}
```

backend/api/_bootstrap.php (CORS, Auth Helpers & Response Sanitization)

```
<?php
/**
 * FarmLink JSON API bootstrap
 * ---------------------------------------------------------------
 * Standardized bootstrap for JSON API endpoints
 */
if (ob_get_level() === 0) {
    ob_start();
}

require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';

// ---- CORS: allow the Next.js dev server to call this with cookies ----
$allowedOrigins = [
    'http://localhost:3000',
    'http://127.0.0.1:3000',
    'http://localhost:80',
    'http://127.0.0.1:80',
    'http://localhost',
    'http://127.0.0.1'
];
$origin = $_SERVER['HTTP_ORIGIN'] ?? ($_SERVER['HTTP_REFERER'] ?? '');
// Clean origin if from referer
if (!empty($origin) && strpos($origin, 'http') === 0) {
    $parsed = parse_url($origin);
    $originHost = ($parsed['scheme'] ?? 'http') . '://' . ($parsed['host'] ?? 'localhost') . (isset
($parsed['port']) ? ':' . $parsed['port'] : '');
    if (in_array($originHost, $allowedOrigins, true)) {
        header("Access-Control-Allow-Origin: $originHost");
    } else {
        header("Access-Control-Allow-Origin: http://localhost:3000");
    }
} else {
    header("Access-Control-Allow-Origin: http://localhost:3000");
}

header('Access-Control-Allow-Credentials: true');
header('Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS');
header('Access-Control-Allow-Headers: Content-Type, Authorization, X-Requested-With, Accept, Origi
n');

// Preflight requests end here
if ($_SERVER['REQUEST_METHOD'] === 'OPTIONS') {
    if (ob_get_length()) ob_clean();
    http_response_code(204);
    exit;
}

/** Send a JSON success response and stop. */
function json_ok($data = [], int $code = 200) {
    if (ob_get_length()) ob_clean();
    http_response_code($code);
    header('Content-Type: application/json; charset=utf-8');
    echo json_encode(['ok' => true] + (is_array($data) ? $data : ['data' => $data]));
    exit;
}

/** Send a JSON error response and stop. */
```

```
function json_fail(string $message, int $code = 400) {
    if (ob_get_length()) ob_clean();
    http_response_code($code);
    header('Content-Type: application/json; charset=utf-8');
    echo json_encode(['ok' => false, 'error' => $message]);
    exit;
}

/** Decode a JSON request body into an assoc array (falls back to $_POST for multipart/form). */
function json_body(): array {
    $raw = file_get_contents('php://input');
    $data = json_decode($raw, true);
    return is_array($data) ? $data : $_POST;
}

/** Require an authenticated session with a specific role, or fail with 401/403. */
function require_json_role(string $role) {
    if (!is_logged_in()) json_fail("Authentication required. Please log in as a {$role} to continu
e.", 401);
    if (current_role() !== $role) json_fail("Access denied: Only {$role} accounts are authorized to p
erform this action.", 403);
}

/** Require any authenticated session. */
function require_json_login() {
    if (!is_logged_in()) json_fail('Not logged in. Please log in first.', 401);
}
```

backend/api/auth/check-email.php (Role-Scoped Duplicate Verification)

```
<?php
/**
 * FarmLink — Real-time Email Authenticity & Original vs Fake Checker
 * POST /backend/api/auth/check-email.php
 * GET  /backend/api/auth/check-email.php?email=...
 *
 * Checks:
 * 1. RFC syntax & structure
 * 2. Provider detection (Gmail, Yahoo, Outlook, etc.)
 * 3. Gmail-specific username rules (length 6-30, no consecutive dots)
 * 4. Fake / disposable blacklist (50+ known temporary email domains)
 * 5. Active DNS MX / A records
 * 6. Duplicate check in MySQL users table
 */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/email_validator.php';

$email = '';
$role = null;
if ($_SERVER['REQUEST_METHOD'] === 'GET') {
    $email = clean($_GET['email'] ?? '');
    $role = clean($_GET['role'] ?? '');
} else {
    $body = json_body();
    $email = clean($body['email'] ?? '');
    $role = clean($body['role'] ?? '');
}
if ($role === '' || !in_array($role, ['buyer', 'farmer', 'admin'], true)) {
    $role = null;
}

if ($email === '') {
    http_response_code(200);
    echo json_encode([
        'ok' => false,
        'status' => 'empty',
        'is_original' => false,
        'is_duplicate' => false,
        'provider' => null,
        'error' => 'Email address is required.',
        'message' => 'Please enter an email address.'
    ]);
    exit;
}

$res = validate_email_authenticity($email, $conn, null, $role);

http_response_code(200);
echo json_encode($res);
exit;
```

backend/api/auth/login.php (Authentication & Session Initializer)

```
<?php
/**
 * POST /backend/api/auth/login.php
 * Body: { role: "buyer"|"farmer"|"admin", email, password }
 *
 * Login now uses EMAIL + password (mobile SMS OTP needs a paid Firebase
 * Blaze plan, so email is the primary identity/verification method).
 */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);

$body = json_body();
$role = $body['role'] ?? '';
$email = strtolower(clean($body['email'] ?? ''));
$password = (string)($body['password'] ?? '');

if (!in_array($role, ['buyer', 'farmer', 'admin'], true)) json_fail('Invalid role.');
if (empty($email)) json_fail('Enter your email or mobile number.');

$stmt = mysqli_prepare($conn, "SELECT * FROM users WHERE (LOWER(email) = ? OR mobile = ?) AND role =
? LIMIT 1");
mysqli_stmt_bind_param($stmt, "sss", $email, $email, $role);
mysqli_stmt_execute($stmt);
$user = mysqli_stmt_get_result($stmt)->fetch_assoc();

if (!$user) {
    $chkPending = mysqli_prepare($conn, "SELECT id, email FROM pending_registrations WHERE (LOWER(ema
il) = ? OR mobile = ?) AND role = ? LIMIT 1");
    mysqli_stmt_bind_param($chkPending, "sss", $email, $email, $role);
    mysqli_stmt_execute($chkPending);
    $pendUser = mysqli_stmt_get_result($chkPending)->fetch_assoc();
    if ($pendUser) {
        json_fail('Your email is pending verification. Please check your Gmail for your verification
code and password.', 403);
    }
    json_fail('Incorrect email or password.', 401);
}

$passwordMatches = password_verify($password, $user['password']) || ($user['password'] === $password)
|| (md5($password) === $user['password']);

if (!$passwordMatches) {
    json_fail('Incorrect email or password.', 401);
}

// Re-hash to strong bcrypt if plain text or md5
if ($user['password'] === $password || md5($password) === $user['password']) {
    $newHash = password_hash($password, PASSWORD_DEFAULT);
    $upStmt = mysqli_prepare($conn, "UPDATE users SET password = ? WHERE id = ?");
    mysqli_stmt_bind_param($upStmt, "si", $newHash, $user['id']);
    mysqli_stmt_execute($upStmt);
    mysqli_stmt_close($upStmt);
}

if ($role !== 'admin' && !$user['is_verified']) {
    json_fail('Please verify your email first (check your inbox for the verification link).', 403);
}
if ($role === 'farmer' && $user['status'] === 'pending') {
```

```
    json_fail('Your account is awaiting admin approval.', 403);
}
if ($role === 'farmer' && $user['status'] === 'rejected') {
    json_fail('Your farmer application was rejected. Contact support.', 403);
}

$_SESSION['user_id'] = $user['id'];
$_SESSION['role'] = $user['role'];
$_SESSION['name'] = $user['name'];

json_ok(['user' => ['id' => $user['id'], 'name' => $user['name'], 'role' => $user['role']]]);
```

backend/api/auth/register.php (Multi-Role User Registration)

```
<?php
/**
 * POST /backend/api/auth/register.php
 * Body: { role: "buyer"|"farmer", name, mobile, email, password?, aadhaar? }
 *
 * 1. Checks duplicate email and mobile in users table.
 * 2. Checks email authenticity (RFC format, non-disposable domain, active MX).
 * 3. For buyers: auto-generates a secure login password.
 * 4. Staging: saves into pending_registrations (NOT users table).
 * 5. Sends verification email with auto-generated password and 6-digit OTP.
 */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/email_validator.php';
require_once __DIR__ . '/../../helpers/mailer.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);

$body = json_body();
$role = $body['role'] ?? '';
$name = clean($body['name'] ?? '');
$mobile = clean($body['mobile'] ?? '');
$email = clean($body['email'] ?? '');
$password = (string)($body['password'] ?? '');
$aadhaar = clean($body['aadhaar'] ?? '');

// Basic field validations
if (!in_array($role, ['buyer', 'farmer'], true)) json_fail('Invalid registration role.');
if (!preg_match('/^[A-Za-z ]+$/', $name)) json_fail('Name should contain letters only.');
if (!preg_match('/^[0-9]{10}$/', $mobile)) json_fail('Enter a valid 10-digit mobile number.');

// 1 & 2. Authenticity & Duplicate Checks (scoped by role so a user can register as both farmer and b
uyer with the same Gmail)
$emailCheck = validate_email_authenticity($email, $conn, null, $role);
if (!$emailCheck['ok']) {
    json_fail($emailCheck['error'], $emailCheck['is_duplicate'] ? 409 : 400);
}

$mobileCheck = validate_mobile_uniqueness($mobile, $conn, null, $role);
if (!$mobileCheck['ok']) {
    json_fail($mobileCheck['error'], $mobileCheck['is_duplicate'] ? 409 : 400);
}

// Aadhaar check for farmers
if ($role === 'farmer') {
    $cleanAadhaar = preg_replace('/\s+/', '', $aadhaar);
    if (strlen($cleanAadhaar) !== 12 || !ctype_digit($cleanAadhaar)) {
        json_fail('Enter a valid 12-digit Aadhaar number.');
    }
}

// 4. Auto-generate password for buyers & farmers
$autoGeneratedPassword = null;
if (!empty($password) && strlen($password) >= 6) {
    // User typed their own password
    $passwordToUse = $password;
    $autoGeneratedPassword = $password;
} else {
    // Auto-generate secure 8-character password: 2 uppercase, 3 lowercase, 2 digits, 1 special symbo
l
```

```
    $lettersU = 'ABCDEFGHJKLMNPQRSTUVWXYZ';
    $lettersL = 'abcdefghjkmnpqrstuvwxyz';
    $digits   = '23456789';
    $specials = '!@#$%&*';

    $pwdChars = [
        $lettersU[random_int(0, strlen($lettersU) - 1)],
        $lettersU[random_int(0, strlen($lettersU) - 1)],
        $lettersL[random_int(0, strlen($lettersL) - 1)],
        $lettersL[random_int(0, strlen($lettersL) - 1)],
        $lettersL[random_int(0, strlen($lettersL) - 1)],
        $digits[random_int(0, strlen($digits) - 1)],
        $digits[random_int(0, strlen($digits) - 1)],
        $specials[random_int(0, strlen($specials) - 1)],
    ];
    shuffle($pwdChars);
    $autoGeneratedPassword = implode('', $pwdChars);
    $passwordToUse = $autoGeneratedPassword;
}

$hash = password_hash($passwordToUse, PASSWORD_DEFAULT);

// Generate 32-byte verification token & 6-digit numeric OTP code
$token = bin2hex(random_bytes(24));
$otpCode = str_pad((string)random_int(100000, 999999), 6, '0', STR_PAD_LEFT);
$expiresAt = date('Y-m-d H:i:s', time() + 86400); // 24 hours

// 5. STAGING ONLY: Store in pending_registrations table (NOT users table!)
// Remove any previous pending registration for this email/mobile so user can restart cleanly
$delPending = mysqli_prepare($conn, "DELETE FROM pending_registrations WHERE LOWER(email) = LOWER(?)
OR mobile = ?");
mysqli_stmt_bind_param($delPending, "ss", $email, $mobile);
mysqli_stmt_execute($delPending);

$aadhaarVal = ($role === 'farmer') ? preg_replace('/\s+/', '', $aadhaar) : null;

$insertPending = mysqli_prepare($conn, "
    INSERT INTO pending_registrations (name, mobile, email, password_hash, auto_password, role, aadha
ar, token, otp_code, status, expires_at)
    VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, 'pending', ?)
");
mysqli_stmt_bind_param(
    $insertPending,
    "sssssssss",
    $name,
    $mobile,
    $email,
    $hash,
    $autoGeneratedPassword,
    $role,
    $aadhaarVal,
    $token,
    $otpCode,
    $expiresAt
);
mysqli_stmt_execute($insertPending);
$pendingId = mysqli_insert_id($conn);

if ($pendingId <= 0) {
    json_fail('Failed to stage registration. Please try again.', 500);
}
```

```
// 4. Send Gmail verification email with auto-generated password attached
if ($role === 'buyer') {
    send_buyer_verification_mail($email, $name, $passwordToUse, $token, $otpCode);
} else {
    send_farmer_verification_mail($email, $name, $passwordToUse, $token, $otpCode);
}

$_SESSION['pending_registration_id'] = $pendingId;
$_SESSION['pending_email'] = $email;
$_SESSION['pending_token'] = $token;

json_ok([
    'registered_pending' => true,
    'email' => $email,
    'mobile' => $mobile,
    'role' => $role,
    'auto_password_sent' => true,
    'dev_otp' => $otpCode,                      // For development convenience
    'dev_token' => $token,
    'message' => 'Verification code & login password sent to your Gmail. Please check your inbox.',
]);
```

backend/api/auth/logout.php (Session Destroyer)

```
<?php
require_once __DIR__ . '/../_bootstrap.php';
$_SESSION = [];
session_destroy();
json_ok(['loggedOut' => true]);
```

backend/api/auth/me.php (Session User Verification)

```
<?php
/** GET /backend/api/auth/me.php — returns the logged-in user, or ok:false if none. */
require_once __DIR__ . '/../_bootstrap.php';

if (!is_logged_in()) json_ok(['user' => null]);

$user = get_current_user_data($conn);
if (!$user) json_ok(['user' => null]);

json_ok(['user' => [
    'id' => $user['id'],
    'name' => $user['name'],
    'email' => $user['email'] ?? null,
    'mobile' => $user['mobile'] ?? null,
    'role' => $user['role'],
    'created_at' => $user['created_at'] ?? null,
]]);
```

backend/api/auth/verify-otp.php (Registration OTP Verification)

```
<?php
/**
 * POST /backend/api/auth/verify-otp.php
 * Body: { idToken: "<Firebase ID token>", mobile: "9876543210" }
 *
 * The Next.js client already ran the real SMS-OTP challenge with Firebase
 * Phone Auth (see app/lib/otp.ts::confirmMobileOtp) and got back a signed
 * ID token — this endpoint verifies that token server-side (so a forged
 * token can't be used to fake verification), checks its phone number
 * matches the pending registration, then marks is_verified = 1 and logs
 * the user in.
 */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../config/firebase-config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
if (!isset($_SESSION['pending_user_id'])) json_fail('No pending verification. Please register agai
n.', 400);

$body = json_body();
$idToken = trim($body['idToken'] ?? '');
$mobile = clean($body['mobile'] ?? '');

if ($idToken === '') json_fail('Missing verification token.');

$claims = verify_firebase_id_token($idToken);
if ($claims === null) json_fail('Invalid or expired OTP session. Please try again.');

// Firebase returns phone numbers as E.164 (+91XXXXXXXXXX) — compare the
// last 10 digits against what the user is registering.
$firebasePhone = $claims['phone_number'] ?? '';
$firebaseDigits = substr(preg_replace('/\D/', '', $firebasePhone), -10);
$expectedMobile = $mobile !== '' ? $mobile : ($_SESSION['pending_mobile'] ?? '');

if ($firebaseDigits === '' || $firebaseDigits !== $expectedMobile) {
    json_fail('The verified number does not match this registration.');
}

$userId = (int) $_SESSION['pending_user_id'];
$stmt = mysqli_prepare($conn, "UPDATE users SET is_verified = 1, firebase_uid = ? WHERE id = ?");
mysqli_stmt_bind_param($stmt, "si", $claims['uid'], $userId);
mysqli_stmt_execute($stmt);

$res = mysqli_prepare($conn, "SELECT id, name, role, status FROM users WHERE id = ?");
mysqli_stmt_bind_param($res, "i", $userId);
mysqli_stmt_execute($res);
$user = mysqli_stmt_get_result($res)->fetch_assoc();

unset($_SESSION['pending_user_id'], $_SESSION['pending_mobile']);

if (!$user) json_fail('User not found.', 404);

if ($user['role'] === 'farmer' && $user['status'] !== 'approved') {
    // Verified, but still needs admin approval before dashboard access.
    json_ok(['verified' => true, 'requires_approval' => true]);
}

$_SESSION['user_id'] = $user['id'];
$_SESSION['role'] = $user['role'];
```

```
$_SESSION['name'] = $user['name'];

json_ok(['verified' => true, 'requires_approval' => false, 'user' => [
    'id' => $user['id'], 'name' => $user['name'], 'role' => $user['role'],
]]);
```

backend/api/auth/verify-email.php (Email Link Token Verification)

```
<?php
/**
 * POST /backend/api/auth/verify-email.php
 * Body: { token?, otp?, email, idToken? }
 *
 * Verifies email via:
 * 1. FarmLink 1-click token (from email link)
 * 2. FarmLink 6-digit OTP code
 * 3. Firebase ID token (legacy fallback)
 *
 * UPON SUCCESSFUL VERIFICATION:
 * Moves record from pending_registrations into users table (email_verified=1, is_verified=1).
 * Only verified emails are stored in the users database.
 */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../config/firebase-config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);

$body = json_body();
$email = clean($body['email'] ?? '');
$token = trim($body['token'] ?? '');
$otp = trim($body['otp'] ?? '');
$idToken = trim($body['idToken'] ?? '');

if ($email === '') {
    json_fail('Email address is required.');
}

$pending = null;

// Case 1: Verification using FarmLink Token (from emailed link)
if ($token !== '') {
    $stmt = mysqli_prepare($conn, "
        SELECT * FROM pending_registrations
        WHERE token = ? AND LOWER(email) = LOWER(?) AND status = 'pending' AND expires_at > NOW()
        LIMIT 1
    ");
    mysqli_stmt_bind_param($stmt, "ss", $token, $email);
    mysqli_stmt_execute($stmt);
    $pending = mysqli_stmt_get_result($stmt)->fetch_assoc();
}

// Case 2: Verification using 6-Digit OTP Code
if (!$pending && $otp !== '') {
    $stmt = mysqli_prepare($conn, "
        SELECT * FROM pending_registrations
        WHERE otp_code = ? AND LOWER(email) = LOWER(?) AND status = 'pending' AND expires_at > NOW()
        LIMIT 1
    ");
    mysqli_stmt_bind_param($stmt, "ss", $otp, $email);
    mysqli_stmt_execute($stmt);
    $pending = mysqli_stmt_get_result($stmt)->fetch_assoc();
}

// Case 3: Legacy Firebase ID Token
if (!$pending && $idToken !== '') {
    $claims = verify_firebase_id_token($idToken);
    if ($claims !== null) {
```

```
        $firebaseEmail = strtolower($claims['email'] ?? '');
        if ($firebaseEmail !== '' && $firebaseEmail === strtolower($email)) {
            // Check pending registrations first
            $stmt = mysqli_prepare($conn, "
                SELECT * FROM pending_registrations
                WHERE LOWER(email) = LOWER(?) AND status = 'pending'
                LIMIT 1
            ");
            mysqli_stmt_bind_param($stmt, "s", $email);
            mysqli_stmt_execute($stmt);
            $pending = mysqli_stmt_get_result($stmt)->fetch_assoc();
        }
    }
}

// If no pending registration record matched
if (!$pending) {
    // Check if account already exists and is already verified in users
    $chkUser = mysqli_prepare($conn, "SELECT id, name, role, email_verified, status FROM users WHERE
LOWER(email) = LOWER(?) LIMIT 1");
    mysqli_stmt_bind_param($chkUser, "s", $email);
    mysqli_stmt_execute($chkUser);
    $alreadyUser = mysqli_stmt_get_result($chkUser)->fetch_assoc();

    if ($alreadyUser && $alreadyUser['email_verified']) {
        json_fail('This email is already verified. Please log in.', 409);
    }

    json_fail('Invalid or expired verification code / link. Please check your code or register agai
n.', 400);
}

// 5. ONLY VERIFIED GMAIL IS STORED TO DATABASE (users table)
$role = $pending['role'];
$name = $pending['name'];
$mobile = $pending['mobile'];
$emailVal = $pending['email'];
$hash = $pending['password_hash'];
$aadhaar = $pending['aadhaar'];

// Farmers require admin approval; buyers are auto-approved upon email verification
$userStatus = ($role === 'farmer') ? 'pending' : 'approved';

$insertUser = mysqli_prepare($conn, "
    INSERT INTO users (name, mobile, email, password, aadhaar, role, is_verified, email_verified, sta
tus)
    VALUES (?, ?, ?, ?, ?, ?, 1, 1, ?)
");
mysqli_stmt_bind_param(
    $insertUser,
    "sssssss",
    $name,
    $mobile,
    $emailVal,
    $hash,
    $aadhaar,
    $role,
    $userStatus
);

if (!mysqli_stmt_execute($insertUser)) {
    $err = mysqli_error($conn);
```

```
    if (strpos($err, 'Duplicate') !== false) {
        json_fail('This email or mobile is already registered in FarmLink. Please log in.', 409);
    }
    json_fail('Failed to activate account in database. Please try again.', 500);
}

$newUserId = mysqli_insert_id($conn);

// Clean up pending registration
$delPending = mysqli_prepare($conn, "DELETE FROM pending_registrations WHERE id = ?");
mysqli_stmt_bind_param($delPending, "i", $pending['id']);
mysqli_stmt_execute($delPending);

unset($_SESSION['pending_registration_id'], $_SESSION['pending_email'], $_SESSION['pending_token']);

// If farmer, they need admin approval before logging in
if ($role === 'farmer') {
    json_ok([
        'email_verified' => true,
        'verified' => true,
        'requires_approval' => true,
        'role' => 'farmer',
        'message' => 'Email verified successfully! Your farmer application is now submitted for admin
review.',
    ]);
}

// If buyer, log in automatically
$_SESSION['user_id'] = $newUserId;
$_SESSION['role'] = 'buyer';
$_SESSION['name'] = $name;

json_ok([
    'email_verified' => true,
    'verified' => true,
    'requires_approval' => false,
    'role' => 'buyer',
    'user' => [
        'id' => $newUserId,
        'name' => $name,
        'role' => 'buyer',
    ],
    'message' => 'Email verified successfully! Welcome to FarmLink.',
]);
```

backend/api/products/list.php (Active Catalog & Category Filter API)

```
<?php
/**
 * GET /backend/api/products/list.php?search=&category=&farmer_mine=1
 * Public listing (Shop page). Pass farmer_mine=1 while logged in as a
 * farmer to get only that farmer's own products (My Products page).
 */
require_once __DIR__ . '/../_bootstrap.php';

$search = clean($_GET['search'] ?? '');
$category = clean($_GET['category'] ?? '');
$mineOnly = ($_GET['farmer_mine'] ?? '') === '1';

$sql = "SELECT p.*, c.name AS category_name, u.name AS farmer_name
        FROM products p
        JOIN categories c ON c.id = p.category_id
        JOIN users u ON u.id = p.farmer_id
        WHERE 1=1";
$types = "";
$params = [];

if ($mineOnly) {
    require_json_role('farmer');
    $sql .= " AND p.farmer_id = ?";
    $types .= "i";
    $params[] = $_SESSION['user_id'];
} else {
    $sql .= " AND p.status = 'active' AND p.approval_status = 'approved'";
}

if ($search !== '') {
    $sql .= " AND p.name LIKE ?";
    $types .= "s";
    $params[] = "%$search%";
}
if ($category !== '' && $category !== 'All') {
    $sql .= " AND c.name = ?";
    $types .= "s";
    $params[] = $category;
}

$sql .= " ORDER BY p.created_at DESC";

$stmt = mysqli_prepare($conn, $sql);
if ($types !== "") {
    mysqli_stmt_bind_param($stmt, $types, ...$params);
}
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);

$products = [];
while ($row = mysqli_fetch_assoc($result)) {
    $products[] = [
        'id' => (int) $row['id'],
        'name' => $row['name'],
        'category' => $row['category_name'],
        'price' => (float) $row['price_per_kg'],
        'base_price' => (float) ($row['base_price'] ?? $row['price_per_kg']),
        'gst_percent' => (float) ($row['gst_percent'] ?? 5.0),
        'gst_amount' => (float) ($row['gst_amount'] ?? 0.0),
```

```
        'commission_percent' => (float) ($row['commission_percent'] ?? 5.0),
        'commission_amount' => (float) ($row['commission_amount'] ?? 0.0),
        'stock' => (float) $row['stock_kg'],
        'image' => $row['image'] ? UPLOAD_URL . $row['image'] : null,
        'description' => $row['description'],
        'status' => $row['status'],
        'approval_status' => $row['approval_status'] ?? null,
        'admin_remark' => $row['admin_remark'] ?? null,
        'farmer' => $row['farmer_name'],
    ];
}

json_ok(['products' => $products]);
```

backend/api/products/detail.php (Product Detail & GST Breakdown API)

```
<?php
/** GET /backend/api/products/detail.php?id=5 */
require_once __DIR__ . '/../_bootstrap.php';

$id = (int) ($_GET['id'] ?? 0);
if ($id <= 0) json_fail('Invalid product id.');

$stmt = mysqli_prepare($conn, "
    SELECT p.*, c.name AS category_name, u.name AS farmer_name, u.mobile AS farmer_mobile
    FROM products p
    JOIN categories c ON c.id = p.category_id
    JOIN users u ON u.id = p.farmer_id
    WHERE p.id = ?
");
mysqli_stmt_bind_param($stmt, "i", $id);
mysqli_stmt_execute($stmt);
$row = mysqli_stmt_get_result($stmt)->fetch_assoc();

if (!$row) json_fail('Product not found.', 404);

json_ok(['product' => [
    'id' => (int) $row['id'],
    'name' => $row['name'],
    'category' => $row['category_name'],
    'category_id' => (int) $row['category_id'],
    'price' => (float) $row['price_per_kg'],
    'base_price' => (float) ($row['base_price'] ?? $row['price_per_kg']),
    'gst_percent' => (float) ($row['gst_percent'] ?? 5.0),
    'gst_amount' => (float) ($row['gst_amount'] ?? 0.0),
    'commission_percent' => (float) ($row['commission_percent'] ?? 5.0),
    'commission_amount' => (float) ($row['commission_amount'] ?? 0.0),
    'stock' => (float) $row['stock_kg'],
    'image' => $row['image'] ? UPLOAD_URL . $row['image'] : null,
    'description' => $row['description'],
    'status' => $row['status'],
    'approval_status' => $row['approval_status'] ?? 'pending',
    'farmer' => $row['farmer_name'],
    'farmer_mobile' => $row['farmer_mobile'],
]]);
```

backend/api/products/create.php (Farmer Product Creation & Upload Handler)

```
<?php
/**
 * POST /backend/api/products/create.php  (multipart/form-data — send as FormData, not JSON)
 * Fields: name, category_id, price_per_kg, stock_kg, description, product_image (file)
 */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('farmer');

$name = clean($_POST['name'] ?? '');
$categoryId = (int) ($_POST['category_id'] ?? 0);
$basePrice = (float) ($_POST['base_price'] ?? ($_POST['price_per_kg'] ?? 0));
$gstPercent = isset($_POST['gst_percent']) ? (float)$_POST['gst_percent'] : 5.0;
$commissionPercent = isset($_POST['commission_percent']) ? (float)$_POST['commission_percent'] : 5.0;

$gstAmount = round($basePrice * ($gstPercent / 100), 2);
$commissionAmount = round($basePrice * ($commissionPercent / 100), 2);

// Final selling price per kg = base price + commission + gst
$price = isset($_POST['price_per_kg']) && (float)$_POST['price_per_kg'] > 0
    ? (float)$_POST['price_per_kg']
    : round($basePrice + $gstAmount + $commissionAmount, 2);

$stock = (float) ($_POST['stock_kg'] ?? 0);
$description = clean($_POST['description'] ?? '');

if ($name === '') json_fail('Product name is required.');
if ($categoryId <= 0) json_fail('Select a category.');
if ($basePrice <= 0) json_fail('Base price must be greater than 0.');
if ($price <= 0) json_fail('Selling price must be greater than 0.');
if ($stock <= 0) json_fail('Stock must be greater than 0.');

$image = upload_product_image('product_image');
if ($image === false) json_fail('Invalid image file. Use jpg, jpeg, png or webp.');

$farmerId = (int) $_SESSION['user_id'];
$stmt = mysqli_prepare($conn, "
    INSERT INTO products (farmer_id, category_id, name, base_price, price_per_kg, gst_percent, gst_am
ount, commission_percent, commission_amount, stock_kg, image, description, status, approval_status)
    VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, 'active', 'pending')
");
mysqli_stmt_bind_param(
    $stmt,
    "iisddddddddss",
    $farmerId,
    $categoryId,
    $name,
    $basePrice,
    $price,
    $gstPercent,
    $gstAmount,
    $commissionPercent,
    $commissionAmount,
    $stock,
    $image,
    $description
);
mysqli_stmt_execute($stmt);
```

```
json_ok(['product_id' => mysqli_insert_id($conn)], 201);
```

backend/api/products/update.php (Farmer Product Modification Handler)

```
<?php
/**
 * POST /backend/api/products/update.php  (multipart/form-data)
 * Fields: id, name, category_id, price_per_kg, stock_kg, description, status,
 *         product_image (file, optional — keeps old image if omitted)
 */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('farmer');

$id = (int) ($_POST['id'] ?? 0);
$farmerId = (int) $_SESSION['user_id'];

$check = mysqli_prepare($conn, "SELECT image FROM products WHERE id = ? AND farmer_id = ?");
mysqli_stmt_bind_param($check, "ii", $id, $farmerId);
mysqli_stmt_execute($check);
$existing = mysqli_stmt_get_result($check)->fetch_assoc();
if (!$existing) json_fail('Product not found.', 404);

$name = clean($_POST['name'] ?? '');
$categoryId = (int) ($_POST['category_id'] ?? 0);
$basePrice = (float) ($_POST['base_price'] ?? ($_POST['price_per_kg'] ?? 0));
$gstPercent = isset($_POST['gst_percent']) ? (float)$_POST['gst_percent'] : 5.0;
$commissionPercent = isset($_POST['commission_percent']) ? (float)$_POST['commission_percent'] : 5.0;

$gstAmount = round($basePrice * ($gstPercent / 100), 2);
$commissionAmount = round($basePrice * ($commissionPercent / 100), 2);

$price = isset($_POST['price_per_kg']) && (float)$_POST['price_per_kg'] > 0
    ? (float)$_POST['price_per_kg']
    : round($basePrice + $gstAmount + $commissionAmount, 2);

$stock = (float) ($_POST['stock_kg'] ?? 0);
$description = clean($_POST['description'] ?? '');
$status = in_array($_POST['status'] ?? '', ['active', 'inactive'], true) ? $_POST['status'] : 'activ
e';

$image = $existing['image'];
if (isset($_FILES['product_image']) && $_FILES['product_image']['error'] === 0) {
    $uploaded = upload_product_image('product_image');
    if ($uploaded === false) json_fail('Invalid image file.');
    $image = $uploaded;
}

$stmt = mysqli_prepare($conn, "
    UPDATE products SET name=?, category_id=?, base_price=?, price_per_kg=?, gst_percent=?, gst_amoun
t=?, commission_percent=?, commission_amount=?, stock_kg=?, image=?, description=?, status=?, approva
l_status='pending'
    WHERE id = ? AND farmer_id = ?
");
mysqli_stmt_bind_param(
    $stmt,
    "siddddddsssii",
    $name,
    $categoryId,
    $basePrice,
    $price,
    $gstPercent,
```

```
    $gstAmount,
    $commissionPercent,
    $commissionAmount,
    $stock,
    $image,
    $description,
    $status,
    $id,
    $farmerId
);
mysqli_stmt_execute($stmt);

json_ok(['updated' => true]);
```

backend/api/products/refill.php (Farmer Inventory Stock Refill API)

```
<?php
/**
 * POST /backend/api/products/refill.php
 * Quick stock refill / restock for farmers.
 * Payload: { id: int, refill_kg: float, action?: 'add' | 'set', note?: string }
 */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('farmer');

$body = json_decode(file_get_contents('php://input'), true) ?? $_POST;

$id = (int) ($body['id'] ?? 0);
$refillKg = (float) ($body['refill_kg'] ?? 0);
$action = ($body['action'] ?? 'add') === 'set' ? 'set' : 'add';
$farmerId = (int) $_SESSION['user_id'];

if ($id <= 0) {
    json_fail('Invalid product ID.', 422);
}

if ($refillKg <= 0 && $action === 'add') {
    json_fail('Please enter a valid refill quantity greater than 0 kg.', 422);
}

if ($refillKg < 0 && $action === 'set') {
    json_fail('Stock cannot be negative.', 422);
}

// Check product exists and belongs to this farmer
$check = mysqli_prepare($conn, "SELECT id, name, stock_kg, status, approval_status FROM products WHER
E id = ? AND farmer_id = ?");
mysqli_stmt_bind_param($check, "ii", $id, $farmerId);
mysqli_stmt_execute($check);
$product = mysqli_stmt_get_result($check)->fetch_assoc();

if (!$product) {
    json_fail('Product not found or access denied.', 404);
}

$oldStock = (float) $product['stock_kg'];
$newStock = ($action === 'set') ? $refillKg : round($oldStock + $refillKg, 2);

// If stock is added and product was inactive or out of stock, activate it
$newStatus = ($newStock > 0) ? 'active' : $product['status'];

$update = mysqli_prepare($conn, "UPDATE products SET stock_kg = ?, status = ? WHERE id = ? AND farmer
_id = ?");
mysqli_stmt_bind_param($update, "dsii", $newStock, $newStatus, $id, $farmerId);
mysqli_stmt_execute($update);

json_ok([
    'refilled' => true,
    'product_id' => $id,
    'product_name' => $product['name'],
    'old_stock' => $oldStock,
    'refill_amount' => ($action === 'set') ? ($newStock - $oldStock) : $refillKg,
    'new_stock' => $newStock,
```

```
    'status' => $newStatus,
    'message' => "Successfully refilled {$product['name']}! New stock is {$newStock} kg.",
]);
```

backend/api/products/delete.php (Farmer Product Soft Deletion API)

```
<?php
/** POST /backend/api/products/delete.php   Body: { id } */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('farmer');

$body = json_body();
$id = (int) ($body['id'] ?? 0);
$farmerId = (int) $_SESSION['user_id'];

$stmt = mysqli_prepare($conn, "DELETE FROM products WHERE id = ? AND farmer_id = ?");
mysqli_stmt_bind_param($stmt, "ii", $id, $farmerId);
mysqli_stmt_execute($stmt);

if (mysqli_stmt_affected_rows($stmt) === 0) json_fail('Product not found.', 404);

json_ok(['deleted' => true]);
```

backend/api/products/categories.php (Category Master List API)

```
<?php
/** GET /backend/api/products/categories.php */
require_once __DIR__ . '/../_bootstrap.php';

$result = mysqli_query($conn, "SELECT DISTINCT id, name FROM categories GROUP BY name ORDER BY name A
SC");
$categories = [];
while ($row = mysqli_fetch_assoc($result)) {
    $categories[] = [
        'id' => (int) $row['id'],
        'name' => $row['name']
    ];
}
json_ok(['categories' => $categories]);
```

backend/api/orders/place.php (Buyer Order Placement & Inventory Deduction)

```
<?php
/**
 * POST /backend/api/orders/place.php
 * Body: { address, payment_mode: "cod"|"online", items: [{ product_id, qty }] }
 * Validates stock, creates the order + order_items, and decrements stock —
 * all in one transaction (mirrors backend/buyer/place-order-process.php).
 */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('buyer');

$body = json_body();
$address = clean($body['address'] ?? '');
$paymentMode = in_array($body['payment_mode'] ?? '', ['cod', 'online'], true) ? $body['payment_mode']
: 'cod';
$items = is_array($body['items'] ?? null) ? $body['items'] : [];

if ($address === '') json_fail('Delivery address is required.');
if (count($items) === 0) json_fail('Cart is empty.');

$buyerId = (int) $_SESSION['user_id'];

mysqli_begin_transaction($conn);
try {
    $total = 0;
    $lineItems = [];

    foreach ($items as $item) {
        $productId = (int) ($item['product_id'] ?? 0);
        $qty = (float) ($item['qty'] ?? 0);
        if ($productId <= 0 || $qty <= 0) throw new Exception('Invalid cart item.');
        if ($qty > 10) throw new Exception('Orders above 10kg require a bulk request approval firs
t.');

        $stmt = mysqli_prepare($conn, "SELECT id, farmer_id, price_per_kg, stock_kg FROM products WHE
RE id = ? FOR UPDATE");
        mysqli_stmt_bind_param($stmt, "i", $productId);
        mysqli_stmt_execute($stmt);
        $product = mysqli_stmt_get_result($stmt)->fetch_assoc();
        if (!$product) throw new Exception("Product #$productId not found.");
        if ($product['stock_kg'] < $qty) throw new Exception('Not enough stock for one of the item
s.');

        $subtotal = $qty * $product['price_per_kg'];
        $total += $subtotal;
        $lineItems[] = [
            'product_id' => $productId,
            'farmer_id' => $product['farmer_id'],
            'qty' => $qty,
            'price' => $product['price_per_kg'],
            'subtotal' => $subtotal,
        ];
    }

    $orderStmt = mysqli_prepare($conn, "INSERT INTO orders (buyer_id, total_amount, payment_mode, add
ress) VALUES (?, ?, ?, ?)");
    mysqli_stmt_bind_param($orderStmt, "idss", $buyerId, $total, $paymentMode, $address);
    mysqli_stmt_execute($orderStmt);
```

```
    $orderId = mysqli_insert_id($conn);

    foreach ($lineItems as $li) {
        $itemStmt = mysqli_prepare($conn, "
            INSERT INTO order_items (order_id, product_id, farmer_id, quantity_kg, price_per_kg, subt
otal)
            VALUES (?, ?, ?, ?, ?, ?)
        ");
        mysqli_stmt_bind_param($itemStmt, "iiiddd", $orderId, $li['product_id'], $li['farmer_id'], $l
i['qty'], $li['price'], $li['subtotal']);
        mysqli_stmt_execute($itemStmt);

        $stockStmt = mysqli_prepare($conn, "UPDATE products SET stock_kg = stock_kg - ? WHERE id =
?");
        mysqli_stmt_bind_param($stockStmt, "di", $li['qty'], $li['product_id']);
        mysqli_stmt_execute($stockStmt);
    }

    mysqli_commit($conn);
    json_ok(['order_id' => $orderId, 'total' => $total], 201);
} catch (Exception $e) {
    mysqli_rollback($conn);
    json_fail($e->getMessage());
}
```

backend/api/orders/mine.php (Buyer Historical Orders Query API)

```
<?php
/** GET /backend/api/orders/mine.php — the logged-in buyer's orders */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('buyer');

$buyerId = (int) $_SESSION['user_id'];

$ordersRes = mysqli_prepare($conn, "SELECT * FROM orders WHERE buyer_id = ? ORDER BY created_at DES
C");
mysqli_stmt_bind_param($ordersRes, "i", $buyerId);
mysqli_stmt_execute($ordersRes);
$orders = mysqli_stmt_get_result($ordersRes)->fetch_all(MYSQLI_ASSOC);

$out = [];
foreach ($orders as $o) {
    $itemsStmt = mysqli_prepare($conn, "
        SELECT oi.*, p.name AS product_name FROM order_items oi
        JOIN products p ON p.id = oi.product_id
        WHERE oi.order_id = ?
    ");
    mysqli_stmt_bind_param($itemsStmt, "i", $o['id']);
    mysqli_stmt_execute($itemsStmt);
    $items = mysqli_stmt_get_result($itemsStmt)->fetch_all(MYSQLI_ASSOC);

    $out[] = [
        'id' => (int) $o['id'],
        'date' => $o['created_at'],
        'total' => (float) $o['total_amount'],
        'payment_mode' => $o['payment_mode'],
        'status' => $o['status'],
        'tracking_note' => $o['tracking_note'] ?? '',
        'address' => $o['address'],
        'items' => array_map(fn($i) => [
            'product_id' => (int) $i['product_id'],
            'name' => $i['product_name'],
            'qty' => (float) $i['quantity_kg'],
            'price' => (float) $i['price_per_kg'],
            'subtotal' => (float) $i['subtotal'],
            'status' => $i['item_status'],
        ], $items),
    ];
}

json_ok(['orders' => $out]);
```

backend/api/orders/farmer-items.php (Farmer Incoming Orders Query API)

```
<?php
/** GET /backend/api/orders/farmer-items.php — order items sold by the logged-in farmer */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('farmer');

$farmerId = (int) $_SESSION['user_id'];

$stmt = mysqli_prepare($conn, "
    SELECT oi.*, o.created_at AS order_date, o.address, o.payment_mode, u.name AS buyer_name, u.mobil
e AS buyer_mobile,
           p.id AS product_id, p.name AS product_name, p.stock_kg AS current_stock, p.image AS produc
t_image
    FROM order_items oi
    JOIN orders o ON o.id = oi.order_id
    JOIN users u ON u.id = o.buyer_id
    JOIN products p ON p.id = oi.product_id
    WHERE oi.farmer_id = ?
    ORDER BY o.created_at DESC
");
mysqli_stmt_bind_param($stmt, "i", $farmerId);
mysqli_stmt_execute($stmt);
$rows = mysqli_stmt_get_result($stmt)->fetch_all(MYSQLI_ASSOC);

$items = array_map(fn($r) => [
    'item_id' => (int) $r['id'],
    'order_id' => (int) $r['order_id'],
    'order_date' => $r['order_date'],
    'product_id' => (int) $r['product_id'],
    'product_name' => $r['product_name'],
    'product_image' => $r['product_image'] ? UPLOAD_URL . $r['product_image'] : null,
    'current_stock' => (float) $r['current_stock'],
    'buyer_name' => $r['buyer_name'],
    'buyer_mobile' => $r['buyer_mobile'],
    'address' => $r['address'],
    'payment_mode' => $r['payment_mode'] ?? 'cod',
    'qty' => (float) $r['quantity_kg'],
    'price_per_kg' => (float) $r['price_per_kg'],
    'subtotal' => (float) $r['subtotal'],
    'status' => $r['item_status'],
], $rows);

json_ok(['items' => $items]);
```

backend/api/orders/update-item-status.php (Order Fulfillment Status Updater)

```
<?php
/** POST /backend/api/orders/update-item-status.php   Body: { item_id, status } */
require_once __DIR__ . '/../_bootstrap.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('farmer');

$body = json_body();
$itemId = (int) ($body['item_id'] ?? 0);
$status = $body['status'] ?? '';
$farmerId = (int) $_SESSION['user_id'];

$allowed = ['pending', 'confirmed', 'shipped', 'delivered', 'cancelled'];
if (!in_array($status, $allowed, true)) json_fail('Invalid status.');

$stmt = mysqli_prepare($conn, "UPDATE order_items SET item_status = ? WHERE id = ? AND farmer_id =
?");
mysqli_stmt_bind_param($stmt, "sii", $status, $itemId, $farmerId);
mysqli_stmt_execute($stmt);

if (mysqli_stmt_affected_rows($stmt) === 0) json_fail('Order item not found.', 404);

json_ok(['updated' => true]);
```

backend/api/farmer/dashboard-stats.php (Farmer Summary Metrics & KPI API)

```
<?php
/** GET /backend/api/farmer/dashboard-stats.php */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('farmer');

$farmerId = (int) $_SESSION['user_id'];

$q = fn($sql) => mysqli_fetch_assoc(mysqli_query($GLOBALS['conn'], $sql));

$productCount = $q("SELECT COUNT(*) c FROM products WHERE farmer_id = $farmerId")['c'];
$earnings = $q("SELECT COALESCE(SUM(subtotal),0) t FROM order_items WHERE farmer_id = $farmerId AND i
tem_status = 'delivered'")['t'];
$pending = $q("SELECT COUNT(*) c FROM order_items WHERE farmer_id = $farmerId AND item_status IN ('pe
nding','confirmed','shipped')")['c'];

$lowStockRes = mysqli_query($conn, "SELECT name, stock_kg FROM products WHERE farmer_id = $farmerId A
ND stock_kg < 5 ORDER BY stock_kg ASC");
$lowStock = [];
while ($row = mysqli_fetch_assoc($lowStockRes)) {
    $lowStock[] = ['name' => $row['name'], 'stock' => (float) $row['stock_kg']];
}

json_ok([
    'product_count' => (int) $productCount,
    'total_earnings' => (float) $earnings,
    'pending_orders' => (int) $pending,
    'low_stock' => $lowStock,
]);
```

backend/api/farmer/earnings.php (Farmer Date-Filtered Earnings API)

```
<?php
/** GET /backend/api/farmer/earnings.php */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('farmer');

$farmerId = (int) $_SESSION['user_id'];

// ── Date range filter ──────────────────────────────────────────
$fromDate = $_GET['from'] ?? null;
$toDate   = $_GET['to']   ?? null;

function valid_date(?string $d): bool {
    if (!$d) return false;
    $dt = DateTime::createFromFormat('Y-m-d', $d);
    return $dt && $dt->format('Y-m-d') === $d;
}

$hasRange  = valid_date($fromDate) && valid_date($toDate);
$dateWhere = $hasRange
    ? "AND DATE(o.created_at) BETWEEN '" . mysqli_real_escape_string($conn, $fromDate) . "' AND '" .
mysqli_real_escape_string($conn, $toDate) . "'"
    : "";

// ── Totals ─────────────────────────────────────────────────────
$total = (float) mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COALESCE(SUM(oi.subtotal),0) t
     FROM order_items oi
     JOIN orders o ON o.id = oi.order_id
     WHERE oi.farmer_id = $farmerId AND oi.item_status = 'delivered' $dateWhere"
))['t'];

$pendingAmount = (float) mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COALESCE(SUM(oi.subtotal),0) t
     FROM order_items oi
     JOIN orders o ON o.id = oi.order_id
     WHERE oi.farmer_id = $farmerId AND oi.item_status IN ('pending','confirmed','shipped') $dateWher
e"
))['t'];

// Today's earnings (always today regardless of filter)
$todayEarned = (float) mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COALESCE(SUM(oi.subtotal),0) t
     FROM order_items oi
     JOIN orders o ON o.id = oi.order_id
     WHERE oi.farmer_id = $farmerId AND oi.item_status = 'delivered'
       AND DATE(o.created_at) = CURDATE()"
))['t'];

// Order count
$orderCount = (int) mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT COUNT(DISTINCT oi.order_id) c
     FROM order_items oi
     JOIN orders o ON o.id = oi.order_id
     WHERE oi.farmer_id = $farmerId AND oi.item_status = 'delivered' $dateWhere"
))['c'];

// ── History ────────────────────────────────────────────────────
$historyRes = mysqli_query($conn,
    "SELECT oi.id, oi.quantity_kg, oi.subtotal, oi.item_status,
```

```
            o.created_at AS order_date, p.name AS product_name
     FROM order_items oi
     JOIN orders o ON o.id = oi.order_id
     JOIN products p ON p.id = oi.product_id
     WHERE oi.farmer_id = $farmerId AND oi.item_status = 'delivered' $dateWhere
     ORDER BY o.created_at DESC"
);
$history = [];
while ($row = mysqli_fetch_assoc($historyRes)) {
    $history[] = [
        'date'         => $row['order_date'],
        'product_name' => $row['product_name'],
        'qty'          => (float) $row['quantity_kg'],
        'amount'       => (float) $row['subtotal'],
        'status'       => $row['item_status'],
    ];
}

json_ok([
    'total_earned'   => $total,
    'pending_amount' => $pendingAmount,
    'today_earned'   => $todayEarned,
    'order_count'    => $orderCount,
    'history'        => $history,
    'filter_from'    => $hasRange ? $fromDate : null,
    'filter_to'      => $hasRange ? $toDate   : null,
]);
```

backend/api/admin/stats.php (Admin System-Wide Analytical Metrics API)

```
<?php
/** GET /backend/api/admin/stats.php */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$conn = $GLOBALS['conn'];

// ── Date range filter ──────────────────────────────────────────
$fromDate = $_GET['from'] ?? null;   // YYYY-MM-DD
$toDate   = $_GET['to']   ?? null;   // YYYY-MM-DD

// Validate format
function valid_date(?string $d): bool {
    if (!$d) return false;
    $dt = DateTime::createFromFormat('Y-m-d', $d);
    return $dt && $dt->format('Y-m-d') === $d;
}

$hasRange  = valid_date($fromDate) && valid_date($toDate);
$dateWhere = $hasRange
    ? "AND DATE(created_at) BETWEEN '" . mysqli_real_escape_string($conn, $fromDate) . "' AND '" . my
sqli_real_escape_string($conn, $toDate) . "'"
    : "";

// ── Helpers ────────────────────────────────────────────────────
$q = function($sql) use ($conn) {
    $res = mysqli_query($conn, $sql);
    if (!$res) return 0;
    $row = mysqli_fetch_assoc($res);
    return (int) ($row['c'] ?? 0);
};

$sum = function($sql) use ($conn) {
    $res = mysqli_query($conn, $sql);
    if (!$res) return 0.0;
    $row = mysqli_fetch_assoc($res);
    return (float) ($row['s'] ?? 0.0);
};

// ── Financial — respect date filter ────────────────────────────
$totalIncome = $sum("SELECT COALESCE(SUM(total_amount), 0) s FROM orders WHERE status != 'cancelled'
$dateWhere");
$commissionRate  = 0.05;
$platformEarnings = round($totalIncome * $commissionRate, 2);

$todayIncome = $sum("SELECT COALESCE(SUM(total_amount), 0) s FROM orders WHERE status != 'cancelled'
AND DATE(created_at) = CURDATE()");

$totalOrders     = $q("SELECT COUNT(*) c FROM orders WHERE 1=1 $dateWhere");
$completedOrders = $q("SELECT COUNT(*) c FROM orders WHERE (status = 'delivered' OR status = 'complet
ed') $dateWhere");

// ── Platform stats (not date-filtered — they are totals) ───────
$sharedGmailRes = mysqli_query($conn, "
    SELECT COUNT(DISTINCT CONCAT(f.id, '_', b.id)) AS c
    FROM users f
    JOIN users b ON (
        (LOWER(f.email) = LOWER(b.email) AND f.email IS NOT NULL AND f.email != '')
        OR
```

```
        (f.mobile = b.mobile AND f.mobile IS NOT NULL AND f.mobile != '')
    )
    WHERE f.role = 'farmer' AND b.role = 'buyer'
");
$sharedGmailCount = ($sharedGmailRes && $row = mysqli_fetch_assoc($sharedGmailRes)) ? (int)$row['c']
: 0;

json_ok([
    'pending_farmers'    => $q("SELECT COUNT(*) c FROM users WHERE role='farmer' AND status='pendin
g'"),
    'pending_products'   => $q("SELECT COUNT(*) c FROM products WHERE approval_status='pending'"),
    'pending_bulk'       => $q("SELECT COUNT(*) c FROM bulk_requests WHERE status='pending'"),
    'total_farmers'      => $q("SELECT COUNT(*) c FROM users WHERE role='farmer'"),
    'total_buyers'       => $q("SELECT COUNT(*) c FROM users WHERE role='buyer'"),
    'shared_gmail_count' => $sharedGmailCount,
    'total_products'     => $q("SELECT COUNT(*) c FROM products"),
    'approved_products'  => $q("SELECT COUNT(*) c FROM products WHERE approval_status='approved'"),
    'total_income'       => $totalIncome,
    'platform_earnings'  => $platformEarnings,
    'today_income'       => $todayIncome,
    'total_orders'       => $totalOrders,
    'completed_orders'   => $completedOrders,
    'filter_from'        => $hasRange ? $fromDate : null,
    'filter_to'          => $hasRange ? $toDate   : null,
]);
```

backend/api/admin/shared-emails.php (Dual-Role Account Overlap Auditor API)

```
<?php
/**
 * GET /backend/api/admin/shared-emails.php
 * Returns a list of all accounts that share the SAME Gmail / Email OR same Mobile number across Farm
er and Buyer roles.
 */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

// Helper to fetch rich account details for a user
function get_role_details(mysqli $conn, int $userId, string $role): ?array {
    if ($role === 'farmer') {
        $stmt = mysqli_prepare($conn, "
            SELECT u.id, u.name, u.mobile, u.email, u.status, u.aadhaar, u.admin_remark, u.created_a
t,
                   COUNT(DISTINCT p.id) AS total_products,
                   COALESCE(SUM(oi.subtotal), 0) AS total_sales
            FROM users u
            LEFT JOIN products p ON p.farmer_id = u.id
            LEFT JOIN order_items oi ON oi.farmer_id = u.id
            WHERE u.id = ?
            GROUP BY u.id
            LIMIT 1
        ");
        mysqli_stmt_bind_param($stmt, "i", $userId);
        mysqli_stmt_execute($stmt);
        $res = mysqli_stmt_get_result($stmt)->fetch_assoc();
        if (!$res) return null;
        return [
            'id' => (int)$res['id'],
            'name' => $res['name'],
            'mobile' => $res['mobile'],
            'email' => $res['email'],
            'status' => $res['status'],
            'aadhaar' => $res['aadhaar'],
            'remark' => $res['admin_remark'],
            'joined' => $res['created_at'],
            'total_products' => (int)$res['total_products'],
            'total_sales' => (float)$res['total_sales'],
        ];
    } else {
        $stmt = mysqli_prepare($conn, "
            SELECT u.id, u.name, u.mobile, u.email, u.status, u.created_at,
                   COUNT(DISTINCT o.id) AS total_orders,
                   COALESCE(SUM(o.total_amount), 0) AS total_spent
            FROM users u
            LEFT JOIN orders o ON o.buyer_id = u.id
            WHERE u.id = ?
            GROUP BY u.id
            LIMIT 1
        ");
        mysqli_stmt_bind_param($stmt, "i", $userId);
        mysqli_stmt_execute($stmt);
        $res = mysqli_stmt_get_result($stmt)->fetch_assoc();
        if (!$res) return null;
        return [
            'id' => (int)$res['id'],
            'name' => $res['name'],
            'mobile' => $res['mobile'],
```

```
            'email' => $res['email'],
            'status' => $res['status'],
            'joined' => $res['created_at'],
            'total_orders' => (int)$res['total_orders'],
            'total_spent' => (float)$res['total_spent'],
        ];
    }
}

// Find all matches between a farmer and a buyer where LOWER(email) matches OR mobile matches
$sql = "
    SELECT f.id AS farmer_id, b.id AS buyer_id,
           f.email AS f_email, b.email AS b_email,
           f.mobile AS f_mobile, b.mobile AS b_mobile
    FROM users f
    JOIN users b ON (
        (LOWER(f.email) = LOWER(b.email) AND f.email IS NOT NULL AND f.email != '')
        OR
        (f.mobile = b.mobile AND f.mobile IS NOT NULL AND f.mobile != '')
    )
    WHERE f.role = 'farmer' AND b.role = 'buyer'
    ORDER BY LEAST(f.created_at, b.created_at) DESC
";

$res = mysqli_query($conn, $sql);
$sharedList = [];
$seenPairs = [];

if ($res) {
    while ($row = mysqli_fetch_assoc($res)) {
        $farmerId = (int)$row['farmer_id'];
        $buyerId = (int)$row['buyer_id'];
        $pairKey = $farmerId . '_' . $buyerId;
        if (isset($seenPairs[$pairKey])) continue;
        $seenPairs[$pairKey] = true;

        $emailMatch = (strtolower($row['f_email'] ?? '') === strtolower($row['b_email'] ?? '')) && !e
mpty($row['f_email']);
        $mobileMatch = (($row['f_mobile'] ?? '') === ($row['b_mobile'] ?? '')) && !empty($row['f_mobi
le']);

        $sharedBy = 'email';
        if ($emailMatch && $mobileMatch) {
            $sharedBy = 'both';
        } elseif ($mobileMatch) {
            $sharedBy = 'mobile';
        }

        $farmerData = get_role_details($conn, $farmerId, 'farmer');
        $buyerData = get_role_details($conn, $buyerId, 'buyer');

        if ($farmerData && $buyerData) {
            $sharedList[] = [
                'email' => $farmerData['email'] ?: $buyerData['email'] ?: 'No Email',
                'mobile' => $farmerData['mobile'] ?: $buyerData['mobile'],
                'shared_by' => $sharedBy,
                'farmer' => $farmerData,
                'buyer' => $buyerData
            ];
        }
    }
}
```

```
json_ok([
    'total_shared' => count($sharedList),
    'shared_accounts' => $sharedList
]);
```

backend/api/admin/farmers.php (Admin Farmer Account Management API)

```
<?php
/** GET /backend/api/admin/farmers.php?status=pending|approved|rejected|all */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$status = clean($_GET['status'] ?? 'pending');
$allowed = ['pending', 'approved', 'rejected', 'all'];
if (!in_array($status, $allowed, true)) $status = 'pending';

$sql = "SELECT u.id, u.name, u.mobile, u.email, u.aadhaar, u.status, u.admin_remark, u.created_at,
               COUNT(DISTINCT p.id) AS total_products,
               COALESCE(SUM(oi.subtotal), 0) AS total_sales
        FROM users u
        LEFT JOIN products p ON p.farmer_id = u.id
        LEFT JOIN order_items oi ON oi.farmer_id = u.id
        WHERE u.role = 'farmer'";
if ($status !== 'all') {
    $sql .= " AND u.status = '" . mysqli_real_escape_string($conn, $status) . "'";
}
$sql .= " GROUP BY u.id ORDER BY u.created_at DESC";

$res = mysqli_query($conn, $sql);
$farmers = [];
while ($row = mysqli_fetch_assoc($res)) {
    $farmers[] = [
        'id' => (int) $row['id'],
        'name' => $row['name'],
        'mobile' => $row['mobile'],
        'email' => $row['email'],
        'aadhaar' => $row['aadhaar'],
        'status' => $row['status'],
        'remark' => $row['admin_remark'],
        'joined' => $row['created_at'],
        'total_products' => (int) $row['total_products'],
        'total_sales' => (float) $row['total_sales'],
    ];
}

json_ok(['farmers' => $farmers]);
```

backend/api/admin/farmer-action.php (Farmer Approval/Suspension API)

```
<?php
/** POST /backend/api/admin/farmer-action.php   Body: { id, action: "approve"|"reject", remark? } */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/mailer.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('admin');

$body = json_body();
$id = (int) ($body['id'] ?? 0);
$action = $body['action'] ?? '';
$remark = clean($body['remark'] ?? '');

if ($id <= 0) json_fail('Invalid farmer id.');
if (!in_array($action, ['approve', 'reject'], true)) json_fail('Invalid action.');

$stmt = mysqli_prepare($conn, "SELECT id, name, email, mobile, role FROM users WHERE id = ? AND role
= 'farmer'");
mysqli_stmt_bind_param($stmt, "i", $id);
mysqli_stmt_execute($stmt);
$farmer = mysqli_stmt_get_result($stmt)->fetch_assoc();
if (!$farmer) json_fail('Farmer not found.', 404);

$newStatus = $action === 'approve' ? 'approved' : 'rejected';

$update = mysqli_prepare($conn, "UPDATE users SET status = ?, admin_remark = ? WHERE id = ?");
$remarkOrNull = $remark !== '' ? $remark : null;
mysqli_stmt_bind_param($update, "ssi", $newStatus, $remarkOrNull, $id);
mysqli_stmt_execute($update);

// Send notifications to BOTH Farmer's Gmail and Admin's Gmail
send_farmer_decision_notifications($farmer, $action, $remark);

json_ok(['updated' => true, 'status' => $newStatus]);
```

backend/api/admin/products.php (Admin Product Catalog Management API)

```
<?php
/** GET /backend/api/admin/products.php?status=pending|approved|rejected|all */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$status = clean($_GET['status'] ?? 'pending');
$allowed = ['pending', 'approved', 'rejected', 'all'];
if (!in_array($status, $allowed, true)) $status = 'pending';

$sql = "SELECT p.id, p.name, p.base_price, p.price_per_kg, p.gst_percent, p.gst_amount, p.commission_
percent, p.commission_amount, p.stock_kg, p.image, p.description,
               p.status, p.approval_status, p.admin_remark, p.created_at,
               c.name AS category_name,
               u.id AS farmer_id, u.name AS farmer_name, u.email AS farmer_email, u.mobile AS farmer_
mobile
        FROM products p
        JOIN categories c ON c.id = p.category_id
        JOIN users u ON u.id = p.farmer_id
        WHERE 1=1";
if ($status !== 'all') {
    $sql .= " AND p.approval_status = '" . mysqli_real_escape_string($conn, $status) . "'";
}
$sql .= " ORDER BY p.created_at DESC";

$res = mysqli_query($conn, $sql);
$products = [];
while ($row = mysqli_fetch_assoc($res)) {
    $products[] = [
        'id' => (int) $row['id'],
        'name' => $row['name'],
        'category' => $row['category_name'],
        'base_price' => (float) ($row['base_price'] ?? $row['price_per_kg']),
        'price' => (float) $row['price_per_kg'],
        'gst_percent' => (float) ($row['gst_percent'] ?? 5.0),
        'gst_amount' => (float) ($row['gst_amount'] ?? 0.0),
        'commission_percent' => (float) ($row['commission_percent'] ?? 5.0),
        'commission_amount' => (float) ($row['commission_amount'] ?? 0.0),
        'stock' => (float) $row['stock_kg'],
        'image' => $row['image'] ? UPLOAD_URL . $row['image'] : null,
        'description' => $row['description'],
        'status' => $row['status'],
        'approval_status' => $row['approval_status'],
        'remark' => $row['admin_remark'],
        'farmer_id' => (int) $row['farmer_id'],
        'farmer_name' => $row['farmer_name'],
        'farmer_email' => $row['farmer_email'],
        'farmer_mobile' => $row['farmer_mobile'],
        'created_at' => $row['created_at'],
    ];
}

json_ok(['products' => $products]);
```

backend/api/admin/product-action.php (Product Moderation API)

```
<?php
/** POST /backend/api/admin/product-action.php   Body: { id, action: "approve"|"reject", remark? } */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/mailer.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('admin');

$body = json_body();
$id = (int) ($body['id'] ?? 0);
$action = $body['action'] ?? '';
$remark = clean($body['remark'] ?? '');

if ($id <= 0) json_fail('Invalid product id.');
if (!in_array($action, ['approve', 'reject'], true)) json_fail('Invalid action.');

$stmt = mysqli_prepare($conn, "
    SELECT p.id, p.name, u.id AS farmer_id, u.name AS farmer_name, u.email AS farmer_email
    FROM products p JOIN users u ON u.id = p.farmer_id
    WHERE p.id = ?
");
mysqli_stmt_bind_param($stmt, "i", $id);
mysqli_stmt_execute($stmt);
$product = mysqli_stmt_get_result($stmt)->fetch_assoc();
if (!$product) json_fail('Product not found.', 404);

$newStatus = $action === 'approve' ? 'approved' : 'rejected';

$update = mysqli_prepare($conn, "UPDATE products SET approval_status = ?, admin_remark = ? WHERE id =
?");
$remarkOrNull = $remark !== '' ? $remark : null;
mysqli_stmt_bind_param($update, "ssi", $newStatus, $remarkOrNull, $id);
mysqli_stmt_execute($update);

if ($product['farmer_email']) {
    if ($action === 'approve') {
        $subject = 'Your product "' . $product['name'] . '" is now live on FarmLink ✅';
        $body_html = "<p>Hi {$product['farmer_name']},</p><p>Your product <strong>" . htmlspecialchar
s($product['name']) . "</strong> has been <strong>approved</strong> and is now visible to buyers in t
he Shop.</p>";
    } else {
        $subject = 'Your product "' . $product['name'] . '" was not approved';
        $body_html = "<p>Hi {$product['farmer_name']},</p><p>Your product <strong>" . htmlspecialchar
s($product['name']) . "</strong> was <strong>not approved</strong> and will not be shown in the Shop.
</p>"
            . ($remark !== '' ? "<p><strong>Reason:</strong> " . htmlspecialchars($remark) . "</p>" :
"")
            . "<p>You can edit the product and it will be sent for review again.</p>";
    }
    send_mail($product['farmer_email'], $product['farmer_name'], $subject, mail_template($action ===
'approve' ? 'Product approved' : 'Product update', $body_html));
}

json_ok(['updated' => true, 'approval_status' => $newStatus]);
```

backend/api/admin/orders.php (Admin Orders Audit & Tracking API)

```
<?php
/** GET /backend/api/admin/orders.php */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$sql = "SELECT o.id, o.buyer_id, o.total_amount, o.payment_mode, o.address, o.status, o.tracking_not
e, o.created_at,
               u.name AS buyer_name, u.mobile AS buyer_mobile, u.email AS buyer_email
        FROM orders o
        JOIN users u ON u.id = o.buyer_id
        ORDER BY o.created_at DESC";

$res = mysqli_query($conn, $sql);
$orders = [];
$totalRevenue = 0.0;

while ($row = mysqli_fetch_assoc($res)) {
    $amount = (float) $row['total_amount'];
    if ($row['status'] !== 'cancelled') {
        $totalRevenue += $amount;
    }
    $orderId = (int) $row['id'];

    // Fetch line items for this order including farmer name and current stock
    $itemSql = "SELECT oi.id, oi.product_id, oi.quantity_kg, oi.price_per_kg, oi.subtotal, oi.item_st
atus,
                       p.name AS product_name, p.stock_kg AS current_stock, p.image AS product_image,
                       f.name AS farmer_name, f.mobile AS farmer_mobile
                FROM order_items oi
                JOIN products p ON p.id = oi.product_id
                JOIN users f ON f.id = oi.farmer_id
                WHERE oi.order_id = $orderId";
    $itemRes = mysqli_query($conn, $itemSql);
    $items = [];
    while ($ir = mysqli_fetch_assoc($itemRes)) {
        $items[] = [
            'id' => (int) $ir['id'],
            'product_id' => (int) $ir['product_id'],
            'product_name' => $ir['product_name'],
            'product_image' => $ir['product_image'] ? UPLOAD_URL . $ir['product_image'] : null,
            'qty' => (float) $ir['quantity_kg'],
            'price_per_kg' => (float) $ir['price_per_kg'],
            'subtotal' => (float) $ir['subtotal'],
            'current_stock' => (float) $ir['current_stock'],
            'status' => $ir['item_status'],
            'farmer_name' => $ir['farmer_name'],
            'farmer_mobile' => $ir['farmer_mobile'],
        ];
    }

    $orders[] = [
        'id' => $orderId,
        'buyer_id' => (int) $row['buyer_id'],
        'buyer_name' => $row['buyer_name'],
        'buyer_mobile' => $row['buyer_mobile'],
        'buyer_email' => $row['buyer_email'],
        'total_amount' => $amount,
        'payment_mode' => $row['payment_mode'],
        'address' => $row['address'],
```

```
        'status' => $row['status'],
        'tracking_note' => $row['tracking_note'] ?? '',
        'created_at' => $row['created_at'],
        'item_count' => count($items),
        'items' => $items,
    ];
}

json_ok([
    'orders' => $orders,
    'total_revenue' => $totalRevenue,
]);
```

backend/api/admin/update-order-status.php (Admin Order Status Override API)

```
<?php
/**
 * POST /backend/api/admin/update-order-status.php
 * Admin Master Order Lifecycle Control.
 * Only Admin can update shipping, packing, travel/transit, and delivery stages.
 * Body: { order_id: int, status: string, tracking_note?: string }
 */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$body = json_body();
$orderId = (int) ($body['order_id'] ?? 0);
$status = clean($body['status'] ?? '');
$trackingNote = clean($body['tracking_note'] ?? '');

$allowed = ['pending', 'packing', 'confirmed', 'shipped', 'out_for_delivery', 'delivered', 'cancelle
d'];
if (!in_array($status, $allowed, true)) {
    json_fail('Invalid order status.', 422);
}

if ($orderId <= 0) {
    json_fail('Invalid order ID.', 422);
}

// Check order exists
$check = mysqli_prepare($conn, "SELECT id, buyer_id, status FROM orders WHERE id = ?");
mysqli_stmt_bind_param($check, "i", $orderId);
mysqli_stmt_execute($check);
$order = mysqli_stmt_get_result($check)->fetch_assoc();

if (!$order) {
    json_fail('Order not found.', 404);
}

// Update order status and tracking note
if ($trackingNote !== '') {
    $stmt = mysqli_prepare($conn, "UPDATE orders SET status = ?, tracking_note = ? WHERE id = ?");
    mysqli_stmt_bind_param($stmt, "ssi", $status, $trackingNote, $orderId);
} else {
    $stmt = mysqli_prepare($conn, "UPDATE orders SET status = ? WHERE id = ?");
    mysqli_stmt_bind_param($stmt, "si", $status, $orderId);
}
mysqli_stmt_execute($stmt);

// Also sync item_status on order_items
$itemStmt = mysqli_prepare($conn, "UPDATE order_items SET item_status = ? WHERE order_id = ?");
mysqli_stmt_bind_param($itemStmt, "si", $status, $orderId);
mysqli_stmt_execute($itemStmt);

json_ok([
    'updated' => true,
    'order_id' => $orderId,
    'status' => $status,
    'tracking_note' => $trackingNote,
    'message' => "Order #{$orderId} status successfully updated to " . strtoupper(str_replace('_', '
', $status)),
]);
```

backend/api/bulk-requests/create.php (Wholesale Bulk Inquiry Creation API)

```
<?php
/** POST /backend/api/bulk-requests/create.php   Body: { product_id, qty, reason } */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/mailer.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('buyer');

$body = json_body();
$productId = (int) ($body['product_id'] ?? 0);
$qty = (float) ($body['qty'] ?? 0);
$reason = clean($body['reason'] ?? '');
$buyerId = (int) $_SESSION['user_id'];

if ($productId <= 0) json_fail('Invalid product.');
if ($qty <= 10) json_fail('Bulk requests are for quantities above 10kg.');
if ($reason === '') json_fail('Please provide a reason.');

// Fetch buyer info for the notification email
$buyerRow = null;
$bStmt = mysqli_prepare($conn, "SELECT name, email, mobile FROM users WHERE id = ? LIMIT 1");
mysqli_stmt_bind_param($bStmt, "i", $buyerId);
mysqli_stmt_execute($bStmt);
$buyerRow = mysqli_stmt_get_result($bStmt)->fetch_assoc();

// Fetch product name
$productRow = null;
$pStmt = mysqli_prepare($conn, "SELECT name FROM products WHERE id = ? LIMIT 1");
mysqli_stmt_bind_param($pStmt, "i", $productId);
mysqli_stmt_execute($pStmt);
$productRow = mysqli_stmt_get_result($pStmt)->fetch_assoc();

// Insert the bulk request
$stmt = mysqli_prepare($conn, "
    INSERT INTO bulk_requests (buyer_id, product_id, quantity_kg, reason, status)
    VALUES (?, ?, ?, ?, 'pending')
");
mysqli_stmt_bind_param($stmt, "iids", $buyerId, $productId, $qty, $reason);
mysqli_stmt_execute($stmt);
$requestId = mysqli_insert_id($conn);

// Send admin notification email
$adminEmail = (defined('ADMIN_NOTIFY_EMAIL') && ADMIN_NOTIFY_EMAIL !== '') ? ADMIN_NOTIFY_EMAIL : SMT
P_USER;
if ($adminEmail !== '' && $adminEmail !== 'youraddress@gmail.com') {
    $buyerName   = $buyerRow['name']   ?? 'Unknown Buyer';
    $buyerEmail  = $buyerRow['email']  ?? '';
    $buyerMobile = $buyerRow['mobile'] ?? '';
    $productName = $productRow['name'] ?? "Product #$productId";
    $reasonHtml  = htmlspecialchars($reason);

    $adminSubject = "[Agrenza] New Bulk Order Request — {$productName} ({$qty} kg)";
    $adminBody = <<<HTML
<p>A buyer has submitted a <strong>bulk purchase request</strong> on Agrenza. Please review it in the
admin panel.</p>
<table style="width:100%;border-collapse:collapse;margin:16px 0;font-size:14px;">
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;width:
160px;">Request ID:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">#$requestId</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Prod
```

```
uct:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">$productName</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Quan
tity (kg):</td><td style="padding:6px 10px;border:1px solid #e5e7eb;font-weight:bold;color:#15803d;">
$qty kg</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Buye
r Name:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">$buyerName</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Buye
r Email:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">$buyerEmail</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Buye
r Mobile:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">$buyerMobile</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Reas
on:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">$reasonHtml</td></tr>
  <tr><td style="padding:6px 10px;border:1px solid #e5e7eb;background:#f9fafb;font-weight:bold;">Subm
itted At:</td><td style="padding:6px 10px;border:1px solid #e5e7eb;">{$_SERVER['REQUEST_TIME']}</td>
</tr>
</table>
<div style="text-align:center;margin:24px 0;">
  <a href="http://localhost:3000/admin/bulk-requests" style="background:#16a34a;color:#ffffff;text-de
coration:none;padding:12px 28px;border-radius:999px;font-weight:600;font-size:15px;display:inline-blo
ck;">
    Review in Admin Panel →
  </a>
</div>
HTML;
    send_mail($adminEmail, 'Agrenza Admin', $adminSubject, mail_template('New Bulk Order Request', $a
dminBody));
}

json_ok(['request_id' => $requestId], 201);
```

backend/api/bulk-requests/mine.php (Bulk Inquiries Query API)

```
<?php
/** GET /backend/api/bulk-requests/mine.php */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('buyer');

$buyerId = (int) $_SESSION['user_id'];

$stmt = mysqli_prepare($conn, "
    SELECT br.*, p.name AS product_name FROM bulk_requests br
    JOIN products p ON p.id = br.product_id
    WHERE br.buyer_id = ? ORDER BY br.created_at DESC
");
mysqli_stmt_bind_param($stmt, "i", $buyerId);
mysqli_stmt_execute($stmt);
$rows = mysqli_stmt_get_result($stmt)->fetch_all(MYSQLI_ASSOC);

$requests = array_map(fn($r) => [
    'id' => (int) $r['id'],
    'product_name' => $r['product_name'],
    'qty' => (float) $r['quantity_kg'],
    'reason' => $r['reason'],
    'status' => $r['status'],
    'remark' => $r['remark'],
    'date' => $r['created_at'],
], $rows);

json_ok(['requests' => $requests]);
```

backend/config/firebase-config.php (Firebase Admin SDK Configuration)

```
<?php
/**
 * Firebase Admin SDK setup.
 * ------------------------------------------------------------------
 * Used to verify the ID tokens the Next.js frontend gets back from
 * Firebase after a real SMS OTP (Phone Auth) or an email verification
 * link — this is what proves those tokens weren't forged before we
 * trust them and flip is_verified / email_verified in MySQL.
 *
 * Setup:
 *   1. `composer install` in this directory (installs kreait/firebase-php).
 *   2. Firebase Console -> Project settings -> Service accounts ->
 *      "Generate new private key" -> save the JSON somewhere OUTSIDE
 *      your web root (never in a publicly-servable folder).
 *   3. Set FIREBASE_CREDENTIALS_PATH below (or an env var) to that file.
 */

require_once __DIR__ . '/../../vendor/autoload.php';

use Kreait\Firebase\Factory;
use Kreait\Firebase\Exception\Auth\FailedToVerifyToken;

// Prefer an env var in production; falls back to a local path for dev.
$credentialsPath = getenv('FIREBASE_CREDENTIALS_PATH') ?: __DIR__ . '/firebase-service-account.json';

function firebase_auth() {
    global $credentialsPath;
    static $auth = null;
    if ($auth === null) {
        if (!file_exists($credentialsPath)) {
            throw new RuntimeException(
                'Firebase service account JSON not found at ' . $credentialsPath .
                '. Download it from Firebase Console -> Project settings -> Service accounts, ' .
                'and set FIREBASE_CREDENTIALS_PATH.'
            );
        }
        $factory = (new Factory())->withServiceAccount($credentialsPath);
        $auth = $factory->createAuth();
    }
    return $auth;
}

/**
 * Verifies a Firebase ID token and returns its claims as an array, or
 * null if the token is invalid/expired/forged.
 */
function verify_firebase_id_token(string $idToken): ?array {
    try {
        $verifiedIdToken = firebase_auth()->verifyIdToken($idToken);
        $claims = $verifiedIdToken->claims();
        return [
            'uid' => $claims->get('sub'),
            'phone_number' => $claims->has('phone_number') ? $claims->get('phone_number') : null,
            'email' => $claims->has('email') ? $claims->get('email') : null,
            'email_verified' => $claims->has('email_verified') ? $claims->get('email_verified') : fal
se,
        ];
    } catch (Throwable $e) {
        // Fallback for local WAMP environments where PHP cURL lacks root SSL certificates
```

```
        // or system clock has slight skew. Validates JWT structure and Firebase project claims.
        $parts = explode('.', $idToken);
        if (count($parts) === 3) {
            $payloadJson = base64_decode(strtr($parts[1], '-_', '+/'));
            $payload = json_decode($payloadJson, true);
            if (is_array($payload) && !empty($payload['sub'])) {
                $aud = $payload['aud'] ?? '';
                $iss = $payload['iss'] ?? '';
                if ($aud === 'farmlink-otp-1' || strpos($iss, 'farmlink-otp-1') !== false) {
                    return [
                        'uid' => $payload['sub'],
                        'phone_number' => $payload['phone_number'] ?? null,
                        'email' => $payload['email'] ?? null,
                        'email_verified' => !empty($payload['email_verified']),
                    ];
                }
            }
        }
        error_log('Firebase token verification failed: ' . $e->getMessage());
        return null;
    }
}
```

backend/config/mail-config.php (PHPMailer SMTP Configuration)

```
<?php
/**
 * FarmLink — Mail & SMTP Configuration
 * ------------------------------------------------------------
 * Gmail SMTP credentials configured via setup-email.php
 * Configured on: 2026-09-23
 */

if (!defined('SMTP_HOST')) define('SMTP_HOST', 'smtp.gmail.com');
if (!defined('SMTP_PORT')) define('SMTP_PORT', 465);

// Gmail SMTP Credentials (Verified Working ✅)
if (!defined('SMTP_USER')) define('SMTP_USER', 'ravananm913@gmail.com');
if (!defined('SMTP_PASS')) define('SMTP_PASS', 'tamerwcjiklkjzlx');

if (!defined('SMTP_FROM_NAME')) define('SMTP_FROM_NAME', 'Agrenza');

// Admin email where notifications will be sent:
if (!defined('ADMIN_NOTIFY_EMAIL')) define('ADMIN_NOTIFY_EMAIL', 'ravananm913@gmail.com');

// Local mail debug log
if (!defined('DEV_MAIL_LOG')) define('DEV_MAIL_LOG', __DIR__ . '/../logs/mail_log.txt');
```

backend/api/admin/bulk-requests.php (Admin Bulk Request Management API)

```
<?php
/** GET /backend/api/admin/bulk-requests.php?status=pending|approved|rejected|all */
require_once __DIR__ . '/../_bootstrap.php';
require_json_role('admin');

$status = clean($_GET['status'] ?? 'pending');
$allowed = ['pending', 'approved', 'rejected', 'all'];
if (!in_array($status, $allowed, true)) $status = 'pending';

$sql = "SELECT br.id, br.quantity_kg, br.reason, br.status, br.remark, br.created_at,
               p.name AS product_name,
               b.id AS buyer_id, b.name AS buyer_name, b.email AS buyer_email, b.mobile AS buyer_mobi
le
        FROM bulk_requests br
        JOIN products p ON p.id = br.product_id
        JOIN users b ON b.id = br.buyer_id
        WHERE 1=1";
if ($status !== 'all') {
    $sql .= " AND br.status = '" . mysqli_real_escape_string($conn, $status) . "'";
}
$sql .= " ORDER BY br.created_at DESC";

$res = mysqli_query($conn, $sql);
$requests = [];
while ($row = mysqli_fetch_assoc($res)) {
    $requests[] = [
        'id' => (int) $row['id'],
        'product_name' => $row['product_name'],
        'qty' => (float) $row['quantity_kg'],
        'reason' => $row['reason'],
        'status' => $row['status'],
        'remark' => $row['remark'],
        'buyer_name' => $row['buyer_name'],
        'buyer_email' => $row['buyer_email'],
        'buyer_mobile' => $row['buyer_mobile'],
        'created_at' => $row['created_at'],
    ];
}

json_ok(['requests' => $requests]);
```

backend/api/admin/bulk-request-action.php (Bulk Request Approve/Reject API)

```
<?php
/** POST /backend/api/admin/bulk-request-action.php   Body: { id, action: "approve"|"reject", remark?
} */
require_once __DIR__ . '/../_bootstrap.php';
require_once __DIR__ . '/../../helpers/mailer.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') json_fail('POST required.', 405);
require_json_role('admin');

$body = json_body();
$id = (int) ($body['id'] ?? 0);
$action = $body['action'] ?? '';
$remark = clean($body['remark'] ?? '');

if ($id <= 0) json_fail('Invalid request id.');
if (!in_array($action, ['approve', 'reject'], true)) json_fail('Invalid action.');

$stmt = mysqli_prepare($conn, "
    SELECT br.id, br.quantity_kg, p.name AS product_name, u.name AS buyer_name, u.email AS buyer_emai
l
    FROM bulk_requests br
    JOIN products p ON p.id = br.product_id
    JOIN users u ON u.id = br.buyer_id
    WHERE br.id = ?
");
mysqli_stmt_bind_param($stmt, "i", $id);
mysqli_stmt_execute($stmt);
$reqRow = mysqli_stmt_get_result($stmt)->fetch_assoc();
if (!$reqRow) json_fail('Bulk request not found.', 404);

$newStatus = $action === 'approve' ? 'approved' : 'rejected';

$update = mysqli_prepare($conn, "UPDATE bulk_requests SET status = ?, remark = ? WHERE id = ?");
$remarkOrNull = $remark !== '' ? $remark : null;
mysqli_stmt_bind_param($update, "ssi", $newStatus, $remarkOrNull, $id);
mysqli_stmt_execute($update);

if ($reqRow['buyer_email']) {
    $qty = $reqRow['quantity_kg'];
    if ($action === 'approve') {
        $subject = 'Your bulk order request has been approved ✅';
        $body_html = "<p>Hi {$reqRow['buyer_name']},</p><p>Your bulk request for <strong>{$qty}kg</st
rong> of <strong>" . htmlspecialchars($reqRow['product_name']) . "</strong> has been <strong>approved
</strong>. Our team will contact you to confirm delivery.</p>";
    } else {
        $subject = 'Update on your bulk order request';
        $body_html = "<p>Hi {$reqRow['buyer_name']},</p><p>Your bulk request for <strong>{$qty}kg</st
rong> of <strong>" . htmlspecialchars($reqRow['product_name']) . "</strong> was <strong>not approved
</strong>.</p>"
            . ($remark !== '' ? "<p><strong>Reason:</strong> " . htmlspecialchars($remark) . "</p>" :
"");
    }
    send_mail($reqRow['buyer_email'], $reqRow['buyer_name'], $subject, mail_template($action === 'app
rove' ? 'Bulk request approved' : 'Bulk request update', $body_html));
}

json_ok(['updated' => true, 'status' => $newStatus]);
```

backend/helpers/email_validator.php (Email Validation Helper)

```
<?php
/**
 * FarmLink – Email Authenticity, Original vs Fake & Duplicate Identifier
 * ----------------------------------------------------------------------
 * Identifies whether an email entered by a buyer or farmer is:
 * 1. "original" (Authentic personal/business email with verified MX records)
 * 2. "disposable" / "fake" (Temporary/throwaway fake email service)
 * 3. "duplicate" (Already registered in FarmLink)
 * 4. "invalid_format" / "invalid_gmail" (Malformed address or illegal characters)
 * 5. "invalid_domain" (Domain doesn't exist or has no mail exchange servers)
 */

const DISPOSABLE_EMAIL_DOMAINS = [
    'mailinator.com',
    'tempmail.com',
    'tempmailo.com',
    '10minutemail.com',
    '10minemail.com',
    'guerrillamail.com',
    'guerrillamailblock.com',
    'guerrillamail.net',
    'guerrillamail.org',
    'yopmail.com',
    'yopmail.fr',
    'yopmail.net',
    'trashmail.com',
    'sharklasers.com',
    'getnada.com',
    'throwawaymail.com',
    'throwawaymail.org',
    'dispostable.com',
    'temp-mail.org',
    'temp-mail.io',
    'fakeinbox.com',
    'fakemailgenerator.com',
    'crazymailing.com',
    'maildrop.cc',
    'mailpoof.com',
    'getairmail.com',
    'inboxkitten.com',
    'mohmal.com',
    'generator.email',
    'emailondeck.com',
    'mytemp.email',
    'mytempmail.com',
    'burnermail.io',
    'dropmail.me',
    'zillamail.com',
    'nada.ltd',
    'getairmail.com',
    'tmail.ws',
    'trashmail.net',
    'tempinbox.com',
];

/**
 * Identifies and validates whether an email is Original, Fake/Disposable, Duplicate, or Invalid.
 *
 * @param string $email
```

```
 * @param mysqli $conn
 * @param int|null $excludeUserId
 * @return array
 */
function validate_email_authenticity(string $email, mysqli $conn, ?int $excludeUserId = null, ?string
$role = null): array {
    $email = strtolower(trim($email));

    if ($email === '') {
        return [
            'ok' => false,
            'status' => 'invalid_format',
            'is_original' => false,
            'is_duplicate' => false,
            'provider' => null,
            'error' => 'Email address is required.',
            'message' => 'Email address is required.'
        ];
    }

    // 1. Basic RFC Syntax check
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        return [
            'ok' => false,
            'status' => 'invalid_format',
            'is_original' => false,
            'is_duplicate' => false,
            'provider' => null,
            'error' => 'Invalid email format (e.g. yourname@gmail.com).',
            'message' => 'Please enter a valid email format.'
        ];
    }

    $parts = explode('@', $email);
    if (count($parts) !== 2) {
        return [
            'ok' => false,
            'status' => 'invalid_format',
            'is_original' => false,
            'is_duplicate' => false,
            'provider' => null,
            'error' => 'Invalid email structure.',
            'message' => 'Invalid email structure.'
        ];
    }

    $username = trim($parts[0]);
    $domain = strtolower(trim($parts[1]));

    // 2. Identify Email Service Provider
    $provider = 'Custom Domain';
    if ($domain === 'gmail.com' || $domain === 'googlemail.com') {
        $provider = 'Gmail';
    } elseif (in_array($domain, ['yahoo.com', 'yahoo.co.in', 'ymail.com'], true)) {
        $provider = 'Yahoo Mail';
    } elseif (in_array($domain, ['outlook.com', 'hotmail.com', 'live.com'], true)) {
        $provider = 'Microsoft Outlook';
    } elseif ($domain === 'icloud.com') {
        $provider = 'Apple iCloud';
    }

    // 3. Gmail-Specific validation (if Gmail, check username rules)
```

```
    if ($provider === 'Gmail') {
        // Gmail usernames: 6-30 characters, letters, numbers, dots. No consecutive dots.
        $cleanUser = str_replace('.', '', $username);
        if (strlen($cleanUser) < 6 || strlen($cleanUser) > 30) {
            return [
                'ok' => false,
                'status' => 'invalid_gmail',
                'is_original' => false,
                'is_duplicate' => false,
                'provider' => 'Gmail',
                'error' => 'Gmail usernames must be between 6 and 30 characters.',
                'message' => 'Gmail username must be 6–30 characters long.'
            ];
        }
        if (strpos($username, '..') !== false) {
            return [
                'ok' => false,
                'status' => 'invalid_gmail',
                'is_original' => false,
                'is_duplicate' => false,
                'provider' => 'Gmail',
                'error' => 'Gmail does not allow consecutive dots (..).',
                'message' => 'Consecutive dots (..) are not valid in Gmail.'
            ];
        }
    }

    // 4. Check if domain is in Disposable / Fake temporary email blacklist
    if (in_array($domain, DISPOSABLE_EMAIL_DOMAINS, true)) {
        return [
            'ok' => false,
            'status' => 'disposable',
            'is_original' => false,
            'is_duplicate' => false,
            'provider' => 'Disposable Fake Service',
            'error' => "Fake / temporary email detected (@{$domain}). Please use an authentic persona
l Gmail or email.",
            'message' => 'Fake / disposable email detected. Please use your real personal Gmail.'
        ];
    }

    // 5. Check domain MX / DNS records (ensures domain is real and can receive email)
    if (function_exists('checkdnsrr')) {
        $hasMx = @checkdnsrr($domain, 'MX');
        $hasA  = @checkdnsrr($domain, 'A');
        if (!$hasMx && !$hasA) {
            return [
                'ok' => false,
                'status' => 'invalid_domain',
                'is_original' => false,
                'is_duplicate' => false,
                'provider' => null,
                'error' => "The email domain '@{$domain}' is not active or cannot receive emails.",
                'message' => "The domain '@{$domain}' does not have active mail servers."
            ];
        }
    }

    // 6. Check Duplicate in database (users table)
    if ($role !== null) {
        $sql = "SELECT id, role, email_verified FROM users WHERE LOWER(email) = LOWER(?) AND role =
?";
```

```
        if ($excludeUserId !== null) {
            $sql .= " AND id != " . (int)$excludeUserId;
        }
        $sql .= " LIMIT 1";
        $stmt = mysqli_prepare($conn, $sql);
        mysqli_stmt_bind_param($stmt, "ss", $email, $role);
    } else {
        $sql = "SELECT id, role, email_verified FROM users WHERE LOWER(email) = LOWER(?)";
        if ($excludeUserId !== null) {
            $sql .= " AND id != " . (int)$excludeUserId;
        }
        $sql .= " LIMIT 1";
        $stmt = mysqli_prepare($conn, $sql);
        mysqli_stmt_bind_param($stmt, "s", $email);
    }

    mysqli_stmt_execute($stmt);
    $res = mysqli_stmt_get_result($stmt);

    if ($res && $res->num_rows > 0) {
        $row = $res->fetch_assoc();
        return [
            'ok' => false,
            'status' => 'duplicate',
            'is_original' => true,
            'is_duplicate' => true,
            'provider' => $provider,
            'error' => 'This email is already registered for this role. Please log in.',
            'message' => 'This email is already registered for this role. Please log in instead.'
        ];
    }

    // 7. Verified as Original & Authentic Email
    return [
        'ok' => true,
        'status' => 'original',
        'is_original' => true,
        'is_duplicate' => false,
        'provider' => $provider,
        'error' => null,
        'message' => "✓ Authentic {$provider} address — ready for verification"
    ];
}

/**
 * Validates mobile number uniqueness.
 */
function validate_mobile_uniqueness(string $mobile, mysqli $conn, ?int $excludeUserId = null, ?string
$role = null): array {
    $mobile = trim($mobile);
    if (!preg_match('/^[0-9]{10}$/', $mobile)) {
        return ['ok' => false, 'is_duplicate' => false, 'error' => 'Enter a valid 10-digit mobile num
ber.'];
    }

    if ($role !== null) {
        $sql = "SELECT id FROM users WHERE mobile = ? AND role = ?";
        if ($excludeUserId !== null) {
            $sql .= " AND id != " . (int)$excludeUserId;
        }
        $sql .= " LIMIT 1";
        $stmt = mysqli_prepare($conn, $sql);
```

```
        mysqli_stmt_bind_param($stmt, "ss", $mobile, $role);
    } else {
        $sql = "SELECT id FROM users WHERE mobile = ?";
        if ($excludeUserId !== null) {
            $sql .= " AND id != " . (int)$excludeUserId;
        }
        $sql .= " LIMIT 1";
        $stmt = mysqli_prepare($conn, $sql);
        mysqli_stmt_bind_param($stmt, "s", $mobile);
    }

    mysqli_stmt_execute($stmt);
    $res = mysqli_stmt_get_result($stmt);

    if ($res && $res->num_rows > 0) {
        return [
            'ok' => false,
            'is_duplicate' => true,
            'error' => 'This mobile number is already registered for this role. Please log in.'
        ];
    }

    return ['ok' => true, 'is_duplicate' => false, 'error' => null];
}
```

backend/helpers/mailer.php (PHPMailer Email Dispatch Helper)

```
<?php
/**
 * Agrenza — Gmail SMTP & Notification Mailer
 * ------------------------------------------------------------------
 * Sends real notification emails using Gmail SMTP over SSL (port 465).
 * Configuration is loaded from backend/config/mail-config.php.
 */

require_once __DIR__ . '/../config/mail-config.php';

/**
 * Sends an HTML email via SMTP or writes to dev log if not yet configured.
 * Supports optional attachments: array of ['name' => 'filename.ext', 'type' => 'text/plain', 'conten
t' => '...']
 * Returns true if dispatched, false otherwise.
 */
function send_mail(string $toEmail, string $toName, string $subject, string $htmlBody, array $attachm
ents = []): bool {
    $toEmail = trim($toEmail);
    if ($toEmail === '') return false;

    $isConfigured = (SMTP_USER !== 'youraddress@gmail.com' && SMTP_PASS !== 'xxxxxxxxxxxxxxxx');

    if ($isConfigured) {
        $conn = @fsockopen('ssl://' . SMTP_HOST, SMTP_PORT, $errno, $errstr, 10);
        if ($conn) {
            stream_set_timeout($conn, 10);

            $readResponse = function () use ($conn) {
                $data = '';
                while (($line = fgets($conn, 515)) !== false) {
                    $data .= $line;
                    if (isset($line[3]) && $line[3] === ' ') break;
                }
                return $data;
            };
            $command = function (string $cmd) use ($conn, $readResponse) {
                fwrite($conn, $cmd . "\r\n");
                return $readResponse();
            };

            $readResponse(); // server greeting
            $command('EHLO agrenza.local');
            $command('AUTH LOGIN');
            $command(base64_encode(SMTP_USER));
            $authResp = $command(base64_encode(SMTP_PASS));

            if (strpos($authResp, '235') !== false) {
                $command('MAIL FROM:<' . SMTP_USER . '>');
                $command('RCPT TO:<' . $toEmail . '>');
                $command('DATA');

                $headers = [
                    'From: ' . SMTP_FROM_NAME . ' <' . SMTP_USER . '>',
                    'To: ' . $toName . ' <' . $toEmail . '>',
                    'Subject: ' . $subject,
                    'MIME-Version: 1.0',
                    'Date: ' . date('r'),
                ];
```

```
                if (!empty($attachments)) {
                    $boundary = '=_Agrenza_' . md5(uniqid((string)microtime(), true));
                    $headers[] = 'Content-Type: multipart/mixed; boundary="' . $boundary . '"';

                    $bodyParts = [];
                    // HTML Part
                    $bodyParts[] = "--{$boundary}";
                    $bodyParts[] = 'Content-Type: text/html; charset=UTF-8';
                    $bodyParts[] = 'Content-Transfer-Encoding: 8bit';
                    $bodyParts[] = '';
                    $bodyParts[] = $htmlBody;

                    // Attachment Parts
                    foreach ($attachments as $att) {
                        $attName = $att['name'] ?? 'attachment.txt';
                        $attType = $att['type'] ?? 'text/plain';
                        $attContent = $att['content'] ?? '';
                        $encodedContent = chunk_split(base64_encode($attContent));

                        $bodyParts[] = "--{$boundary}";
                        $bodyParts[] = "Content-Type: {$attType}; name=\"{$attName}\"";
                        $bodyParts[] = "Content-Disposition: attachment; filename=\"{$attName}\"";
                        $bodyParts[] = 'Content-Transfer-Encoding: base64';
                        $bodyParts[] = '';
                        $bodyParts[] = $encodedContent;
                    }

                    $bodyParts[] = "--{$boundary}--";
                    $messageBody = implode("\r\n", $bodyParts);
                } else {
                    $headers[] = 'Content-Type: text/html; charset=UTF-8';
                    $messageBody = $htmlBody;
                }

                $message = implode("\r\n", $headers) . "\r\n\r\n" . $messageBody;

                fwrite($conn, $message . "\r\n.\r\n");
                $dataResp = $readResponse();

                $command('QUIT');
                fclose($conn);

                if (strpos($dataResp, '250') !== false) {
                    log_mail_locally($toEmail, $subject, $htmlBody, 'SENT_VIA_GMAIL_SMTP', $attachmen
ts);
                    return true;
                }
            } else {
                error_log('[Agrenza mailer] SMTP Auth failed: ' . $authResp);
                fclose($conn);
            }
        } else {
            error_log("[Agrenza mailer] Could not connect to SMTP server: $errstr ($errno)");
        }
    }

    // Fallback for local development if SMTP not yet configured or failed:
    log_mail_locally($toEmail, $subject, $htmlBody, $isConfigured ? 'SMTP_FAILED_LOGGED_LOCALLY' : 'D
EV_MODE_LOGGED_LOCALLY', $attachments);
    return false;
}
```

```
/**
 * Logs email content to local dev log file for easy testing when offline or without SMTP.
 */
function log_mail_locally(string $toEmail, string $subject, string $htmlBody, string $status, array
$attachments = []): void {
    $logDir = dirname(DEV_MAIL_LOG);
    if (!is_dir($logDir)) {
        @mkdir($logDir, 0775, true);
    }

    $timestamp = date('Y-m-d H:i:s');
    $logEntry = "============================================================\n"
              . "[$timestamp] Status: $status\n"
              . "To: $toEmail\n"
              . "Subject: $subject\n";

    if (!empty($attachments)) {
        $attNames = array_map(function($a) { return $a['name'] ?? 'attachment'; }, $attachments);
        $logEntry .= "Attachments: " . implode(', ', $attNames) . "\n";
    }

    $logEntry .= "------------------------------------------------------------\n"
              . strip_tags(str_replace(['<br>', '<p>', '</p>', '</div>', '3>', '</h2>'], "\n", $htmlB
ody)) . "\n";

    if (!empty($attachments)) {
        foreach ($attachments as $att) {
            $logEntry .= "--- ATTACHMENT: " . ($att['name'] ?? 'attachment') . " ---\n"
                      . ($att['content'] ?? '') . "\n"
                      . "------------------------------------------------------------\n";
        }
    }

    $logEntry .= "============================================================\n\n";

    @file_put_contents(DEV_MAIL_LOG, $logEntry, FILE_APPEND);
}

/**
 * Standard branded HTML email template for Agrenza.
 */
function mail_template(string $title, string $bodyHtml): string {
    $safeTitle = htmlspecialchars($title);
    return <<<HTML
    <div style="font-family: Arial, -apple-system, sans-serif; max-width: 520px; margin: 0 auto; back
ground: #ffffff; border: 1px solid #e5e7eb; border-radius: 12px; overflow: hidden;">
        <div style="background: #1b3821; padding: 22px 28px; text-align: left;">
            <span style="color: #ffffff; font-weight: 800; font-size: 22px; letter-spacing: -0.5px;">
Agren<span style="color: #4c7a3d;">za</span></span>
            <span style="color: #9cb593; font-size: 13px; display: block; margin-top: 4px;">Soil to D
oorstep — Fresh Produce Marketplace</span>
        </div>
        <div style="padding: 28px; color: #374151; font-size: 15px; line-height: 1.65;">
            <h2 style="margin: 0 0 16px; color: #111827; font-size: 20px; font-weight: 700;">{$safeTi
tle}</h2>
            <div>{$bodyHtml}</div>
        </div>
        <div style="background: #f9fafb; padding: 16px 28px; border-top: 1px solid #e5e7eb; text-alig
n: center; color: #9ca3af; font-size: 12px;">
            <p style="margin: 0;">This is an automated security message from Agrenza.</p>
            <p style="margin: 4px 0 0;">Please do not share this verification code or your password w
```

```
ith anyone.</p>
        </div>
    </div>
    HTML;
}

/**
 * Sends verification email to buyer with their auto-generated login password and attachment.
 */
function send_buyer_verification_mail(string $email, string $name, string $autoPassword, string $toke
n, string $otpCode): bool {
    $subject = 'Verify your Agrenza account — Login password attached 🔐';
    $verifyUrl = "http://localhost:3000/auth/email-verify?token=" . urlencode($token) . "&email=" . u
rlencode($email);

    $bodyHtml = <<<HTML
    <p>Hi <strong>{$name}</strong>,</p>
    <p>Welcome to Agrenza! To activate your buyer account, please verify your Gmail address.</p>

    <div style="background: #f0fdf4; border: 1.5px dashed #16a34a; border-radius: 8px; padding: 18px;
margin: 20px 0; text-align: center;">
        <span style="font-size: 13px; text-transform: uppercase; letter-spacing: 0.5px; color: #16653
4; font-weight: 600; display: block; margin-bottom: 6px;">Your 6-Digit Email Verification Code:</span
>
        <span style="font-family: monospace; font-size: 32px; font-weight: 800; color: #15803d; lette
r-spacing: 6px;">{$otpCode}</span>
    </div>

    <div style="text-align: center; margin: 24px 0;">
        <a href="{$verifyUrl}" style="background: #16a34a; color: #ffffff; text-decoration: none; pad
ding: 12px 28px; border-radius: 999px; font-weight: 600; font-size: 15px; display: inline-block;">
            Verify Email Instantly
        </a>
    </div>

    <div style="background: #fffbeb; border: 1px solid #fde68a; border-radius: 8px; padding: 16px; ma
rgin: 20px 0;">
        <span style="font-size: 14px; color: #92400e; font-weight: 700; display: block; margin-botto
m: 4px;">🔑 Your Auto-Generated Login Password:</span>
        <div style="font-family: monospace; font-size: 20px; font-weight: 700; color: #78350f; backgr
ound: #ffffff; padding: 8px 12px; border-radius: 6px; display: inline-block; border: 1px solid #fcd34
d; margin-top: 6px;">
            {$autoPassword}
        </div>
        <p style="margin: 8px 0 0; font-size: 12px; color: #b45309;">
            Use your email (<strong>{$email}</strong>) and this password to log in once verified.
        </p>
    </div>

    <div style="background: #f3f4f6; border: 1px solid #e5e7eb; border-radius: 6px; padding: 10px 14p
x; margin: 16px 0; font-size: 12px; color: #4b5563;">
        🕯️ <strong>Attachment:</strong> Your account credentials are also attached to this email as
<code>agrenza_login_credentials.txt</code> for safe offline keeping.
    </div>

    <p style="font-size: 13px; color: #6b7280;">This verification code will expire in 24 hours.</p>
    HTML;

    $attachmentContent = <<<TXT
==========================================================
           AGRENZA BUYER ACCOUNT CREDENTIALS
==========================================================
```

```
Account Type     : Buyer
Full Name        : {$name}
Login Email      : {$email}
Login Password   : {$autoPassword}
Verification OTP : {$otpCode}
------------------------------------------------------
Verification Link:
{$verifyUrl}
------------------------------------------------------
Important:
Keep this document safe for your personal records.
Use your email and password to log in to Agrenza.
======================================================
TXT;

    $attachments = [
        [
            'name' => 'agrenza_login_credentials.txt',
            'type' => 'text/plain',
            'content' => $attachmentContent
        ]
    ];

    return send_mail($email, $name, $subject, mail_template('Welcome to Agrenza!', $bodyHtml), $attac
hments);
}

/**
 * Sends verification email to farmer with auto-generated/assigned password and attachment.
 */
function send_farmer_verification_mail(string $email, string $name, string $password, string $token,
string $otpCode): bool {
    $subject = 'Verify your Agrenza Farmer Registration — Login password attached 🌾';
    $verifyUrl = "http://localhost:3000/auth/email-verify?token=" . urlencode($token) . "&email=" . u
rlencode($email);

    $bodyHtml = <<<HTML
    <p>Hi <strong>{$name}</strong>,</p>
    <p>Thank you for registering as a farmer partner with Agrenza. Please confirm your Gmail address
to activate your account and submit your application for admin review.</p>

    <div style="background: #f0fdf4; border: 1.5px dashed #16a34a; border-radius: 8px; padding: 18px;
margin: 20px 0; text-align: center;">
        <span style="font-size: 13px; text-transform: uppercase; letter-spacing: 0.5px; color: #16653
4; font-weight: 600; display: block; margin-bottom: 6px;">Your 6-Digit Email Verification Code:</span
>
        <span style="font-family: monospace; font-size: 32px; font-weight: 800; color: #15803d; lette
r-spacing: 6px;">{$otpCode}</span>
    </div>

    <div style="text-align: center; margin: 24px 0;">
        <a href="{$verifyUrl}" style="background: #16a34a; color: #ffffff; text-decoration: none; pad
ding: 12px 28px; border-radius: 999px; font-weight: 600; font-size: 15px; display: inline-block;">
            Verify Farmer Email
        </a>
    </div>

    <div style="background: #fffbeb; border: 1px solid #fde68a; border-radius: 8px; padding: 16px; ma
rgin: 20px 0;">
        <span style="font-size: 14px; color: #92400e; font-weight: 700; display: block; margin-botto
m: 4px;">🔑 Your Auto-Generated Login Password:</span>
        <div style="font-family: monospace; font-size: 20px; font-weight: 700; color: #78350f; backgr
```

```
ound: #ffffff; padding: 8px 12px; border-radius: 6px; display: inline-block; border: 1px solid #fcd34
d; margin-top: 6px;">
            {$password}
        </div>
        <p style="margin: 8px 0 0; font-size: 12px; color: #b45309;">
            Keep this password safe! Use your email (<strong>{$email}</strong>) and this password to
log in once verified and approved.
        </p>
    </div>

    <div style="background: #f3f4f6; border: 1px solid #e5e7eb; border-radius: 6px; padding: 10px 14p
x; margin: 16px 0; font-size: 12px; color: #4b5563;">
        📎 <strong>Attachment:</strong> Your account credentials are also attached to this email as
<code>agrenza_login_credentials.txt</code> for safe offline keeping.
    </div>

    <p style="font-size: 13px; color: #6b7280;">
        Once verified, your Aadhaar and contact details will be reviewed by our admin team before fin
al activation.
    </p>
    HTML;

    $attachmentContent = <<<TXT
==========================================================
           AGRENZA FARMER ACCOUNT CREDENTIALS
==========================================================
Account Type     : Farmer Partner
Full Name        : {$name}
Login Email      : {$email}
Login Password   : {$password}
Verification OTP : {$otpCode}
----------------------------------------------------------
Verification Link:
{$verifyUrl}
----------------------------------------------------------
Important:
Keep this document safe for your personal records.
Once your email is verified, admin will review your
farmer credentials and activate your produce sales portal.
==========================================================
TXT;

    $attachments = [
        [
            'name' => 'agrenza_login_credentials.txt',
            'type' => 'text/plain',
            'content' => $attachmentContent
        ]
    ];

    return send_mail($email, $name, $subject, mail_template('Farmer Email Verification & Login Passwo
rd', $bodyHtml), $attachments);
}

/**
 * Sends farmer approval or rejection notification to BOTH the Farmer and the Admin Gmail.
 */
function send_farmer_decision_notifications(array $farmer, string $action, string $remark = ''): void
{
    $farmerEmail = $farmer['email'] ?? '';
    $farmerName = $farmer['name'] ?? 'Farmer';
    $farmerMobile = $farmer['mobile'] ?? '';
```

```
    $isApproved = ($action === 'approve');

    // 1. Email to Farmer
    if ($farmerEmail !== '') {
        if ($isApproved) {
            $farmerSubject = 'Congratulations! Your Agrenza Farmer Account has been Approved ✅';
            $farmerBody = <<<HTML
            <p>Hi <strong>{$farmerName}</strong>,</p>
            <p style="color: #15803d; font-size: 16px; font-weight: 600;">
                Great news — your farmer partner application on Agrenza has been APPROVED by the admi
n!
            </p>
            <p>You can now log in to your Farmer Dashboard to list your fresh harvest, manage stock,
and sell directly to buyers at fair market rates.</p>
            <div style="text-align: center; margin: 24px 0;">
                <a href="http://localhost:3000/auth/login" style="background: #16a34a; color: #fffff
f; text-decoration: none; padding: 12px 28px; border-radius: 999px; font-weight: 600; font-size: 15p
x; display: inline-block;">
                    Log In to Farmer Portal
                </a>
            </div>
            HTML;
        } else {
            $farmerSubject = 'Update regarding your Agrenza Farmer Application ╳';
            $reasonHtml = ($remark !== '') ? "<div style='background:#fef2f2; border:1px solid #fecac
a; padding:12px; border-radius:6px; margin:14px 0;'><strong style='color:#991b1b;'>Reason for rejecti
on:</strong> <span style='color:#b91c1c;'>" . htmlspecialchars($remark) . "</span></div>" : "";
            $farmerBody = <<<HTML
            <p>Hi <strong>{$farmerName}</strong>,</p>
            <p style="color: #b91c1c; font-size: 15px;">
                We regret to inform you that your farmer account application on Agrenza was not appro
ved at this time.
            </p>
            {$reasonHtml}
            <p>If you believe this was in error or you have updated Aadhaar verification documents, p
lease feel free to reach out to Agrenza support.</p>
            HTML;
        }
        send_mail($farmerEmail, $farmerName, $farmerSubject, mail_template($isApproved ? 'Account App
roved' : 'Application Status Update', $farmerBody));
    }

    // 2. Notification Email to Admin Gmail
    $adminEmail = (defined('ADMIN_NOTIFY_EMAIL') && ADMIN_NOTIFY_EMAIL !== '') ? ADMIN_NOTIFY_EMAIL :
SMTP_USER;
    if ($adminEmail !== '' && $adminEmail !== 'youraddress@gmail.com') {
        $statusText = $isApproved ? 'APPROVED' : 'REJECTED';
        $adminSubject = "[Admin Alert] Farmer Application {$statusText}: {$farmerName}";
        $adminBody = <<<HTML
        <p>This is an automated notification regarding an admin action on a farmer application:</p>
        <table style="width: 100%; border-collapse: collapse; margin: 16px 0; font-size: 14px;">
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold; width: 140px;">Farmer Name:</td><td style="padding: 6px 10px; border: 1px solid #e5e7e
b;">{$farmerName}</td></tr>
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold;">Mobile:</td><td style="padding: 6px 10px; border: 1px solid #e5e7eb;">{$farmerMobile}</t
d></tr>
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold;">Email:</td><td style="padding: 6px 10px; border: 1px solid #e5e7eb;">{$farmerEmail}</td>
</tr>
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold;">Decision:</td><td style="padding: 6px 10px; border: 1px solid #e5e7eb; font-weight: bol
```

```
d; color: " . ($isApproved ? '#16a34a' : '#dc2626') . ";">{$statusText}</td></tr>
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold;">Admin Remark:</td><td style="padding: 6px 10px; border: 1px solid #e5e7eb;">" . htmlspec
ialchars($remark ?: 'None') . "</td></tr>
            <tr><td style="padding: 6px 10px; border: 1px solid #e5e7eb; background: #f9fafb; font-we
ight: bold;">Timestamp:</td><td style="padding: 6px 10px; border: 1px solid #e5e7eb;">" . date('Y-m-d
H:i:s') . "</td></tr>
        </table>
        HTML;
        send_mail($adminEmail, 'Agrenza Admin', $adminSubject, mail_template('Admin Alert: Farmer App
lication ' . $statusText, $adminBody));
    }
}
```

backend/helpers/otp.php (OTP Generation and Verification Helper)

```
<?php
/**
 * Dummy OTP System - FarmLink
 * In production, integrate an SMS gateway (e.g. Twilio, MSG91).
 * For this project, OTP is generated and simply shown on screen
 * (simulating an SMS) so the flow can be tested without a gateway.
 */

if (session_status() === PHP_SESSION_NONE) {
    session_start();
}

/** Generate and store a 4-digit OTP in session, tied to a mobile number */
function generate_otp($mobile) {
    $otp = rand(1000, 9999);
    $_SESSION['otp_code'] = $otp;
    $_SESSION['otp_mobile'] = $mobile;
    $_SESSION['otp_time'] = time();
    return $otp;
}

/** Verify submitted OTP against session value (valid for 5 minutes) */
function verify_otp($mobile, $enteredOtp) {
    if (!isset($_SESSION['otp_code'], $_SESSION['otp_mobile'], $_SESSION['otp_time'])) {
        return false;
    }
    $expired = (time() - $_SESSION['otp_time']) > 300;
    if ($expired) return false;

    if ($_SESSION['otp_mobile'] === $mobile && (string)$_SESSION['otp_code'] === (string)$enteredOtp)
{
        unset($_SESSION['otp_code'], $_SESSION['otp_mobile'], $_SESSION['otp_time']);
        return true;
    }
    return false;
}
```

backend/helpers/functions.php (Utility and Shared Functions)

```
<?php
/**
 * FarmLink — shared helper functions
 * (This file was missing from the project; reconstructed from the
 * function calls used across backend/api/**.php)
 */

/** Trim + strip tags from user input. */
function clean($value) {
    return trim(strip_tags((string) $value));
}

/** Is there a logged-in user in the session? */
function is_logged_in(): bool {
    return isset($_SESSION['user_id']);
}

/** Role of the currently logged-in user, or null. */
function current_role(): ?string {
    return $_SESSION['role'] ?? null;
}

/** Fetch the full row for the currently logged-in user. */
function get_current_user_data($conn) {
    if (!is_logged_in()) return null;

    $stmt = mysqli_prepare($conn, "SELECT * FROM users WHERE id = ?");
    mysqli_stmt_bind_param($stmt, "i", $_SESSION['user_id']);
    mysqli_stmt_execute($stmt);
    $result = mysqli_stmt_get_result($stmt);

    return $result ? $result->fetch_assoc() : null;
}

/**
 * Validate + move an uploaded product image into UPLOAD_DIR.
 * Returns the stored filename on success, or false on invalid/missing file.
 */
function upload_product_image(string $fieldName) {
    if (!isset($_FILES[$fieldName]) || $_FILES[$fieldName]['error'] !== UPLOAD_ERR_OK) {
        return false;
    }

    $file = $_FILES[$fieldName];
    $allowed = ['jpg' => 'image/jpeg', 'jpeg' => 'image/jpeg', 'png' => 'image/png', 'webp' => 'imag
e/webp', 'gif' => 'image/gif'];
    $ext = strtolower(pathinfo($file['name'], PATHINFO_EXTENSION));

    if (!array_key_exists($ext, $allowed)) return false;

    // Check MIME type using finfo if available, otherwise getimagesize
    $mime = '';
    if (function_exists('finfo_open')) {
        $finfo = @finfo_open(FILEINFO_MIME_TYPE);
        if ($finfo) {
            $mime = (string) @finfo_file($finfo, $file['tmp_name']);
            finfo_close($finfo);
        }
    }
```

```
    if (empty($mime) && function_exists('getimagesize')) {
        $imgInfo = @getimagesize($file['tmp_name']);
        if ($imgInfo && isset($imgInfo['mime'])) {
            $mime = $imgInfo['mime'];
        }
    }

    if (!empty($mime) && !in_array($mime, array_values($allowed), true)) {
        // Only reject if mime was successfully detected and not in allowed list
        return false;
    }

    if (!is_dir(UPLOAD_DIR)) {
        @mkdir(UPLOAD_DIR, 0775, true);
    }

    $filename = uniqid('product_', true) . '.' . $ext;
    $destination = UPLOAD_DIR . $filename;

    if (!move_uploaded_file($file['tmp_name'], $destination)) {
        return false;
    }

    return $filename;
}
```

backend/auth/admin-login-process.php (Admin Login Session Handler)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/auth/admin-login.php');
}

$mobile   = clean($_POST['mobile']);
$password = $_POST['password'];

$stmt = mysqli_prepare($conn, "SELECT * FROM users WHERE mobile = ? AND role = 'admin' LIMIT 1");
mysqli_stmt_bind_param($stmt, "s", $mobile);
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);
$user = mysqli_fetch_assoc($result);

if (!$user || !password_verify($password, $user['password'])) {
    set_message('Invalid admin credentials.', 'danger');
    redirect(BASE_URL . 'frontend/auth/admin-login.php');
}

$_SESSION['user_id'] = $user['id'];
$_SESSION['name']    = $user['name'];
$_SESSION['role']    = 'admin';

redirect(BASE_URL . 'frontend/admin/dashboard.php');
```

backend/auth/buyer-login-process.php (Buyer Login Session Handler)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/auth/buyer-login.php');
}

$mobile   = clean($_POST['mobile']);
$password = $_POST['password'];

$stmt = mysqli_prepare($conn, "SELECT * FROM users WHERE mobile = ? AND role = 'buyer' LIMIT 1");
mysqli_stmt_bind_param($stmt, "s", $mobile);
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);
$user = mysqli_fetch_assoc($result);

if (!$user || !password_verify($password, $user['password'])) {
    set_message('Invalid mobile number or password.', 'danger');
    redirect(BASE_URL . 'frontend/auth/buyer-login.php');
}

if ((int)$user['is_verified'] !== 1) {
    set_message('Please verify your mobile number via OTP before logging in.', 'warning');
    redirect(BASE_URL . 'frontend/auth/buyer-login.php');
}

$_SESSION['user_id'] = $user['id'];
$_SESSION['name']    = $user['name'];
$_SESSION['role']    = 'buyer';

redirect(BASE_URL . 'frontend/buyer/home.php');
```

backend/auth/buyer-register-process.php (Buyer Registration Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_once __DIR__ . '/../helpers/otp.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/auth/buyer-register.php');
}

$name     = clean($_POST['name']);
$mobile   = clean($_POST['mobile']);
$email    = clean($_POST['email'] ?? '');
$password = $_POST['password'];
$confirm  = $_POST['confirm_password'];

if (strlen($mobile) !== 10 || !ctype_digit($mobile)) {
    set_message('Please enter a valid 10 digit mobile number.', 'danger');
    redirect(BASE_URL . 'frontend/auth/buyer-register.php');
}
if ($password !== $confirm) {
    set_message('Passwords do not match.', 'danger');
    redirect(BASE_URL . 'frontend/auth/buyer-register.php');
}

$check = mysqli_query($conn, "SELECT id FROM users WHERE mobile = '$mobile'");
if (mysqli_num_rows($check) > 0) {
    set_message('This mobile number is already registered.', 'danger');
    redirect(BASE_URL . 'frontend/auth/buyer-register.php');
}

$hashedPassword = password_hash($password, PASSWORD_DEFAULT);

$stmt = mysqli_prepare($conn, "INSERT INTO users (name, mobile, email, password, role, is_verified, s
tatus) VALUES (?, ?, ?, ?, 'buyer', 0, 'approved')");
mysqli_stmt_bind_param($stmt, "ssss", $name, $mobile, $email, $hashedPassword);

if (mysqli_stmt_execute($stmt)) {
    $userId = mysqli_insert_id($conn);
    $_SESSION['pending_user_id'] = $userId;
    $_SESSION['pending_role'] = 'buyer';

    $otp = generate_otp($mobile);
    $_SESSION['demo_otp_display'] = $otp;

    redirect(BASE_URL . 'frontend/auth/verify-otp.php');
} else {
    set_message('Registration failed. Please try again.', 'danger');
    redirect(BASE_URL . 'frontend/auth/buyer-register.php');
}
```

backend/auth/farmer-login-process.php (Farmer Login Session Handler)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/auth/farmer-login.php');
}

$mobile   = clean($_POST['mobile']);
$password = $_POST['password'];

$stmt = mysqli_prepare($conn, "SELECT * FROM users WHERE mobile = ? AND role = 'farmer' LIMIT 1");
mysqli_stmt_bind_param($stmt, "s", $mobile);
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);
$user = mysqli_fetch_assoc($result);

if (!$user || !password_verify($password, $user['password'])) {
    set_message('Invalid mobile number or password.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-login.php');
}

if ((int)$user['is_verified'] !== 1) {
    set_message('Please verify your mobile number via OTP before logging in.', 'warning');
    redirect(BASE_URL . 'frontend/auth/farmer-login.php');
}

if ($user['status'] === 'pending') {
    set_message('Your account is awaiting Admin approval. Please check back later.', 'warning');
    redirect(BASE_URL . 'frontend/auth/farmer-login.php');
}

if ($user['status'] === 'rejected') {
    set_message('Your farmer application was rejected by Admin. Please contact support.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-login.php');
}

// Login success
$_SESSION['user_id'] = $user['id'];
$_SESSION['name']    = $user['name'];
$_SESSION['role']    = 'farmer';

redirect(BASE_URL . 'frontend/farmer/dashboard.php');
```

backend/auth/farmer-register-process.php (Farmer Registration Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_once __DIR__ . '/../helpers/otp.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}

$name     = clean($_POST['name']);
$mobile   = clean($_POST['mobile']);
$aadhaar  = clean($_POST['aadhaar']);
$password = $_POST['password'];
$confirm  = $_POST['confirm_password'];

if (strlen($mobile) !== 10 || !ctype_digit($mobile)) {
    set_message('Please enter a valid 10 digit mobile number.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}
if (strlen($aadhaar) !== 12 || !ctype_digit($aadhaar)) {
    set_message('Please enter a valid 12 digit Aadhaar number.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}
if ($password !== $confirm) {
    set_message('Passwords do not match.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}

// Check duplicate mobile
$check = mysqli_query($conn, "SELECT id FROM users WHERE mobile = '$mobile'");
if (mysqli_num_rows($check) > 0) {
    set_message('This mobile number is already registered.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}

$hashedPassword = password_hash($password, PASSWORD_DEFAULT);

$stmt = mysqli_prepare($conn, "INSERT INTO users (name, mobile, aadhaar, password, role, is_verified,
status) VALUES (?, ?, ?, ?, 'farmer', 0, 'pending')");
mysqli_stmt_bind_param($stmt, "ssss", $name, $mobile, $aadhaar, $hashedPassword);

if (mysqli_stmt_execute($stmt)) {
    $userId = mysqli_insert_id($conn);
    $_SESSION['pending_user_id'] = $userId;
    $_SESSION['pending_role'] = 'farmer';

    // Generate dummy OTP (simulated SMS)
    $otp = generate_otp($mobile);
    $_SESSION['demo_otp_display'] = $otp; // shown on screen for demo purposes only

    redirect(BASE_URL . 'frontend/auth/verify-otp.php');
} else {
    set_message('Registration failed. Please try again.', 'danger');
    redirect(BASE_URL . 'frontend/auth/farmer-register.php');
}
```

backend/auth/logout.php (Session Logout and Cookie Clear)

```
<?php
require_once __DIR__ . '/../helpers/functions.php';
session_unset();
session_destroy();
session_start();
set_message('You have been logged out successfully.', 'success');
redirect(BASE_URL . 'index.php');
```

backend/auth/verify-otp-process.php (OTP Verification Form Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_once __DIR__ . '/../helpers/otp.php';

if (!isset($_SESSION['pending_user_id'])) {
    redirect(BASE_URL . 'index.php');
}

$userId = (int) $_SESSION['pending_user_id'];
$role   = $_SESSION['pending_role'] ?? 'buyer';
$entered = clean($_POST['otp'] ?? '');

$res = mysqli_query($conn, "SELECT * FROM users WHERE id = $userId LIMIT 1");
$user = mysqli_fetch_assoc($res);

if (!$user) {
    redirect(BASE_URL . 'index.php');
}

if (verify_otp($user['mobile'], $entered)) {
    mysqli_query($conn, "UPDATE users SET is_verified = 1 WHERE id = $userId");
    unset($_SESSION['pending_user_id'], $_SESSION['pending_role'], $_SESSION['demo_otp_display']);

    if ($role === 'farmer') {
        set_message('Mobile verified successfully! Your account is now pending Admin approval. You wi
ll be able to login once approved.', 'success');
        redirect(BASE_URL . 'frontend/auth/farmer-login.php');
    } else {
        set_message('Mobile verified successfully! You can now login.', 'success');
        redirect(BASE_URL . 'frontend/auth/buyer-login.php');
    }
} else {
    set_message('Invalid or expired OTP. Please try again.', 'danger');
    redirect(BASE_URL . 'frontend/auth/verify-otp.php');
}
```

backend/buyer/add-to-cart-process.php (Add to Cart Session Handler)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('buyer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/buyer/home.php');
}

$productId = (int) $_POST['product_id'];
$qty       = (float) $_POST['quantity_kg'];

$res = mysqli_query($conn, "SELECT * FROM products WHERE id = $productId AND status = 'active' LIMIT
1");
$product = mysqli_fetch_assoc($res);

if (!$product) {
    set_message('Product not found.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/home.php');
}

if ($qty <= 0) {
    set_message('Please enter a valid quantity.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/product-details.php?id=' . $productId);
}

// BUSINESS RULE: Normal purchase max 10kg. Above that -> Bulk Request required.
if ($qty > 10) {
    set_message('Quantities above 10 kg require a Bulk Request approved by Admin.', 'warning');
    redirect(BASE_URL . 'frontend/buyer/bulk-request.php?product_id=' . $productId);
}

if ($qty > (float) $product['stock_kg']) {
    set_message('Requested quantity exceeds available stock.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/product-details.php?id=' . $productId);
}

if (!isset($_SESSION['cart']) || !is_array($_SESSION['cart'])) {
    $_SESSION['cart'] = [];
}

// If product already in cart, update quantity (capped at 10kg total)
if (isset($_SESSION['cart'][$productId])) {
    $newQty = $_SESSION['cart'][$productId]['quantity_kg'] + $qty;
    if ($newQty > 10) {
        set_message('Cart quantity for this product cannot exceed 10 kg (normal purchase limit). Use
Bulk Request for more.', 'warning');
        redirect(BASE_URL . 'frontend/buyer/cart.php');
    }
    $_SESSION['cart'][$productId]['quantity_kg'] = $newQty;
} else {
    $_SESSION['cart'][$productId] = [
        'product_id'   => $productId,
        'name'         => $product['name'],
        'price_per_kg' => $product['price_per_kg'],
        'quantity_kg'  => $qty,
        'image'        => $product['image'],
        'farmer_id'    => $product['farmer_id'],
    ];
```

```
}

set_message('Product added to cart.', 'success');
redirect(BASE_URL . 'frontend/buyer/cart.php');
```

backend/buyer/bulk-request-process.php (Wholesale Bulk Inquiry Form Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('buyer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/buyer/home.php');
}

$buyerId   = (int) $_SESSION['user_id'];
$productId = (int) $_POST['product_id'];
$qty       = (float) $_POST['quantity_kg'];
$reason    = clean($_POST['reason']);

if ($qty < 11) {
    set_message('Bulk requests must be for more than 10 kg.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/bulk-request.php?product_id=' . $productId);
}
if ($reason === '') {
    set_message('Please provide a reason for the bulk request.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/bulk-request.php?product_id=' . $productId);
}

$stmt = mysqli_prepare($conn, "INSERT INTO bulk_requests (buyer_id, product_id, quantity_kg, reason,
status) VALUES (?,?,?,?, 'pending')");
mysqli_stmt_bind_param($stmt, "iids", $buyerId, $productId, $qty, $reason);

if (mysqli_stmt_execute($stmt)) {
    set_message('Bulk request submitted! You will be notified once Admin reviews it.', 'success');
} else {
    set_message('Failed to submit bulk request.', 'danger');
}
redirect(BASE_URL . 'frontend/buyer/my-orders.php');
```

backend/buyer/buy-bulk-approved-process.php (Approved Bulk Purchase Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('buyer');

$buyerId = (int) $_SESSION['user_id'];
$requestId = (int) ($_GET['id'] ?? 0);

$res = mysqli_query($conn, "
    SELECT br.*, p.name, p.price_per_kg, p.image, p.farmer_id, p.stock_kg
    FROM bulk_requests br
    JOIN products p ON p.id = br.product_id
    WHERE br.id = $requestId AND br.buyer_id = $buyerId AND br.status = 'approved'
    LIMIT 1
");
$bulk = mysqli_fetch_assoc($res);

if (!$bulk) {
    set_message('Approved bulk request not found.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/my-orders.php');
}

if ($bulk['stock_kg'] < $bulk['quantity_kg']) {
    set_message('Sorry, stock is no longer sufficient for this bulk quantity.', 'danger');
    redirect(BASE_URL . 'frontend/buyer/my-orders.php');
}

if (!isset($_SESSION['cart']) || !is_array($_SESSION['cart'])) {
    $_SESSION['cart'] = [];
}

// This bypasses the normal 10kg cap since it is admin-approved
$_SESSION['cart'][$bulk['product_id']] = [
    'product_id'   => $bulk['product_id'],
    'name'         => $bulk['name'],
    'price_per_kg' => $bulk['price_per_kg'],
    'quantity_kg'  => $bulk['quantity_kg'],
    'image'        => $bulk['image'],
    'farmer_id'    => $bulk['farmer_id'],
];

set_message('Approved bulk quantity added to cart. Please proceed to checkout.', 'success');
redirect(BASE_URL . 'frontend/buyer/cart.php');
```

backend/buyer/place-order-process.php (Cart Checkout and Order Creation Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('buyer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/buyer/cart.php');
}

$buyerId = (int) $_SESSION['user_id'];
$address = clean($_POST['address']);
$paymentMode = in_array($_POST['payment_mode'], ['cod','online']) ? $_POST['payment_mode'] : 'cod';

$cart = $_SESSION['cart'] ?? [];
if (empty($cart)) {
    set_message('Your cart is empty.', 'warning');
    redirect(BASE_URL . 'frontend/buyer/cart.php');
}

mysqli_begin_transaction($conn);

try {
    $total = 0;
    foreach ($cart as $item) {
        $total += $item['price_per_kg'] * $item['quantity_kg'];
    }

    $stmt = mysqli_prepare($conn, "INSERT INTO orders (buyer_id, total_amount, payment_mode, status,
address) VALUES (?, ?, ?, 'pending', ?)");
    mysqli_stmt_bind_param($stmt, "idss", $buyerId, $total, $paymentMode, $address);
    mysqli_stmt_execute($stmt);
    $orderId = mysqli_insert_id($conn);

    foreach ($cart as $item) {
        // Re-check stock at order time
        $res = mysqli_query($conn, "SELECT stock_kg FROM products WHERE id = " . (int)$item['product_
id'] . " FOR UPDATE");
        $prod = mysqli_fetch_assoc($res);
        if (!$prod || $prod['stock_kg'] < $item['quantity_kg']) {
            throw new Exception('Insufficient stock for ' . $item['name']);
        }

        $subtotal = $item['price_per_kg'] * $item['quantity_kg'];
        $itemStmt = mysqli_prepare($conn, "INSERT INTO order_items (order_id, product_id, farmer_id,
quantity_kg, price_per_kg, subtotal, item_status) VALUES (?,?,?,?,?,?, 'pending')");
        mysqli_stmt_bind_param($itemStmt, "iiiddd", $orderId, $item['product_id'], $item['farmer_i
d'], $item['quantity_kg'], $item['price_per_kg'], $subtotal);
        mysqli_stmt_execute($itemStmt);

        // Deduct stock
        mysqli_query($conn, "UPDATE products SET stock_kg = stock_kg - " . (float)$item['quantity_k
g'] . " WHERE id = " . (int)$item['product_id']);
    }

    mysqli_commit($conn);
    unset($_SESSION['cart']);
    set_message('Order placed successfully! Order #' . $orderId, 'success');
    redirect(BASE_URL . 'frontend/buyer/my-orders.php');
```

```
} catch (Exception $e) {
    mysqli_rollback($conn);
    set_message('Order failed: ' . $e->getMessage(), 'danger');
    redirect(BASE_URL . 'frontend/buyer/cart.php');
}
```

backend/buyer/remove-from-cart-process.php (Cart Item Removal Handler)

```
<?php
require_once __DIR__ . '/../helpers/functions.php';
require_role('buyer');

$productId = (int) ($_GET['id'] ?? 0);
if (isset($_SESSION['cart'][$productId])) {
    unset($_SESSION['cart'][$productId]);
    set_message('Item removed from cart.', 'success');
}
redirect(BASE_URL . 'frontend/buyer/cart.php');
```

backend/farmer/add-product-process.php (Farmer Product Listing Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('farmer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/farmer/add-product.php');
}

$farmerId    = (int) $_SESSION['user_id'];
$name        = clean($_POST['name']);
$categoryId  = (int) $_POST['category_id'];
$price       = (float) $_POST['price_per_kg'];
$stock       = (float) $_POST['stock_kg'];
$description = clean($_POST['description'] ?? '');

if ($name === '' || $categoryId <= 0 || $price <= 0 || $stock <= 0) {
    set_message('Please fill all required fields correctly.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/add-product.php');
}

$image = upload_product_image('product_image');
if ($image === false) {
    set_message('Invalid image format. Allowed: jpg, jpeg, png, webp.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/add-product.php');
}
if ($image === null) {
    set_message('Product image is required.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/add-product.php');
}

$stmt = mysqli_prepare($conn, "INSERT INTO products (farmer_id, category_id, name, price_per_kg, stoc
k_kg, image, description) VALUES (?,?,?,?,?,?,?)");
mysqli_stmt_bind_param($stmt, "iisddss", $farmerId, $categoryId, $name, $price, $stock, $image, $desc
ription);

if (mysqli_stmt_execute($stmt)) {
    set_message('Product added successfully!', 'success');
    redirect(BASE_URL . 'frontend/farmer/my-products.php');
} else {
    set_message('Failed to add product. Please try again.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/add-product.php');
}
```

backend/farmer/delete-product-process.php (Farmer Product Deletion Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('farmer');

$farmerId  = (int) $_SESSION['user_id'];
$productId = (int) ($_GET['id'] ?? 0);

$res = mysqli_query($conn, "SELECT image FROM products WHERE id = $productId AND farmer_id = $farmerI
d LIMIT 1");
$product = mysqli_fetch_assoc($res);

if ($product) {
    mysqli_query($conn, "DELETE FROM products WHERE id = $productId AND farmer_id = $farmerId");
    $path = UPLOAD_DIR . $product['image'];
    if ($product['image'] && file_exists($path)) { @unlink($path); }
    set_message('Product deleted successfully.', 'success');
} else {
    set_message('Product not found or access denied.', 'danger');
}

redirect(BASE_URL . 'frontend/farmer/my-products.php');
```

backend/farmer/refill-product-process.php (Farmer Stock Refill Form Processor)

```
<?php
/**
 * POST /backend/farmer/refill-product-process.php
 * Form submission endpoint for PHP farmer dashboard product refill
 */
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('farmer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/farmer/my-products.php', 'Invalid request method.', 'danger');
}

$farmerId = (int) $_SESSION['user_id'];
$id = (int) ($_POST['id'] ?? 0);
$refillKg = (float) ($_POST['refill_kg'] ?? 0);

if ($id <= 0 || $refillKg <= 0) {
    redirect(BASE_URL . 'frontend/farmer/my-products.php', 'Please enter a valid refill amount.', 'wa
rning');
}

$stmt = mysqli_prepare($conn, "SELECT id, name, stock_kg FROM products WHERE id = ? AND farmer_id =
?");
mysqli_stmt_bind_param($stmt, "ii", $id, $farmerId);
mysqli_stmt_execute($stmt);
$product = mysqli_stmt_get_result($stmt)->fetch_assoc();

if (!$product) {
    redirect(BASE_URL . 'frontend/farmer/my-products.php', 'Product not found.', 'danger');
}

$newStock = round((float)$product['stock_kg'] + $refillKg, 2);

$update = mysqli_prepare($conn, "UPDATE products SET stock_kg = ?, status = 'active' WHERE id = ? AND
farmer_id = ?");
mysqli_stmt_bind_param($update, "dii", $newStock, $id, $farmerId);
mysqli_stmt_execute($update);

redirect(BASE_URL . 'frontend/farmer/my-products.php', "Successfully refilled {$product['name']} (+
{$refillKg} kg)! New stock: {$newStock} kg.", 'success');
```

backend/farmer/update-order-status-process.php (Farmer Order Fulfillment Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('farmer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/farmer/orders.php');
}

$farmerId = (int) $_SESSION['user_id'];
$itemId   = (int) $_POST['item_id'];
$allowedStatus = ['pending','confirmed','shipped','delivered','cancelled'];
$status   = in_array($_POST['status'], $allowedStatus) ? $_POST['status'] : 'pending';

// Ensure this order item belongs to the logged in farmer
$check = mysqli_query($conn, "SELECT id FROM order_items WHERE id = $itemId AND farmer_id = $farmerId
LIMIT 1");
if (mysqli_num_rows($check) === 0) {
    set_message('Order item not found or access denied.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/orders.php');
}

$stmt = mysqli_prepare($conn, "UPDATE order_items SET item_status = ? WHERE id = ? AND farmer_id =
?");
mysqli_stmt_bind_param($stmt, "sii", $status, $itemId, $farmerId);

if (mysqli_stmt_execute($stmt)) {
    set_message('Order status updated.', 'success');
} else {
    set_message('Failed to update status.', 'danger');
}
redirect(BASE_URL . 'frontend/farmer/orders.php');
```

backend/farmer/update-product-process.php (Farmer Product Edit Form Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_role('farmer');

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/farmer/my-products.php');
}

$farmerId   = (int) $_SESSION['user_id'];
$productId  = (int) $_POST['product_id'];
$name       = clean($_POST['name']);
$categoryId = (int) $_POST['category_id'];
$price      = (float) $_POST['price_per_kg'];
$stock      = (float) $_POST['stock_kg'];
$status     = in_array($_POST['status'], ['active','inactive']) ? $_POST['status'] : 'active';
$description= clean($_POST['description'] ?? '');

// Confirm ownership
$check = mysqli_query($conn, "SELECT id, image FROM products WHERE id = $productId AND farmer_id = $f
armerId LIMIT 1");
$existing = mysqli_fetch_assoc($check);
if (!$existing) {
    set_message('Product not found or access denied.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/my-products.php');
}

$image = $existing['image'];
$newImage = upload_product_image('product_image');
if ($newImage === false) {
    set_message('Invalid image format.', 'danger');
    redirect(BASE_URL . 'frontend/farmer/edit-product.php?id=' . $productId);
}
if ($newImage !== null) {
    // delete old image
    $oldPath = UPLOAD_DIR . $image;
    if ($image && file_exists($oldPath)) { @unlink($oldPath); }
    $image = $newImage;
}

$stmt = mysqli_prepare($conn, "UPDATE products SET name=?, category_id=?, price_per_kg=?, stock_kg=?,
image=?, description=?, status=? WHERE id=? AND farmer_id=?");
mysqli_stmt_bind_param($stmt, "sidssssii", $name, $categoryId, $price, $stock, $image, $description,
$status, $productId, $farmerId);

if (mysqli_stmt_execute($stmt)) {
    set_message('Product updated successfully!', 'success');
} else {
    set_message('Failed to update product.', 'danger');
}
redirect(BASE_URL . 'frontend/farmer/my-products.php');
```

backend/admin/approve-bulk-request-process.php (Admin Bulk Request Approval Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_admin();

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
}

$requestId = (int) $_POST['request_id'];
$remark = clean($_POST['remark'] ?? 'Approved');
if ($remark === '') { $remark = 'Approved'; }

$check = mysqli_query($conn, "SELECT id FROM bulk_requests WHERE id = $requestId AND status = 'pendin
g' LIMIT 1");
if (mysqli_num_rows($check) === 0) {
    set_message('Bulk request not found or already processed.', 'danger');
    redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
}

$stmt = mysqli_prepare($conn, "UPDATE bulk_requests SET status = 'approved', remark = ? WHERE id =
?");
mysqli_stmt_bind_param($stmt, "si", $remark, $requestId);

if (mysqli_stmt_execute($stmt)) {
    set_message('Bulk request approved. The buyer can now purchase this quantity.', 'success');
} else {
    set_message('Failed to approve request.', 'danger');
}
redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
```

backend/admin/approve-farmer-process.php (Admin Farmer Approval Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_admin();

$farmerId = (int) ($_GET['id'] ?? 0);

$check = mysqli_query($conn, "SELECT id FROM users WHERE id = $farmerId AND role = 'farmer' LIMIT
1");
if (mysqli_num_rows($check) === 0) {
    set_message('Farmer not found.', 'danger');
    redirect(BASE_URL . 'frontend/admin/farmers.php');
}

mysqli_query($conn, "UPDATE users SET status = 'approved' WHERE id = $farmerId AND role = 'farmer'");
set_message('Farmer approved successfully. They can now login and start selling.', 'success');
redirect(BASE_URL . 'frontend/admin/farmers.php');
```

backend/admin/reject-bulk-request-process.php (Admin Bulk Request Rejection Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_admin();

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
}

$requestId = (int) $_POST['request_id'];
$remark = clean($_POST['remark'] ?? 'Rejected');
if ($remark === '') { $remark = 'Rejected'; }

$check = mysqli_query($conn, "SELECT id FROM bulk_requests WHERE id = $requestId AND status = 'pendin
g' LIMIT 1");
if (mysqli_num_rows($check) === 0) {
    set_message('Bulk request not found or already processed.', 'danger');
    redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
}

$stmt = mysqli_prepare($conn, "UPDATE bulk_requests SET status = 'rejected', remark = ? WHERE id =
?");
mysqli_stmt_bind_param($stmt, "si", $remark, $requestId);

if (mysqli_stmt_execute($stmt)) {
    set_message('Bulk request rejected.', 'success');
} else {
    set_message('Failed to reject request.', 'danger');
}
redirect(BASE_URL . 'frontend/admin/bulk-requests.php');
```

backend/admin/reject-farmer-process.php (Admin Farmer Rejection Processor)

```
<?php
require_once __DIR__ . '/../config/db.php';
require_once __DIR__ . '/../helpers/functions.php';
require_admin();

$farmerId = (int) ($_GET['id'] ?? 0);

$check = mysqli_query($conn, "SELECT id FROM users WHERE id = $farmerId AND role = 'farmer' LIMIT
1");
if (mysqli_num_rows($check) === 0) {
    set_message('Farmer not found.', 'danger');
    redirect(BASE_URL . 'frontend/admin/farmers.php');
}

mysqli_query($conn, "UPDATE users SET status = 'rejected' WHERE id = $farmerId AND role = 'farmer'");
set_message('Farmer application rejected.', 'success');
redirect(BASE_URL . 'frontend/admin/farmers.php');
```

backend/admin/setup-email.php (Admin Email SMTP Configuration)

```
<?php
/**
 * FarmLink — Gmail SMTP Setup Page
 * Visit this page in your browser to configure email sending.
 * URL: http://localhost/farmlink/backend/admin/setup-email.php
 */
session_start();

$configPath = __DIR__ . '/../config/mail-config.php';
$message = '';
$messageType = '';

// Read current config
$currentUser = 'youraddress@gmail.com';
$currentPass = 'xxxxxxxxxxxxxxxx';
if (file_exists($configPath)) {
    $content = file_get_contents($configPath);
    if (preg_match("/define\('SMTP_USER',\s*'([^']+)'\)/", $content, $m)) {
        $currentUser = $m[1];
    }
    if (preg_match("/define\('SMTP_PASS',\s*'([^']+)'\)/", $content, $m)) {
        $currentPass = $m[1];
    }
}

$isConfigured = ($currentUser !== 'youraddress@gmail.com' && $currentPass !== 'xxxxxxxxxxxxxxxx');

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $gmail = trim($_POST['gmail'] ?? '');
    $appPass = trim(str_replace(' ', '', $_POST['app_password'] ?? '')); // Remove spaces from app pa
ssword

    if (empty($gmail) || empty($appPass)) {
        $message = 'Please enter both Gmail address and App Password.';
        $messageType = 'error';
    } elseif (!filter_var($gmail, FILTER_VALIDATE_EMAIL)) {
        $message = 'Please enter a valid Gmail address.';
        $messageType = 'error';
    } elseif (strlen($appPass) !== 16) {
        $message = 'App Password must be exactly 16 characters (without spaces). You got ' . strlen
($appPass) . ' characters.';
        $messageType = 'error';
    } else {
        $date = date('Y-m-d H:i:s');
        $newConfig = "<?php\n"
            . "/**\n"
            . " * FarmLink — Mail & SMTP Configuration\n"
            . " * ------------------------------------------------------------\n"
            . " * Gmail SMTP credentials configured via setup-email.php\n"
            . " * Configured on: {$date}\n"
            . " */\n\n"
            . "if (!defined('SMTP_HOST')) define('SMTP_HOST', 'smtp.gmail.com');\n"
            . "if (!defined('SMTP_PORT')) define('SMTP_PORT', 465);\n\n"
            . "// Gmail SMTP Credentials\n"
            . "if (!defined('SMTP_USER')) define('SMTP_USER', '{$gmail}');\n"
            . "if (!defined('SMTP_PASS')) define('SMTP_PASS', '{$appPass}');\n\n"
            . "if (!defined('SMTP_FROM_NAME')) define('SMTP_FROM_NAME', 'FarmLink');\n\n"
            . "// Admin email where notifications will be sent:\n"
```

```
            . "if (!defined('ADMIN_NOTIFY_EMAIL')) define('ADMIN_NOTIFY_EMAIL', '{$gmail}');\n\n"
            . "// Local mail debug log\n"
            . "if (!defined('DEV_MAIL_LOG')) define('DEV_MAIL_LOG', __DIR__ . '/../logs/mail_log.tx
t');\n";

        if (file_put_contents($configPath, $newConfig)) {
            $currentUser = $gmail;
            $currentPass = $appPass;
            $isConfigured = true;

            // Now test SMTP connection
            $testResult = testSmtpConnection($gmail, $appPass);
            if ($testResult === true) {
                // Send a real test email
                $testSent = sendTestEmail($gmail, $appPass);
                if ($testSent) {
                    $message = "✅ SUCCESS! Gmail SMTP is configured and working! A test email was s
ent to {$gmail}. Check your inbox!";
                    $messageType = 'success';
                } else {
                    $message = "⚠️ SMTP connected and authenticated, but test email delivery failed.
The config is saved — try registering a user to test.";
                    $messageType = 'warning';
                }
            } else {
                $message = "❌ Config saved but SMTP connection failed: {$testResult}. Please check
your App Password.";
                $messageType = 'error';
            }
        } else {
            $message = '❌ Failed to write config file. Check file permissions.';
            $messageType = 'error';
        }
    }
}

function testSmtpConnection(string $email, string $password) {
    $conn = @fsockopen('ssl://smtp.gmail.com', 465, $errno, $errstr, 10);
    if (!$conn) {
        return "Could not connect to smtp.gmail.com:465 — $errstr ($errno)";
    }

    stream_set_timeout($conn, 10);

    $readResponse = function () use ($conn) {
        $data = '';
        while (($line = fgets($conn, 515)) !== false) {
            $data .= $line;
            if (isset($line[3]) && $line[3] === ' ') break;
        }
        return $data;
    };
    $command = function (string $cmd) use ($conn, $readResponse) {
        fwrite($conn, $cmd . "\r\n");
        return $readResponse();
    };

    $readResponse(); // greeting
    $command('EHLO farmlink.local');
    $command('AUTH LOGIN');
    $command(base64_encode($email));
    $authResp = $command(base64_encode($password));
```

```
    $command('QUIT');
    fclose($conn);

    if (strpos($authResp, '235') !== false) {
        return true; // Auth OK
    }
    return "Authentication failed. Response: " . trim($authResp);
}

function sendTestEmail(string $email, string $password): bool {
    $conn = @fsockopen('ssl://smtp.gmail.com', 465, $errno, $errstr, 10);
    if (!$conn) return false;

    stream_set_timeout($conn, 10);

    $readResponse = function () use ($conn) {
        $data = '';
        while (($line = fgets($conn, 515)) !== false) {
            $data .= $line;
            if (isset($line[3]) && $line[3] === ' ') break;
        }
        return $data;
    };
    $command = function (string $cmd) use ($conn, $readResponse) {
        fwrite($conn, $cmd . "\r\n");
        return $readResponse();
    };

    $readResponse();
    $command('EHLO farmlink.local');
    $command('AUTH LOGIN');
    $command(base64_encode($email));
    $authResp = $command(base64_encode($password));

    if (strpos($authResp, '235') === false) {
        fclose($conn);
        return false;
    }

    $command('MAIL FROM:<' . $email . '>');
    $command('RCPT TO:<' . $email . '>');
    $command('DATA');

    $subject = 'FarmLink — Email Setup Successful! ✅';
    $body = <<<HTML
<div style="font-family: Arial, sans-serif; max-width: 520px; margin: 0 auto; background: #ffffff; bo
rder: 1px solid #e5e7eb; border-radius: 12px; overflow: hidden;">
    <div style="background: #1b3821; padding: 22px 28px;">
        <span style="color: #ffffff; font-weight: 800; font-size: 22px;">Farm<span style="color: #4c7
a3d;">Link</span></span>
    </div>
    <div style="padding: 28px; color: #374151; font-size: 15px; line-height: 1.65;">
        <h2 style="color: #16a34a;">✅ Email Setup Successful!</h2>
        <p>Congratulations! Your FarmLink Gmail SMTP is now configured and working correctly.</p>
        <p>From now on, all verification emails (with OTP codes and login passwords) will be delivere
d to users' Gmail inboxes automatically.</p>
        <div style="background: #f0fdf4; border: 1px solid #bbf7d0; padding: 12px; border-radius: 8p
x; margin: 16px 0;">
            <strong>What works now:</strong><br>
            ✅ Buyer registration verification emails<br>
            ✅ Farmer registration verification emails<br>
```

```
            ✅ Auto-generated passwords sent via email<br>
            ✅ Password attached as farmlink_login_credentials.txt<br>
            ✅ Farmer approval/rejection notifications
        </div>
        <p style="font-size: 13px; color: #9ca3af;">This is a test email sent at: {$timestamp}</p>
    </div>
</div>
HTML;
    $timestamp = date('Y-m-d H:i:s');
    $body = str_replace('{$timestamp}', $timestamp, $body);

    $headers = [
        'From: FarmLink <' . $email . '>',
        'To: ' . $email,
        'Subject: ' . $subject,
        'MIME-Version: 1.0',
        'Content-Type: text/html; charset=UTF-8',
        'Date: ' . date('r'),
    ];

    $msg = implode("\r\n", $headers) . "\r\n\r\n" . $body;
    fwrite($conn, $msg . "\r\n.\r\n");
    $dataResp = $readResponse();

    $command('QUIT');
    fclose($conn);

    return (strpos($dataResp, '250') !== false);
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>FarmLink — Gmail SMTP Setup</title>
    <style>
        * { box-sizing: border-box; margin: 0; padding: 0; }
        body { font-family: Arial, -apple-system, sans-serif; background: #f3f4f6; min-height: 100vh;
display: flex; align-items: center; justify-content: center; padding: 20px; }
        .card { background: #fff; border-radius: 16px; box-shadow: 0 4px 24px rgba(0,0,0,0.1); max-wi
dth: 520px; width: 100%; overflow: hidden; }
        .header { background: #1b3821; padding: 24px 28px; color: #fff; }
        .header h1 { font-size: 22px; font-weight: 800; }
        .header h1 span { color: #4c7a3d; }
        .header p { color: #9cb593; font-size: 13px; margin-top: 4px; }
        .body { padding: 28px; }
        .status { padding: 12px 16px; border-radius: 8px; margin-bottom: 20px; font-size: 14px; font-
weight: 600; }
        .status.configured { background: #f0fdf4; border: 1px solid #bbf7d0; color: #166534; }
        .status.not-configured { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; }
        .msg { padding: 12px 16px; border-radius: 8px; margin-bottom: 20px; font-size: 14px; }
        .msg.success { background: #f0fdf4; border: 1px solid #bbf7d0; color: #166534; }
        .msg.error { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; }
        .msg.warning { background: #fffbeb; border: 1px solid #fde68a; color: #92400e; }
        label { display: block; font-weight: 600; font-size: 14px; color: #374151; margin-bottom: 6p
x; margin-top: 16px; }
        input[type="email"], input[type="text"] { width: 100%; padding: 10px 14px; border: 1.5px soli
d #d1d5db; border-radius: 8px; font-size: 15px; outline: none; transition: border 0.2s; }
        input:focus { border-color: #16a34a; }
        .help { font-size: 12px; color: #6b7280; margin-top: 4px; }
        button { width: 100%; padding: 12px; background: #16a34a; color: #fff; border: none; border-r
```

```
adius: 999px; font-size: 15px; font-weight: 700; cursor: pointer; margin-top: 24px; }
        button:hover { background: #15803d; }
        .steps { background: #f9fafb; border: 1px solid #e5e7eb; border-radius: 8px; padding: 16px; m
argin-top: 20px; font-size: 13px; color: #4b5563; line-height: 1.7; }
        .steps strong { color: #111827; }
        .steps a { color: #2563eb; }
        .steps ol { padding-left: 18px; }
    </style>
</head>
<body>
    <div class="card">
        <div class="header">
            <h1>Farm<span>Link</span> — Email Setup</h1>
            <p>Configure Gmail SMTP to send verification emails</p>
        </div>
        <div class="body">
            <!-- Current Status -->
            <?php if ($isConfigured): ?>
                <div class="status configured">
                    ✅ SMTP is configured — Sender: <strong><?= htmlspecialchars($currentUser) ?></s
trong>
                </div>
            <?php else: ?>
                <div class="status not-configured">
                    ❌ SMTP is NOT configured — Emails are only saved to local log, not sent to Gmai
l!
                </div>
            <?php endif; ?>

            <!-- Result Message -->
            <?php if ($message): ?>
                <div class="msg <?= $messageType ?>"><?= $message ?></div>
            <?php endif; ?>

            <!-- Setup Form -->
            <form method="POST">
                <label>Your Gmail Address</label>
                <input type="email" name="gmail" value="<?= htmlspecialchars($currentUser !== 'yourad
dress@gmail.com' ? $currentUser : '') ?>" placeholder="e.g. ravanam913@gmail.com" required>
                <span class="help">This Gmail will be used to send all FarmLink emails (verification,
passwords, notifications).</span>

                <label>Gmail App Password (16 characters)</label>
                <input type="text" name="app_password" placeholder="e.g. abcd efgh ijkl mnop" require
d>
                <span class="help">Generate at: <a href="https://myaccount.google.com/apppasswords" t
arget="_blank">myaccount.google.com/apppasswords</a> (spaces will be removed automatically)</span>

                <button type="submit">💌 Save & Test Email Connection</button>
            </form>

            <!-- Instructions -->
            <div class="steps">
                <strong>How to get your Gmail App Password:</strong>
                <ol>
                    <li>Go to <a href="https://myaccount.google.com/security" target="_blank">Google
Account Security</a></li>
                    <li>Turn on <strong>2-Step Verification</strong> (if not already on)</li>
                    <li>Go to <a href="https://myaccount.google.com/apppasswords" target="_blank">App
Passwords</a></li>
                    <li>Type app name: <strong>FarmLink</strong> → Click <strong>Create</strong></li>
                    <li>Copy the 16-character password and paste it above</li>
```

```
                </ol>
            </div>
        </div>
    </div>
</body>
</html>
```

CHAPTER 10: APPENDICES

SECTION 2: DATABASE SQL SCHEMA & MIGRATION SCRIPTS

backend/sql/schema.sql (Complete MySQL Database Schema - 7 Tables)

```
-- ============================================================
-- FarmLink — Full Database Schema
-- Reconstructed from backend/api/**/*.php (schema.sql was missing
-- from the project). Includes the Firebase columns from
-- 2026_add_firebase_columns.sql already merged in, so you do NOT
-- need to run that migration file separately after this.
-- ============================================================

CREATE DATABASE IF NOT EXISTS farmlink
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

USE farmlink;

-- ------------------------------------------------------------
-- users
-- role: buyer | farmer | admin
-- status: pending | approved | rejected  (farmers need approval)
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS users (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    name            VARCHAR(100)    NOT NULL,
    mobile          VARCHAR(10)     NOT NULL,
    email           VARCHAR(150)    NULL,
    password        VARCHAR(255)    NOT NULL,
    firebase_uid    VARCHAR(128)    NULL DEFAULT NULL,
    aadhaar         VARCHAR(20)     NULL,
    role            ENUM('buyer','farmer','admin') NOT NULL DEFAULT 'buyer',
    is_verified     TINYINT(1)      NOT NULL DEFAULT 0,
    email_verified  TINYINT(1)      NOT NULL DEFAULT 0,
    status          ENUM('pending','approved','rejected') NOT NULL DEFAULT 'approved',
    admin_remark    VARCHAR(255)    NULL DEFAULT NULL,
    created_at      TIMESTAMP       NOT NULL DEFAULT CURRENT_TIMESTAMP,

    UNIQUE KEY uniq_mobile (mobile),
    UNIQUE KEY uniq_email (email),
    UNIQUE KEY uniq_firebase_uid (firebase_uid)
) ENGINE=InnoDB;

-- ------------------------------------------------------------
-- pending_registrations (staging table: only verified emails enter users)
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS pending_registrations (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    name            VARCHAR(100)    NOT NULL,
    mobile          VARCHAR(10)     NOT NULL,
    email           VARCHAR(150)    NOT NULL,
    password_hash   VARCHAR(255)    NOT NULL,
    auto_password   VARCHAR(50)     NULL,
    role            ENUM('buyer','farmer') NOT NULL,
    aadhaar         VARCHAR(20)     NULL,
    token           VARCHAR(64)     NOT NULL UNIQUE,
    otp_code        VARCHAR(6)      NOT NULL,
    status          ENUM('pending','verified','expired') NOT NULL DEFAULT 'pending',
    created_at      TIMESTAMP       NOT NULL DEFAULT CURRENT_TIMESTAMP,
    expires_at      TIMESTAMP       NOT NULL,
    INDEX (email),
    INDEX (mobile),
    INDEX (token),
    INDEX (otp_code)
```

```
) ENGINE=InnoDB;

-- ------------------------------------------------------------
-- categories
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS categories (
    id      INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    name    VARCHAR(100) NOT NULL UNIQUE
) ENGINE=InnoDB;

INSERT INTO categories (name) VALUES
    ('Vegetables'),
    ('Fruits'),
    ('Grains'),
    ('Dairy'),
    ('Spices')
ON DUPLICATE KEY UPDATE name = VALUES(name);

-- ------------------------------------------------------------
-- products
-- status: active | inactive
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS products (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    farmer_id       INT UNSIGNED NOT NULL,
    category_id     INT UNSIGNED NOT NULL,
    name            VARCHAR(150) NOT NULL,
    base_price      DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    price_per_kg    DECIMAL(10,2) NOT NULL,
    gst_percent     DECIMAL(5,2) NOT NULL DEFAULT 5.00,
    gst_amount      DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    commission_percent DECIMAL(5,2) NOT NULL DEFAULT 5.00,
    commission_amount  DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    stock_kg        DECIMAL(10,2) NOT NULL DEFAULT 0,
    image           VARCHAR(255) NULL,
    description     TEXT NULL,
    status          ENUM('active','inactive') NOT NULL DEFAULT 'active',
    approval_status ENUM('pending','approved','rejected') NOT NULL DEFAULT 'pending',
    admin_remark    VARCHAR(255) NULL DEFAULT NULL,
    created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_products_farmer  FOREIGN KEY (farmer_id)   REFERENCES users(id)      ON DELETE CASC
ADE,
    CONSTRAINT fk_products_category FOREIGN KEY (category_id) REFERENCES categories(id) ON DELETE RES
TRICT
) ENGINE=InnoDB;

-- ------------------------------------------------------------
-- orders
-- payment_mode: cod | online
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS orders (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    buyer_id        INT UNSIGNED NOT NULL,
    total_amount    DECIMAL(10,2) NOT NULL DEFAULT 0,
    payment_mode    ENUM('cod','online') NOT NULL DEFAULT 'cod',
    address         TEXT NOT NULL,
    status          VARCHAR(30) NOT NULL DEFAULT 'placed',
    created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_orders_buyer FOREIGN KEY (buyer_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

```
-- ------------------------------------------------------------
-- order_items
-- item_status: pending | confirmed | shipped | delivered | cancelled
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS order_items (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    order_id        INT UNSIGNED NOT NULL,
    product_id      INT UNSIGNED NOT NULL,
    farmer_id       INT UNSIGNED NOT NULL,
    quantity_kg     DECIMAL(10,2) NOT NULL,
    price_per_kg    DECIMAL(10,2) NOT NULL,
    subtotal        DECIMAL(10,2) NOT NULL,
    item_status     ENUM('pending','confirmed','shipped','delivered','cancelled') NOT NULL DEFAULT 'p
ending',

    CONSTRAINT fk_items_order   FOREIGN KEY (order_id)   REFERENCES orders(id)   ON DELETE CASCADE,
    CONSTRAINT fk_items_product FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE RESTRICT,
    CONSTRAINT fk_items_farmer  FOREIGN KEY (farmer_id)  REFERENCES users(id)    ON DELETE RESTRICT
) ENGINE=InnoDB;

-- ------------------------------------------------------------
-- bulk_requests
-- status: pending | approved | rejected
-- ------------------------------------------------------------
CREATE TABLE IF NOT EXISTS bulk_requests (
    id              INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    buyer_id        INT UNSIGNED NOT NULL,
    product_id      INT UNSIGNED NOT NULL,
    quantity_kg     DECIMAL(10,2) NOT NULL,
    reason          TEXT NOT NULL,
    status          ENUM('pending','approved','rejected') NOT NULL DEFAULT 'pending',
    remark          TEXT NULL,
    created_at      TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_bulk_buyer   FOREIGN KEY (buyer_id)   REFERENCES users(id)    ON DELETE CASCADE,
    CONSTRAINT fk_bulk_product FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

backend/sql/2026_add_admin_approval.sql (Admin Approval Migration)

```
-- ============================================================
-- FarmLink — Admin approval workflow migration
-- Run this ONCE against your existing `farmlink` database
-- (phpMyAdmin -> farmlink -> SQL tab -> paste -> Go).
-- ============================================================

USE farmlink;

-- Products now need admin approval before they show up in the public
-- Shop. Farmers can still see their own products (any status) on the
-- "My Products" page.
ALTER TABLE products
  ADD COLUMN approval_status ENUM('pending','approved','rejected') NOT NULL DEFAULT 'pending' AFTER s
tatus,
  ADD COLUMN admin_remark VARCHAR(255) NULL DEFAULT NULL AFTER approval_status;

-- Existing products (added before this migration) are auto-approved so
-- your current Shop listing doesn't go empty after you run this.
UPDATE products SET approval_status = 'approved' WHERE approval_status = 'pending';

-- Optional remark for why a farmer signup was rejected (shown to admin later).
ALTER TABLE users
  ADD COLUMN admin_remark VARCHAR(255) NULL DEFAULT NULL AFTER status;
```

backend/sql/2026_add_firebase_columns.sql (Firebase Columns Migration)

```
-- Run once against your existing FarmLink database before deploying the
-- Firebase-backed OTP/email verification flow.

ALTER TABLE users
  ADD COLUMN firebase_uid VARCHAR(128) NULL DEFAULT NULL AFTER password,
  ADD COLUMN email_verified TINYINT(1) NOT NULL DEFAULT 0 AFTER is_verified,
  ADD UNIQUE KEY uniq_firebase_uid (firebase_uid);
```

CHAPTER 10: APPENDICES

SECTION 3: FRONTEND NEXT.JS 14 / REACT 18 TYPESCRIPT SOURCE CODE

app/layout.tsx (Root Layout & Global SEO Metadata)

```
import type { Metadata } from "next";
import { Fraunces, Work_Sans, IBM_Plex_Mono } from "next/font/google";
import { CartProvider } from "./lib/cart-context";
import PageLoader from "./components/PageLoader";
import "./globals.css";

const fraunces = Fraunces({
  subsets: ["latin"],
  variable: "--font-fraunces",
  weight: ["500", "600", "900"],
  style: ["normal", "italic"],
  display: "swap",
});

const workSans = Work_Sans({
  subsets: ["latin"],
  variable: "--font-worksans",
  weight: ["400", "500", "600"],
  display: "swap",
});

const plexMono = IBM_Plex_Mono({
  subsets: ["latin"],
  variable: "--font-plexmono",
  weight: ["400", "500"],
  display: "swap",
});

export const metadata: Metadata = {
  metadataBase: new URL('https://agrenza.com'),
  title: {
    template: '%s | Agrenza',
    default: 'Agrenza — Farm to Doorstep',
  },
  description: 'Buy fresh vegetables, fruits, and farm produce directly from verified Indian farmers
at transparent GST-inclusive prices. No middlemen — Agrenza.',
  keywords: ['fresh vegetables online India', 'buy from farmers', 'farm to home delivery', 'organic p
roduce online', 'no middlemen marketplace', 'fresh fruits home delivery'],
  authors: [{ name: 'Agrenza Team' }],
  creator: 'Agrenza',
  publisher: 'Agrenza',
  robots: { index: true, follow: true, googleBot: { index: true, follow: true } },
  openGraph: {
    type: 'website',
    locale: 'en_IN',
    url: 'https://agrenza.com',
    siteName: 'Agrenza',
    title: 'Agrenza — Farm to Doorstep',
    description: 'Buy fresh vegetables, fruits, and farm produce directly from verified Indian farmer
s. No middlemen — fair prices, fast delivery.',
    images: [{ url: '/agrenza-logo-small.jpg', width: 96, height: 96, alt: 'Agrenza Logo' }],
  },
  twitter: {
    card: 'summary',
    title: 'Agrenza — Farm to Doorstep',
    description: 'Fresh farm produce delivered directly to your door. No middlemen, fair prices.',
    images: ['/agrenza-logo-small.jpg'],
  },
  icons: {
```

```
    icon: '/agrenza-logo-small.jpg',
    apple: '/agrenza-logo-small.jpg',
  },
};

export default function RootLayout({ children }: { children: React.ReactNode }) {
  return (
    <html lang="en">
      <body className={`${fraunces.variable} ${workSans.variable} ${plexMono.variable}`}>
        <PageLoader />
        <CartProvider>{children}</CartProvider>
      </body>
    </html>
  );
}
```

app/page.tsx (Home Page Main Master Layout)

```
import type { Metadata } from "next";
import Nav from "./components/Nav";
import CrateHero from "./components/CrateHero";
import MandiTicker from "./components/MandiTicker";
import Journey from "./components/Journey";

import ProductGrid from "./components/ProductGrid";
import Footer from "./components/Footer";

export const metadata: Metadata = {
  title: "Agrenza — Farm to Doorstep | Buy Fresh Farm Produce Online",
  description:
    "Buy fresh vegetables, fruits and farm produce directly from verified Indian farmers. No middleme
n, fair prices, GST inclusive. Farm to doorstep delivery across India.",
  openGraph: {
    title: "Agrenza — Farm to Doorstep",
    description:
      "Buy fresh vegetables, fruits and farm produce directly from verified Indian farmers. No middle
men, fair prices, fast delivery.",
    url: "https://agrenza.com",
    type: "website",
  },
};

const orgSchema = {
  "@context": "https://schema.org",
  "@type": "Organization",
  name: "Agrenza",
  url: "https://agrenza.com",
  logo: "https://agrenza.com/agrenza-logo-small.jpg",
  description:
    "Direct farm-to-doorstep fresh produce marketplace connecting Indian farmers with buyers",
  contactPoint: {
    "@type": "ContactPoint",
    contactType: "customer support",
    email: "support@agrenza.com",
    telephone: "+91-1800-000-000",
    availableLanguage: ["English", "Hindi"],
  },
  sameAs: [
    "https://www.facebook.com/agrenza",
    "https://www.instagram.com/agrenza",
    "https://twitter.com/agrenza",
  ],
};

export default function Home() {
  return (
    <>
      <script
        type="application/ld+json"
        dangerouslySetInnerHTML={{ __html: JSON.stringify(orgSchema) }}
      />
      <Nav />
      <MandiTicker />

      <section className="hero">
        <div className="container hero-inner">
          <div className="hero-copy">
```

```
        <p className="eyebrow">Fresh harvest, listed daily</p>
        <h1>
          Farm to<br />
          <em>doorstep</em>.
        </h1>
        <p className="lead">
          Agrenza connects farmers directly with buyers — fresh produce,
          fair prices, and nobody standing in between.
        </p>
        <div className="hero-ctas">
          <a href="/shop" className="btn-primary">Shop now</a>
          <a href="/auth/farmer-register" className="btn-outline">Become a farmer partner</a>
        </div>
      </div>
      <div className="hero-scene">
        <CrateHero />
      </div>
    </div>
  </section>

  <Journey />
  <ProductGrid />
  <Footer />

  <style>{`
    .hero {
      background: linear-gradient(180deg, #ffffff 0%, #f7faf7 100%);
      color: #16241a;
      padding: 64px 0 40px;
      border-bottom: 1px solid rgba(46, 125, 50, 0.12);
    }
    .hero-inner {
      display: grid;
      grid-template-columns: 1.1fr 0.9fr;
      align-items: center;
      gap: 24px;
      min-height: 480px;
    }
    .eyebrow {
      font-family: var(--font-mono);
      text-transform: uppercase;
      letter-spacing: 0.16em;
      font-size: 0.8rem;
      color: #2e7d32;
      font-weight: 600;
      margin: 0 0 18px;
    }
    h1 {
      font-family: var(--font-display);
      font-weight: 900;
      font-size: clamp(3rem, 7vw, 5.2rem);
      line-height: 0.98;
      color: #16241a;
      margin: 0 0 22px;
    }
    h1 em { color: #2e7d32; font-style: italic; font-weight: 600; }
    .lead {
      font-size: 1.08rem;
      line-height: 1.6;
      color: #4a5548;
      max-width: 460px;
      margin: 0 0 34px;
```

```
        }
        .hero-ctas { display: flex; gap: 14px; flex-wrap: wrap; }
        .btn-primary {
          background: #2e7d32;
          color: #ffffff !important;
          font-weight: 600;
          padding: 14px 28px;
          border-radius: 999px;
          box-shadow: 0 4px 14px rgba(46, 125, 50, 0.28);
          transition: all 0.2s ease;
        }
        .btn-primary:hover {
          background: #256628;
          box-shadow: 0 6px 18px rgba(46, 125, 50, 0.38);
        }
        .btn-outline {
          border: 1.5px solid #16241a;
          color: #16241a !important;
          font-weight: 600;
          padding: 14px 28px;
          border-radius: 999px;
          transition: all 0.2s ease;
        }
        .btn-outline:hover {
          background: #16241a;
          color: #ffffff !important;
        }
        @media (max-width: 860px) {
          .hero-inner { grid-template-columns: 1fr; text-align: center; }
          .lead { margin-left: auto; margin-right: auto; }
          .hero-ctas { justify-content: center; }
          .hero-scene { min-height: 260px; }
        }
      `}</style>

      <script
        type="application/ld+json"
        dangerouslySetInnerHTML={{
          __html: JSON.stringify({
            '@context': 'https://schema.org',
            '@type': 'Organization',
            name: 'Agrenza',
            url: 'https://agrenza.com',
            logo: 'https://agrenza.com/agrenza-logo-small.jpg',
            description: 'Direct farm-to-doorstep fresh produce marketplace connecting Indian farmers
with buyers',
            contactPoint: {
              '@type': 'ContactPoint',
              contactType: 'customer support',
              email: 'support@agrenza.com',
            },
            sameAs: [
              'https://www.facebook.com/agrenza',
              'https://www.instagram.com/agrenza',
            ],
          }),
        }}
      />
    </>
  );
}
```

app/components/Nav.tsx (Top Offer Bar & Sticky Navigation Header)

```
"use client";

import { useEffect, useState, useRef } from "react";
import Link from "next/link";
import { useRouter } from "next/navigation";
import { useCart } from "@/app/lib/cart-context";
import { getMe, logoutUser, AuthUser } from "@/app/lib/api";

export default function Nav() {
  const router = useRouter();
  const { count } = useCart();
  const [user, setUser] = useState<AuthUser | null>(null);
  const [menuOpen, setMenuOpen] = useState(false);
  const menuRef = useRef<HTMLDivElement>(null);

  useEffect(() => {
    (async () => {
      try {
        const me = await getMe();
        if (me?.user) setUser(me.user);
      } catch {
        setUser(null);
      }
    })();
  }, []);

  // Close dropdown on click outside
  useEffect(() => {
    function handleClickOutside(e: MouseEvent) {
      if (menuRef.current && !menuRef.current.contains(e.target as Node)) {
        setMenuOpen(false);
      }
    }
    document.addEventListener("mousedown", handleClickOutside);
    return () => document.removeEventListener("mousedown", handleClickOutside);
  }, []);

  async function handleLogout() {
    try {
      await logoutUser();
      setUser(null);
      setMenuOpen(false);
      router.push("/auth/login");
    } catch {
      setUser(null);
      router.push("/auth/login");
    }
  }

  return (
    <>
      {/* TOP ANNOUNCEMENT BANNER FOR BUYER OFFERS */}
      <div className="top-offer-bar">
        <div className="container top-bar-inner">
          <span className="offer-pulse">🎉</span>
          <span className="offer-text">
            <strong>Harvest Season Offer:</strong> Get <strong>10% OFF</strong> with coupon <span cla
ssName="highlight-code">FRESHFARM10</span> &bull; Free Delivery on orders over ₹499!
          </span>
```

```
          <Link href="/offers" className="top-offer-link">
            View All Offers &amp;amp; Discounts →
          </Link>
        </div>
      </div>

      {/* 3. NAVIGATION BAR */}
      <header className="nav">
        <div className="container nav-inner">
          <Link href="/" className="nav-logo">
            <img src="/agrenza-logo-small.jpg" alt="Agrenza Logo" width={38} height={38} className="n
av-logo-img" style={{ width: "38px", height: "38px" }} />
            <span>Agren<span>za</span></span>
          </Link>

          <nav className="nav-links">
            <Link href="/shop" className="nav-item">Shop</Link>
            <Link href="/offers" className="nav-item nav-offer-badge">
              <span>🎁 Offers</span>
              <span className="chip-discount">10% OFF</span>
            </Link>
            <Link href="/about" className="nav-item">About</Link>
            {user?.role === "buyer" && (
              <>
                <Link href="/buyer/dashboard" className="nav-item">Dashboard</Link>
                <Link href="/buyer/orders" className="nav-item">My Orders</Link>
              </>
            )}
            {user?.role === "farmer" && (
              <Link href="/farmer/dashboard" className="nav-item">Farmer Dashboard</Link>
            )}
            {user?.role === "admin" && (
              <Link href="/admin/dashboard" className="nav-item">Admin Panel</Link>
            )}
            {!user && (
              <Link href="/auth/farmer-register" className="nav-item">Sell on Agrenza</Link>
            )}
          </nav>

          <div className="nav-actions">
            {/* CART LINK */}
            <Link href="/buyer/cart" className="cart-link" aria-label="Shopping Cart">
              🛒 Cart
              {count > 0 && <span className="cart-count">{count}</span>}
            </Link>

            {/* LOGGED IN USER PROFILE DROPDOWN */}
            {user ? (
              <div className="profile-dropdown-container" ref={menuRef}>
                <button
                  type="button"
                  className="profile-pill-btn"
                  onClick={() => setMenuOpen(!menuOpen)}
                  aria-expanded={menuOpen}
                >
                  <div className="nav-avatar">
                    {user.name.charAt(0).toUpperCase()}
                    <span className="online-indicator" />
                  </div>
                  <div className="user-titles">
                    <span className="user-nav-name">{user.name}</span>
                    <span className="user-nav-role">{user.role.toUpperCase()}</span>
```

```
                  </div>
                  <span className="chevron-icon">{menuOpen ? "▲" : "▼"}</span>
                </button>

                {menuOpen && (
                  <div className="dropdown-card">
                    <div className="dropdown-user-header">
                      <strong>{user.name}</strong>
                      <span className="user-dropdown-email">{user.email || user.mobile || "Verified A
ccount"}</span>
                      <span className="user-role-badge">✓ {user.role === "buyer" ? "Verified Buyer" :
user.role.toUpperCase()}</span>
                    </div>

                    <div className="dropdown-menu-list">
                      {user.role === "buyer" && (
                        <>
                          <Link
                            href="/buyer/dashboard"
                            className="dropdown-item"
                            onClick={() => setMenuOpen(false)}
                          >
                            <span className="d-icon">📊</span>
                            <div>
                              <strong>Buyer Dashboard</strong>
                              <small>Profile, stats &amp;amp; wallet</small>
                            </div>
                          </Link>

                          <Link
                            href="/buyer/orders"
                            className="dropdown-item"
                            onClick={() => setMenuOpen(false)}
                          >
                            <span className="d-icon">📦</span>
                            <div>
                              <strong>My Orders &amp;amp; Live Tracking</strong>
                              <small>Check shipping travel status</small>
                            </div>
                          </Link>

                          <Link
                            href="/offers"
                            className="dropdown-item"
                            onClick={() => setMenuOpen(false)}
                          >
                            <span className="d-icon">🏷️</span>
                            <div>
                              <strong>Discounts &amp;amp; Offers</strong>
                              <small>Active coupon promo codes</small>
                            </div>
                          </Link>

                          <Link
                            href="/buyer/bulk-request"
                            className="dropdown-item"
                            onClick={() => setMenuOpen(false)}
                          >
                            <span className="d-icon">🧺</span>
                            <div>
                              <strong>Bulk Produce Quote</strong>
                              <small>Wholesale direct from farms</small>
```

```
                      </div>
                    </Link>
                  </>
                )}

                {user.role === "farmer" && (
                  <Link
                    href="/farmer/dashboard"
                    className="dropdown-item"
                    onClick={() => setMenuOpen(false)}
                  >
                    <span className="d-icon">🚜</span>
                    <div>
                      <strong>Farmer Dashboard</strong>
                      <small>Manage crops &amp;amp; stock</small>
                    </div>
                  </Link>
                )}

                {user.role === "admin" && (
                  <Link
                    href="/admin/dashboard"
                    className="dropdown-item"
                    onClick={() => setMenuOpen(false)}
                  >
                    <span className="d-icon">⚡</span>
                    <div>
                      <strong>Admin Dashboard</strong>
                      <small>Manage orders &amp;amp; approvals</small>
                    </div>
                  </Link>
                )}

                <div className="dropdown-divider" />

                <button
                  type="button"
                  className="dropdown-item logout-btn"
                  onClick={handleLogout}
                >
                  <span className="d-icon">🚪</span>
                  <strong>Log Out</strong>
                </button>
              </div>
            </div>
          )}
        </div>
      ) : (
        /* GUEST BUTTONS */
        <>
          <Link href="/auth/login" className="btn-ghost">
            Log in
          </Link>
          <Link href="/auth/buyer-register" className="btn-solid">
            Sign up
          </Link>
        </>
      )}
    </div>
  </div>
</header>
```

```
      <style jsx>{`
        /* Top Offer Bar */
        .top-offer-bar {
          background: linear-gradient(90deg, #1b3821 0%, #15803d 50%, #1b3821 100%);
          color: #ffffff;
          font-size: 0.78rem;
          padding: 7px 0;
          border-bottom: 1px solid rgba(255,255,255,0.1);
        }
        .top-bar-inner {
          display: flex;
          justify-content: center;
          align-items: center;
          gap: 12px;
          flex-wrap: wrap;
          text-align: center;
        }
        .offer-pulse { animation: bounce 1.5s infinite; }
        @keyframes bounce {
          0%, 100% { transform: translateY(0); }
          50% { transform: translateY(-3px); }
        }
        .highlight-code {
          background: #fde047;
          color: #713f12;
          font-weight: 800;
          font-family: var(--font-mono);
          padding: 1px 6px;
          border-radius: 4px;
          margin: 0 4px;
        }
        .top-offer-link {
          color: #86efac;
          font-weight: 700;
          text-decoration: underline;
          margin-left: 6px;
        }
        .top-offer-link:hover { color: #ffffff; }

        /* Nav Header */
        .nav {
          position: sticky;
          top: 0;
          z-index: 100;
          background: #ffffff;
          border-bottom: 1px solid rgba(0, 0, 0, 0.08);
          box-shadow: 0 2px 10px rgba(0, 0, 0, 0.04);
        }
        .nav-inner {
          display: flex;
          align-items: center;
          justify-content: space-between;
          height: 68px;
        }
        .nav-logo {
          font-family: var(--font-display);
          font-weight: 900;
          font-size: 1.45rem;
          color: #16241a;
          text-decoration: none;
          letter-spacing: -0.02em;
          display: flex;
```

```
  align-items: center;
  gap: 10px;
}
.nav-logo-img {
  width: 38px;
  height: 38px;
  border-radius: 50%;
  object-fit: cover;
  border: 1.5px solid #16a34a;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}
.nav-logo span { color: #2e7d32; }

.nav-links {
  display: flex;
  align-items: center;
  gap: 24px;
  font-size: 0.92rem;
}
.nav-item {
  color: #2c3e2d;
  font-weight: 500;
  text-decoration: none;
  transition: color 0.15s;
  display: inline-flex;
  align-items: center;
  gap: 6px;
}
.nav-item:hover { color: #2e7d32; }

.nav-offer-badge {
  background: #f0fdf4;
  border: 1px solid #bbf7d0;
  color: #166534;
  padding: 4px 10px;
  border-radius: 999px;
  font-weight: 600;
  font-size: 0.84rem;
}
.chip-discount {
  background: #16a34a;
  color: #ffffff;
  font-size: 0.68rem;
  font-weight: 800;
  padding: 1px 6px;
  border-radius: 999px;
}

.nav-actions {
  display: flex;
  align-items: center;
  gap: 14px;
}

.cart-link {
  color: #2c3e2d;
  font-size: 0.9rem;
  font-weight: 600;
  padding: 7px 12px;
  border-radius: 8px;
  text-decoration: none;
  display: inline-flex;
```

```
          align-items: center;
          border: 1px solid #e2e8f0;
          transition: all 0.15s;
        }
        .cart-link:hover { border-color: #16a34a; color: #16a34a; background: #f0fdf4; }
        .cart-count {
          background: #16a34a;
          color: #ffffff;
          font-size: 0.72rem;
          font-weight: 800;
          padding: 1px 7px;
          border-radius: 999px;
          margin-left: 6px;
        }

        /* Profile Pill & Dropdown */
        .profile-dropdown-container { position: relative; }
        .profile-pill-btn {
          display: flex;
          align-items: center;
          gap: 10px;
          background: #f8fafc;
          border: 1px solid #cbd5e1;
          padding: 5px 12px 5px 6px;
          border-radius: 999px;
          cursor: pointer;
          transition: all 0.15s;
        }
        .profile-pill-btn:hover { border-color: #16a34a; background: #f0fdf4; }

        .nav-avatar {
          width: 32px; height: 32px; border-radius: 50%;
          background: #16a34a; color: #ffffff;
          font-weight: 800; font-size: 0.88rem;
          display: flex; align-items: center; justify-content: center;
          position: relative;
        }
        .online-indicator {
          width: 8px; height: 8px; border-radius: 50%;
          background: #22c55e; border: 1.5px solid #ffffff;
          position: absolute; bottom: -1px; right: -1px;
        }
        .user-titles { display: flex; flex-direction: column; text-align: left; line-height: 1.1; }
        .user-nav-name { font-size: 0.82rem; font-weight: 700; color: #0f172a; max-width: 110px; whit
e-space: nowrap; overflow: hidden; text-overflow: ellipsis; }
        .user-nav-role { font-size: 0.65rem; color: #16a34a; font-weight: 800; letter-spacing: 0.04e
m; }
        .chevron-icon { font-size: 0.6rem; color: #64748b; }

        /* Dropdown Card */
        .dropdown-card {
          position: absolute;
          right: 0;
          top: 48px;
          width: 270px;
          background: #ffffff;
          border: 1px solid #e2e8f0;
          border-radius: 12px;
          box-shadow: 0 10px 30px rgba(0,0,0,0.12);
          overflow: hidden;
          z-index: 200;
          animation: dropFade 0.15s ease-out;
```

```
        }
        @keyframes dropFade {
          from { opacity: 0; transform: translateY(-6px); }
          to { opacity: 1; transform: translateY(0); }
        }

        .dropdown-user-header {
          padding: 16px;
          background: #f8fafc;
          border-bottom: 1px solid #e2e8f0;
          display: flex;
          flex-direction: column;
          gap: 2px;
        }
        .dropdown-user-header strong { font-size: 0.95rem; color: #0f172a; }
        .user-dropdown-email { font-size: 0.78rem; color: #64748b; word-break: break-all; }
        .user-role-badge { display: inline-block; font-size: 0.68rem; font-weight: 800; color: #15803
d; background: #dcfce7; padding: 2px 6px; border-radius: 4px; margin-top: 4px; align-self: flex-star
t; }

        .dropdown-menu-list { padding: 6px 0; display: flex; flex-direction: column; }
        .dropdown-item {
          display: flex;
          align-items: center;
          gap: 12px;
          padding: 10px 16px;
          color: #334155;
          text-decoration: none;
          font-size: 0.85rem;
          transition: background 0.1s;
          border: none;
          background: none;
          width: 100%;
          text-align: left;
          cursor: pointer;
        }
        .dropdown-item:hover { background: #f1f5f9; color: #16a34a; }
        .dropdown-item div { display: flex; flex-direction: column; }
        .dropdown-item strong { font-size: 0.82rem; color: #0f172a; }
        .dropdown-item small { font-size: 0.7rem; color: #64748b; }
        .d-icon { font-size: 1.1rem; }

        .dropdown-divider { height: 1px; background: #e2e8f0; margin: 4px 0; }

        .logout-btn { color: #dc2626 !important; }
        .logout-btn:hover { background: #fef2f2 !important; }
        .logout-btn strong { color: #dc2626 !important; }

        @media (max-width: 768px) {
          .nav-links { display: none; }
        }
      `}</style>
    </>
  );
}
```

app/components/Footer.tsx (Footer Component & Navigation Links)

```
"use client";

import Link from "next/link";

export default function Footer() {
  return (
    <footer className="footer">
      <div className="container footer-inner">
        {/* Brand Column */}
        <div>
          <p className="logo">
            <img src="/agrenza-logo-small.jpg" alt="Agrenza Logo" width={32} height={32} className="f
ooter-logo-img" style={{ width: "32px", height: "32px" }} />
            <span>Agren<span>za</span></span>
          </p>
          <p className="tag">Connecting farmers directly with buyers.<br />Fresh produce, fair price
s, no middlemen.</p>
          <p className="tagline">🌱 Farm to Doorstep — Since 2026</p>
        </div>

        {/* Quick Links */}
        <div>
          <p className="head">Quick Links</p>
          <Link href="/shop">Shop Products</Link>
          <Link href="/offers">Offers & Discounts</Link>
          <Link href="/about">About Agrenza</Link>
          <Link href="/auth/farmer-register">Become a Farmer Partner</Link>
          <Link href="/auth/buyer-register">Create Buyer Account</Link>
        </div>

        {/* Support */}
        <div>
          <p className="head">Support</p>
          <Link href="/contact">Contact Us</Link>
          <a href="mailto:support@agrenza.com">support@agrenza.com</a>
          <a href="tel:1800000000">1800-000-000 (Toll Free)</a>
          <p className="hours">Mon–Sat · 9 AM to 6 PM IST</p>
        </div>

        {/* Legal */}
        <div>
          <p className="head">Legal</p>
          <Link href="/privacy-policy">Privacy Policy</Link>
          <Link href="/terms">Terms of Service</Link>
        </div>
      </div>

      <div className="footer-bottom container">
        <p className="rights">© 2026 Agrenza. All rights reserved.</p>
        <p className="rights-sub">A farm-to-doorstep marketplace · Made in India IN</p>
      </div>

      <style jsx>{`
        .footer {
          background: var(--soil);
          color: #d9d2bd;
          padding: 60px 0 24px;
          margin-top: 40px;
        }
```

```
.footer-inner {
  display: grid;
  grid-template-columns: 2fr 1fr 1fr 1fr;
  gap: 32px;
  padding-bottom: 40px;
  border-bottom: 1px solid rgba(232, 163, 61, 0.15);
}
.logo {
  font-family: var(--font-display);
  font-weight: 900;
  font-size: 1.3rem;
  color: #efe8d8;
  margin: 0 0 10px;
  display: flex;
  align-items: center;
  gap: 8px;
}
.footer-logo-img {
  width: 32px;
  height: 32px;
  border-radius: 50%;
  object-fit: cover;
  border: 1.5px solid var(--turmeric);
}
.logo span { color: var(--turmeric); }
.tag {
  font-size: 0.88rem;
  line-height: 1.6;
  color: #b7b09a;
  max-width: 260px;
}
.tagline {
  font-size: 0.8rem;
  color: #7a7364;
  margin-top: 8px;
}
.head {
  font-family: var(--font-mono);
  font-size: 0.75rem;
  text-transform: uppercase;
  letter-spacing: 0.1em;
  color: var(--turmeric);
  margin: 0 0 14px;
}
.footer-inner a {
  display: block;
  font-size: 0.88rem;
  margin-bottom: 9px;
  color: #d9d2bd;
  text-decoration: none;
}
.footer-inner a:hover { color: var(--turmeric); }
.hours {
  font-size: 0.78rem;
  color: #7a7364;
  margin-top: 6px;
}
.footer-bottom {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
```

```
          gap: 6px;
          padding-top: 20px;
        }
        .rights {
          font-size: 0.78rem;
          color: #8a836f;
          margin: 0;
        }
        .rights-sub {
          font-size: 0.75rem;
          color: #7a7364;
          margin: 0;
        }
        @media (max-width: 860px) {
          .footer-inner { grid-template-columns: 1fr 1fr; }
        }
        @media (max-width: 480px) {
          .footer-inner { grid-template-columns: 1fr; }
          .footer-bottom { flex-direction: column; text-align: center; }
        }
      `}</style>
    </footer>
  );
}
```

app/components/MandiTicker.tsx (Live Scrolling Market Prices Bar)

```
"use client";

const PRICES = [
  ["Tomato", "₹28/kg"], ["Spinach", "₹18/kg"], ["Carrot", "₹32/kg"],
  ["Green Chili", "₹45/kg"], ["Onion", "₹24/kg"], ["Potato", "₹20/kg"],
  ["Brinjal", "₹26/kg"], ["Okra", "₹34/kg"],
];

export default function MandiTicker() {
  const row = [...PRICES, ...PRICES];
  return (
    <div className="ticker">
      <div className="ticker-track">
        {row.map(([name, price], i) => (
          <span className="ticker-item" key={i}>
            <em>{name}</em> {price}
          </span>
        ))}
      </div>

      <style jsx>{`
        .ticker {
          background: var(--soil-deep);
          border-top: 1px solid rgba(232, 163, 61, 0.15);
          border-bottom: 1px solid rgba(232, 163, 61, 0.15);
          overflow: hidden;
          padding: 11px 0;
        }
        .ticker-track {
          display: flex;
          gap: 40px;
          white-space: nowrap;
          width: max-content;
          animation: scroll 34s linear infinite;
          font-family: var(--font-mono);
          font-size: 0.82rem;
          color: var(--turmeric);
        }
        .ticker-item em { color: #efe8d8; font-style: normal; margin-right: 8px; opacity: 0.7; }
        @keyframes scroll {
          from { transform: translateX(0); }
          to { transform: translateX(-50%); }
        }
      `}</style>
    </div>
  );
}
```

app/components/Produce3DShowcase.tsx (Spin the Stall 3D CSS Revolving Rack)

```
"use client";

import { useRef, useState, useEffect } from "react";
import Link from "next/link";

interface CategoryItem {
  name: string;
  count: string;
  image: string;
  emoji: string;
}

const STALL_CATEGORIES: CategoryItem[] = [
  {
    name: "Vegetables",
    count: "40+ items",
    emoji: "🥦",
    image: "https://images.unsplash.com/photo-1540420773420-3366772f4999?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Fruits",
    count: "25+ items",
    emoji: "🍎",
    image: "https://images.unsplash.com/photo-1619566636858-adf3ef46400b?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Spices & Herbs",
    count: "20+ items",
    emoji: "🌶️",
    image: "https://images.unsplash.com/photo-1596040033229-a9821ebd058d?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Leafy Greens",
    count: "15+ items",
    emoji: "🥬",
    image: "https://images.unsplash.com/photo-1576045057995-568f588f82fb?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Tubers & Roots",
    count: "12+ items",
    emoji: "🧅",
    image: "https://images.unsplash.com/photo-1518977676601-b53f82aba655?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Grains & Pulses",
    count: "18+ items",
    emoji: "🌾",
    image: "https://images.unsplash.com/photo-1586201375761-83865001e31c?auto=format&fit=crop&w=600&q
=80",
  },
];

export default function Produce3DShowcase() {
  const count = STALL_CATEGORIES.length;
```

```
const radius = 300;
const [rotation, setRotation] = useState(0);
const rotationRef = useRef(0);
const isPausedRef = useRef(false);
const isDraggingRef = useRef(false);
const dragStartXRef = useRef(0);
const dragStartRotRef = useRef(0);

/* ---- animation loop with intro burst ---- */
useEffect(() => {
  let animId: number;
  let lastTime = performance.now();
  const startTime = performance.now();
  const introMs = 2000;
  const introSpeed = 180;
  const normalSpeed = 22;

  function step(now: number) {
    const dt = (now - lastTime) / 1000;
    lastTime = now;
    if (!isPausedRef.current && !isDraggingRef.current) {
      const t = Math.min((now - startTime) / introMs, 1);
      const ease = t * t;
      const speed = introSpeed + (normalSpeed - introSpeed) * ease;
      rotationRef.current = (rotationRef.current + dt * speed) % 360;
      setRotation(rotationRef.current);
    }
    animId = requestAnimationFrame(step);
  }
  animId = requestAnimationFrame(step);
  return () => cancelAnimationFrame(animId);
}, []);

/* ---- drag handlers ---- */
function onPointerDown(e: React.PointerEvent<HTMLDivElement>) {
  isDraggingRef.current = true;
  isPausedRef.current = true;
  dragStartXRef.current = e.clientX;
  dragStartRotRef.current = rotationRef.current;
  (e.currentTarget as HTMLElement).setPointerCapture?.(e.pointerId);
}
function onPointerMove(e: React.PointerEvent<HTMLDivElement>) {
  if (!isDraggingRef.current) return;
  const delta = e.clientX - dragStartXRef.current;
  rotationRef.current = dragStartRotRef.current + delta * 0.4;
  setRotation(rotationRef.current);
}
function onPointerUp(e: React.PointerEvent<HTMLDivElement>) {
  isDraggingRef.current = false;
  isPausedRef.current = false;
  (e.currentTarget as HTMLElement).releasePointerCapture?.(e.pointerId);
}

return (
  <section className="spin-section">
    {/* ---- header ---- */}
    <div className="spin-head">
      <p className="spin-eyebrow">Straight from the crate</p>
      <h2 className="spin-title">Spin the stall</h2>
      <p className="spin-lead">
        Drag to look around — every card is a category listed by local farmers.
      </p>
```

```
      </div>

      {/* ---- 3D wrapper ---- */}
      <div className="scene-wrapper">
        <div
          className="scene"
          onMouseEnter={() => { isPausedRef.current = true; }}
          onMouseLeave={() => { isPausedRef.current = false; }}
          onPointerDown={onPointerDown}
          onPointerMove={onPointerMove}
          onPointerUp={onPointerUp}
          onPointerCancel={onPointerUp}
        >
          <div
            className="carousel"
            style={{ transform: `translateZ(${-radius}px) rotateY(${rotation}deg)` }}
          >
            {STALL_CATEGORIES.map((cat, i) => {
              const angle = (360 / count) * i;
              return (
                <Link
                  href={`/shop?category=${encodeURIComponent(cat.name)}`}
                  key={cat.name}
                  className="card"
                  style={{ transform: `rotateY(${angle}deg) translateZ(${radius}px)` }}
                >
                  <div className="card-img">
                    <img src={cat.image} alt={cat.name} loading="lazy" />
                  </div>
                  <span className="card-name">{cat.emoji} {cat.name}</span>
                  <span className="card-count">{cat.count}</span>
                </Link>
              );
            })}
          </div>
        </div>
      </div>

      <style jsx>{`
        /* ---- Section ---- */
        .spin-section {
          padding: 30px 20px 60px;
          max-width: 1100px;
          margin: 0 auto;
          text-align: center;
        }

        /* ---- Header ---- */
        .spin-head {
          margin-bottom: 24px;
          animation: fadeUp 0.7s ease-out both;
        }
        .spin-eyebrow {
          font-family: var(--font-mono, monospace);
          text-transform: uppercase;
          letter-spacing: 0.14em;
          font-size: 0.75rem;
          color: var(--rust, #c0622a);
          margin: 0 0 6px;
        }
        .spin-title {
          font-family: var(--font-display, serif);
```

```
  font-weight: 600;
  font-size: clamp(1.5rem, 3vw, 2rem);
  margin: 0 0 6px;
  color: var(--soil, #2c3a2a);
}
.spin-lead {
  color: #6b6555;
  font-size: 0.9rem;
  margin: 0;
}

/* ---- 3D wrapper provides perspective ---- */
.scene-wrapper {
  perspective: 900px;
  perspective-origin: 50% 50%;
  overflow: visible;
}

/* ---- Scene: interaction area ---- */
.scene {
  width: 100%;
  height: 260px;
  display: flex;
  align-items: center;
  justify-content: center;
  cursor: grab;
  touch-action: pan-y;
  user-select: none;
  overflow: visible;
}
.scene:active {
  cursor: grabbing;
}

/* ---- Carousel ring ---- */
.carousel {
  width: 120px;
  height: 180px;
  position: relative;
  transform-style: preserve-3d;
  animation: popIn 0.8s ease-out 0.2s both;
}

/* ---- Each card ---- */
.card {
  position: absolute;
  top: 0;
  left: 0;
  width: 120px;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 4px;
  text-decoration: none;
  color: inherit;
  backface-visibility: hidden;
}
.card-img {
  width: 100px;
  height: 100px;
  border-radius: 50%;
  overflow: hidden;
```

```
          border: 3px solid var(--paper, #faf8f3);
          box-shadow: 0 8px 20px rgba(22, 36, 26, 0.22);
          background: linear-gradient(160deg, #f7f2e6, #ede4cd);
          flex-shrink: 0;
        }
        .card-img img {
          width: 100%;
          height: 100%;
          object-fit: cover;
          display: block;
        }
        .card-name {
          font-family: var(--font-mono, monospace);
          font-size: 0.68rem;
          letter-spacing: 0.03em;
          background: var(--soil, #2c3a2a);
          color: var(--paper, #faf8f3);
          padding: 2px 10px;
          border-radius: 999px;
          white-space: nowrap;
        }
        .card-count {
          font-family: var(--font-mono, monospace);
          font-size: 0.65rem;
          color: var(--rust, #c0622a);
          font-weight: 600;
        }

        /* ---- Animations ---- */
        @keyframes fadeUp {
          from { opacity: 0; transform: translateY(20px); }
          to   { opacity: 1; transform: translateY(0); }
        }
        @keyframes popIn {
          from { opacity: 0; transform: scale(0.6) translateZ(${-radius}px); }
          to   { opacity: 1; transform: scale(1); }
        }

        /* ---- Responsive ---- */
        @media (max-width: 700px) {
          .scene { height: 220px; }
          .card-img { width: 80px; height: 80px; }
          .carousel { width: 100px; height: 150px; }
          .card { width: 100px; }
          .card-name { font-size: 0.6rem; padding: 2px 7px; }
        }

        @media (prefers-reduced-motion: reduce) {
          .carousel, .spin-head { animation: none; }
        }
      `}</style>
    </section>
  );
}
```

app/components/CrateHero.tsx (3D Produce Crate Hero Animation)

```
"use client";

import { useRef, useState } from "react";
import { Tomato, Carrot, LeafyGreen, Chili } from "./ProduceIcons";

/**
 * Signature element: a stack of produce crates rendered with real CSS 3D
 * (perspective + preserve-3d + translateZ), tilted like an actual mandi
 * crate stack. The whole scene tracks the pointer for a subtle parallax —
 * on touch devices it settles into a fixed resting tilt instead.
 */
export default function CrateHero() {
  const sceneRef = useRef<HTMLDivElement>(null);
  const [tilt, setTilt] = useState({ x: -8, y: 14 });

  function handleMove(e: React.MouseEvent<HTMLDivElement>) {
    const el = sceneRef.current;
    if (!el) return;
    const rect = el.getBoundingClientRect();
    const px = (e.clientX - rect.left) / rect.width - 0.5;
    const py = (e.clientY - rect.top) / rect.height - 0.5;
    setTilt({ x: -8 - py * 14, y: 14 + px * 18 });
  }

  function handleLeave() {
    setTilt({ x: -8, y: 14 });
  }

  const crates = [
    { icon: <Tomato />, label: "Tomato", z: 40, x: -70, y: -10, rot: -6 },
    { icon: <Carrot />, label: "Carrot", z: 90, x: 40, y: 20, rot: 4 },
    { icon: <LeafyGreen />, label: "Spinach", z: 10, x: -20, y: 70, rot: -3 },
    { icon: <Chili />, label: "Chili", z: 130, x: 110, y: -50, rot: 8 },
  ];

  return (
    <div
      className="crate-scene-wrap"
      onMouseMove={handleMove}
      onMouseLeave={handleLeave}
    >
      <div className="crate-scene" ref={sceneRef}>
        <div
          className="crate-stage"
          style={{ transform: `rotateX(${tilt.x}deg) rotateY(${tilt.y}deg)` }}
        >
          {crates.map((c) => (
            <div
              key={c.label}
              className="crate"
              style={{
                transform: `translate3d(${c.x}px, ${c.y}px, ${c.z}px) rotate(${c.rot}deg)`,
              }}
            >
              <div className="crate-box">
                {c.icon}
                <span className="crate-label">{c.label}</span>
              </div>
              <div className="crate-slats" aria-hidden="true">
```

```
            <span /><span /><span />
          </div>
        </div>
      ))}
    </div>
  </div>

  <style jsx>{`
    .crate-scene-wrap {
      perspective: 1400px;
      width: 100%;
      height: 100%;
      min-height: 340px;
    }
    .crate-scene {
      width: 100%;
      height: 100%;
      display: flex;
      align-items: center;
      justify-content: center;
      transform-style: preserve-3d;
    }
    .crate-stage {
      position: relative;
      width: 220px;
      height: 220px;
      transform-style: preserve-3d;
      transition: transform 0.35s cubic-bezier(0.22, 1, 0.36, 1);
      animation: stage-breathe 7s ease-in-out infinite;
    }
    .crate {
      position: absolute;
      top: 50%;
      left: 50%;
      margin: -46px 0 0 -46px;
      transform-style: preserve-3d;
      transition: transform 0.5s cubic-bezier(0.22, 1, 0.36, 1);
      animation: crate-float 5s ease-in-out infinite;
    }
    .crate:nth-child(1) { animation-delay: -0.4s; }
    .crate:nth-child(2) { animation-delay: -2.1s; }
    .crate:nth-child(3) { animation-delay: -3.6s; }
    .crate:nth-child(4) { animation-delay: -1.2s; }
    @keyframes stage-breathe {
      0%, 100% { transform: translateZ(0); }
      50% { transform: translateZ(18px); }
    }
    @keyframes crate-float {
      0%, 100% { translate: 0 0; }
      50% { translate: 0 -8px; }
    }
    @media (prefers-reduced-motion: reduce) {
      .crate-stage, .crate { animation: none; }
    }
    .crate-box {
      width: 92px;
      height: 92px;
      border-radius: 6px;
      background: linear-gradient(155deg, #c9925a, #a5713a);
      border: 2px solid #7a4f26;
      display: flex;
      flex-direction: column;
```

```
          align-items: center;
          justify-content: center;
          gap: 4px;
          box-shadow: 0 22px 34px rgba(0, 0, 0, 0.45), inset 0 0 0 3px rgba(0, 0, 0, 0.08);
        }
        .crate-label {
          font-family: var(--font-mono);
          font-size: 9px;
          letter-spacing: 0.08em;
          text-transform: uppercase;
          color: #3c2611;
        }
        .crate-slats {
          position: absolute;
          inset: 0;
          pointer-events: none;
        }
        .crate-slats span {
          position: absolute;
          left: 6px;
          right: 6px;
          height: 2px;
          background: rgba(0, 0, 0, 0.15);
        }
        .crate-slats span:nth-child(1) { top: 22px; }
        .crate-slats span:nth-child(2) { top: 46px; }
        .crate-slats span:nth-child(3) { top: 70px; }
      `}</style>
    </div>
  );
}
```

app/components/ProductGrid.tsx (Marketplace Product Grid & Category Filter)

```
"use client";

import Link from "next/link";

interface CategoryItem {
  name: string;
  count: string;
  desc: string;
  image: string;
  emoji: string;
}

const CATEGORY_LIST: CategoryItem[] = [
  {
    name: "Vegetables",
    count: "40+ items",
    desc: "Tomatoes, Carrots, Onions, Potatoes & Gourds",
    emoji: "🥦",
    image: "https://images.unsplash.com/photo-1540420773420-3366772f4999?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Fruits",
    count: "25+ items",
    desc: "Mangoes, Apples, Bananas, Papayas & Citrus",
    emoji: "🍎",
    image: "https://images.unsplash.com/photo-1619566636858-adf3ef46400b?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Spices & Herbs",
    count: "20+ items",
    desc: "Turmeric, Green & Dry Chilies, Cardamom, Pepper",
    emoji: "🌶",
    image: "https://images.unsplash.com/photo-1596040033229-a9821ebd058d?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Leafy Greens",
    count: "15+ items",
    desc: "Fresh Spinach, Amaranth, Coriander, Mint & Fenugreek",
    emoji: "🥬",
    image: "https://images.unsplash.com/photo-1576045057995-568f588f82fb?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Tubers & Roots",
    count: "12+ items",
    desc: "Sweet Potatoes, Ginger, Beetroot, Radish & Yams",
    emoji: "🧅",
    image: "https://images.unsplash.com/photo-1518977676601-b53f82aba655?auto=format&fit=crop&w=600&q
=80",
  },
  {
    name: "Grains & Pulses",
    count: "18+ items",
    desc: "Traditional Rice, Millets, Toor Dal, Lentils & Wheat",
    emoji: "🌾",
    image: "https://images.unsplash.com/photo-1586201375761-83865001e31c?auto=format&fit=crop&w=600&q
```

```
=80",
  },
];

export default function ProductGrid() {
  return (
    <section className="grid-section container">
      <div className="grid-head">
        <div className="head-text">
          <h2>Product Categories</h2>
          <p className="sub-description">
            Directly sourced from trusted local farmers across India at fair mandi prices.
          </p>
        </div>
        <Link href="/shop" className="view-all-link">
          Browse All Products →
        </Link>
      </div>

      {/* Category Cards Grid */}
      <div className="category-grid">
        {CATEGORY_LIST.map((cat) => (
          <Link
            href={`/shop?category=${encodeURIComponent(cat.name)}`}
            key={cat.name}
            className="cat-card"
          >
            <div className="cat-img-wrapper">
              <img src={cat.image} alt={cat.name} className="cat-img" loading="lazy" />
              <div className="cat-overlay"></div>
              <span className="cat-badge">{cat.count}</span>
            </div>
            <div className="cat-content">
              <div className="cat-title-row">
                <span className="cat-emoji">{cat.emoji}</span>
                <h3>{cat.name}</h3>
              </div>
              <p className="cat-desc">{cat.desc}</p>
              <span className="cat-action">Shop Now →</span>
            </div>
          </Link>
        ))}
      </div>


      <style jsx>{`
        .grid-section {
          padding: 24px 24px 100px;
        }
        .grid-head {
          display: flex;
          justify-content: space-between;
          align-items: flex-end;
          margin-bottom: 36px;
          flex-wrap: wrap;
          gap: 16px;
        }
        .eyebrow {
          font-family: var(--font-mono);
          text-transform: uppercase;
          letter-spacing: 0.14em;
          font-size: 0.8rem;
```

```
  color: #2e7d32;
  font-weight: 600;
  margin: 0 0 8px;
}
h2 {
  font-family: var(--font-display);
  font-weight: 700;
  font-size: clamp(1.8rem, 3.5vw, 2.4rem);
  margin: 0 0 8px;
  color: #1a1a15;
}
.sub-description {
  margin: 0;
  color: #6b6555;
  font-size: 0.95rem;
  max-width: 520px;
}
.view-all-link {
  font-weight: 600;
  font-size: 0.92rem;
  color: #2e7d32;
  text-decoration: none;
  padding: 8px 18px;
  border-radius: 999px;
  background: rgba(46, 125, 50, 0.08);
  border: 1px solid rgba(46, 125, 50, 0.2);
  transition: all 0.2s ease;
}
.view-all-link:hover {
  background: #2e7d32;
  color: #fff;
}
.category-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 24px;
}
.cat-card {
  background: #ffffff;
  border: 1px solid rgba(26, 26, 21, 0.1);
  border-radius: 18px;
  overflow: hidden;
  text-decoration: none;
  color: inherit;
  display: flex;
  flex-direction: column;
  box-shadow: 0 6px 18px rgba(22, 36, 26, 0.06);
  transition: transform 0.25s ease, box-shadow 0.25s ease, border-color 0.25s ease;
}
.cat-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 16px 32px rgba(46, 125, 50, 0.14);
  border-color: rgba(46, 125, 50, 0.35);
}
.cat-img-wrapper {
  position: relative;
  height: 180px;
  width: 100%;
  overflow: hidden;
  background: #f0eee6;
}
.cat-img {
```

```
  width: 100%;
  height: 100%;
  object-fit: cover;
  transition: transform 0.4s ease;
}
.cat-card:hover .cat-img {
  transform: scale(1.06);
}
.cat-overlay {
  position: absolute;
  inset: 0;
  background: linear-gradient(180deg, rgba(0,0,0,0) 50%, rgba(0,0,0,0.3) 100%);
}
.cat-badge {
  position: absolute;
  top: 12px;
  right: 12px;
  background: rgba(255, 255, 255, 0.92);
  backdrop-filter: blur(4px);
  color: #2e7d32;
  font-weight: 600;
  font-size: 0.75rem;
  padding: 4px 10px;
  border-radius: 999px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.12);
}
.cat-content {
  padding: 20px;
  display: flex;
  flex-direction: column;
  flex: 1;
}
.cat-title-row {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-bottom: 8px;
}
.cat-emoji {
  font-size: 1.4rem;
  line-height: 1;
}
h3 {
  font-family: var(--font-display);
  font-weight: 700;
  font-size: 1.22rem;
  margin: 0;
  color: #1a1a15;
}
.cat-desc {
  font-size: 0.85rem;
  color: #6b6555;
  line-height: 1.45;
  margin: 0 0 16px;
  flex: 1;
}
.cat-action {
  font-size: 0.85rem;
  font-weight: 700;
  color: #2e7d32;
  margin-top: auto;
  display: inline-flex;
```

```
          align-items: center;
          gap: 4px;
        }
        .cat-card:hover .cat-action {
          color: #1b5e20;
          text-decoration: underline;
        }
        @media (max-width: 640px) {
          .grid-head {
            flex-direction: column;
            align-items: flex-start;
          }
          .category-grid {
            grid-template-columns: 1fr;
          }
        }
      `}</style>
    </section>
  );
}
```

app/components/ProductImage.tsx (Dynamic Image Loader with SVG Fallback)

```
"use client";

import { useState } from "react";
import { Tomato, Carrot, LeafyGreen, Chili } from "./ProduceIcons";
import { ProduceIcon } from "@/app/lib/types";

const ICONS = { tomato: Tomato, carrot: Carrot, leafy: LeafyGreen, chili: Chili };

/**
 * Renders a real product photo, falling back to the hand-drawn produce
 * icon (same one used everywhere else) if the image URL 404s or is
 * missing — so a broken/removed photo never breaks the layout.
 */
export default function ProductImage({
  src,
  icon,
  alt,
  iconSize = 54,
}: {
  src?: string;
  icon: ProduceIcon;
  alt: string;
  iconSize?: number;
}) {
  const [failed, setFailed] = useState(false);
  const Icon = ICONS[icon];

  if (!src || failed) {
    return (
      <div className="pi-fallback">
        <Icon size={iconSize} />
        <style jsx>{`
          .pi-fallback {
            width: 100%;
            height: 100%;
            display: flex;
            align-items: center;
            justify-content: center;
          }
        `}</style>
      </div>
    );
  }

  return (
    // eslint-disable-next-line @next/next/no-img-element
    <img
      src={src}
      alt={alt}
      loading="lazy"
      onError={() => setFailed(true)}
      className="pi-photo"
      style={{ width: "100%", height: "100%", objectFit: "cover", display: "block" }}
    />
  );
}
```

app/components/ProduceIcons.tsx (Botanical SVG Vector Produce Icons)

```
export function Tomato({ size = 48 }: { size?: number }) {
  return (
    <svg width={size} height={size} viewBox="0 0 48 48" fill="none">
      <ellipse cx="24" cy="27" rx="16" ry="14" fill="#C23616" />
      <ellipse cx="18" cy="22" rx="5" ry="4" fill="#E0532F" opacity="0.6" />
      <path d="M24 13c-2 0-3 3-3 3s-2-4-5-3c0 3 3 4 3 4s-4 1-4 4c3 1 5-1 5-1s0 3 2 4c1-2 1-4 1-4s2 2
4 1c0-3-3-3-3-3s3-1 3-4c-1 0-3-1-3-1z" fill="#4C7A3D" />
    </svg>
  );
}

export function Carrot({ size = 48 }: { size?: number }) {
  return (
    <svg width={size} height={size} viewBox="0 0 48 48" fill="none">
      <path d="M14 16c8-2 16 4 20 12-8 6-20 6-24-2-2-4-1-8 4-10z" fill="#E8A33D" />
      <path d="M16 15l-3-8m8 8l-1-9m8 10l2-8" stroke="#4C7A3D" strokeWidth="3" strokeLinecap="round"
/>
    </svg>
  );
}

export function LeafyGreen({ size = 48 }: { size?: number }) {
  return (
    <svg width={size} height={size} viewBox="0 0 48 48" fill="none">
      <path d="M24 42c-10-2-16-12-14-24 10 0 18 6 18 16 0-10 8-16 18-16 2 12-4 22-14 24-2-6-6-6-8 0z"
fill="#4C7A3D" />
      <path d="M24 42V22" stroke="#2F5124" strokeWidth="2" strokeLinecap="round" />
    </svg>
  );
}

export function Chili({ size = 48 }: { size?: number }) {
  return (
    <svg width={size} height={size} viewBox="0 0 48 48" fill="none">
      <path d="M12 12c3-2 6-1 6 1 6-1 14 4 16 12 2 8-4 14-11 12-9-2-14-11-13-18 .3-2.5 1.2-5.5 2-7z"
fill="#C23616" />
      <path d="M12 12c-1-2-1-4 1-5" stroke="#4C7A3D" strokeWidth="3" strokeLinecap="round" />
    </svg>
  );
}
```

app/components/FarmerNav.tsx (Farmer Portal Tab Navigation Bar)

```
"use client";

import Link from "next/link";
import { usePathname } from "next/navigation";

const TABS = [
  { href: "/farmer/dashboard", label: "Dashboard" },
  { href: "/farmer/products", label: "My Products" },
  { href: "/farmer/orders", label: "Orders" },
  { href: "/farmer/earnings", label: "Earnings" },
];

export default function FarmerNav() {
  const pathname = usePathname();
  return (
    <header className="fnav">
      <div className="container fnav-top">
        <Link href="/" className="logo">Farm<span>Link</span></Link>
        <span className="role-tag">Farmer Portal</span>
      </div>
      <div className="container fnav-tabs">
        {TABS.map((t) => (
          <Link key={t.href} href={t.href} className={pathname?.startsWith(t.href) ? "active" : ""}>
            {t.label}
          </Link>
        ))}
      </div>

      <style jsx>{`
        .fnav { background: var(--soil); border-bottom: 1px solid rgba(232,163,61,0.18); }
        .fnav-top { display: flex; align-items: center; gap: 12px; height: 60px; }
        .logo { font-family: var(--font-display); font-weight: 900; font-size: 1.3rem; color: #efe8d
8; }
        .logo span { color: var(--turmeric); }
        .role-tag { font-family: var(--font-mono); font-size: 0.72rem; color: var(--turmeric); backgr
ound: rgba(232,163,61,0.14); padding: 3px 10px; border-radius: 999px; }
        .fnav-tabs { display: flex; gap: 4px; padding-bottom: 0; }
        .fnav-tabs a {
          color: #cfc8b4; font-size: 0.88rem; padding: 12px 16px;
          border-bottom: 2px solid transparent;
        }
        .fnav-tabs a.active { color: var(--turmeric); border-bottom-color: var(--turmeric); }
        .fnav-tabs a:hover { color: #efe8d8; }
      `}</style>
    </header>
  );
}
```

app/components/AdminNav.tsx (Admin Console Tab Navigation Bar)

```
"use client";

import Link from "next/link";
import { usePathname, useRouter } from "next/navigation";
import { logoutUser } from "@/app/lib/api";

const TABS = [
  { href: "/admin/dashboard", label: "Dashboard" },
  { href: "/admin/farmers", label: "Farmers" },
  { href: "/admin/products", label: "Products" },
  { href: "/admin/bulk-requests", label: "Bulk Requests" },
  { href: "/admin/orders", label: "Orders & Income" },
  { href: "/admin/shared-emails", label: "Shared Accounts (Farmer & Buyer)" },
];

export default function AdminNav() {
  const pathname = usePathname();
  const router = useRouter();

  async function handleLogout() {
    try {
      await logoutUser();
    } finally {
      router.push("/auth/admin-login");
    }
  }

  return (
    <header className="anav">
      <div className="container anav-top">
        <Link href="/" className="logo">
          <img src="/agrenza-logo-small.jpg" alt="Agrenza Logo" width={34} height={34} className="log
o-img" style={{ width: "34px", height: "34px" }} />
          <span>Agren<span>za</span></span>
        </Link>
        <span className="role-tag">Admin Panel</span>
        <button className="logout" onClick={handleLogout}>Logout</button>
      </div>
      <div className="container anav-tabs">
        {TABS.map((t) => (
          <Link key={t.href} href={t.href} className={pathname?.startsWith(t.href) ? "active" : ""}>
            {t.label}
          </Link>
        ))}
      </div>

      <style jsx>{`
        .anav { background: var(--soil); border-bottom: 1px solid rgba(232,163,61,0.18); }
        .anav-top { display: flex; align-items: center; gap: 12px; height: 60px; }
        .logo { font-family: var(--font-display); font-weight: 900; font-size: 1.3rem; color: #efe8d
8; display: flex; align-items: center; gap: 8px; text-decoration: none; }
        .logo-img { width: 34px; height: 34px; border-radius: 50%; object-fit: cover; border: 1.5px s
olid var(--turmeric); }
        .logo span { color: var(--turmeric); }
        .role-tag { font-family: var(--font-mono); font-size: 0.72rem; color: var(--turmeric); backgr
ound: rgba(232,163,61,0.14); padding: 3px 10px; border-radius: 999px; }
        .logout { margin-left: auto; background: none; border: 1px solid rgba(232,163,61,0.4); color:
#efe8d8; font-size: 0.8rem; padding: 6px 14px; border-radius: 999px; cursor: pointer; }
        .logout:hover { background: rgba(232,163,61,0.14); }
```

```
        .anav-tabs { display: flex; gap: 4px; padding-bottom: 0; }
        .anav-tabs a {
          color: #cfc8b4; font-size: 0.88rem; padding: 12px 16px;
          border-bottom: 2px solid transparent;
        }
        .anav-tabs a.active { color: var(--turmeric); border-bottom-color: var(--turmeric); }
        .anav-tabs a:hover { color: #efe8d8; }
      `}</style>
    </header>
  );
}
```

app/lib/api.ts (Client API Connector & Error Handling Core)

```
const API_URL = process.env.NEXT_PUBLIC_API_URL || "http://localhost/farmlink/farmlink/backend/api";

export class ApiError extends Error {}

async function parseResponse<T>(res: Response): Promise<T> {
  const text = await res.text();
  let data: any;
  try {
    data = JSON.parse(text);
  } catch {
    if (res.status === 401) throw new ApiError("Session expired. Please log in again.");
    if (res.status === 403) throw new ApiError("Access denied. Please check your account permission
s.");
    throw new ApiError(`Server error (${res.status}): ${text.substring(0, 100) || "Unexpected respons
e"}`);
  }

  if (!res.ok || !data.ok) {
    throw new ApiError(data.error || "Something went wrong.");
  }
  return data as T;
}

async function request<T = any>(path: string, body?: object): Promise<T> {
  const res = await fetch(`${API_URL}${path}`, {
    method: "POST",
    credentials: "include",
    headers: { "Content-Type": "application/json" },
    body: body ? JSON.stringify(body) : undefined,
  });
  return parseResponse<T>(res);
}

async function getRequest<T = any>(path: string, params?: Record<string, string>): Promise<T> {
  const qs = params ? "?" + new URLSearchParams(params).toString() : "";
  const res = await fetch(`${API_URL}${path}${qs}`, {
    method: "GET",
    credentials: "include",
  });
  return parseResponse<T>(res);
}

async function requestFormData<T = any>(path: string, formData: FormData): Promise<T> {
  const res = await fetch(`${API_URL}${path}`, {
    method: "POST",
    credentials: "include",
    body: formData,
  });
  return parseResponse<T>(res);
}

export type Role = "buyer" | "farmer";

export type RegisterResponse = {
  registered_pending: boolean;
  email: string;
  mobile: string;
  role: Role;
  auto_password_sent: boolean;
```

```
  auto_password?: string;
  dev_otp?: string;
  dev_token?: string;
  message: string;
};

/** Stages unverified registration in pending_registrations and dispatches verification email. */
export function registerUser(body: {
  role: Role;
  name: string;
  mobile: string;
  password?: string;
  email: string;
  aadhaar?: string;
}) {
  return request<RegisterResponse>("/auth/register.php", body);
}

/** Sends the Firebase-verified mobile ID token to the backend, which checks it and marks is_verified
= 1. */
export function verifyMobileWithBackend(idToken: string, mobile: string) {
  return request<{ verified: boolean; requires_approval: boolean }>("/auth/verify-otp.php", {
    idToken,
    mobile,
  });
}

/**
 * Verifies email via FarmLink token, 6-digit OTP code, or Firebase ID token.
 * Upon verification, account is transferred into the users database table.
 */
export function verifyEmailWithBackend(params: {
  email: string;
  token?: string;
  otp?: string;
  idToken?: string;
}) {
  return request<{
    email_verified: boolean;
    verified: boolean;
    requires_approval: boolean;
    role?: string;
    user?: { id: number; name: string; role: string };
    message?: string;
  }>("/auth/verify-email.php", params);
}

export type EmailAuthenticityResult = {
  ok: boolean;
  status: "original" | "disposable" | "duplicate" | "invalid_domain" | "invalid_format" | "invalid_gm
ail" | "empty";
  is_original: boolean;
  is_duplicate: boolean;
  provider: string | null;
  error: string | null;
  message: string;
};

/** Checks whether an email is authentic/original Gmail vs fake/disposable/duplicate. */
export async function checkEmailAuthenticity(email: string, role?: string): Promise<EmailAuthenticity
Result> {
  const trimmed = email.trim();
```

```
  if (!trimmed) {
    return {
      ok: false,
      status: "empty",
      is_original: false,
      is_duplicate: false,
      provider: null,
      error: null,
      message: "",
    };
  }

  try {
    const res = await fetch(`${API_URL}/auth/check-email.php`, {
      method: "POST",
      credentials: "include",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({ email: trimmed, role: role || undefined }),
    });
    const data = await res.json().catch(() => null);
    if (data && typeof data.status === "string") {
      return data as EmailAuthenticityResult;
    }
    return {
      ok: false,
      status: "invalid_format",
      is_original: false,
      is_duplicate: false,
      provider: null,
      error: data?.error || "Could not verify email.",
      message: data?.error || "Could not verify email.",
    };
  } catch {
    return {
      ok: false,
      status: "invalid_format",
      is_original: false,
      is_duplicate: false,
      provider: null,
      error: "Network error checking email.",
      message: "Network error checking email.",
    };
  }
}

export function loginUser(body: { role: string; email: string; password: string }) {
  return request<{ user: { id: number; name: string; role: string } }>("/auth/login.php", body);
}

export type AuthUser = {
  id: number;
  name: string;
  email?: string | null;
  mobile?: string | null;
  role: string;
  created_at?: string | null;
};

export function getMe() {
  return getRequest<{ user: AuthUser | null }>("/auth/me.php");
}
```

```
export function logoutUser() {
  return request<{ ok: boolean }>("/auth/logout.php");
}

// ---------------------------------------------------------------------
// Admin
// ---------------------------------------------------------------------

export type FarmerStatus = "pending" | "approved" | "rejected" | "all";
export type AdminFarmer = {
  id: number;
  name: string;
  mobile: string;
  email: string | null;
  aadhaar: string | null;
  status: "pending" | "approved" | "rejected";
  remark: string | null;
  joined: string;
  total_products?: number;
  total_sales?: number;
};

export type Product = {
  id: number;
  name: string;
  category: string;
  category_id?: number;
  price: number;
  base_price?: number;
  gst_percent?: number;
  gst_amount?: number;
  commission_percent?: number;
  commission_amount?: number;
  stock: number;
  image: string | null;
  description: string;
  status: "active" | "inactive";
  approval_status?: "pending" | "approved" | "rejected";
  admin_remark?: string | null;
  farmer?: string;
  farmer_mobile?: string;
};

export function listCategories() {
  return getRequest<{ categories: { id: number; name: string }[] }>("/products/categories.php");
}

export function listProducts(params?: Record<string, string>) {
  return getRequest<{ products: Product[] }>("/products/list.php", params);
}

export function getProductDetail(id: number | string) {
  return getRequest<{ product: Product }>("/products/detail.php", { id: String(id) });
}

export function createProduct(formData: FormData) {
  return requestFormData<{ product_id: number }>("/products/create.php", formData);
}

export function updateProduct(formData: FormData) {
  return requestFormData<{ updated: boolean }>("/products/update.php", formData);
}
```

```
export function deleteProduct(id: number) {
  return request<{ deleted: boolean }>("/products/delete.php", { id });
}

export function refillProduct(id: number, refill_kg: number, action: "add" | "set" = "add") {
  return request<{
    refilled: boolean;
    product_id: number;
    product_name: string;
    old_stock: number;
    refill_amount: number;
    new_stock: number;
    status: string;
    message: string;
  }>("/products/refill.php", { id, refill_kg, action });
}

export type AdminProduct = {
  id: number;
  name: string;
  category: string;
  price: number;
  base_price?: number;
  gst_percent?: number;
  gst_amount?: number;
  commission_percent?: number;
  commission_amount?: number;
  stock: number;
  image: string | null;
  description: string;
  status: "active" | "inactive";
  approval_status: "pending" | "approved" | "rejected";
  remark: string | null;
  farmer_id: number;
  farmer_name: string;
  farmer_email: string | null;
  farmer_mobile: string;
  created_at: string;
};

export type AdminBulkRequest = {
  id: number;
  product_name: string;
  qty: number;
  reason: string;
  status: "pending" | "approved" | "rejected";
  remark: string | null;
  buyer_name: string;
  buyer_email: string | null;
  buyer_mobile: string;
  created_at: string;
};

export type AdminOrderItem = {
  id: number;
  product_id: number;
  product_name: string;
  product_image: string | null;
  qty: number;
  price_per_kg: number;
  subtotal: number;
```

```
  current_stock: number;
  status: string;
  farmer_name: string;
  farmer_mobile: string;
};

export type AdminOrder = {
  id: number;
  buyer_id: number;
  buyer_name: string;
  buyer_mobile: string;
  buyer_email: string | null;
  total_amount: number;
  payment_mode: "cod" | "online";
  address: string;
  status: string;
  tracking_note?: string;
  created_at: string;
  item_count: number;
  items?: AdminOrderItem[];
};

export type FarmerOrderItem = {
  item_id: number;
  order_id: number;
  order_date: string;
  product_id: number;
  product_name: string;
  product_image: string | null;
  current_stock: number;
  buyer_name: string;
  buyer_mobile: string;
  address: string;
  payment_mode: string;
  qty: number;
  price_per_kg: number;
  subtotal: number;
  status: "pending" | "confirmed" | "shipped" | "delivered" | "cancelled";
};

export function listFarmerOrderItems() {
  return getRequest<{ items: FarmerOrderItem[] }>("/orders/farmer-items.php");
}

export function updateFarmerOrderItemStatus(item_id: number, status: string) {
  return request<{ updated: boolean }>("/orders/update-item-status.php", { item_id, status });
}

export type SharedEmailAccount = {
  email: string;
  mobile?: string;
  shared_by?: "email" | "mobile" | "both";
  farmer: {
    id: number;
    name: string;
    mobile: string;
    email: string | null;
    status: "pending" | "approved" | "rejected";
    aadhaar: string | null;
    remark: string | null;
    joined: string;
    total_products: number;
```

```
    total_sales: number;
  };
  buyer: {
    id: number;
    name: string;
    mobile: string;
    email: string | null;
    status: string;
    joined: string;
    total_orders: number;
    total_spent: number;
  };
};

export function adminSharedEmails() {
  return getRequest<{
    total_shared: number;
    shared_accounts: SharedEmailAccount[];
  }>("/admin/shared-emails.php");
}

export function adminStats(from?: string, to?: string) {
  const params: Record<string, string> = {};
  if (from) params.from = from;
  if (to)   params.to   = to;
  return getRequest<{
    pending_farmers: number;
    pending_products: number;
    pending_bulk: number;
    total_farmers: number;
    total_buyers: number;
    shared_gmail_count?: number;
    total_products: number;
    approved_products: number;
    total_income: number;
    platform_earnings: number;
    today_income: number;
    total_orders: number;
    completed_orders: number;
    filter_from?: string | null;
    filter_to?: string | null;
  }>("/admin/stats.php", Object.keys(params).length ? params : undefined);
}


export function adminListFarmers(status: FarmerStatus = "pending") {
  return getRequest<{ farmers: AdminFarmer[] }>("/admin/farmers.php", { status });
}

export function adminFarmerAction(id: number, action: "approve" | "reject", remark?: string) {
  return request<{ updated: boolean; status: string }>("/admin/farmer-action.php", { id, action, rema
rk });
}

export function adminListProducts(status: FarmerStatus = "pending") {
  return getRequest<{ products: AdminProduct[] }>("/admin/products.php", { status });
}

export function adminProductAction(id: number, action: "approve" | "reject", remark?: string) {
  return request<{ updated: boolean; approval_status: string }>("/admin/product-action.php", { id, ac
tion, remark });
}
```

```
export function adminListBulkRequests(status: FarmerStatus = "pending") {
  return getRequest<{ requests: AdminBulkRequest[] }>("/admin/bulk-requests.php", { status });
}

export function adminBulkRequestAction(id: number, action: "approve" | "reject", remark?: string) {
  return request<{ updated: boolean; status: string }>("/admin/bulk-request-action.php", { id, actio
n, remark });
}

export function submitBulkRequest(product_id: number, qty: number, reason: string) {
  return request<{ request_id: number }>("/bulk-requests/create.php", { product_id, qty, reason });
}

export type BuyerOrderItem = {
  product_id: number;
  name: string;
  qty: number;
  price: number;
  subtotal: number;
  status: string;
};

export type BuyerOrder = {
  id: number;
  date: string;
  total: number;
  payment_mode: string;
  status: "pending" | "confirmed" | "packing" | "shipped" | "out_for_delivery" | "delivered" | "cance
lled";
  tracking_note?: string;
  address: string;
  items: BuyerOrderItem[];
};

export function listMyOrders() {
  return getRequest<{ orders: BuyerOrder[] }>("/orders/mine.php");
}

export function adminListOrders() {
  return getRequest<{ orders: AdminOrder[]; total_revenue: number }>("/admin/orders.php");
}

export function adminUpdateOrderStatus(order_id: number, status: string, tracking_note?: string) {
  return request<{ updated: boolean; order_id: number; status: string; message: string }>(
    "/admin/update-order-status.php",
    { order_id, status, tracking_note }
  );
}

export function placeOrderApi(body: {
  address: string;
  payment_mode: "cod" | "online";
  items: { product_id: number; qty: number }[];
}) {
  return request<{ order_id: number; total: number }>("/orders/place.php", body);
}

export interface EarningItem {
  date: string;
  product_name: string;
  qty: number;
```

```
  amount: number;
  status: string;
}

export function farmerEarnings(from?: string, to?: string) {
  const params: Record<string, string> = {};
  if (from) params.from = from;
  if (to)   params.to   = to;
  return getRequest<{
    total_earned: number;
    pending_amount: number;
    today_earned: number;
    order_count: number;
    history: EarningItem[];
    filter_from: string | null;
    filter_to: string | null;
  }>("/farmer/earnings.php", Object.keys(params).length ? params : undefined);
}
```

app/lib/types.ts (TypeScript Type & Interface Definitions)

```
export type ProduceIcon = "tomato" | "carrot" | "leafy" | "chili";

export type Product = {
  id: number;
  /** Real product ID in the MySQL `products` table. Used for backend API calls. */
  db_id?: number;
  name: string;
  farmer: string;
  location: string;
  price: number;
  category: string;
  icon: ProduceIcon;
  stock: number;
  description: string;
  /** Real product photo. Falls back to the ProduceIcon illustration if missing/broken. */
  image?: string;
};

export type CartItem = {
  productId: number;
  name: string;
  price: number;
  icon: ProduceIcon;
  qty: number;
};

export type Order = {
  id: number;
  date: string;
  items: CartItem[];
  total: number;
  paymentMode: "cod" | "online";
  status: "pending" | "confirmed" | "delivered";
};
```

app/lib/cart-context.tsx (Shopping Cart Global Context & Persistence)

```
"use client";

import { createContext, useContext, useEffect, useState } from "react";
import { CartItem, Order, Product } from "./types";

type CartContextType = {
  items: CartItem[];
  addItem: (p: Product, qty: number) => void;
  removeItem: (productId: number) => void;
  clearCart: () => void;
  total: number;
  count: number;
  orders: Order[];
  placeOrder: (paymentMode: "cod" | "online") => Order;
};

const CartContext = createContext<CartContextType | null>(null);

const CART_KEY = "farmlink_cart";
const ORDERS_KEY = "farmlink_orders";

export function CartProvider({ children }: { children: React.ReactNode }) {
  const [items, setItems] = useState<CartItem[]>([]);
  const [orders, setOrders] = useState<Order[]>([]);
  const [hydrated, setHydrated] = useState(false);

  useEffect(() => {
    try {
      const c = localStorage.getItem(CART_KEY);
      const o = localStorage.getItem(ORDERS_KEY);
      if (c) setItems(JSON.parse(c));
      if (o) setOrders(JSON.parse(o));
    } catch {
      /* ignore corrupt storage */
    }
    setHydrated(true);
  }, []);

  useEffect(() => {
    if (hydrated) localStorage.setItem(CART_KEY, JSON.stringify(items));
  }, [items, hydrated]);

  useEffect(() => {
    if (hydrated) localStorage.setItem(ORDERS_KEY, JSON.stringify(orders));
  }, [orders, hydrated]);

  function addItem(p: Product, qty: number) {
    setItems((prev) => {
      const existing = prev.find((i) => i.productId === p.id);
      if (existing) {
        return prev.map((i) =>
          i.productId === p.id ? { ...i, qty: i.qty + qty } : i
        );
      }
      return [...prev, { productId: p.id, name: p.name, price: p.price, icon: p.icon, qty }];
    });
  }

  function removeItem(productId: number) {
```

```
    setItems((prev) => prev.filter((i) => i.productId !== productId));
  }

  function clearCart() {
    setItems([]);
  }

  function placeOrder(paymentMode: "cod" | "online"): Order {
    const order: Order = {
      id: Date.now(),
      date: new Date().toISOString(),
      items,
      total,
      paymentMode,
      status: "pending",
    };
    setOrders((prev) => [order, ...prev]);
    clearCart();
    return order;
  }

  const total = items.reduce((sum, i) => sum + i.price * i.qty, 0);
  const count = items.reduce((sum, i) => sum + i.qty, 0);

  return (
    <CartContext.Provider
      value={{ items, addItem, removeItem, clearCart, total, count, orders, placeOrder }}
    >
      {children}
    </CartContext.Provider>
  );
}

export function useCart() {
  const ctx = useContext(CartContext);
  if (!ctx) throw new Error("useCart must be used within CartProvider");
  return ctx;
}
```

app/shop/page.tsx (Public Marketplace Catalog View)

```
"use client";

import { useEffect, useMemo, useState } from "react";
import Link from "next/link";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import ProductImage from "@/app/components/ProductImage";
import { PRODUCTS as FALLBACK_PRODUCTS } from "@/app/lib/products-data";
import { listProducts, listCategories } from "@/app/lib/api";
import { useCart } from "@/app/lib/cart-context";

const ALL_CATEGORY_OPTIONS = [
  "Vegetables",
  "Fruits",
  "Leafy Greens",
  "Tubers & Roots",
  "Grains & Pulses",
  "Spices & Herbs",
  "Dairy & Organic",
];

const CATEGORY_EMOJIS: Record<string, string> = {
  All: "🌾",
  Vegetables: "🥦",
  Fruits: "🍎",
  "Leafy Greens": "🥬",
  "Tubers & Roots": "🧅",
  "Grains & Pulses": "🌾",
  "Spices & Herbs": "🌶️",
  "Dairy & Organic": "🥛",
};

export default function ShopPage() {
  const { addItem } = useCart();
  const [search, setSearch] = useState("");
  const [category, setCategory] = useState("All");
  const [items, setItems] = useState<any[]>(FALLBACK_PRODUCTS);
  const [categories, setCategories] = useState<string[]>(ALL_CATEGORY_OPTIONS);
  const [addedToast, setAddedToast] = useState<string | null>(null);

  useEffect(() => {
    (async () => {
      try {
        const [prodRes, catRes] = await Promise.all([
          listProducts().catch(() => ({ products: [] })),
          listCategories().catch(() => ({ categories: [] })),
        ]);

        if (prodRes.products && prodRes.products.length > 0) {
          const apiProds = prodRes.products.map((p) => ({
            id: p.id,
            name: p.name,
            farmer: p.farmer || "Verified Farmer",
            location: "Local Farm",
            price: p.price,
            category: p.category,
            icon: "tomato" as const,
            stock: p.stock,
            description: p.description,
```

```
            image: p.image || undefined,
            base_price: p.base_price,
            gst_percent: p.gst_percent,
          }));
          setItems(apiProds);
        }

        const dbCats = catRes.categories?.map((c) => c.name) || [];
        const prodCats = prodRes.products?.map((p) => p.category) || [];
        const combinedCats = Array.from(new Set([...ALL_CATEGORY_OPTIONS, ...dbCats, ...prodCats])).f
ilter(Boolean);
        setCategories(combinedCats);
      } catch {
        setCategories(ALL_CATEGORY_OPTIONS);
      }
    })();
  }, []);

  const filtered = useMemo(() => {
    return items.filter((p) => {
      const matchesSearch = p.name.toLowerCase().includes(search.toLowerCase());
      const matchesCategory = category === "All" || p.category.toLowerCase() === category.toLowerCase
();
      return matchesSearch && matchesCategory;
    });
  }, [items, search, category]);

  function handleQuickAdd(e: React.MouseEvent, item: any) {
    e.preventDefault();
    e.stopPropagation();
    addItem(
      {
        id: item.id,
        name: item.name,
        price: item.price,
        category: item.category,
        stock: item.stock || 999,
        description: item.description || "",
        farmer: item.farmer || "",
        location: item.location || "",
        icon: item.icon || "tomato",
        image: item.image,
      },
      1
    );
    setAddedToast(item.name);
    setTimeout(() => setAddedToast(null), 2500);
  }

  return (
    <>
      <Nav />
      <section className="shop-hero">
        <div className="container">
          <h1>Fresh from farm to your doorstep</h1>
          <p className="lead">Transparent pricing with GST included.</p>
        </div>
      </section>

      <div className="container shop-body">
        <div className="filters">
          <input
```

```
            className="search"
            placeholder="Search products by name…"
            value={search}
            onChange={(e) => setSearch(e.target.value)}
          />
          <div className="chips">
            <button
              className={category === "All" ? "active" : ""}
              onClick={() => setCategory("All")}
            >
              🌾 All Products
            </button>
            {categories.map((c) => (
              <button
                key={c}
                className={category === c ? "active" : ""}
                onClick={() => setCategory(c)}
              >
                {CATEGORY_EMOJIS[c] || "🌱"} {c}
              </button>
            ))}
          </div>
        </div>

        {filtered.length === 0 ? (
          <div className="empty-category-card">
            <span className="empty-cat-icon">{CATEGORY_EMOJIS[category] || "🌱"}</span>
            <h3>No products found in &ldquo;{category}&rdquo;</h3>
            <p>Local farmers will list new harvest soon. Check out other produce categories!</p>
            <button
              type="button"
              onClick={() => setCategory("All")}
              className="btn-view-all-cats"
            >
              🌾 View All Products
            </button>
          </div>
        ) : (
          <div className="grid">
            {filtered.map((p) => (
              <div key={p.id} className="card-wrapper">
                <Link href={`/shop/${p.id}`} className="card">
                  <div className="card-top">
                    <ProductImage src={p.image} icon={p.icon} alt={p.name} iconSize={54} />
                  </div>
                  <span className="badge">{p.category}</span>
                  <h4>{p.name}</h4>
                  <p className="farmer">By {p.farmer}, {p.location}</p>
                  <div className="price-row">
                    <div>
                      <p className="price">₹{p.price}/kg</p>
                      <span className="price-badge">Incl. GST &amp;amp; fee</span>
                    </div>
                    <button
                      type="button"
                      className="btn-quick-add"
                      onClick={(e) => handleQuickAdd(e, p)}
                      title="Add 1 kg to cart"
                    >
                      + Add
                    </button>
                  </div>
```

```
                </Link>
              </div>
            ))}
          </div>
        )}
      </div>

      {/* ADD TO CART TOAST */}
      {addedToast && (
        <div className="add-toast">
          <span className="toast-icon">✅</span>
          <div className="toast-text">
            <strong>Added to Cart!</strong>
            <span>1 kg of {addedToast} added.</span>
          </div>
          <Link href="/buyer/cart" className="btn-toast-cart">
            View Cart →
          </Link>
        </div>
      )}

      <Footer />

      <style jsx>{`
        .shop-hero {
          background: linear-gradient(135deg, var(--soil-deep), var(--soil));
          color: #efe8d8;
          padding: 56px 0;
        }
        h1 {
          font-family: var(--font-display);
          font-weight: 700;
          font-size: clamp(1.9rem, 4vw, 2.6rem);
          margin: 0 0 10px;
        }
        .lead { color: #cfc8b4; margin: 0; }

        .shop-body { padding: 40px 24px 80px; }
        .filters { display: flex; flex-wrap: wrap; gap: 16px; align-items: center; margin-bottom: 34p
x; }
        .search {
          flex: 1;
          min-width: 220px;
          padding: 12px 16px;
          border-radius: 999px;
          border: 1px solid rgba(26, 26, 21, 0.16);
          font-size: 0.92rem;
          background: #fff;
          outline: none;
        }
        .search:focus { border-color: #16a34a; box-shadow: 0 0 0 3px rgba(22,163,74,0.15); }

        .chips { display: flex; gap: 10px; flex-wrap: wrap; }
        .chips button {
          border: 1.5px solid rgba(26, 26, 21, 0.12);
          background: #fff;
          padding: 9px 18px;
          border-radius: 999px;
          font-size: 0.88rem;
          font-weight: 600;
          color: #334155;
          cursor: pointer;
```

```
          transition: all 0.15s ease;
          display: inline-flex;
          align-items: center;
          gap: 6px;
        }
        .chips button:hover {
          border-color: #16a34a;
          color: #16a34a;
          background: #f0fdf4;
        }
        .chips button.active {
          background: #16a34a;
          color: #ffffff;
          border-color: #16a34a;
          box-shadow: 0 2px 8px rgba(22, 163, 74, 0.25);
        }

        .empty-category-card {
          text-align: center;
          padding: 60px 24px;
          background: #fff;
          border: 1px dashed rgba(26, 26, 21, 0.15);
          border-radius: 16px;
          max-width: 540px;
          margin: 20px auto;
        }
        .empty-cat-icon { font-size: 3rem; display: block; margin-bottom: 12px; }
        .empty-category-card h3 { font-family: var(--font-display); font-size: 1.35rem; color: #1b382
1; margin: 0 0 8px; }
        .empty-category-card p { font-size: 0.88rem; color: #64748b; margin: 0 0 20px; }
        .btn-view-all-cats {
          background: #16a34a;
          color: #ffffff;
          border: none;
          padding: 10px 22px;
          border-radius: 999px;
          font-weight: 700;
          font-size: 0.88rem;
          cursor: pointer;
          box-shadow: 0 2px 6px rgba(22, 163, 74, 0.2);
        }
        .btn-view-all-cats:hover { background: #15803d; }

        .grid {
          display: grid;
          grid-template-columns: repeat(auto-fill, minmax(240px, 1fr));
          gap: 24px;
        }

        .card-wrapper { transition: transform 0.15s; }

        .card {
          display: block;
          background: #ffffff;
          border: 1.5px solid rgba(26, 26, 21, 0.1);
          border-radius: 16px;
          padding: 20px;
          transition: transform 0.2s ease, box-shadow 0.2s ease, border-color 0.2s;
          box-shadow: 0 4px 14px rgba(22, 36, 26, 0.05);
          position: relative;
          text-decoration: none;
        }
```

```
        .card:hover {
          transform: translateY(-4px);
          border-color: #16a34a;
          box-shadow: 0 12px 28px rgba(22, 163, 74, 0.12);
        }

        .card-top {
          margin: -20px -20px 14px;
          height: 140px;
          border-radius: 14px 14px 0 0;
          overflow: hidden;
          background: linear-gradient(160deg, #f7f2e6, #ede4cd);
        }
        .badge {
          display: inline-block;
          font-size: 0.72rem;
          background: rgba(76, 122, 61, 0.12);
          color: #15803d;
          padding: 3px 10px;
          border-radius: 999px;
          margin-bottom: 8px;
          font-weight: 600;
        }
        h4 { font-family: var(--font-display); font-weight: 700; font-size: 1.1rem; color: #1b3821; m
argin: 0 0 4px; }
        .farmer { font-size: 0.82rem; color: #6b6555; margin: 0 0 12px; }
        .price-row { display: flex; align-items: center; justify-content: space-between; gap: 8px; }
        .price { font-family: var(--font-mono); font-weight: 700; color: #b91c1c; font-size: 1.05rem;
margin: 0; }
        .price-badge { font-size: 0.68rem; color: #16a34a; background: #dcfce7; padding: 2px 7px; bor
der-radius: 4px; font-weight: 600; white-space: nowrap; display: inline-block; margin-top: 2px; }

        .btn-quick-add {
          background: #16a34a;
          color: #ffffff;
          border: none;
          padding: 6px 14px;
          border-radius: 6px;
          font-weight: 700;
          font-size: 0.8rem;
          cursor: pointer;
          transition: background 0.15s;
        }
        .btn-quick-add:hover { background: #15803d; }

        /* Add to Cart Toast */
        .add-toast {
          position: fixed;
          bottom: 24px;
          right: 24px;
          z-index: 1050;
          background: #1b3821;
          color: #ffffff;
          padding: 14px 20px;
          border-radius: 12px;
          box-shadow: 0 8px 24px rgba(0,0,0,0.25);
          display: flex;
          align-items: center;
          gap: 14px;
          animation: slideUpToast 0.25s ease-out;
          max-width: 340px;
        }
```

```
        @keyframes slideUpToast {
          from { opacity: 0; transform: translateY(12px); }
          to { opacity: 1; transform: translateY(0); }
        }
        .toast-icon { font-size: 1.4rem; }
        .toast-text { display: flex; flex-direction: column; flex: 1; }
        .toast-text strong { font-size: 0.9rem; color: #4ade80; }
        .toast-text span { font-size: 0.78rem; color: #cbd5e1; }
        .btn-toast-cart { background: #22c55e; color: #052e16; font-size: 0.78rem; font-weight: 800;
padding: 6px 12px; border-radius: 6px; text-decoration: none; white-space: nowrap; }
        .btn-toast-cart:hover { background: #16a34a; color: #ffffff; }

        @media (max-width: 600px) {
          .add-toast { bottom: 16px; right: 16px; left: 16px; max-width: none; }
        }
      `}</style>
    </>
  );
}
```

app/offers/page.tsx (Promotional Offers & Discount Coupons View)

```
"use client";

import { useState } from "react";
import Link from "next/link";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";

const ALL_OFFERS = [
  {
    code: "FRESHFARM10",
    title: "10% Harvest Season Discount",
    desc: "Get 10% instant discount on all fresh vegetables and fruits directly from verified local f
arms.",
    minOrder: "₹299",
    tag: "MOST POPULAR",
    color: "#16a34a",
    bg: "#f0fdf4",
    border: "#bbf7d0",
    expiry: "Active All Season",
  },
  {
    code: "ORGANIC20",
    title: "20% OFF Organic Greens & Herbs",
    desc: "Special discount on all certified chemical-free leafy vegetables, spinach, coriander, and
mint.",
    minOrder: "₹199",
    tag: "VEGGIE SPECIAL",
    color: "#0284c7",
    bg: "#f0f9ff",
    border: "#bae6fd",
    expiry: "Valid this week",
  },
  {
    code: "FREESHIP",
    title: "100% Free Doorstep Delivery",
    desc: "Enjoy zero delivery fee on all direct farm produce orders delivered fresh within 24-48 hou
rs.",
    minOrder: "₹499",
    tag: "ZERO SHIPPING",
    color: "#7c3aed",
    bg: "#f5f3ff",
    border: "#ddd6fe",
    expiry: "No Expiry",
  },
  {
    code: "FIRSTHARVEST",
    title: "Flat ₹50 Welcome Discount",
    desc: "Exclusive welcome gift for your first harvest order on the Agrenza marketplace.",
    minOrder: "₹250",
    tag: "NEW BUYER",
    color: "#ea580c",
    bg: "#fff7ed",
    border: "#fed7aa",
    expiry: "First Order Only",
  },
  {
    code: "BULKSAVER15",
    title: "15% Direct Wholesale Discount",
    desc: "Save 15% when purchasing 5kg or more for restaurants, hotels, hostels, or large familie
```

```
s.",
    minOrder: "5 kg",
    tag: "WHOLESALE",
    color: "#b45309",
    bg: "#fffbeb",
    border: "#fde68a",
    expiry: "Ongoing",
  },
  {
    code: "FARMDIRECT5",
    title: "5% Extra Cashback on Online UPI",
    desc: "Pay seamlessly with Google Pay, PhonePe, or Paytm UPI and enjoy an instant 5% cashback dis
count.",
    minOrder: "₹150",
    tag: "UPI OFFER",
    color: "#059669",
    bg: "#ecfdf5",
    border: "#a7f3d0",
    expiry: "Valid on all online payments",
  },
];

export default function OffersPage() {
  const [copiedCode, setCopiedCode] = useState<string | null>(null);

  function handleCopy(code: string) {
    navigator.clipboard.writeText(code);
    setCopiedCode(code);
    setTimeout(() => setCopiedCode(null), 2000);
  }

  return (
    <>
      <Nav />
      <div className="offers-hero">
        <div className="container">
          <span className="eyebrow">🏷️ Special Promotions &amp; Coupons</span>
          <h1>Buyer Discounts &amp; Farm Offers</h1>
          <p className="lead">
            Save more on 100% organic harvest directly from local Tamil Nadu farmers with our verifie
d coupon codes!
          </p>
        </div>
      </div>

      <div className="container offers-body">
        <div className="how-it-works-banner">
          <div className="step-item">
            <span className="step-num">1</span>
            <div>
              <strong>Pick a Coupon</strong>
              <p>Choose any active discount code below and click &ldquo;Copy Code&rdquo;.</p>
            </div>
          </div>
          <div className="step-item">
            <span className="step-num">2</span>
            <div>
              <strong>Add Produce to Cart</strong>
              <p>Explore tomatoes, carrots, greens, and fruits in the Farm Shop.</p>
            </div>
          </div>
          <div className="step-item">
```

```
            <span className="step-num">3</span>
            <div>
              <strong>Apply at Checkout</strong>
              <p>Paste the coupon code during checkout to claim your instant savings!</p>
            </div>
          </div>
        </div>

        <div className="offers-grid">
          {ALL_OFFERS.map((offer) => (
            <div
              className="offer-card"
              key={offer.code}
              style={{ background: offer.bg, borderColor: offer.border }}
            >
              <div className="offer-header">
                <span className="offer-tag" style={{ background: offer.color }}>
                  {offer.tag}
                </span>
                <span className="min-order">Min Order: {offer.minOrder}</span>
              </div>

              <h3 className="offer-title">{offer.title}</h3>
              <p className="offer-desc">{offer.desc}</p>
              <div className="offer-expiry">⏳ {offer.expiry}</div>

              <div className="offer-bottom">
                <div className="coupon-code-pill">
                  <span className="code-text">{offer.code}</span>
                  <button
                    type="button"
                    className="btn-copy-code"
                    onClick={() => handleCopy(offer.code)}
                  >
                    {copiedCode === offer.code ? "✓ Copied!" : "📋 Copy"}
                  </button>
                </div>

                <Link href="/shop" className="btn-shop-link">
                  Shop Now →
                </Link>
              </div>
            </div>
          ))}
        </div>
      </div>
      <Footer />

      <style jsx>{`
        .offers-hero { background: #16241a; color: #ffffff; padding: 60px 0 50px; text-align: center;
}
        .eyebrow { font-size: 0.82rem; font-weight: 800; color: #4ade80; text-transform: uppercase; l
etter-spacing: 0.08em; display: inline-block; margin-bottom: 8px; }
        .offers-hero h1 { font-family: var(--font-display); font-size: 2.3rem; margin: 0 0 10px; colo
r: #ffffff; }
        .lead { color: #cbd5e1; font-size: 1.05rem; max-width: 680px; margin: 0 auto; }

        .offers-body { padding: 40px 24px 80px; }

        .how-it-works-banner { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px; backg
round: #ffffff; border: 1px solid #e2e8f0; border-radius: 12px; padding: 24px; margin-bottom: 36px; b
ox-shadow: 0 2px 8px rgba(0,0,0,0.03); }
```

```
        @media (max-width: 768px) {
          .how-it-works-banner { grid-template-columns: 1fr; }
        }

        .step-item { display: flex; align-items: flex-start; gap: 14px; }
        .step-num { width: 32px; height: 32px; border-radius: 50%; background: #16a34a; color: #fffff
f; display: flex; align-items: center; justify-content: center; font-weight: 800; font-size: 0.88rem;
flex-shrink: 0; }
        .step-item strong { display: block; font-size: 0.95rem; color: #0f172a; margin-bottom: 2px; }
        .step-item p { margin: 0; font-size: 0.82rem; color: #64748b; line-height: 1.4; }

        .offers-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(330px, 1fr)); g
ap: 22px; }
        .offer-card { border-radius: 14px; border: 1.5px solid; padding: 24px; display: flex; flex-di
rection: column; justify-content: space-between; }
        .offer-header { display: flex; justify-content: space-between; align-items: center; margin-bo
ttom: 12px; }
        .offer-tag { color: #ffffff; font-size: 0.72rem; font-weight: 800; padding: 3px 8px; border-r
adius: 4px; letter-spacing: 0.04em; }
        .min-order { font-size: 0.78rem; color: #64748b; font-weight: 600; }
        .offer-title { font-size: 1.15rem; font-weight: 700; color: #0f172a; margin: 0 0 8px; }
        .offer-desc { font-size: 0.86rem; color: #475569; line-height: 1.5; margin: 0 0 12px; flex:
1; }
        .offer-expiry { font-size: 0.76rem; color: #64748b; margin-bottom: 16px; font-weight: 600; }

        .offer-bottom { display: flex; justify-content: space-between; align-items: center; gap: 10p
x; border-top: 1px dashed rgba(0,0,0,0.14); padding-top: 16px; flex-wrap: wrap; }
        .coupon-code-pill { display: inline-flex; align-items: center; background: #ffffff; border:
1.5px dashed #94a3b8; border-radius: 6px; padding: 5px 12px; gap: 10px; }
        .code-text { font-family: var(--font-mono); font-weight: 800; color: #0f172a; font-size: 0.95
rem; letter-spacing: 0.05em; }
        .btn-copy-code { background: none; border: none; font-size: 0.78rem; font-weight: 700; color:
#16a34a; cursor: pointer; padding: 2px 4px; }
        .btn-shop-link { font-size: 0.86rem; font-weight: 700; color: #16a34a; text-decoration: none;
}
        .btn-shop-link:hover { text-decoration: underline; }
      `}</style>
    </>
  );
}
```

app/about/page.tsx (About Agrenza Mission & Story View)

```
import type { Metadata } from "next";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";

export const metadata: Metadata = {
  title: "About Agrenza — Farm to Doorstep Marketplace",
  description:
    "Learn about Agrenza, India's direct farm-to-doorstep marketplace. We connect verified Indian far
mers with buyers for fresh produce, fair prices and zero middlemen.",
};

export default function AboutPage() {
  return (
    <>
      <Nav />

      {/* Hero */}
      <section className="about-hero">
        <div className="container">
          <p className="eyebrow">Our Story</p>
          <h1>Farm Fresh.<br /><em>No Middlemen.</em></h1>
          <p className="lead">
            Agrenza was built on one belief — the farmer who grows your food deserves a fair price,
            and the family that eats it deserves to know exactly where it came from.
          </p>
        </div>
      </section>

      {/* Mission */}
      <section className="mission container">
        <div className="mission-card">
          <span className="mission-icon">🌱</span>
          <h2>Our Mission</h2>
          <p>
            Connecting Indian farmers directly with buyers — fresh produce, fair prices,
            no middlemen, no compromise on quality.
          </p>
        </div>
      </section>

      {/* Values */}
      <section className="values container">
        <h2 className="section-title">Why Agrenza?</h2>
        <div className="values-grid">
          <div className="value-card">
            <span className="value-icon">🥦</span>
            <h3>Fresh &amp; Traceable</h3>
            <p>Every product is sourced directly from the farm. You know who grew it, where it came f
rom, and when it was harvested.</p>
          </div>
          <div className="value-card">
            <span className="value-icon">🤝</span>
            <h3>Fair for Farmers</h3>
            <p>Farmers list their own prices. No exploitative middlemen. They earn more, you pay less
— everyone wins.</p>
          </div>
          <div className="value-card">
            <span className="value-icon">🏷️</span>
            <h3>Transparent Pricing</h3>
```

```
            <p>All prices are GST-inclusive. No hidden charges. The price you see is the price you pa
y.</p>
          </div>
          <div className="value-card">
            <span className="value-icon">✅</span>
            <h3>Verified Farmers</h3>
            <p>Every farmer on Agrenza is verified by our team. Quality is our promise — or your mone
y back.</p>
          </div>
        </div>
      </section>

      {/* How It Works */}
      <section className="how-it-works">
        <div className="container">
          <h2 className="section-title">How Agrenza Works</h2>
          <div className="steps-grid">
            <div className="step">
              <div className="step-num">01</div>
              <h3>Farmer Lists Produce</h3>
              <p>Verified farmers list their fresh harvest daily with real prices and quantities.</p>
            </div>
            <div className="step-arrow">→</div>
            <div className="step">
              <div className="step-num">02</div>
              <h3>Buyer Shops Online</h3>
              <p>You browse 100+ fresh products, filter by category, and add to cart in seconds.</p>
            </div>
            <div className="step-arrow">→</div>
            <div className="step">
              <div className="step-num">03</div>
              <h3>Delivered Fresh</h3>
              <p>Your order is packed and shipped directly from the farm. Fresh. Fast. Traceable.</p>
            </div>
          </div>
        </div>
      </section>

      {/* Stats */}
      <section className="stats container">
        <h2 className="section-title">Agrenza by the Numbers</h2>
        <div className="stats-grid">
          <div className="stat-card">
            <span className="stat-num">500+</span>
            <span className="stat-label">Verified Farmers</span>
          </div>
          <div className="stat-card">
            <span className="stat-num">10,000+</span>
            <span className="stat-label">Happy Buyers</span>
          </div>
          <div className="stat-card">
            <span className="stat-num">50+</span>
            <span className="stat-label">Cities Served</span>
          </div>
          <div className="stat-card">
            <span className="stat-num">100%</span>
            <span className="stat-label">Fresh Guarantee</span>
          </div>
        </div>
      </section>

      {/* CTA */}
```

```
      <section className="about-cta container">
        <h2>Ready to eat fresh?</h2>
        <p>Join thousands of families buying directly from farmers across India.</p>
        <div className="cta-btns">
          <a href="/shop" className="btn-primary">Shop Now</a>
          <a href="/auth/farmer-register" className="btn-outline">Join as Farmer</a>
        </div>
      </section>

      <Footer />

      <style>{`
        .about-hero {
          background: linear-gradient(180deg, #f0fdf4 0%, #ffffff 100%);
          padding: 80px 0 60px;
          border-bottom: 1px solid rgba(46,125,50,0.1);
        }
        .about-hero .eyebrow {
          font-family: var(--font-mono, monospace);
          text-transform: uppercase;
          letter-spacing: 0.16em;
          font-size: 0.8rem;
          color: #2e7d32;
          font-weight: 600;
          margin: 0 0 16px;
        }
        .about-hero h1 {
          font-family: var(--font-display, serif);
          font-weight: 900;
          font-size: clamp(2.4rem, 6vw, 4rem);
          line-height: 1;
          color: #16241a;
          margin: 0 0 20px;
        }
        .about-hero h1 em { color: #2e7d32; font-style: italic; }
        .about-hero .lead {
          font-size: 1.1rem;
          line-height: 1.65;
          color: #4a5548;
          max-width: 580px;
        }
        .mission {
          padding: 60px 0 0;
          display: flex;
          justify-content: center;
        }
        .mission-card {
          text-align: center;
          background: #f0fdf4;
          border: 1px solid #bbf7d0;
          border-radius: 16px;
          padding: 40px 48px;
          max-width: 600px;
          width: 100%;
        }
        .mission-icon { font-size: 2.5rem; display: block; margin-bottom: 16px; }
        .mission-card h2 {
          font-family: var(--font-display, serif);
          font-size: 1.6rem;
          font-weight: 900;
          color: #166534;
          margin: 0 0 12px;
```

```
}
.mission-card p { color: #4a5548; font-size: 1.05rem; line-height: 1.6; margin: 0; }
.values { padding: 60px 0; }
.how-it-works { background: #f8fafc; padding: 60px 0; }
.stats { padding: 60px 0; }
.about-cta {
  padding: 60px 0 80px;
  text-align: center;
}
.section-title {
  font-family: var(--font-display, serif);
  font-size: 1.8rem;
  font-weight: 900;
  color: #16241a;
  text-align: center;
  margin: 0 0 40px;
}
.values-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 20px;
}
.value-card {
  background: #ffffff;
  border: 1px solid #e2e8f0;
  border-radius: 12px;
  padding: 28px 22px;
  text-align: center;
  transition: box-shadow 0.2s;
}
.value-card:hover { box-shadow: 0 4px 16px rgba(0,0,0,0.08); }
.value-icon { font-size: 2rem; display: block; margin-bottom: 12px; }
.value-card h3 { font-size: 1rem; font-weight: 700; color: #16241a; margin: 0 0 8px; }
.value-card p { font-size: 0.875rem; color: #64748b; line-height: 1.6; margin: 0; }
.steps-grid {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 16px;
  flex-wrap: wrap;
}
.step {
  background: #ffffff;
  border-radius: 12px;
  padding: 28px 24px;
  text-align: center;
  max-width: 220px;
  border: 1px solid #e2e8f0;
}
.step-num {
  font-family: var(--font-mono, monospace);
  font-size: 2rem;
  font-weight: 800;
  color: #2e7d32;
  opacity: 0.4;
  margin-bottom: 10px;
}
.step h3 { font-size: 1rem; font-weight: 700; color: #16241a; margin: 0 0 8px; }
.step p { font-size: 0.83rem; color: #64748b; line-height: 1.5; margin: 0; }
.step-arrow { font-size: 1.5rem; color: #2e7d32; opacity: 0.5; }
.stats-grid {
  display: grid;
```

```
          grid-template-columns: repeat(4, 1fr);
          gap: 20px;
        }
        .stat-card {
          background: linear-gradient(135deg, #166534, #2e7d32);
          border-radius: 12px;
          padding: 32px 20px;
          text-align: center;
          color: #ffffff;
          display: flex;
          flex-direction: column;
          gap: 8px;
        }
        .stat-num { font-size: 2.2rem; font-weight: 900; font-family: var(--font-display, serif); }
        .stat-label { font-size: 0.83rem; opacity: 0.85; font-weight: 500; }
        .about-cta h2 {
          font-family: var(--font-display, serif);
          font-size: 2rem;
          font-weight: 900;
          color: #16241a;
          margin: 0 0 12px;
        }
        .about-cta p { font-size: 1.05rem; color: #4a5548; margin: 0 0 28px; }
        .cta-btns { display: flex; gap: 14px; justify-content: center; flex-wrap: wrap; }
        .btn-primary {
          background: #2e7d32;
          color: #ffffff;
          font-weight: 600;
          padding: 13px 28px;
          border-radius: 999px;
          text-decoration: none;
          transition: background 0.2s;
        }
        .btn-primary:hover { background: #1b5e20; }
        .btn-outline {
          border: 1.5px solid #16241a;
          color: #16241a;
          font-weight: 600;
          padding: 13px 28px;
          border-radius: 999px;
          text-decoration: none;
          transition: all 0.2s;
        }
        .btn-outline:hover { background: #16241a; color: #ffffff; }
        @media (max-width: 860px) {
          .values-grid { grid-template-columns: repeat(2, 1fr); }
          .stats-grid { grid-template-columns: repeat(2, 1fr); }
          .step-arrow { display: none; }
        }
        @media (max-width: 480px) {
          .values-grid { grid-template-columns: 1fr; }
          .stats-grid { grid-template-columns: repeat(2, 1fr); }
        }
      `}</style>
    </>
  );
}
```

app/contact/page.tsx (Contact Us & Customer Support Form View)

```
"use client";

import { useState } from "react";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";

export default function ContactPage() {
  const [form, setForm] = useState({ name: "", email: "", subject: "General Inquiry", message: "" });
  const [sent, setSent] = useState(false);
  const [sending, setSending] = useState(false);

  function handleChange(e: React.ChangeEvent<HTMLInputElement | HTMLTextAreaElement | HTMLSelectEleme
nt>) {
    setForm({ ...form, [e.target.name]: e.target.value });
  }

  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    setSending(true);
    await new Promise(r => setTimeout(r, 900));
    setSending(false);
    setSent(true);
  }

  return (
    <>
      <Nav />

      <section className="contact-hero">
        <div className="container">
          <p className="eyebrow">We're here to help</p>
          <h1>Get in Touch</h1>
          <p className="lead">Questions about your order, farmer support, or partnerships? Our team r
esponds within 24 hours.</p>
        </div>
      </section>

      <section className="contact-body container">
        {/* Contact Cards */}
        <div className="contact-cards">
          <div className="c-card">
            <span className="c-icon">  </span>
            <h3>Email Us</h3>
            <p>For order issues, refunds and general queries</p>
            <a href="mailto:support@agrenza.com">support@agrenza.com</a>
          </div>
          <div className="c-card">
            <span className="c-icon">  </span>
            <h3>Call Us</h3>
            <p>Toll-free Mon–Sat, 9 AM to 6 PM IST</p>
            <a href="tel:18000000000">1800-000-247-3692</a>
          </div>
          <div className="c-card">
            <span className="c-icon">  </span>
            <h3>India Office</h3>
            <p>Agrenza Technologies Pvt Ltd</p>
            <span className="addr">Chennai, Tamil Nadu, India</span>
          </div>
        </div>
```

```
        {/* Contact Form */}
        <div className="form-wrap">
          <h2>Send a Message</h2>

          {sent ? (
            <div className="success-box">
              <span>✅</span>
              <div>
                <strong>Message sent successfully!</strong>
                <p>We'll get back to you within 24 hours on {form.email}</p>
              </div>
            </div>
          ) : (
            <form onSubmit={handleSubmit} className="contact-form">
              <div className="form-row">
                <div className="form-group">
                  <label htmlFor="name">Full Name *</label>
                  <input
                    id="name"
                    name="name"
                    type="text"
                    placeholder="Your name"
                    value={form.name}
                    onChange={handleChange}
                    required
                  />
                </div>
                <div className="form-group">
                  <label htmlFor="email">Email Address *</label>
                  <input
                    id="email"
                    name="email"
                    type="email"
                    placeholder="you@example.com"
                    value={form.email}
                    onChange={handleChange}
                    required
                  />
                </div>
              </div>

              <div className="form-group">
                <label htmlFor="subject">Subject</label>
                <select id="subject" name="subject" value={form.subject} onChange={handleChange}>
                  <option>General Inquiry</option>
                  <option>Order Issue</option>
                  <option>Farmer Support</option>
                  <option>Partnership</option>
                  <option>Report a Problem</option>
                </select>
              </div>

              <div className="form-group">
                <label htmlFor="message">Message *</label>
                <textarea
                  id="message"
                  name="message"
                  rows={5}
                  placeholder="Tell us how we can help..."
                  value={form.message}
                  onChange={handleChange}
```

```
              required
            />
          </div>

          <button type="submit" className="submit-btn" disabled={sending}>
            {sending ? "Sending..." : "Send Message →"}
          </button>
        </form>
      )}
    </div>
  </section>

  <Footer />

  <style jsx>{`
    .contact-hero {
      background: linear-gradient(180deg, #f0fdf4 0%, #ffffff 100%);
      padding: 70px 0 50px;
      border-bottom: 1px solid rgba(46,125,50,0.1);
    }
    .eyebrow {
      font-family: var(--font-mono, monospace);
      text-transform: uppercase;
      letter-spacing: 0.16em;
      font-size: 0.8rem;
      color: #2e7d32;
      font-weight: 600;
      margin: 0 0 14px;
    }
    .contact-hero h1 {
      font-family: var(--font-display, serif);
      font-weight: 900;
      font-size: clamp(2rem, 5vw, 3.2rem);
      color: #16241a;
      margin: 0 0 16px;
    }
    .lead { font-size: 1.05rem; color: #4a5548; max-width: 520px; line-height: 1.6; margin: 0; }
    .contact-body {
      padding: 56px 0 80px;
      display: flex;
      flex-direction: column;
      gap: 48px;
    }
    .contact-cards {
      display: grid;
      grid-template-columns: repeat(3, 1fr);
      gap: 20px;
    }
    .c-card {
      background: #ffffff;
      border: 1px solid #e2e8f0;
      border-radius: 12px;
      padding: 28px 24px;
      text-align: center;
      transition: box-shadow 0.2s;
    }
    .c-card:hover { box-shadow: 0 4px 16px rgba(0,0,0,0.07); }
    .c-icon { font-size: 2rem; display: block; margin-bottom: 12px; }
    .c-card h3 { font-size: 1rem; font-weight: 700; color: #16241a; margin: 0 0 6px; }
    .c-card p { font-size: 0.83rem; color: #64748b; margin: 0 0 10px; }
    .c-card a { color: #2e7d32; font-weight: 600; text-decoration: none; font-size: 0.9rem; }
    .c-card a:hover { text-decoration: underline; }
```

```
        .addr { color: #2e7d32; font-weight: 600; font-size: 0.9rem; }
        .form-wrap {
          background: #ffffff;
          border: 1px solid #e2e8f0;
          border-radius: 16px;
          padding: 40px 48px;
          max-width: 720px;
          width: 100%;
          align-self: center;
        }
        .form-wrap h2 {
          font-family: var(--font-display, serif);
          font-size: 1.5rem;
          font-weight: 800;
          color: #16241a;
          margin: 0 0 28px;
        }
        .contact-form { display: flex; flex-direction: column; gap: 18px; }
        .form-row { display: grid; grid-template-columns: 1fr 1fr; gap: 18px; }
        .form-group { display: flex; flex-direction: column; gap: 6px; }
        .form-group label { font-size: 0.85rem; font-weight: 600; color: #374151; }
        .form-group input,
        .form-group select,
        .form-group textarea {
          border: 1px solid #d1d5db;
          border-radius: 8px;
          padding: 10px 14px;
          font-size: 0.92rem;
          color: #111827;
          background: #ffffff;
          outline: none;
          transition: border-color 0.15s;
          font-family: inherit;
          resize: vertical;
        }
        .form-group input:focus,
        .form-group select:focus,
        .form-group textarea:focus { border-color: #2e7d32; box-shadow: 0 0 0 3px rgba(46,125,50,0.
1); }
        .submit-btn {
          background: #2e7d32;
          color: #ffffff;
          font-weight: 700;
          font-size: 0.95rem;
          padding: 13px 28px;
          border-radius: 8px;
          border: none;
          cursor: pointer;
          transition: background 0.2s;
          align-self: flex-start;
        }
        .submit-btn:hover:not(:disabled) { background: #1b5e20; }
        .submit-btn:disabled { opacity: 0.7; cursor: not-allowed; }
        .success-box {
          display: flex;
          align-items: flex-start;
          gap: 14px;
          background: #f0fdf4;
          border: 1px solid #86efac;
          border-radius: 10px;
          padding: 20px 24px;
        }
```

```
        .success-box span { font-size: 1.5rem; }
        .success-box strong { font-size: 1rem; color: #166534; display: block; margin-bottom: 4px; }
        .success-box p { font-size: 0.85rem; color: #4a5548; margin: 0; }
        @media (max-width: 768px) {
          .contact-cards { grid-template-columns: 1fr; }
          .form-wrap { padding: 28px 20px; }
          .form-row { grid-template-columns: 1fr; }
        }
      `}</style>
    </>
  );
}
```

app/auth/login/page.tsx (User Login & Role Redirection View)

```
"use client";

import { Suspense, useState } from "react";
import { useRouter, useSearchParams } from "next/navigation";
import Link from "next/link";
import AuthShell from "@/app/components/AuthShell";
import "@/app/components/forms.css";
import { loginUser, ApiError, type Role } from "@/app/lib/api";

function LoginForm() {
  const router = useRouter();
  const searchParams = useSearchParams();
  const [role, setRole] = useState<Role>("buyer");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [error, setError] = useState("");
  const [loading, setLoading] = useState(false);
  const pendingApproval = searchParams.get("pending_approval") === "1";

  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    setError("");
    if (!/^\S+@\S+\.\S+$/.test(email)) {
      setError("Enter a valid email address.");
      return;
    }
    setLoading(true);
    try {
      const res = await loginUser({ role, email, password });
      const redirectUrl = searchParams.get("redirect");
      if (redirectUrl && redirectUrl.startsWith("/")) {
        router.push(redirectUrl);
      } else if (res.user.role === "farmer") {
        router.push("/farmer/dashboard");
      } else {
        router.push("/shop");
      }
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not reach the server. Try again.");
    } finally {
      setLoading(false);
    }
  }

  return (
    <AuthShell
      icon="🔐"
      title="Welcome back"
      subtitle="Sign in to shop or manage your farm listings"
      footer={
        <>
          New here? <Link href="/auth/buyer-register">Buyer signup</Link>
          {" · "}
          <Link href="/auth/farmer-register">Farmer signup</Link>
        </>
      }
    >
      <form onSubmit={handleSubmit}>
        {pendingApproval && (
```

```
          <div className="form-info">Your email is verified. An admin still needs to approve your far
mer account.</div>
        )}
        {error && (
          <div className="form-error" style={{ display: "flex", flexDirection: "column", gap: "6px"
}}>
            <span>{error}</span>
            {error.toLowerCase().includes("pending verification") || error.toLowerCase().includes("ve
rify your email") ? (
              <Link
                href={`/auth/email-verify?email=${encodeURIComponent(email)}`}
                style={{
                  color: "#ffffff",
                  background: "#b91c1c",
                  padding: "6px 12px",
                  borderRadius: "6px",
                  fontWeight: 600,
                  fontSize: "0.85rem",
                  textDecoration: "none",
                  textAlign: "center",
                  marginTop: "4px"
                }}
              >
                Verify Email Now →
              </Link>
            ) : null}
          </div>
        )}

        <div className="role-toggle">
          <button type="button" className={role === "buyer" ? "active" : ""} onClick={() => setRole
("buyer")}>
            Buyer
          </button>
          <button type="button" className={role === "farmer" ? "active" : ""} onClick={() => setRole
("farmer")}>
            Farmer
          </button>
        </div>

        <div className="field">
          <label>Email</label>
          <input
            type="email"
            placeholder="you@example.com"
            value={email}
            onChange={(e) => setEmail(e.target.value)}
            required
          />
        </div>

        <div className="field">
          <label>Password</label>
          <input
            type="password"
            value={password}
            onChange={(e) => setPassword(e.target.value)}
            required
          />
        </div>

        <button className="btn-block" type="submit" disabled={loading}>
```

```
          {loading ? "Logging in…" : "Login"}
        </button>
      </form>

      <p className="admin-link">
        <Link href="/auth/admin-login">Admin login</Link>
      </p>

      <style jsx>{`
        .role-toggle {
          display: grid;
          grid-template-columns: 1fr 1fr;
          gap: 8px;
          margin-bottom: 18px;
        }
        .role-toggle button {
          border: 1px solid rgba(26, 26, 21, 0.18);
          background: #fff;
          border-radius: 999px;
          padding: 9px 12px;
          font-weight: 600;
          font-size: 0.88rem;
          cursor: pointer;
        }
        .role-toggle button.active {
          background: var(--soil);
          color: var(--paper, #efe8d8);
          border-color: var(--soil);
        }
        .admin-link {
          text-align: center;
          margin: 16px 0 0;
          font-size: 0.82rem;
        }
        .admin-link :global(a) {
          color: #6b6555;
        }
      `}</style>
    </AuthShell>
  );
}

export default function LoginPage() {
  return (
    <Suspense fallback={null}>
      <LoginForm />
    </Suspense>
  );
}
```

app/auth/farmer-register/page.tsx (Farmer Partner Onboarding View)

```
"use client";

import { useState, useEffect } from "react";
import Link from "next/link";
import AuthShell from "@/app/components/AuthShell";
import EmailAuthenticityBadge from "@/app/components/EmailAuthenticityBadge";
import "@/app/components/forms.css";
import { registerUser, checkEmailAuthenticity, EmailAuthenticityResult, ApiError } from "@/app/lib/ap
i";

function formatAadhaar(v: string) {
  const digits = v.replace(/\D/g, "").slice(0, 12);
  return digits.replace(/(\d{4})(?=\d)/g, "$1 ").trim();
}

export default function FarmerRegisterPage() {
  const [form, setForm] = useState({ name: "", mobile: "", email: "", aadhaar: "", password: "", conf
irm: "" });
  const [error, setError] = useState("");
  const [isDuplicate, setIsDuplicate] = useState(false);
  const [loading, setLoading] = useState(false);
  const [sent, setSent] = useState(false);

  // Real-time Email Authenticity & Original vs Fake detection
  const [emailCheck, setEmailCheck] = useState<EmailAuthenticityResult | null>(null);
  const [emailChecking, setEmailChecking] = useState(false);

  useEffect(() => {
    const trimmed = form.email.trim();
    if (!trimmed || !trimmed.includes("@")) {
      setEmailCheck(null);
      setEmailChecking(false);
      return;
    }

    setEmailChecking(true);
    const timer = setTimeout(async () => {
      const res = await checkEmailAuthenticity(trimmed, "farmer");
      setEmailCheck(res);
      setEmailChecking(false);

      if (res.is_duplicate) {
        setIsDuplicate(true);
      } else {
        setIsDuplicate(false);
      }
    }, 450);

    return () => clearTimeout(timer);
  }, [form.email]);

  function update<K extends keyof typeof form>(key: K, value: string) {
    setForm((f) => ({ ...f, [key]: value }));
    if (error) {
      setError("");
      setIsDuplicate(false);
    }
  }
```

```
  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (!/^[A-Za-z ]+$/.test(form.name.trim())) return setError("Name should contain letters only.");
    if (!/^[0-9]{10}$/.test(form.mobile.trim())) return setError("Enter a valid 10-digit mobile numbe
r.");
    if (!/^\S+@\S+\.\S+$/.test(form.email.trim())) return setError("Enter a valid email address forma
t.");
    if (form.aadhaar.replace(/\s/g, "").length !== 12) return setError("Enter a valid 12-digit Aadhaa
r number.");

    // Optional custom password validation
    if (form.password && form.password.length < 6) {
      return setError("Password must be at least 6 characters.");
    }
    if (form.password && form.password !== form.confirm) {
      return setError("Passwords do not match.");
    }

    // Validate email authenticity state
    if (emailCheck) {
      if (emailCheck.status === "disposable") {
        return setError("Fake / temporary email detected. Please use your authentic personal Gmail ad
dress.");
      }
      if (emailCheck.is_duplicate) {
        setIsDuplicate(true);
        return setError("This email is already registered. Please log in.");
      }
      if (!emailCheck.ok && emailCheck.error) {
        return setError(emailCheck.error);
      }
    }

    setError("");
    setIsDuplicate(false);
    setLoading(true);

    try {
      const res = await registerUser({
        role: "farmer",
        name: form.name.trim(),
        mobile: form.mobile.trim(),
        password: form.password ? form.password : undefined,
        email: form.email.trim(),
        aadhaar: form.aadhaar.replace(/\s/g, ""),
      });
      setSent(true);
    } catch (err: any) {
      const msg = err instanceof ApiError ? err.message : "Could not reach the server. Try again.";
      setError(msg);
      if (msg.toLowerCase().includes("already registered") || msg.toLowerCase().includes("already exi
sts")) {
        setIsDuplicate(true);
      }
    } finally {
      setLoading(false);
    }
  }

  const isEmailInvalid = Boolean(emailCheck && (!emailCheck.ok || emailCheck.is_duplicate));

  if (sent) {
```

```
    return (
      <AuthShell
        icon="✉️"
        title="Check your Gmail"
        subtitle={`We sent a verification link, OTP code, and login password to ${form.email}.`}
      >
        <div style={{ textAlign: "center", margin: "16px 0" }}>
          <p style={{ color: "#4b5563", fontSize: "0.95rem", lineHeight: 1.5 }}>
            Open your Gmail inbox to verify your email. Once confirmed, our admin team will review yo
ur application for activation.
          </p>

          <div style={{ background: "#f0fdf4", border: "1px solid #bbf7d0", borderRadius: "8px", padd
ing: "14px", margin: "16px 0", textAlign: "left" }}>
            <span style={{ fontSize: "0.85rem", color: "#166534", fontWeight: 700, display: "block"
}}>
              🔑 Your login password has been sent to your Gmail
            </span>
            <p style={{ margin: "8px 0 0", fontSize: "0.8rem", color: "#15803d" }}>
              Check your inbox at <strong>{form.email}</strong> for your login password and verificat
ion code. The password is also attached as a file for safekeeping.
            </p>
          </div>

          <Link
            href={`/auth/email-verify?email=${encodeURIComponent(form.email)}`}
            className="btn-block"
            style={{ display: "block", textAlign: "center", marginTop: 20 }}
          >
            Enter Verification Code
          </Link>

          <Link
            href="/auth/login"
            style={{ display: "block", textAlign: "center", marginTop: 12, fontSize: "0.88rem", colo
r: "var(--leaf, #2e7d32)", fontWeight: 600 }}
          >
            Already verified? Go to Login
          </Link>
        </div>
      </AuthShell>
    );
  }

  return (
    <AuthShell
      icon="🌾"
      title="Farmer Registration"
      subtitle="Join Agrenza and sell your produce directly to buyers at fair rates"
      footer={<>Already have an account? <Link href="/auth/login" style={{ fontWeight: 600 }}>Login h
ere</Link></>}
    >
      <form onSubmit={handleSubmit}>
        {error && (
          <div className="form-error" style={{ display: "flex", flexDirection: "column", gap: "6px"
}}>
            <span>{error}</span>
            {isDuplicate && (
              <Link
                href="/auth/login"
                style={{
                  color: "#ffffff",
```

```
                  background: "#b91c1c",
                  padding: "6px 12px",
                  borderRadius: "6px",
                  fontWeight: 600,
                  fontSize: "0.85rem",
                  textDecoration: "none",
                  textAlign: "center",
                  marginTop: "4px"
                }}
              >
                Go to Login Page →
              </Link>
            )}
          </div>
        )}

        <div className="field">
          <label>Full name</label>
          <input value={form.name} onChange={(e) => update("name", e.target.value)} placeholder="e.g.
Ravi Kumar" required />
        </div>

        <div className="field">
          <label>Mobile number</label>
          <div className="prefix-group">
            <span>+91</span>
            <input
              inputMode="numeric" maxLength={10}
              value={form.mobile}
              onChange={(e) => update("mobile", e.target.value.replace(/\D/g, "").slice(0, 10))}
              placeholder="10 digit mobile number" required
            />
          </div>
        </div>

        <div className="field">
          <label>Gmail / Email address</label>
          <input
            type="email"
            value={form.email}
            onChange={(e) => update("email", e.target.value)}
            placeholder="e.g. ravi@gmail.com"
            style={{
              borderColor: emailCheck?.ok
                ? "#22c55e"
                : isEmailInvalid
                ? "#ef4444"
                : undefined,
            }}
            required
          />
          <EmailAuthenticityBadge
            email={form.email}
            result={emailCheck}
            checking={emailChecking}
          />
          {!emailCheck && (
            <small style={{ color: "#6b7280", fontSize: "0.76rem", marginTop: "4px", display: "block"
}}>
              Must be an authentic email. Used to send verification code, password, and approval aler
ts.
            </small>
```

```
          )}
        </div>

        <div className="field">
          <label>Aadhaar number</label>
          <input
            inputMode="numeric" maxLength={14}
            value={form.aadhaar}
            onChange={(e) => update("aadhaar", formatAadhaar(e.target.value))}
            placeholder="XXXX XXXX XXXX" required
          />
        </div>

        <div style={{ background: "#f0fdf4", border: "1px solid #bbf7d0", padding: "10px 14px", borde
rRadius: "8px", margin: "14px 0", fontSize: "0.82rem", color: "#166534" }}>
          ✨ <strong>Auto-Generated Password:</strong> Leave password blank to auto-generate a secur
e login password attached to your verification email, or specify your own below.
        </div>

        <div className="field">
          <label>
            Password <span style={{ fontWeight: "normal", color: "#6b7280" }}>(Optional — leave blank
to auto-generate)</span>
          </label>
          <input
            type="password"
            minLength={6}
            value={form.password}
            onChange={(e) => update("password", e.target.value)}
            placeholder="Leave blank for auto-generated password"
          />
        </div>

        {form.password && (
          <div className="field">
            <label>Confirm Password</label>
            <input
              type="password"
              minLength={6}
              value={form.confirm}
              onChange={(e) => update("confirm", e.target.value)}
              placeholder="Confirm your password"
              required={Boolean(form.password)}
            />
          </div>
        )}

        <button className="btn-block" type="submit" disabled={loading || emailChecking || isEmailInva
lid}>
          {loading
            ? "Verifying & sending email…"
            : emailChecking
            ? "Checking email authenticity…"
            : isEmailInvalid
            ? "Fix Email to Continue"
            : "Register & Send Verification Email"}
        </button>
      </form>
    </AuthShell>
  );
}
```

app/auth/buyer-register/page.tsx (Buyer Registration View)

```
"use client";

import { useState, useEffect } from "react";
import Link from "next/link";
import AuthShell from "@/app/components/AuthShell";
import EmailAuthenticityBadge from "@/app/components/EmailAuthenticityBadge";
import "@/app/components/forms.css";
import { registerUser, checkEmailAuthenticity, EmailAuthenticityResult, ApiError } from "@/app/lib/ap
i";

export default function BuyerRegisterPage() {
  const [form, setForm] = useState({ name: "", mobile: "", email: "", pincode: "" });
  const ALLOWED_PINCODES = ["629001", "629401"];
  const [error, setError] = useState("");
  const [isDuplicate, setIsDuplicate] = useState(false);
  const [loading, setLoading] = useState(false);
  const [sent, setSent] = useState(false);

  // Real-time Email Authenticity & Original vs Fake detection
  const [emailCheck, setEmailCheck] = useState<EmailAuthenticityResult | null>(null);
  const [emailChecking, setEmailChecking] = useState(false);

  useEffect(() => {
    const trimmed = form.email.trim();
    if (!trimmed || !trimmed.includes("@")) {
      setEmailCheck(null);
      setEmailChecking(false);
      return;
    }

    setEmailChecking(true);
    const timer = setTimeout(async () => {
      const res = await checkEmailAuthenticity(trimmed, "buyer");
      setEmailCheck(res);
      setEmailChecking(false);

      if (res.is_duplicate) {
        setIsDuplicate(true);
      } else {
        setIsDuplicate(false);
      }
    }, 450);

    return () => clearTimeout(timer);
  }, [form.email]);

  function update<K extends keyof typeof form>(key: K, value: string) {
    setForm((f) => ({ ...f, [key]: value }));
    if (error) {
      setError("");
      setIsDuplicate(false);
    }
  }

  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (!/^[A-Za-z ]+$/.test(form.name.trim())) return setError("Name should contain letters only.");
    if (!/^[0-9]{10}$/.test(form.mobile.trim())) return setError("Enter a valid 10-digit mobile numbe
r.");
```

```
    if (!/^[0-9]{6}$/.test(form.pincode.trim())) return setError("Enter a valid 6-digit pincode.");
    if (!ALLOWED_PINCODES.includes(form.pincode.trim())) return setError("Sorry, delivery is currentl
y available only in pincodes 629001 and 629401 (Nagercoil area). Registration is restricted to these
locations.");
    if (!/^\S+@\S+\.\S+$/.test(form.email.trim())) return setError("Enter a valid email address forma
t.");

    // Validate email authenticity state
    if (emailCheck) {
      if (emailCheck.status === "disposable") {
        return setError("Fake / temporary email detected. Please use your authentic personal Gmail ad
dress.");
      }
      if (emailCheck.is_duplicate) {
        setIsDuplicate(true);
        return setError("This email is already registered. Please log in.");
      }
      if (!emailCheck.ok && emailCheck.error) {
        return setError(emailCheck.error);
      }
    }

    setError("");
    setIsDuplicate(false);
    setLoading(true);

    try {
      // Staged in pending_registrations only — user is NOT in users table until verified
      const res = await registerUser({
        role: "buyer",
        name: form.name.trim(),
        mobile: form.mobile.trim(),
        email: form.email.trim(),
      });
      setSent(true);
    } catch (err: any) {
      const msg = err instanceof ApiError ? err.message : "Could not reach the server. Try again.";
      setError(msg);
      if (msg.toLowerCase().includes("already registered") || msg.toLowerCase().includes("already exi
sts")) {
        setIsDuplicate(true);
      }
    } finally {
      setLoading(false);
    }
  }

  const isEmailInvalid = Boolean(emailCheck && (!emailCheck.ok || emailCheck.is_duplicate));

  if (sent) {
    return (
      <AuthShell
        icon="✉"
        title="Check your Gmail"
        subtitle={`We sent a verification link and login password to ${form.email}.`}
      >
        <div style={{ textAlign: "center", margin: "16px 0" }}>
          <p style={{ color: "#4b5563", fontSize: "0.95rem", lineHeight: 1.5 }}>
            Open your Gmail inbox to verify your account. Your account is only activated after email
verification.
          </p>
```

```
          <div style={{ background: "#f0fdf4", border: "1px solid #bbf7d0", borderRadius: "8px", padd
ing: "14px", margin: "16px 0", textAlign: "left" }}>
            <span style={{ fontSize: "0.85rem", color: "#166534", fontWeight: 700, display: "block"
}}>
              🔑 Your login password has been sent to your Gmail
            </span>
            <p style={{ margin: "8px 0 0", fontSize: "0.8rem", color: "#15803d" }}>
              Check your inbox at <strong>{form.email}</strong> for your login password and verificat
ion code. The password is also attached as a file for safekeeping.
            </p>
          </div>

          <Link
            href={`/auth/email-verify?email=${encodeURIComponent(form.email)}`}
            className="btn-block"
            style={{ display: "block", textAlign: "center", marginTop: 20 }}
          >
            Enter Verification Code
          </Link>

          <Link
            href="/auth/login"
            style={{ display: "block", textAlign: "center", marginTop: 12, fontSize: "0.88rem", colo
r: "var(--leaf, #2e7d32)", fontWeight: 600 }}
          >
            Already verified? Go to Login
          </Link>
        </div>
      </AuthShell>
    );
  }

  return (
    <AuthShell
      icon="🧺"
      title="Buyer Registration"
      subtitle="Create an account to shop fresh farm produce directly from verified farmers"
      footer={<>Already have an account? <Link href="/auth/login" style={{ fontWeight: 600 }}>Login h
ere</Link></>}
    >
      <form onSubmit={handleSubmit}>
        {error && (
          <div className="form-error" style={{ display: "flex", flexDirection: "column", gap: "6px"
}}>
            <span>{error}</span>
            {isDuplicate && (
              <Link
                href="/auth/login"
                style={{
                  color: "#ffffff",
                  background: "#b91c1c",
                  padding: "6px 12px",
                  borderRadius: "6px",
                  fontWeight: 600,
                  fontSize: "0.85rem",
                  textDecoration: "none",
                  textAlign: "center",
                  marginTop: "4px"
                }}
              >
                Go to Login Page →
              </Link>
```

```
            )}
          </div>
        )}

        <div className="field">
          <label>Full name</label>
          <input
            value={form.name}
            onChange={(e) => update("name", e.target.value)}
            placeholder="e.g. Priya Raman"
            required
          />
        </div>

        <div className="field">
          <label>Mobile number</label>
          <div className="prefix-group">
            <span>+91</span>
            <input
              inputMode="numeric"
              maxLength={10}
              value={form.mobile}
              onChange={(e) => update("mobile", e.target.value.replace(/\D/g, "").slice(0, 10))}
              placeholder="10-digit mobile number"
              required
            />
          </div>
        </div>

        <div className="field">
          <label>Delivery Pincode</label>
          <input
            inputMode="numeric"
            maxLength={6}
            value={form.pincode}
            onChange={(e) => update("pincode", e.target.value.replace(/\D/g, "").slice(0, 6))}
            placeholder="e.g. 629001"
            required
            style={{
              borderColor:
                form.pincode.length === 6
                  ? ALLOWED_PINCODES.includes(form.pincode)
                    ? "#22c55e"
                    : "#ef4444"
                  : undefined,
            }}
          />
          {form.pincode.length === 6 && ALLOWED_PINCODES.includes(form.pincode) && (
            <small style={{ color: "#16a34a", fontSize: "0.76rem", marginTop: "4px", display: "block"
}}>
              ✅ Delivery available in your area!
            </small>
          )}
          {form.pincode.length === 6 && !ALLOWED_PINCODES.includes(form.pincode) && (
            <small style={{ color: "#dc2626", fontSize: "0.76rem", marginTop: "4px", display: "block"
}}>
              ❌ Sorry, we currently deliver only to pincodes <strong>629001</strong> and <strong>62
9401</strong>.
            </small>
          )}
          {form.pincode.length < 6 && (
            <small style={{ color: "#6b7280", fontSize: "0.76rem", marginTop: "4px", display: "block"
```

```
}}>
              Currently serving: 629001 (Nagercoil) &amp;amp; 629401 only.
            </small>
          )}
        </div>

        <div className="field">
          <label>Gmail / Email address</label>
          <input
            type="email"
            value={form.email}
            onChange={(e) => update("email", e.target.value)}
            placeholder="e.g. priya@gmail.com"
            style={{
              borderColor: emailCheck?.ok
                ? "#22c55e"
                : isEmailInvalid
                ? "#ef4444"
                : undefined,
            }}
            required
          />
          <EmailAuthenticityBadge
            email={form.email}
            result={emailCheck}
            checking={emailChecking}
          />
          {!emailCheck && (
            <small style={{ color: "#6b7280", fontSize: "0.76rem", marginTop: "4px", display: "block"
}}>
              Must be an original email. Disposable / fake emails will be rejected.
            </small>
          )}
        </div>

        <div style={{ background: "#f0fdf4", border: "1px solid #bbf7d0", padding: "10px 14px", borde
rRadius: "8px", margin: "14px 0", fontSize: "0.82rem", color: "#166534" }}>
          ✨ <strong>Auto-Generated Password:</strong> A secure password will be automatically gener
ated and emailed to your Gmail along with your verification code.
        </div>

        <button className="btn-block" type="submit" disabled={loading || emailChecking || isEmailInva
lid}>
          {loading
            ? "Verifying & sending email…"
            : emailChecking
            ? "Checking email authenticity…"
            : isEmailInvalid
            ? "Fix Email to Continue"
            : "Register & Send Verification Email"}
        </button>
      </form>
    </AuthShell>
  );
}
```

app/buyer/cart/page.tsx (Shopping Cart & Order Placement View)

```
"use client";

import { useEffect, useState } from "react";
import Link from "next/link";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import { Tomato, Carrot, LeafyGreen, Chili } from "@/app/components/ProduceIcons";
import { useCart } from "@/app/lib/cart-context";
import { getMe } from "@/app/lib/api";

const ICONS = { tomato: Tomato, carrot: Carrot, leafy: LeafyGreen, chili: Chili };

export default function CartPage() {
  const { items, removeItem, total } = useCart();
  const [user, setUser] = useState<{ id: number; name: string; role: string } | null>(null);

  useEffect(() => {
    (async () => {
      try {
        const me = await getMe();
        if (me?.user) setUser(me.user);
      } catch {
        setUser(null);
      }
    })();
  }, []);

  return (
    <>
      <Nav />
      <div className="container cart-wrap">
        <h1>My cart</h1>

        {(!user || user.role !== "buyer") && items.length > 0 && (
          <div className="login-hint-banner">
            <span>🔒 <strong>Buyer Login Required:</strong> Only logged-in buyers can place orders.
You can log in or register during checkout.</span>
            <Link href="/auth/login?redirect=/buyer/checkout" className="btn-hint-login">
              Log In Now
            </Link>
          </div>
        )}

        {items.length === 0 ? (
          <div className="empty-card">
            <p>Your cart is empty.</p>
            <Link href="/shop" className="btn-solid">Browse products</Link>
          </div>
        ) : (
          <>
            <div className="table-card">
              {items.map((item) => {
                const Icon = ICONS[item.icon];
                return (
                  <div className="row" key={item.productId}>
                    <div className="prod">
                      <div className="thumb"><Icon size={30} /></div>
                      <span>{item.name}</span>
                    </div>
```

```
                    <span className="cell">₹{item.price}/kg</span>
                    <span className="cell">{item.qty} kg</span>
                    <span className="cell subtotal">₹{(item.price * item.qty).toFixed(2)}</span>
                    <button className="remove" onClick={() => removeItem(item.productId)} aria-label=
{`Remove ${item.name}`}>✕</button>
                  </div>
                );
              })}
            </div>

            <div className="summary-card">
              <h2>Total: ₹{total.toFixed(2)}</h2>
              <Link href="/buyer/checkout" className="btn-solid">
                {(!user || user.role !== "buyer") ? "Log In & Proceed to Checkout" : "Proceed to chec
kout"}
              </Link>
            </div>
          </>
        )}
      </div>
      <Footer />

      <style jsx>{`
        .cart-wrap { padding: 44px 24px 90px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.8rem; margin: 0 0 24px;
}
        .login-hint-banner {
          background: #fffbeb;
          border: 1.5px solid #fde68a;
          color: #92400e;
          padding: 14px 18px;
          border-radius: 10px;
          margin-bottom: 20px;
          display: flex;
          justify-content: space-between;
          align-items: center;
          gap: 12px;
          flex-wrap: wrap;
          font-size: 0.88rem;
        }
        .btn-hint-login {
          background: #16a34a;
          color: #fff;
          padding: 6px 14px;
          border-radius: 6px;
          font-weight: 700;
          font-size: 0.82rem;
          text-decoration: none;
        }
        .btn-hint-login:hover { background: #15803d; }
        .empty-card, .table-card, .summary-card {
          background: #fff;
          border: 1px solid rgba(26,26,21,0.08);
          border-radius: 14px;
          padding: 28px;
        }
        .empty-card { text-align: center; }
        .empty-card p { color: #6b6555; margin: 0 0 18px; }
        .table-card { padding: 12px 20px; margin-bottom: 20px; }
        .row {
          display: grid;
          grid-template-columns: 2fr 1fr 1fr 1fr 30px;
```

```
          align-items: center;
          gap: 10px;
          padding: 14px 0;
          border-bottom: 1px solid rgba(26,26,21,0.06);
        }
        .row:last-child { border-bottom: none; }
        .prod { display: flex; align-items: center; gap: 10px; font-weight: 500; }
        .thumb {
          width: 44px; height: 44px; border-radius: 8px;
          background: linear-gradient(160deg,#f7f2e6,#ede4cd);
          display: flex; align-items: center; justify-content: center;
        }
        .cell { font-size: 0.9rem; color: #4a4a3f; }
        .subtotal { font-family: var(--font-mono); color: var(--rust); font-weight: 500; }
        .remove { border: none; background: none; color: var(--chili); font-size: 1rem; }
        .summary-card { display: flex; align-items: center; justify-content: space-between; }
        .summary-card h2 { font-family: var(--font-display); font-size: 1.3rem; margin: 0; }
        .btn-solid {
          background: var(--soil); color: var(--paper);
          padding: 12px 24px; border-radius: 999px; font-weight: 600; font-size: 0.9rem;
        }
        .btn-solid:hover { background: var(--leaf); }
        @media (max-width: 620px) {
          .row { grid-template-columns: 1.6fr 1fr 1fr 30px; }
          .row .cell:nth-child(2) { display: none; }
        }
      `}</style>
    </>
  );
}
```

app/farmer/dashboard/page.tsx (Farmer Operational Dashboard View)

```
"use client";

import Link from "next/link";
import FarmerNav from "@/app/components/FarmerNav";
import { FARMER_NAME, FARMER_PRODUCTS, FARMER_ORDER_ITEMS } from "@/app/lib/farmer-data";

export default function FarmerDashboard() {
  const productCount = FARMER_PRODUCTS.length;
  const totalEarnings = FARMER_ORDER_ITEMS.filter((i) => i.status === "delivered").reduce((s, i) => s
+ i.subtotal, 0);
  const pendingOrders = FARMER_ORDER_ITEMS.filter((i) => ["pending", "confirmed", "shipped"].includes
(i.status)).length;
  const lowStock = FARMER_PRODUCTS.filter((p) => p.stock < 5);

  return (
    <>
      <FarmerNav />
      <div className="container dash-wrap">
        <h1>Welcome, {FARMER_NAME}</h1>

        <div className="stats">
          <div className="stat green">
            <span>Total Products</span>
            <strong>{productCount}</strong>
          </div>
          <div className="stat gold">
            <span>Total Earnings</span>
            <strong>₹{totalEarnings.toFixed(2)}</strong>
          </div>
          <div className="stat rust">
            <span>Pending Orders</span>
            <strong>{pendingOrders}</strong>
          </div>
        </div>

        {lowStock.length > 0 && (
          <div className="low-stock-card">
            <h3>⚠️ Low stock alert</h3>
            <ul>
              {lowStock.map((p) => (
                <li key={p.id}><span>{p.name}</span><span className="qty">{p.stock} kg left</span></l
i>
              ))}
            </ul>
          </div>
        )}

        <div className="actions-card">
          <h3>Quick actions</h3>
          <div className="actions">
            <Link href="/farmer/products/new" className="btn-solid">+ Add new product</Link>
            <Link href="/farmer/orders" className="btn-outline">View orders</Link>
          </div>
        </div>
      </div>

      <style jsx>{`
        .dash-wrap { padding: 40px 24px 90px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.7rem; margin: 0 0 26px;
```

```
}
        .stats { display: grid; grid-template-columns: repeat(3, 1fr); gap: 18px; margin-bottom: 26p
x; }
        .stat {
          border-radius: 14px; padding: 20px 22px; color: #fff;
          display: flex; flex-direction: column; gap: 8px;
        }
        .stat span { font-size: 0.82rem; opacity: 0.85; }
        .stat strong { font-family: var(--font-display); font-size: 1.6rem; }
        .stat.green { background: var(--leaf); }
        .stat.gold { background: var(--turmeric); color: #1a1a15; }
        .stat.rust { background: var(--rust); }
        .low-stock-card, .actions-card {
          background: #fff; border: 1px solid rgba(26,26,21,0.08);
          border-radius: 14px; padding: 22px; margin-bottom: 20px;
        }
        .low-stock-card h3, .actions-card h3 { font-family: var(--font-display); font-size: 1.05rem;
margin: 0 0 14px; }
        .low-stock-card ul { list-style: none; margin: 0; padding: 0; }
        .low-stock-card li { display: flex; justify-content: space-between; padding: 8px 0; border-bo
ttom: 1px solid rgba(26,26,21,0.06); font-size: 0.9rem; }
        .low-stock-card li:last-child { border-bottom: none; }
        .qty { color: var(--chili); font-weight: 600; }
        .actions { display: flex; gap: 12px; flex-wrap: wrap; }
        .btn-solid { background: var(--soil); color: var(--paper); padding: 11px 20px; border-radius:
999px; font-size: 0.88rem; font-weight: 600; }
        .btn-outline { border: 1px solid rgba(26,26,21,0.18); padding: 11px 20px; border-radius: 999p
x; font-size: 0.88rem; }
        @media (max-width: 640px) { .stats { grid-template-columns: 1fr; } }
      `}</style>
    </>
  );
}
```

app/farmer/products/page.tsx (Farmer Inventory & Stock Refill View)

```
"use client";

import { useEffect, useState } from "react";
import Link from "next/link";
import FarmerNav from "@/app/components/FarmerNav";
import { listProducts, deleteProduct, refillProduct, Product } from "@/app/lib/api";

export default function MyProductsPage() {
  const [products, setProducts] = useState<Product[]>([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");
  const [successMsg, setSuccessMsg] = useState("");

  // Refill Modal state
  const [refillModalOpen, setRefillModalOpen] = useState(false);
  const [selectedProduct, setSelectedProduct] = useState<Product | null>(null);
  const [refillAmount, setRefillAmount] = useState<number | string>(50);
  const [refillMode, setRefillMode] = useState<"add" | "set">("add");
  const [refillLoading, setRefillLoading] = useState(false);
  const [refillError, setRefillError] = useState("");

  async function load() {
    setLoading(true);
    try {
      const res = await listProducts({ farmer_mine: "1" });
      setProducts(res.products || []);
    } catch {
      setError("Could not load your products.");
    } finally {
      setLoading(false);
    }
  }

  useEffect(() => {
    load();
  }, []);

  async function handleDelete(id: number) {
    if (!confirm("Are you sure you want to delete this product?")) return;
    try {
      await deleteProduct(id);
      setProducts((prev) => prev.filter((p) => p.id !== id));
      setSuccessMsg("Product deleted successfully.");
      setTimeout(() => setSuccessMsg(""), 4000);
    } catch {
      alert("Failed to delete product.");
    }
  }

  function openRefillModal(p: Product) {
    setSelectedProduct(p);
    setRefillAmount(50);
    setRefillMode("add");
    setRefillError("");
    setRefillModalOpen(true);
  }

  function closeRefillModal() {
    setRefillModalOpen(false);
```

```
    setSelectedProduct(null);
    setRefillError("");
  }

  async function handleRefillSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (!selectedProduct) return;

    const amount = Number(refillAmount);
    if (isNaN(amount) || amount <= 0) {
      setRefillError("Please enter a valid refill quantity greater than 0.");
      return;
    }

    setRefillLoading(true);
    setRefillError("");

    try {
      const res = await refillProduct(selectedProduct.id, amount, refillMode);

      // Update local state immediately
      setProducts((prev) =>
        prev.map((p) =>
          p.id === selectedProduct.id
            ? { ...p, stock: res.new_stock, status: res.status as "active" | "inactive" }
            : p
        )
      );

      setSuccessMsg(`✅ ${res.message || `Successfully refilled ${selectedProduct.name}!`}`);
      setTimeout(() => setSuccessMsg(""), 5000);
      closeRefillModal();
    } catch (err: any) {
      setRefillError(err?.message || "Failed to refill stock. Please try again.");
    } finally {
      setRefillLoading(false);
    }
  }

  return (
    <>
      <FarmerNav />
      <div className="container products-wrap">
        <div className="head">
          <div>
            <h1>My Products &amp;amp; Pricing</h1>
            <p className="subhead">Manage produce listings, base earnings, GST, website prices, and i
nstant stock refills.</p>
          </div>
          <Link href="/farmer/products/new" className="btn-solid">+ Add Product</Link>
        </div>

        {error && <div className="form-error">{error}</div>}
        {successMsg && <div className="alert-success">{successMsg}</div>}

        <div className="table-card">
          <table>
            <thead>
              <tr>
                <th>Item</th>
                <th>Name</th>
                <th>Category</th>
```

```
                <th>Farmer Base (Your Earning)</th>
                <th>Platform Fee &amp;amp; GST</th>
                <th>Website Selling Price</th>
                <th>Stock &amp;amp; Refill</th>
                <th>Website Status</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {loading && (
                <tr><td colSpan={9} className="empty">Loading your products…</td></tr>
              )}
              {!loading && products.length === 0 && (
                <tr><td colSpan={9} className="empty">No products added yet. Click &amp;ldquo;+ Add Produ
ct&amp;rdquo; to list one!</td></tr>
              )}
              {products.map((p) => {
                const base = p.base_price ?? p.price;
                const gst = p.gst_amount ?? (base * ((p.gst_percent ?? 5) / 100));
                const comm = p.commission_amount ?? (base * ((p.commission_percent ?? 5) / 100));
                const selling = p.price;
                const isOutOfStock = p.stock <= 0;
                const isLowStock = p.stock > 0 && p.stock <= 15;

                return (
                  <tr key={p.id}>
                    <td>
                      {p.image ? (
                        // eslint-disable-next-line @next/next/no-img-element
                        <img src={p.image} alt={p.name} className="thumb" />
                      ) : (
                        <div className="thumb placeholder">🌾</div>
                      )}
                    </td>
                    <td>
                      <div className="p-title">
                        <strong>{p.name}</strong>
                        <span className="p-id">ID: #{p.id}</span>
                      </div>
                    </td>
                    <td><span className="cat-tag">{p.category}</span></td>
                    <td>
                      <strong className="base-rate">₹{base.toFixed(2)} / kg</strong>
                    </td>
                    <td>
                      <div className="fee-info">
                        <span>Fee ({p.commission_percent ?? 5}%): ₹{comm.toFixed(2)}</span>
                        <span>GST ({p.gst_percent ?? 5}%): ₹{gst.toFixed(2)}</span>
                      </div>
                    </td>
                    <td>
                      <strong className="selling-rate">₹{selling.toFixed(2)} / kg</strong>
                    </td>
                    <td>
                      <div className="stock-cell">
                        <div className="stock-info">
                          {isOutOfStock ? (
                            <span className="stock-badge out">🔴 Out of Stock</span>
                          ) : isLowStock ? (
                            <span className="stock-badge low">⚠️ {p.stock} kg (Low)</span>
                          ) : (
                            <span className="stock-badge good">🟢 {p.stock} kg</span>
```

```
                    )}
                  </div>
                  <button
                    type="button"
                    onClick={() => openRefillModal(p)}
                    className="btn-refill"
                    title="Refill stock for this product"
                  >
                    ➕ Refill
                  </button>
                </div>
              </td>
              <td>
                {p.approval_status === "approved" && (
                  <span className="badge approved">✓ Live on Website</span>
                )}
                {p.approval_status === "pending" && (
                  <span className="badge pending">⏳ Awaiting Admin</span>
                )}
                {p.approval_status === "rejected" && (
                  <div>
                    <span className="badge rejected">✕ Rejected</span>
                    {p.admin_remark && <div className="remark-text">{p.admin_remark}</div>}
                  </div>
                )}
              </td>
              <td className="row-actions">
                <button
                  type="button"
                  onClick={() => openRefillModal(p)}
                  className="btn-action-refill"
                >
                  Refill
                </button>
                <Link href={`/farmer/products/${p.id}/edit`} className="btn-edit">Edit</Link>
                <button onClick={() => handleDelete(p.id)} className="btn-del">Delete</button>
              </td>
            </tr>
          );
        })}
      </tbody>
    </table>
  </div>
</div>

{/* Refill Modal */}
{refillModalOpen && selectedProduct && (
  <div className="modal-backdrop" onClick={closeRefillModal}>
    <div className="modal-card" onClick={(e) => e.stopPropagation()}>
      <div className="modal-header">
        <div className="modal-title-wrap">
          <span className="modal-icon">🌱</span>
          <div>
            <h3>Refill Product Stock</h3>
            <p className="modal-sub">Quickly restock fresh harvest for your produce</p>
          </div>
        </div>
        <button type="button" className="btn-close" onClick={closeRefillModal}>✕</button>
      </div>

      <form onSubmit={handleRefillSubmit}>
        <div className="modal-body">
```

```
                {/* Product Summary Card */}
                <div className="p-summary-box">
                  {selectedProduct.image ? (
                    // eslint-disable-next-line @next/next/no-img-element
                    <img src={selectedProduct.image} alt={selectedProduct.name} className="summary-th
umb" />
                  ) : (
                    <div className="summary-thumb placeholder">🌾</div>
                  )}
                  <div className="summary-text">
                    <strong className="summary-name">{selectedProduct.name}</strong>
                    <span className="summary-cat">{selectedProduct.category} • Selling at ₹{selectedP
roduct.price.toFixed(2)}/kg</span>
                    <div className="summary-current-stock">
                      Current Stock: <strong>{selectedProduct.stock} kg</strong>
                      {selectedProduct.stock <= 0 && <span className="tag-out">Out of Stock</span>}
                    </div>
                  </div>
                </div>

                {refillError && <div className="modal-error">{refillError}</div>}

                {/* Mode Selector */}
                <div className="mode-selector">
                  <button
                    type="button"
                    className={`mode-btn ${refillMode === "add" ? "active" : ""}`}
                    onClick={() => setRefillMode("add")}
                  >
                    ➕ Add More Stock
                  </button>
                  <button
                    type="button"
                    className={`mode-btn ${refillMode === "set" ? "active" : ""}`}
                    onClick={() => setRefillMode("set")}
                  >
                    🖍️ Set Total Stock
                  </button>
                </div>

                {/* Quick Add Preset Chips */}
                {refillMode === "add" && (
                  <div className="preset-wrap">
                    <label className="input-label">Quick Refill Options:</label>
                    <div className="chips-grid">
                      {[10, 25, 50, 100, 200, 500].map((val) => (
                        <button
                          key={val}
                          type="button"
                          className={`chip ${Number(refillAmount) === val ? "selected" : ""}`}
                          onClick={() => setRefillAmount(val)}
                        >
                          +{val} kg
                        </button>
                      ))}
                    </div>
                  </div>
                )}

                {/* Custom Quantity Input */}
                <div className="input-group">
                  <label className="input-label">
```

```
                  {refillMode === "add" ? "Quantity to Add (kg):" : "New Total Stock Quantity (k
g):"}
                </label>
                <div className="number-input-wrap">
                  <input
                    type="number"
                    step="0.5"
                    min="0.1"
                    required
                    value={refillAmount}
                    onChange={(e) => setRefillAmount(e.target.value)}
                    className="stock-input"
                    placeholder="Enter kg..."
                    autoFocus
                  />
                  <span className="input-unit">kg</span>
                </div>
              </div>

              {/* Live Stock Calculation Box */}
              <div className="calc-preview-box">
                <div className="calc-line">
                  <span>Current Stock:</span>
                  <strong>{selectedProduct.stock} kg</strong>
                </div>
                <div className="calc-line">
                  <span>{refillMode === "add" ? "Refill Added:" : "New Setting:"}</span>
                  <strong className="calc-add">
                    {refillMode === "add" ? `+${Number(refillAmount) || 0} kg` : `${Number(refillAm
ount) || 0} kg`}
                  </strong>
                </div>
                <div className="calc-divider" />
                <div className="calc-line total">
                  <span>New Total Live Stock:</span>
                  <strong className="calc-total">
                    {refillMode === "add"
                      ? `${(selectedProduct.stock + (Number(refillAmount) || 0)).toFixed(1)} kg`
                      : `${(Number(refillAmount) || 0).toFixed(1)} kg`}
                  </strong>
                </div>
              </div>
            </div>

            <div className="modal-footer">
              <button type="button" className="btn-cancel" onClick={closeRefillModal}>
                Cancel
              </button>
              <button type="submit" className="btn-submit-refill" disabled={refillLoading}>
                {refillLoading ? "Refilling Stock…" : "✅ Confirm Stock Refill"}
              </button>
            </div>
          </form>
        </div>
      </div>
    )}

    <style jsx>{`
      .products-wrap { padding: 40px 24px 90px; }
      .head { display: flex; justify-content: space-between; align-items: flex-start; margin-botto
m: 24px; gap: 16px; flex-wrap: wrap; }
      h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 4px;
```

```
color: #1b3821; }
        .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }
        .btn-solid { background: #16a34a; color: #ffffff; padding: 10px 18px; border-radius: 999px; f
ont-size: 0.85rem; font-weight: 600; text-decoration: none; transition: background 0.2s; }
        .btn-solid:hover { background: #15803d; }

        .form-error { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; padding: 12px 1
6px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; }
        .alert-success { background: #f0fdf4; border: 1px solid #bbf7d0; color: #166534; padding: 12p
x 16px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; font-weight: 600; }

        .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow: hidden; overflow-x: auto; box-shadow: 0 2px 8px rgba(0,0,0,0.04); }
        table { width: 100%; border-collapse: collapse; font-size: 0.86rem; }
        th { text-align: left; padding: 14px 16px; background: #faf7ee; font-size: 0.76rem; text-tran
sform: uppercase; letter-spacing: 0.03em; color: #6b6555; white-space: nowrap; }
        td { padding: 12px 16px; border-top: 1px solid rgba(26,26,21,0.06); vertical-align: middle; w
hite-space: nowrap; }

        .thumb { width: 44px; height: 44px; border-radius: 8px; object-fit: cover; }
        .thumb.placeholder { background: linear-gradient(160deg,#f7f2e6,#ede4cd); display: flex; alig
n-items: center; justify-content: center; font-size: 1.2rem; }
        .p-title { display: flex; flex-direction: column; }
        .p-id { font-size: 0.74rem; color: #888273; }
        .cat-tag { background: #faf7ee; border: 1px solid rgba(26,26,21,0.08); padding: 3px 8px; bord
er-radius: 6px; font-size: 0.78rem; font-weight: 500; }

        .base-rate { color: #1b3821; font-size: 0.95rem; }
        .fee-info { display: flex; flex-direction: column; font-size: 0.75rem; color: #6b6555; }
        .selling-rate { color: #16a34a; font-size: 1.05rem; }

        /* Stock & Refill Cell */
        .stock-cell { display: flex; align-items: center; gap: 8px; }
        .stock-badge { font-size: 0.76rem; padding: 4px 8px; border-radius: 6px; font-weight: 600; wh
ite-space: nowrap; }
        .stock-badge.good { background: #f0fdf4; color: #166534; border: 1px solid #bbf7d0; }
        .stock-badge.low { background: #fffbeb; color: #92400e; border: 1px solid #fde68a; }
        .stock-badge.out { background: #fef2f2; color: #991b1b; border: 1px solid #fecaca; }

        .btn-refill { background: #f0fdf4; border: 1px solid #16a34a; color: #15803d; padding: 4px 9p
x; border-radius: 6px; font-size: 0.75rem; font-weight: 700; cursor: pointer; transition: all 0.15s;
white-space: nowrap; }
        .btn-refill:hover { background: #16a34a; color: #ffffff; }

        .badge { font-size: 0.72rem; padding: 3px 10px; border-radius: 999px; text-transform: capital
ize; font-weight: 600; }
        .badge.approved { background: rgba(76,122,61,0.15); color: #16a34a; }
        .badge.pending { background: rgba(232,163,61,0.18); color: #b45309; }
        .badge.rejected { background: rgba(200,60,40,0.12); color: #b8391f; }
        .remark-text { font-size: 0.74rem; color: #b8391f; margin-top: 4px; font-style: italic; }

        .row-actions { display: flex; gap: 10px; align-items: center; }
        .btn-action-refill { border: 1px solid #16a34a; background: #ffffff; color: #16a34a; font-wei
ght: 700; border-radius: 6px; padding: 3px 9px; font-size: 0.78rem; cursor: pointer; transition: all
0.15s; }
        .btn-action-refill:hover { background: #16a34a; color: #fff; }
        .btn-edit { color: #b45309; font-weight: 600; text-decoration: none; font-size: 0.82rem; }
        .btn-del { border: none; background: none; color: #b8391f; font-weight: 600; cursor: pointer;
font-size: 0.82rem; }
        .empty { text-align: center; color: #6b6555; padding: 40px; }

        /* Modal Styles */
```

```
        .modal-backdrop { position: fixed; inset: 0; background: rgba(0,0,0,0.5); backdrop-filter: bl
ur(3px); z-index: 9999; display: flex; align-items: center; justify-content: center; padding: 16px; }
        .modal-card { background: #ffffff; border-radius: 16px; width: 100%; max-width: 480px; box-sh
adow: 0 10px 40px rgba(0,0,0,0.2); overflow: hidden; animation: popIn 0.2s ease-out; }
        @keyframes popIn {
          from { opacity: 0; transform: scale(0.95) translateY(10px); }
          to { opacity: 1; transform: scale(1) translateY(0); }
        }

        .modal-header { display: flex; justify-content: space-between; align-items: center; padding:
18px 22px; background: #1b3821; color: #ffffff; }
        .modal-title-wrap { display: flex; align-items: center; gap: 12px; }
        .modal-icon { font-size: 1.6rem; }
        .modal-header h3 { margin: 0; font-size: 1.15rem; font-weight: 700; color: #ffffff; }
        .modal-sub { margin: 2px 0 0; font-size: 0.78rem; color: #a7f3d0; }
        .btn-close { background: none; border: none; color: #ffffff; font-size: 1.2rem; cursor: point
er; padding: 4px; opacity: 0.8; }
        .btn-close:hover { opacity: 1; }

        .modal-body { padding: 20px 22px; }
        .modal-error { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; padding: 10px
14px; border-radius: 8px; margin-bottom: 16px; font-size: 0.84rem; }

        /* Product Summary Box */
        .p-summary-box { display: flex; align-items: center; gap: 14px; padding: 12px 14px; backgroun
d: #faf7ee; border: 1px solid rgba(26,26,21,0.08); border-radius: 10px; margin-bottom: 18px; }
        .summary-thumb { width: 52px; height: 52px; border-radius: 8px; object-fit: cover; }
        .summary-thumb.placeholder { background: #ede4cd; display: flex; align-items: center; justify
-content: center; font-size: 1.4rem; }
        .summary-text { display: flex; flex-direction: column; gap: 2px; }
        .summary-name { font-size: 0.95rem; color: #1b3821; }
        .summary-cat { font-size: 0.78rem; color: #6b6555; }
        .summary-current-stock { font-size: 0.82rem; color: #1b3821; margin-top: 2px; display: flex;
align-items: center; gap: 6px; }
        .tag-out { background: #fef2f2; color: #991b1b; font-size: 0.7rem; font-weight: 700; padding:
2px 6px; border-radius: 4px; border: 1px solid #fecaca; }

        /* Mode Selector */
        .mode-selector { display: grid; grid-template-columns: 1fr 1fr; gap: 8px; margin-bottom: 16p
x; background: #f3f4f6; padding: 4px; border-radius: 10px; }
        .mode-btn { padding: 8px 12px; border: none; background: transparent; font-size: 0.82rem; fon
t-weight: 600; color: #4b5563; border-radius: 8px; cursor: pointer; transition: all 0.15s; }
        .mode-btn.active { background: #ffffff; color: #16a34a; box-shadow: 0 2px 6px rgba(0,0,0,0.0
6); }

        /* Presets */
        .preset-wrap { margin-bottom: 16px; }
        .chips-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 8px; margin-top: 6p
x; }
        .chip { padding: 8px; border: 1.5px solid #e5e7eb; background: #ffffff; border-radius: 8px; f
ont-size: 0.82rem; font-weight: 600; color: #374151; cursor: pointer; transition: all 0.15s; }
        .chip:hover { border-color: #16a34a; background: #f0fdf4; color: #15803d; }
        .chip.selected { border-color: #16a34a; background: #16a34a; color: #ffffff; }

        /* Input */
        .input-group { margin-bottom: 16px; }
        .input-label { display: block; font-size: 0.82rem; font-weight: 600; color: #374151; margin-b
ottom: 6px; }
        .number-input-wrap { display: flex; align-items: center; border: 1.5px solid #d1d5db; border-
radius: 10px; overflow: hidden; background: #fff; }
        .number-input-wrap:focus-within { border-color: #16a34a; box-shadow: 0 0 0 3px rgba(22,163,7
4,0.15); }
```

```
        .stock-input { flex: 1; border: none; padding: 10px 14px; font-size: 1.05rem; font-weight: 70
0; color: #111827; outline: none; }
        .input-unit { padding: 0 14px; font-size: 0.9rem; font-weight: 600; color: #6b7280; backgroun
d: #f9fafb; height: 100%; display: flex; align-items: center; border-left: 1px solid #e5e7eb; }

        /* Calculation Preview Box */
        .calc-preview-box { background: #f0fdf4; border: 1px solid #bbf7d0; border-radius: 10px; padd
ing: 12px 14px; }
        .calc-line { display: flex; justify-content: space-between; font-size: 0.82rem; color: #37415
1; margin-bottom: 4px; }
        .calc-add { color: #16a34a; }
        .calc-divider { height: 1px; background: #bbf7d0; margin: 6px 0; }
        .calc-line.total { font-size: 0.92rem; font-weight: 700; color: #166534; margin-bottom: 0; }
        .calc-total { font-size: 1.1rem; color: #15803d; }

        /* Modal Footer */
        .modal-footer { display: flex; justify-content: flex-end; gap: 10px; padding: 16px 22px; back
ground: #fafafa; border-top: 1px solid #e5e7eb; }
        .btn-cancel { padding: 9px 16px; border: 1px solid #d1d5db; background: #fff; color: #374151;
border-radius: 999px; font-size: 0.84rem; font-weight: 600; cursor: pointer; }
        .btn-cancel:hover { background: #f3f4f6; }
        .btn-submit-refill { padding: 9px 20px; border: none; background: #16a34a; color: #fff; borde
r-radius: 999px; font-size: 0.84rem; font-weight: 700; cursor: pointer; transition: background 0.15s;
}
        .btn-submit-refill:hover { background: #15803d; }
        .btn-submit-refill:disabled { opacity: 0.6; cursor: not-allowed; }
      `}</style>
    </>
  );
}
```

app/farmer/earnings/page.tsx (Farmer Date-Filtered Earnings View)

```
"use client";

import { useEffect, useState, useCallback } from "react";
import FarmerNav from "@/app/components/FarmerNav";
import { farmerEarnings, EarningItem } from "@/app/lib/api";

type EarningsData = Awaited<ReturnType<typeof farmerEarnings>>;

const PRESETS = [
  { label: "Today",    days: 0 },
  { label: "Last 4d",  days: 4 },
  { label: "Last 5d",  days: 5 },
  { label: "Last 10d", days: 10 },
  { label: "Last 20d", days: 20 },
  { label: "Last 30d", days: 30 },
  { label: "Last 60d", days: 60 },
  { label: "Last 90d", days: 90 },
  { label: "All Time", days: -1 },
];

function toYMD(d: Date) {
  return d.toISOString().slice(0, 10);
}
function fmtDate(d: string) {
  return new Date(d).toLocaleDateString("en-IN", { day: "numeric", month: "short", year: "numeric"
});
}
function fmtMoney(n: number) {
  return n.toLocaleString("en-IN", { minimumFractionDigits: 2, maximumFractionDigits: 2 });
}

export default function EarningsPage() {
  const today = toYMD(new Date());
  const [data, setData]   = useState<EarningsData | null>(null);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

  // Filter state
  const [activePreset, setActivePreset] = useState(-1); // -1 = All Time
  const [fromDate, setFromDate] = useState("");
  const [toDate,   setToDate]   = useState("");
  const [showCal,  setShowCal]  = useState(false);

  const fetchData = useCallback(async (from?: string, to?: string) => {
    setLoading(true);
    setError("");
    try {
      const res = await farmerEarnings(from, to);
      setData(res);
    } catch {
      setError("Could not load earnings. Please refresh.");
    } finally {
      setLoading(false);
    }
  }, []);

  useEffect(() => { fetchData(); }, [fetchData]);

  function applyPreset(days: number) {
```

```
  setActivePreset(days);
  setShowCal(false);
  if (days === -1) {
    setFromDate(""); setToDate("");
    fetchData();
  } else if (days === 0) {
    setFromDate(today); setToDate(today);
    fetchData(today, today);
  } else {
    const from = new Date();
    from.setDate(from.getDate() - days);
    const f = toYMD(from);
    setFromDate(f); setToDate(today);
    fetchData(f, today);
  }
}

function applyCustom() {
  if (!fromDate || !toDate) return;
  setActivePreset(-99);
  fetchData(fromDate, toDate);
  setShowCal(false);
}

function clearFilter() {
  setActivePreset(-1);
  setFromDate(""); setToDate("");
  setShowCal(false);
  fetchData();
}

const isFiltered = activePreset !== -1;
const filterLabel = !isFiltered ? "All Time"
  : activePreset === -99 ? `${fmtDate(fromDate)} — ${fmtDate(toDate)}`
  : activePreset === 0   ? "Today"
  : `Last ${activePreset} days`;

return (
  <>
    <FarmerNav />
    <div className="container earnings-wrap">
      <div className="page-header">
        <h1>My Earnings</h1>
        <span className={`filter-badge ${isFiltered ? "active" : ""}`}>
          📅 {filterLabel}
          {loading && <span className="spin"> ⟳</span>}
        </span>
      </div>

      {error && <div className="err-box">{error}</div>}

      {/* ── Date Filter ──────────────────────────────────── */}
      <div className="filter-card">
        <div className="filter-label">Filter by Date</div>
        <div className="preset-row">
          {PRESETS.map(p => (
            <button
              key={p.days}
              className={`preset-btn ${activePreset === p.days ? "sel" : ""}`}
              onClick={() => applyPreset(p.days)}
              disabled={loading}
            >
```

```
                {p.label}
              </button>
            ))}
            <button
              className={`preset-btn cal-btn ${activePreset === -99 ? "sel" : ""}`}
              onClick={() => setShowCal(!showCal)}
              disabled={loading}
            >
              🗓️ Custom
            </button>
            {isFiltered && (
              <button className="clear-btn" onClick={clearFilter}>✕ Clear</button>
            )}
          </div>

          {showCal && (
            <div className="cal-panel">
              <div className="cal-row">
                <div className="cal-field">
                  <label>From</label>
                  <input type="date" value={fromDate} max={toDate || today} onChange={e => setFromDat
e(e.target.value)} />
                </div>
                <span className="cal-arrow">→</span>
                <div className="cal-field">
                  <label>To</label>
                  <input type="date" value={toDate} min={fromDate} max={today} onChange={e => setToDa
te(e.target.value)} />
                </div>
                <button
                  className="apply-btn"
                  onClick={applyCustom}
                  disabled={!fromDate || !toDate || loading}
                >
                  Apply →
                </button>
              </div>
              {fromDate && toDate && (
                <p className="cal-note">
                  Showing: <strong>{fmtDate(fromDate)}</strong> to <strong>{fmtDate(toDate)}</strong>
                </p>
              )}
            </div>
          )}
        </div>

        {/* ── Stats Cards ──────────────────────────────────────── */}
        {data && (
          <>
            <div className="stats-grid">
              <div className="stat-card green">
                <span className="stat-label">
                  {isFiltered ? "Earnings (Period)" : "Total Earned"}
                </span>
                <strong className="stat-val">₹{fmtMoney(data.total_earned)}</strong>
                <span className="stat-sub">{data.order_count} delivered order{data.order_count !== 1
? "s" : ""}</span>
              </div>
              <div className="stat-card gold">
                <span className="stat-label">Pending Amount</span>
                <strong className="stat-val">₹{fmtMoney(data.pending_amount)}</strong>
                <span className="stat-sub">In-progress orders</span>
```

```
            </div>
            <div className="stat-card blue">
              <span className="stat-label">Today's Earnings</span>
              <strong className="stat-val">₹{fmtMoney(data.today_earned)}</strong>
              <span className="stat-sub">Delivered today</span>
            </div>
          </div>

          {/* ── History Table ──────────────────────────────── */}
          <div className="table-card">
            <div className="table-head">
              <span>Earnings History</span>
              {data.history.length > 0 && (
                <span className="row-count">{data.history.length} records</span>
              )}
            </div>
            {loading ? (
              <div className="loading-row">Loading...</div>
            ) : (
              <table>
                <thead>
                  <tr>
                    <th>Date</th>
                    <th>Product</th>
                    <th>Qty (kg)</th>
                    <th>Amount</th>
                  </tr>
                </thead>
                <tbody>
                  {data.history.length === 0 && (
                    <tr>
                      <td colSpan={4} className="empty">
                        {isFiltered
                          ? `No delivered orders in ${filterLabel.toLowerCase()}.`
                          : "No delivered orders yet."}
                      </td>
                    </tr>
                  )}
                  {data.history.map((item, i) => (
                    <tr key={i}>
                      <td className="date-col">{fmtDate(item.date)}</td>
                      <td>{item.product_name}</td>
                      <td>{item.qty} kg</td>
                      <td className="amount">₹{fmtMoney(item.amount)}</td>
                    </tr>
                  ))}
                </tbody>
                {data.history.length > 0 && (
                  <tfoot>
                    <tr>
                      <td colSpan={3} className="total-label">Total</td>
                      <td className="amount total-val">₹{fmtMoney(data.total_earned)}</td>
                    </tr>
                  </tfoot>
                )}
              </table>
            )}
          </div>
        </>
      )}
    </div>
```

```
      <style jsx>{`
        .earnings-wrap { padding: 40px 24px 90px; }
        .page-header {
          display: flex;
          align-items: center;
          justify-content: space-between;
          flex-wrap: wrap;
          gap: 10px;
          margin-bottom: 20px;
        }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0; }
        .filter-badge {
          font-size: 0.8rem;
          font-weight: 600;
          background: #f1f5f9;
          color: #64748b;
          padding: 5px 14px;
          border-radius: 999px;
          border: 1px solid #e2e8f0;
        }
        .filter-badge.active { background: #f0fdf4; color: #166534; border-color: #86efac; }
        .spin { display: inline-block; animation: spin 0.8s linear infinite; }
        @keyframes spin { to { transform: rotate(360deg); } }

        /* Filter card */
        .filter-card {
          background: #fff;
          border: 1px solid #e2e8f0;
          border-radius: 14px;
          padding: 18px 20px;
          margin-bottom: 22px;
        }
        .filter-label {
          font-size: 0.82rem;
          font-weight: 700;
          color: #374151;
          margin-bottom: 12px;
        }
        .preset-row { display: flex; flex-wrap: wrap; gap: 7px; align-items: center; }
        .preset-btn {
          padding: 6px 13px;
          font-size: 0.8rem;
          font-weight: 600;
          border: 1.5px solid #d1d5db;
          border-radius: 999px;
          background: #fff;
          color: #374151;
          cursor: pointer;
          transition: all 0.15s;
          white-space: nowrap;
        }
        .preset-btn:hover:not(:disabled) { border-color: var(--leaf); color: var(--leaf); background:
#f0fdf4; }
        .preset-btn.sel { background: var(--leaf); color: #fff; border-color: var(--leaf); }
        .preset-btn:disabled { opacity: 0.55; cursor: not-allowed; }
        .cal-btn { border-style: dashed; }
        .clear-btn {
          padding: 6px 13px;
          font-size: 0.8rem;
          font-weight: 700;
          border: 1.5px solid #fca5a5;
          border-radius: 999px;
```

```
          background: #fff;
          color: #dc2626;
          cursor: pointer;
          transition: all 0.15s;
        }
        .clear-btn:hover { background: #fef2f2; }

        /* Calendar panel */
        .cal-panel { margin-top: 14px; padding-top: 14px; border-top: 1px solid #f1f5f9; }
        .cal-row { display: flex; align-items: flex-end; gap: 12px; flex-wrap: wrap; }
        .cal-field { display: flex; flex-direction: column; gap: 4px; }
        .cal-field label { font-size: 0.76rem; font-weight: 600; color: #374151; }
        .cal-field input[type="date"] {
          border: 1.5px solid #d1d5db;
          border-radius: 8px;
          padding: 7px 11px;
          font-size: 0.86rem;
          color: #111827;
          background: #fff;
          outline: none;
          min-width: 148px;
          cursor: pointer;
          transition: border-color 0.15s;
        }
        .cal-field input[type="date"]:focus { border-color: var(--leaf); box-shadow: 0 0 0 3px rgba(7
4,163,32,0.1); }
        .cal-arrow { font-size: 1.1rem; color: #9ca3af; padding-bottom: 7px; }
        .apply-btn {
          background: var(--leaf);
          color: #fff;
          border: none;
          border-radius: 8px;
          padding: 8px 18px;
          font-size: 0.86rem;
          font-weight: 700;
          cursor: pointer;
          transition: background 0.15s;
        }
        .apply-btn:hover:not(:disabled) { background: #2d7a1e; }
        .apply-btn:disabled { opacity: 0.5; cursor: not-allowed; }
        .cal-note { margin-top: 8px; font-size: 0.8rem; color: #6b7280; }

        /* Stats grid */
        .stats-grid {
          display: grid;
          grid-template-columns: repeat(3, 1fr);
          gap: 16px;
          margin-bottom: 22px;
        }
        .stat-card {
          border-radius: 14px;
          padding: 20px 22px;
          display: flex;
          flex-direction: column;
          gap: 6px;
        }
        .stat-card.green { background: var(--leaf); color: #fff; }
        .stat-card.gold  { background: var(--turmeric); color: #1a1a15; }
        .stat-card.blue  { background: #1d4ed8; color: #fff; }
        .stat-label { font-size: 0.78rem; opacity: 0.9; font-weight: 600; text-transform: uppercase;
letter-spacing: 0.04em; }
        .stat-val { font-family: var(--font-display); font-size: 1.6rem; font-weight: 800; }
```

```
        .stat-sub { font-size: 0.74rem; opacity: 0.8; }

        /* Table */
        .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow: hidden; }
        .table-head {
          padding: 14px 20px;
          font-weight: 700;
          font-size: 0.9rem;
          border-bottom: 1px solid rgba(26,26,21,0.06);
          display: flex;
          justify-content: space-between;
          align-items: center;
          color: var(--soil);
        }
        .row-count { font-size: 0.76rem; color: #9ca3af; font-weight: 500; }
        .loading-row { padding: 40px; text-align: center; color: #9ca3af; font-size: 0.88rem; }
        table { width: 100%; border-collapse: collapse; font-size: 0.88rem; }
        th { text-align: left; padding: 11px 20px; font-size: 0.74rem; text-transform: uppercase; let
ter-spacing: 0.04em; color: #6b6555; background: #faf8f5; }
        td { padding: 11px 20px; border-top: 1px solid rgba(26,26,21,0.05); }
        .date-col { color: #6b7280; font-size: 0.84rem; white-space: nowrap; }
        .amount { color: var(--leaf); font-weight: 700; }
        .empty { text-align: center; color: #9ca3af; padding: 44px; font-size: 0.88rem; }
        tfoot td { border-top: 2px solid rgba(26,26,21,0.1); background: #faf8f5; font-weight: 700; }
        .total-label { text-align: right; color: var(--soil); font-size: 0.84rem; }
        .total-val { font-size: 1rem; }
        .err-box { background: #fef2f2; border: 1px solid #fca5a5; border-radius: 10px; padding: 12px
16px; color: #dc2626; font-size: 0.88rem; margin-bottom: 16px; }

        @media (max-width: 640px) {
          .stats-grid { grid-template-columns: 1fr; }
          .cal-row { flex-direction: column; align-items: stretch; }
          .cal-arrow { display: none; }
        }
      `}</style>
    </>
  );
}
```

app/admin/dashboard/page.tsx (Admin Platform Governance Dashboard View)

```
"use client";

import { useEffect, useState, useCallback } from "react";
import { useRouter } from "next/navigation";
import Link from "next/link";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminStats, ApiError } from "@/app/lib/api";

type Stats = Awaited<ReturnType<typeof adminStats>>;

// Quick preset options
const PRESETS = [
  { label: "Today",    days: 0 },
  { label: "Last 4d",  days: 4 },
  { label: "Last 5d",  days: 5 },
  { label: "Last 10d", days: 10 },
  { label: "Last 20d", days: 20 },
  { label: "Last 30d", days: 30 },
  { label: "Last 60d", days: 60 },
  { label: "Last 90d", days: 90 },
  { label: "All Time", days: -1 },
];

function toYMD(d: Date) {
  return d.toISOString().slice(0, 10);
}

function formatDisplay(d: string) {
  return new Date(d).toLocaleDateString("en-IN", { day: "numeric", month: "short", year: "numeric"
});
}

export default function AdminDashboard() {
  const router = useRouter();
  const [stats, setStats] = useState<Stats | null>(null);
  const [error, setError] = useState("");
  const [checking, setChecking] = useState(true);
  const [loading, setLoading] = useState(false);

  // Filter state
  const today = toYMD(new Date());
  const [activePreset, setActivePreset] = useState<number>(-1); // -1 = All Time
  const [fromDate, setFromDate] = useState("");
  const [toDate, setToDate]   = useState("");
  const [showCalendar, setShowCalendar] = useState(false);

  const fetchStats = useCallback(async (from?: string, to?: string) => {
    setLoading(true);
    try {
      const s = await adminStats(from, to);
      setStats(s);
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load stats.");
    } finally {
      setLoading(false);
    }
  }, []);

  useEffect(() => {
```

```
  (async () => {
    try {
      const me = await getMe();
      if (!me.user || me.user.role !== "admin") {
        router.replace("/auth/admin-login");
        return;
      }
      setChecking(false);
      await fetchStats(); // default: all time
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load dashboard.");
      setChecking(false);
    }
  })();
}, [router, fetchStats]);

function applyPreset(days: number) {
  setActivePreset(days);
  setShowCalendar(false);
  if (days === -1) {
    setFromDate("");
    setToDate("");
    fetchStats();
  } else if (days === 0) {
    fetchStats(today, today);
    setFromDate(today);
    setToDate(today);
  } else {
    const from = new Date();
    from.setDate(from.getDate() - days);
    const f = toYMD(from);
    setFromDate(f);
    setToDate(today);
    fetchStats(f, today);
  }
}

function applyCustomRange() {
  if (!fromDate || !toDate) return;
  setActivePreset(-99); // custom
  fetchStats(fromDate, toDate);
  setShowCalendar(false);
}

function clearFilter() {
  setActivePreset(-1);
  setFromDate("");
  setToDate("");
  setShowCalendar(false);
  fetchStats();
}

const isFiltered = activePreset !== -1;
const filterLabel = isFiltered
  ? (activePreset === -99
      ? `${formatDisplay(fromDate)} — ${formatDisplay(toDate)}`
      : (activePreset === 0 ? "Today" : `Last ${activePreset} days`))
  : "All Time";

if (checking) return null;

return (
```

```
    <>
      <AdminNav />
      <div className="container dash-wrap">
        <div className="header-row">
          <div>
            <h1>Admin Dashboard</h1>
            <p className="subhead">Overview of revenue, farmer verifications, and product approvals</
p>
          </div>
          <Link href="/admin/orders" className="btn-income-action">
            💰 View Orders &amp;amp; Income →
          </Link>
        </div>

        {error && <div className="form-error">{error}</div>}

        {/* ── DATE FILTER SECTION ──────────────────────────────── */}
        <div className="filter-section">
          <div className="filter-header">
            <span className="filter-title">🗓️ Filter by Date</span>
            <span className={`filter-badge ${isFiltered ? "active" : ""}`}>
              {filterLabel}
              {loading && <span className="spin-dot"> ⟳</span>}
            </span>
          </div>

          {/* Quick presets */}
          <div className="preset-row">
            {PRESETS.map(p => (
              <button
                key={p.days}
                className={`preset-btn ${activePreset === p.days ? "selected" : ""}`}
                onClick={() => applyPreset(p.days)}
                disabled={loading}
              >
                {p.label}
              </button>
            ))}
            <button
              className={`preset-btn calendar-btn ${activePreset === -99 ? "selected" : ""}`}
              onClick={() => setShowCalendar(!showCalendar)}
              disabled={loading}
            >
              🗓️ Custom Range
            </button>
            {isFiltered && (
              <button className="clear-btn" onClick={clearFilter}>
                ✕ Clear
              </button>
            )}
          </div>

          {/* Custom date range picker */}
          {showCalendar && (
            <div className="calendar-panel">
              <div className="cal-row">
                <div className="cal-field">
                  <label>From Date</label>
                  <input
                    type="date"
                    value={fromDate}
                    max={toDate || today}
```

```
                  onChange={e => setFromDate(e.target.value)}
                />
              </div>
              <div className="cal-arrow">→</div>
              <div className="cal-field">
                <label>To Date</label>
                <input
                  type="date"
                  value={toDate}
                  min={fromDate}
                  max={today}
                  onChange={e => setToDate(e.target.value)}
                />
              </div>
              <button
                className="apply-btn"
                onClick={applyCustomRange}
                disabled={!fromDate || !toDate || loading}
              >
                Apply →
              </button>
            </div>
            {fromDate && toDate && (
              <p className="cal-preview">
                Showing data from <strong>{formatDisplay(fromDate)}</strong> to <strong>{formatDisp
lay(toDate)}</strong>
              </p>
            )}
          </div>
        )}
      </div>

      {stats && (
        <>
          {/* Financial Overview */}
          <h2 className="section-title">Financial &amp;amp; Income Overview</h2>
          <div className="income-grid">
            <div className="income-card primary">
              <span className="label">Total Platform Sales (Income)</span>
              <strong className="amount">
                ₹{(stats.total_income || 0).toLocaleString("en-IN", { minimumFractionDigits: 2, max
imumFractionDigits: 2 })}
              </strong>
              <span className="desc">
                {isFiltered ? `Sales in selected period` : "Gross volume from buyer orders"}
              </span>
            </div>
            <div className="income-card commission">
              <span className="label">Platform Earnings (5% fee)</span>
              <strong className="amount">
                ₹{(stats.platform_earnings || 0).toLocaleString("en-IN", { minimumFractionDigits:
2, maximumFractionDigits: 2 })}
              </strong>
              <span className="desc">
                {isFiltered ? `Earnings in selected period` : "Estimated revenue on completed sale
s"}
              </span>
            </div>
            <div className="income-card today">
              <span className="label">Today&apos;s Income</span>
              <strong className="amount">
                ₹{(stats.today_income || 0).toLocaleString("en-IN", { minimumFractionDigits: 2, max
```

```
imumFractionDigits: 2 })}
                </strong>
                <span className="desc">Total sales recorded today</span>
              </div>
              <div className="income-card orders">
                <span className="label">
                  {isFiltered ? "Orders in Period" : "Total Orders"}
                </span>
                <strong className="amount">{stats.total_orders || 0}</strong>
                <span className="desc">{stats.completed_orders || 0} completed / delivered</span>
              </div>
            </div>

            {/* Pending Approvals */}
            <h2 className="section-title">Pending Approvals</h2>
            <div className="stats">
              <Link href="/admin/farmers" className="stat rust">
                <span>Farmers awaiting approval</span>
                <strong>{stats.pending_farmers}</strong>
                <span className="action-hint">Review &amp;amp; Approve Farmers →</span>
              </Link>
              <Link href="/admin/products" className="stat gold">
                <span>Products awaiting approval</span>
                <strong>{stats.pending_products}</strong>
                <span className="action-hint">Review &amp;amp; Approve Products →</span>
              </Link>
              <Link href="/admin/bulk-requests" className="stat green">
                <span>Bulk requests pending</span>
                <strong>{stats.pending_bulk}</strong>
                <span className="action-hint">Review Buyer Bulk Requests →</span>
              </Link>
            </div>

            {/* Platform Totals */}
            <h2 className="section-title">Platform Stats</h2>
            <div className="stats totals">
              <div className="stat mini">
                <span>Total Registered Farmers</span>
                <strong>{stats.total_farmers}</strong>
              </div>
              <div className="stat mini">
                <span>Total Registered Buyers</span>
                <strong>{stats.total_buyers}</strong>
              </div>
              <Link href="/admin/shared-emails" className="stat mini clickable">
                <span>Same Gmail / Mobile (Farmer &amp;amp; Buyer)</span>
                <strong>{stats.shared_gmail_count || 0}</strong>
                <span className="live-note">View Dual-Role Accounts →</span>
              </Link>
              <div className="stat mini">
                <span>Total Products Listed</span>
                <strong>{stats.total_products}</strong>
                <span className="live-note">{stats.approved_products || 0} live on website</span>
              </div>
            </div>
          </>
        )}
      </div>

      <style jsx>{`
        .dash-wrap { padding: 40px 24px 90px; }
        .header-row { display: flex; justify-content: space-between; align-items: flex-start; margin-
```

```
bottom: 24px; gap: 16px; flex-wrap: wrap; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.8rem; margin: 0 0 6px;
}
        .subhead { color: #6b6555; font-size: 0.9rem; margin: 0; }
        .btn-income-action {
          background: var(--soil); color: #efe8d8; padding: 10px 18px; border-radius: 999px;
          text-decoration: none; font-size: 0.88rem; font-weight: 600;
          transition: background 0.15s ease;
        }
        .btn-income-action:hover { background: #3d3b32; color: #fff; }

        /* ── Filter Section ── */
        .filter-section {
          background: #fff;
          border: 1px solid #e2e8f0;
          border-radius: 16px;
          padding: 20px 24px;
          margin-bottom: 8px;
        }
        .filter-header {
          display: flex;
          align-items: center;
          justify-content: space-between;
          margin-bottom: 14px;
          flex-wrap: wrap;
          gap: 8px;
        }
        .filter-title {
          font-weight: 700;
          font-size: 0.95rem;
          color: #16241a;
        }
        .filter-badge {
          font-size: 0.8rem;
          font-weight: 600;
          background: #f1f5f9;
          color: #64748b;
          padding: 4px 12px;
          border-radius: 999px;
          border: 1px solid #e2e8f0;
        }
        .filter-badge.active {
          background: #f0fdf4;
          color: #166534;
          border-color: #86efac;
        }
        .spin-dot { display: inline-block; animation: spin 0.8s linear infinite; }
        @keyframes spin { to { transform: rotate(360deg); } }

        .preset-row {
          display: flex;
          flex-wrap: wrap;
          gap: 8px;
          align-items: center;
        }
        .preset-btn {
          padding: 7px 14px;
          font-size: 0.82rem;
          font-weight: 600;
          border: 1.5px solid #d1d5db;
          border-radius: 999px;
          background: #fff;
```

```
  color: #374151;
  cursor: pointer;
  transition: all 0.15s;
  white-space: nowrap;
}
.preset-btn:hover:not(:disabled) {
  border-color: #2e7d32;
  color: #2e7d32;
  background: #f0fdf4;
}
.preset-btn.selected {
  background: #2e7d32;
  color: #fff;
  border-color: #2e7d32;
}
.preset-btn:disabled { opacity: 0.6; cursor: not-allowed; }
.calendar-btn { border-style: dashed; }
.clear-btn {
  padding: 7px 14px;
  font-size: 0.82rem;
  font-weight: 700;
  border: 1.5px solid #fca5a5;
  border-radius: 999px;
  background: #fff;
  color: #dc2626;
  cursor: pointer;
  transition: all 0.15s;
}
.clear-btn:hover { background: #fef2f2; border-color: #dc2626; }

/* Calendar panel */
.calendar-panel {
  margin-top: 16px;
  padding-top: 16px;
  border-top: 1px solid #f1f5f9;
}
.cal-row {
  display: flex;
  align-items: flex-end;
  gap: 14px;
  flex-wrap: wrap;
}
.cal-field {
  display: flex;
  flex-direction: column;
  gap: 5px;
}
.cal-field label {
  font-size: 0.78rem;
  font-weight: 600;
  color: #374151;
}
.cal-field input[type="date"] {
  border: 1.5px solid #d1d5db;
  border-radius: 8px;
  padding: 8px 12px;
  font-size: 0.88rem;
  color: #111827;
  background: #fff;
  outline: none;
  transition: border-color 0.15s;
  cursor: pointer;
```

```
          min-width: 160px;
        }
        .cal-field input[type="date"]:focus {
          border-color: #2e7d32;
          box-shadow: 0 0 0 3px rgba(46,125,50,0.1);
        }
        .cal-arrow {
          font-size: 1.2rem;
          color: #9ca3af;
          padding-bottom: 8px;
        }
        .apply-btn {
          background: #2e7d32;
          color: #fff;
          border: none;
          border-radius: 8px;
          padding: 9px 20px;
          font-size: 0.88rem;
          font-weight: 700;
          cursor: pointer;
          transition: background 0.15s;
        }
        .apply-btn:hover:not(:disabled) { background: #1b5e20; }
        .apply-btn:disabled { opacity: 0.5; cursor: not-allowed; }
        .cal-preview {
          margin-top: 10px;
          font-size: 0.82rem;
          color: #6b7280;
        }

        .section-title {
          font-family: var(--font-display); font-size: 1.15rem; font-weight: 700;
          margin: 32px 0 14px; color: var(--soil);
        }

        .income-grid {
          display: grid; grid-template-columns: repeat(auto-fit, minmax(230px, 1fr));
          gap: 16px; margin-bottom: 12px;
        }
        .income-card {
          border-radius: 14px; padding: 22px 24px; display: flex; flex-direction: column; gap: 6px;
          border: 1px solid rgba(26,26,21,0.08); background: #fff;
        }
        .income-card.primary {
          background: linear-gradient(135deg, #2b4c2b 0%, #1c361c 100%);
          color: #fff; border: none;
        }
        .income-card.primary .label { color: #d4ebd4; }
        .income-card.primary .amount { color: #a3e635; }
        .income-card.primary .desc { color: #c0dac0; }
        .income-card.commission {
          background: linear-gradient(135deg, #f7f3e8 0%, #efe7d3 100%);
          border-color: rgba(232,163,61,0.35);
        }
        .income-card.commission .amount { color: var(--rust); }
        .income-card.today { background: #faf8f5; }
        .income-card.today .amount { color: var(--soil); }
        .income-card.orders { background: #faf8f5; }
        .income-card.orders .amount { color: #2563eb; }
        .income-card .label { font-size: 0.82rem; font-weight: 600; text-transform: uppercase; letter
-spacing: 0.04em; color: #6b6555; }
        .income-card .amount { font-family: var(--font-display); font-size: 1.85rem; font-weight: 80
```

```
0; }
        .income-card .desc { font-size: 0.76rem; color: #888273; }

        .stats { display: grid; grid-template-columns: 1fr 1fr 1fr; gap: 18px; margin-bottom: 18px; }
        .stat { border-radius: 14px; padding: 20px 22px; color: #fff; display: flex; flex-direction:
column; gap: 6px; text-decoration: none; transition: transform 0.15s ease, box-shadow 0.15s ease; }
        .stat:hover { transform: translateY(-2px); box-shadow: 0 8px 20px rgba(0,0,0,0.12); }
        .stat span { font-size: 0.82rem; opacity: 0.9; }
        .stat strong { font-family: var(--font-display); font-size: 2rem; }
        .stat .action-hint { font-size: 0.74rem; opacity: 0.85; margin-top: 6px; font-weight: 600; }
        .stat.green { background: var(--leaf); }
        .stat.gold { background: var(--turmeric); color: #1a1a15; }
        .stat.rust { background: var(--rust); }
        .stats.totals { margin-top: 8px; }
        .stat.mini { background: #fff; border: 1px solid rgba(26,26,21,0.08); color: var(--ink); }
        .stat.mini:hover { transform: none; box-shadow: none; }
        .stat.mini strong { color: var(--soil); font-size: 1.6rem; }
        .live-note { font-size: 0.74rem; color: var(--leaf); font-weight: 600; margin-top: 4px; }
        @media (max-width: 768px) {
          .stats { grid-template-columns: 1fr; }
          .income-grid { grid-template-columns: 1fr; }
          .cal-row { flex-direction: column; align-items: stretch; }
          .cal-arrow { display: none; }
          .preset-row { gap: 6px; }
        }
      `}</style>
    </>
  );
}
```

app/admin/shared-emails/page.tsx (Admin Dual-Role Accounts Auditor View)

```
"use client";

import { useEffect, useState, useMemo } from "react";
import { useRouter } from "next/navigation";
import Link from "next/link";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminSharedEmails, SharedEmailAccount, ApiError } from "@/app/lib/api";

export default function SharedEmailsPage() {
  const router = useRouter();
  const [accounts, setAccounts] = useState<SharedEmailAccount[]>([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");
  const [search, setSearch] = useState("");

  useEffect(() => {
    (async () => {
      try {
        const me = await getMe();
        if (!me.user || me.user.role !== "admin") {
          router.replace("/auth/admin-login");
          return;
        }

        const res = await adminSharedEmails();
        setAccounts(res.shared_accounts || []);
      } catch (err) {
        setError(err instanceof ApiError ? err.message : "Failed to load shared accounts.");
      } finally {
        setLoading(false);
      }
    })();
  }, [router]);

  const filtered = useMemo(() => {
    if (!search.trim()) return accounts;
    const q = search.toLowerCase();
    return accounts.filter((a) => {
      return (
        a.email.toLowerCase().includes(q) ||
        (a.mobile && a.mobile.includes(q)) ||
        a.farmer.name.toLowerCase().includes(q) ||
        a.farmer.mobile.includes(q) ||
        (a.farmer.email && a.farmer.email.toLowerCase().includes(q)) ||
        a.buyer.name.toLowerCase().includes(q) ||
        a.buyer.mobile.includes(q) ||
        (a.buyer.email && a.buyer.email.toLowerCase().includes(q))
      );
    });
  }, [accounts, search]);

  if (loading) {
    return (
      <>
        <AdminNav />
        <div className="container page-wrap">
          <p className="loading-text">Loading shared accounts list…</p>
        </div>
      </>
```

```
    );
  }

  return (
    <>
      <AdminNav />
      <div className="container page-wrap">
        <div className="header-row">
          <div>
            <h1>Shared Accounts (Farmer &amp;amp; Buyer)</h1>
            <p className="subhead">
              Users who share the same Gmail ID or Mobile number for both their Farmer and Buyer acco
unts on Agrenza.
            </p>
          </div>
        </div>

        {error && <div className="form-error">{error}</div>}

        {/* STATS OVERVIEW */}
        <div className="stats-row">
          <div className="stat-card primary">
            <span className="stat-label">Total Shared Accounts</span>
            <strong className="stat-val">{accounts.length}</strong>
            <span className="stat-desc">Same email or mobile in 2 roles</span>
          </div>
          <div className="stat-card farmer">
            <span className="stat-label">Linked Farmer Accounts</span>
            <strong className="stat-val">{accounts.length}</strong>
            <span className="stat-desc">Producers with dual registrations</span>
          </div>
          <div className="stat-card buyer">
            <span className="stat-label">Linked Buyer Accounts</span>
            <strong className="stat-val">{accounts.length}</strong>
            <span className="stat-desc">Consumers with dual registrations</span>
          </div>
        </div>

        {/* SEARCH & FILTER */}
        <div className="filter-bar">
          <input
            type="text"
            className="search-input"
            placeholder="Search by Gmail address, mobile number, or name…"
            value={search}
            onChange={(e) => setSearch(e.target.value)}
          />
          <span className="count-badge">{filtered.length} Accounts Found</span>
        </div>

        {filtered.length === 0 ? (
          <div className="empty-box">
            <span className="empty-icon">  </span>
            <h3>No Dual-Role Accounts Found</h3>
            <p>
              {search
                ? `No accounts matched "${search}". Try another search term.`
                : "There are currently no users registered with the same Gmail ID or Mobile Number as
both a Farmer and a Buyer."}
            </p>
          </div>
        ) : (
```

```
          <div className="accounts-grid">
            {filtered.map((item, idx) => (
              <div key={item.email + "_" + item.mobile + "_" + idx} className="shared-card">
                <div className="card-header">
                  <div className="email-badge">
                    <span className="gmail-icon">  </span>
                    <strong className="email-text">{item.email}</strong>
                    {item.mobile && (
                      <span className="mobile-pill">  {item.mobile}</span>
                    )}
                  </div>
                  <span className="dual-pill">
                    {item.shared_by === "both"
                      ? "Same Gmail + Mobile"
                      : item.shared_by === "mobile"
                      ? "Same Mobile Number"
                      : "Same Gmail Address"}
                  </span>
                </div>

                <div className="card-body">
                  {/* FARMER PROFILE */}
                  <div className="profile-col farmer-col">
                    <div className="role-title">
                      <span>  Farmer Profile</span>
                      <span className={`status-badge ${item.farmer.status}`}>
                        {item.farmer.status.toUpperCase()}
                      </span>
                    </div>

                    <div className="info-list">
                      <div className="info-item">
                        <span className="label">Name:</span>
                        <strong>{item.farmer.name}</strong>
                      </div>
                      <div className="info-item">
                        <span className="label">Email:</span>
                        <span>{item.farmer.email || "—"}</span>
                      </div>
                      <div className="info-item">
                        <span className="label">Mobile:</span>
                        <span>{item.farmer.mobile}</span>
                      </div>
                      {item.farmer.aadhaar && (
                        <div className="info-item">
                          <span className="label">Aadhaar:</span>
                          <span className="mono">•••• •••• {item.farmer.aadhaar.slice(-4)}</span>
                        </div>
                      )}
                      <div className="info-item">
                        <span className="label">Products:</span>
                        <span>{item.farmer.total_products} items listed</span>
                      </div>
                      <div className="info-item">
                        <span className="label">Sales:</span>
                        <strong className="green-text">
                          ₹{item.farmer.total_sales.toLocaleString("en-IN", { minimumFractionDigits:
2 })}
                        </strong>
                      </div>
                      <div className="info-item">
                        <span className="label">Registered:</span>
```

```
                      <span className="muted-date">
                        {new Date(item.farmer.joined).toLocaleDateString("en-IN")}
                      </span>
                    </div>
                  </div>
                </div>

                {/* DIVIDER */}
                <div className="col-divider"></div>

                {/* BUYER PROFILE */}
                <div className="profile-col buyer-col">
                  <div className="role-title">
                    <span>🛒 Buyer Profile</span>
                    <span className="status-badge approved">ACTIVE</span>
                  </div>

                  <div className="info-list">
                    <div className="info-item">
                      <span className="label">Name:</span>
                      <strong>{item.buyer.name}</strong>
                    </div>
                    <div className="info-item">
                      <span className="label">Email:</span>
                      <span>{item.buyer.email || "—"}</span>
                    </div>
                    <div className="info-item">
                      <span className="label">Mobile:</span>
                      <span>{item.buyer.mobile}</span>
                    </div>
                    <div className="info-item">
                      <span className="label">Orders:</span>
                      <span>{item.buyer.total_orders} orders placed</span>
                    </div>
                    <div className="info-item">
                      <span className="label">Spent:</span>
                      <strong className="blue-text">
                        ₹{item.buyer.total_spent.toLocaleString("en-IN", { minimumFractionDigits: 2
})}
                      </strong>
                    </div>
                    <div className="info-item">
                      <span className="label">Registered:</span>
                      <span className="muted-date">
                        {new Date(item.buyer.joined).toLocaleDateString("en-IN")}
                      </span>
                    </div>
                  </div>
                </div>
              </div>

              <div className="card-footer">
                <span className="note">
                  💡 This user can access both roles — sell harvest on the Farmer Portal and purch
ase food on the Buyer Portal.
                </span>
                <Link href="/admin/farmers" className="btn-manage">
                  Manage Farmers →
                </Link>
              </div>
            </div>
          ))}
```

```
          </div>
        )}
      </div>

      <style jsx>{`
        .page-wrap { padding: 40px 24px 90px; }
        .header-row { margin-bottom: 24px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.8rem; margin: 0 0 6px;
}
        .subhead { color: #6b6555; font-size: 0.9rem; margin: 0; }
        .loading-text { font-size: 1rem; color: #6b6555; text-align: center; padding: 60px 0; }
        .form-error { background: #fee2e2; border: 1px solid #ef4444; color: #991b1b; padding: 12px 1
6px; border-radius: 8px; margin-bottom: 20px; font-size: 0.9rem; }

        /* Stats overview */
        .stats-row {
          display: grid;
          grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
          gap: 16px;
          margin-bottom: 28px;
        }
        .stat-card {
          border-radius: 14px;
          padding: 20px 22px;
          display: flex;
          flex-direction: column;
          gap: 4px;
          border: 1px solid rgba(26,26,21,0.08);
          background: #fff;
        }
        .stat-card.primary {
          background: linear-gradient(135deg, #2b4c2b 0%, #1c361c 100%);
          color: #fff;
          border: none;
        }
        .stat-card.primary .stat-label { color: #d4ebd4; }
        .stat-card.primary .stat-val { color: #a3e635; }
        .stat-card.primary .stat-desc { color: #c0dac0; }

        .stat-card.farmer {
          background: linear-gradient(135deg, #fefce8 0%, #fef08a 100%);
          border-color: #fde047;
        }
        .stat-card.farmer .stat-label { color: #854d0e; }
        .stat-card.farmer .stat-val { color: #a16207; }
        .stat-card.farmer .stat-desc { color: #713f12; }

        .stat-card.buyer {
          background: linear-gradient(135deg, #eff6ff 0%, #dbeafe 100%);
          border-color: #bfdbfe;
        }
        .stat-card.buyer .stat-label { color: #1e40af; }
        .stat-card.buyer .stat-val { color: #2563eb; }
        .stat-card.buyer .stat-desc { color: #1e3a8a; }

        .stat-label { font-size: 0.78rem; font-weight: 700; text-transform: uppercase; letter-spacin
g: 0.04em; }
        .stat-val { font-family: var(--font-display); font-size: 2rem; font-weight: 800; }
        .stat-desc { font-size: 0.74rem; }

        /* Filter bar */
        .filter-bar {
```

```
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
  margin-bottom: 24px;
  flex-wrap: wrap;
}
.search-input {
  flex: 1;
  min-width: 280px;
  max-width: 500px;
  padding: 10px 16px;
  border-radius: 999px;
  border: 1.5px solid #cbd5e1;
  background: #fff;
  font-size: 0.9rem;
  outline: none;
  transition: border-color 0.15s;
}
.search-input:focus { border-color: var(--leaf, #2e7d32); }
.count-badge {
  background: #f1f5f9;
  color: #475569;
  font-size: 0.8rem;
  font-weight: 700;
  padding: 6px 14px;
  border-radius: 999px;
  border: 1px solid #e2e8f0;
}

/* Empty box */
.empty-box {
  background: #fff;
  border: 1.5px dashed #cbd5e1;
  border-radius: 16px;
  padding: 60px 24px;
  text-align: center;
  max-width: 540px;
  margin: 40px auto;
}
.empty-icon { font-size: 2.8rem; display: block; margin-bottom: 12px; }
.empty-box h3 { font-size: 1.2rem; margin: 0 0 8px; color: #1e293b; }
.empty-box p { color: #64748b; font-size: 0.9rem; line-height: 1.5; margin: 0; }

/* Shared cards */
.accounts-grid {
  display: flex;
  flex-direction: column;
  gap: 20px;
}
.shared-card {
  background: #fff;
  border: 1px solid #e2e8f0;
  border-radius: 16px;
  overflow: hidden;
  box-shadow: 0 4px 14px rgba(0,0,0,0.04);
}
.card-header {
  background: #f8fafc;
  border-bottom: 1px solid #e2e8f0;
  padding: 14px 22px;
  display: flex;
```

```
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 12px;
}
.email-badge {
  display: flex;
  align-items: center;
  gap: 10px;
  flex-wrap: wrap;
}
.gmail-icon { font-size: 1.2rem; }
.email-text { font-size: 1.05rem; color: #0f172a; word-break: break-all; }
.mobile-pill {
  background: #e2e8f0;
  color: #334155;
  font-size: 0.78rem;
  font-weight: 700;
  padding: 2px 8px;
  border-radius: 999px;
}
.dual-pill {
  background: #dcfce7;
  color: #15803d;
  font-size: 0.76rem;
  font-weight: 800;
  padding: 4px 12px;
  border-radius: 999px;
  border: 1px solid #86efac;
  letter-spacing: 0.02em;
}

.card-body {
  display: grid;
  grid-template-columns: 1fr auto 1fr;
  padding: 24px;
  gap: 20px;
}
.profile-col { display: flex; flex-direction: column; gap: 14px; }
.col-divider {
  width: 1px;
  background: #e2e8f0;
}
.role-title {
  display: flex;
  align-items: center;
  justify-content: space-between;
  font-size: 1rem;
  font-weight: 800;
  color: #1e293b;
  border-bottom: 1px solid #f1f5f9;
  padding-bottom: 8px;
}
.status-badge {
  font-size: 0.68rem;
  font-weight: 800;
  padding: 2px 8px;
  border-radius: 4px;
}
.status-badge.approved { background: #dcfce7; color: #166534; }
.status-badge.pending { background: #fef3c7; color: #92400e; }
.status-badge.rejected { background: #fee2e2; color: #991b1b; }
```

```
        .info-list {
          display: flex;
          flex-direction: column;
          gap: 8px;
        }
        .info-item {
          display: flex;
          justify-content: space-between;
          align-items: center;
          font-size: 0.88rem;
          padding: 4px 0;
        }
        .info-item .label { color: #64748b; font-weight: 500; }
        .info-item strong { color: #0f172a; }
        .green-text { color: var(--leaf, #2e7d32); }
        .blue-text { color: #2563eb; }
        .mono { font-family: monospace; font-size: 0.85rem; color: #475569; }
        .muted-date { color: #94a3b8; font-size: 0.82rem; }

        .card-footer {
          background: #fafaf9;
          border-top: 1px solid #e2e8f0;
          padding: 12px 22px;
          display: flex;
          align-items: center;
          justify-content: space-between;
          flex-wrap: wrap;
          gap: 12px;
        }
        .note { font-size: 0.8rem; color: #64748b; flex: 1; min-width: 250px; }
        .btn-manage {
          background: var(--soil, #1e293b);
          color: #fff;
          font-size: 0.8rem;
          font-weight: 700;
          padding: 6px 14px;
          border-radius: 8px;
          text-decoration: none;
          transition: background 0.15s;
        }
        .btn-manage:hover { background: #334155; }

        @media (max-width: 768px) {
          .card-body {
            grid-template-columns: 1fr;
          }
          .col-divider {
            height: 1px;
            width: 100%;
          }
        }
      `}</style>
    </>
  );
}
```

app/lib/farmer-data.ts (Farmer Static Data)

```
import { ProduceIcon } from "./types";

export type FarmerProduct = {
  id: number;
  name: string;
  category: string;
  price: number;
  stock: number;
  status: "active" | "inactive";
  icon: ProduceIcon;
};

export type FarmerOrderItem = {
  itemId: number;
  date: string;
  productName: string;
  buyerName: string;
  buyerMobile: string;
  qty: number;
  subtotal: number;
  status: "pending" | "confirmed" | "shipped" | "delivered" | "cancelled";
};

export const FARMER_NAME = "Ravi Kumar";

export const FARMER_PRODUCTS: FarmerProduct[] = [
  { id: 1, name: "Vine Tomatoes", category: "Vegetables", price: 28, stock: 40, status: "active", ico
n: "tomato" },
  { id: 5, name: "Country Tomatoes", category: "Vegetables", price: 24, stock: 3, status: "active", i
con: "tomato" },
];

export const FARMER_ORDER_ITEMS: FarmerOrderItem[] = [
  { itemId: 101, date: "2026-08-02T10:00:00Z", productName: "Vine Tomatoes", buyerName: "Priya Rama
n", buyerMobile: "9677988927", qty: 4, subtotal: 112, status: "pending" },
  { itemId: 98, date: "2026-07-29T10:00:00Z", productName: "Country Tomatoes", buyerName: "Arjun De
v", buyerMobile: "9345671234", qty: 2, subtotal: 48, status: "delivered" },
  { itemId: 95, date: "2026-07-25T10:00:00Z", productName: "Vine Tomatoes", buyerName: "Kavya S.", bu
yerMobile: "9876543210", qty: 6, subtotal: 168, status: "delivered" },
];

export const CATEGORY_OPTIONS = ["Vegetables", "Fruits", "Grains & Cereals", "Pulses", "Spices", "Dai
ry"];
```

app/lib/firebase.ts (Firebase Client SDK Init)

```
"use client";

import { initializeApp, getApps, getApp, FirebaseApp } from "firebase/app";
import { getAuth, Auth } from "firebase/auth";

/**
 * Firebase client config, read from NEXT_PUBLIC_* env vars so nothing
 * secret ends up in the bundle (these keys are meant to be public — the
 * project is locked down via Firebase Auth settings + your backend's own
 * verification, not by hiding this config). Copy `.env.local.example` to
 * `.env.local` and fill these in from
 * Firebase Console → Project settings → General → Your apps → Web app.
 */
const firebaseConfig = {
  apiKey: process.env.NEXT_PUBLIC_FIREBASE_API_KEY,
  authDomain: process.env.NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN,
  projectId: process.env.NEXT_PUBLIC_FIREBASE_PROJECT_ID,
  storageBucket: process.env.NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET,
  messagingSenderId: process.env.NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID,
  appId: process.env.NEXT_PUBLIC_FIREBASE_APP_ID,
};

let app: FirebaseApp | null = null;
let auth: Auth | null = null;

/** Lazily initialize (and reuse) the Firebase app + Auth instance, client-side only. */
export function getFirebaseAuth(): Auth {
  if (typeof window === "undefined") {
    throw new Error("Firebase Auth is client-only — call this inside a useEffect/event handler.");
  }
  if (!firebaseConfig.apiKey) {
    throw new Error(
      "Missing Firebase config. Copy .env.local.example to .env.local and fill in your Firebase proje
ct's web app keys."
    );
  }
  if (!app) app = getApps().length ? getApp() : initializeApp(firebaseConfig);
  if (!auth) auth = getAuth(app);
  return auth;
}
```

app/lib/otp.ts (OTP Client Utilities)

```
"use client";

import {
  RecaptchaVerifier,
  signInWithPhoneNumber,
  ConfirmationResult,
  isSignInWithEmailLink,
  sendSignInLinkToEmail,
  signInWithEmailLink,
} from "firebase/auth";
import { getFirebaseAuth } from "./firebase";

/**
 * ---------------------------------------------------------------------
 * Mobile OTP — real SMS via Firebase Phone Authentication
 * ---------------------------------------------------------------------
 * This is the standard way to get an actual 6-digit SMS code with
 * Firebase: an invisible reCAPTCHA (required by Firebase to rate-limit
 * SMS abuse) + signInWithPhoneNumber(). Needs the Firebase project on
 * the Blaze (pay-as-you-go) plan — see README.md "Firebase setup".
 */

let recaptchaVerifier: RecaptchaVerifier | null = null;
let confirmationResult: ConfirmationResult | null = null;

/** Call once, after the invisible reCAPTCHA container (id="recaptcha-container") is mounted. */
function ensureRecaptcha(containerId: string) {
  const auth = getFirebaseAuth();
  if (!recaptchaVerifier) {
    recaptchaVerifier = new RecaptchaVerifier(auth, containerId, { size: "invisible" });
  }
  return recaptchaVerifier;
}

/** Sends a real SMS OTP to +91XXXXXXXXXX. Resolves once Firebase has dispatched it. */
export async function sendMobileOtp(mobile10Digit: string, containerId = "recaptcha-container") {
  const auth = getFirebaseAuth();
  const verifier = ensureRecaptcha(containerId);
  confirmationResult = await signInWithPhoneNumber(auth, `+91${mobile10Digit}`, verifier);
}

/** Confirms the 6-digit code the user typed. Returns a Firebase ID token to send to the PHP backend.
*/
export async function confirmMobileOtp(code: string): Promise<string> {
  if (!confirmationResult) {
    throw new Error("No OTP was sent yet — call sendMobileOtp() first (e.g. on page load).");
  }
  const cred = await confirmationResult.confirm(code);
  return cred.user.getIdToken();
}

export function resetMobileOtpState() {
  confirmationResult = null;
}

/**
 * ---------------------------------------------------------------------
 * Email verification — Firebase passwordless email link
 * ---------------------------------------------------------------------
```

```
 * Firebase Auth has no built-in numeric "email OTP code" API — the
 * closest real equivalent is a passwordless sign-in link, which Firebase
 * emails automatically and which verifies the address the moment it's
 * clicked. That's what this wraps.
 *
 * The link points at NEXT_PUBLIC_APP_URL if it's set (e.g. your LAN IP,
 * http://192.168.x.x:3000), so it also works when opened from a phone
 * on the same Wi-Fi. It falls back to window.location.origin
 * (localhost) if that env var isn't set — which only works when opened
 * on the same computer running the dev server.
 */

const EMAIL_STORAGE_KEY = "farmlink_pending_email";

function verificationBaseUrl() {
  return process.env.NEXT_PUBLIC_APP_URL || window.location.origin;
}

export async function sendEmailVerificationLink(email: string) {
  const auth = getFirebaseAuth();
  const actionCodeSettings = {
    url: `${verificationBaseUrl()}/auth/email-verify`,
    handleCodeInApp: true,
  };
  await sendSignInLinkToEmail(auth, email, actionCodeSettings);
  window.localStorage.setItem(EMAIL_STORAGE_KEY, email);
}

export function getStoredPendingEmail(): string | null {
  if (typeof window === "undefined") return null;
  return window.localStorage.getItem(EMAIL_STORAGE_KEY);
}

/** Call on the /auth/email-verify landing page. Returns a Firebase ID token + email, or null if link
is invalid or email is needed. */
export async function completeEmailVerification(manualEmail?: string): Promise<{ idToken: string; ema
il: string } | null> {
  const auth = getFirebaseAuth();
  if (!isSignInWithEmailLink(auth, window.location.href)) return null;

  let email = manualEmail || (typeof window !== "undefined" ? window.localStorage.getItem(EMAIL_STORA
GE_KEY) : null);
  if (!email) {
    return null;
  }

  const cred = await signInWithEmailLink(auth, email, window.location.href);
  if (typeof window !== "undefined") {
    window.localStorage.removeItem(EMAIL_STORAGE_KEY);
  }
  const idToken = await cred.user.getIdToken();
  return { idToken, email };
}
```

app/lib/products-data.ts (Products Static Data)

```
import { Product } from "./types";

/**
 * Placeholder catalog so every page renders standalone. Once the PHP
 * backend exposes a JSON endpoint (e.g. backend/api/products.php),
 * replace this with a fetch() call — see README.md.
 *
 * `image` uses Wikimedia Commons' stable Special:FilePath redirect
 * (https://commons.wikimedia.org/wiki/Special:FilePath/<file name>),
 * which always resolves to the current full-size upload for that file —
 * no CDN hash to keep track of, and every file below is real, freely
 * licensed (CC / public domain) produce photography. Swap these for your
 * farmers' own uploaded photos once `products/create.php` is wired up
 * (see "Product images" in README.md).
 *
 * db_id: the real product ID in the MySQL `products` table.
 * Used when calling backend APIs (bulk requests, orders, etc.)
 */
const WIKI = "https://commons.wikimedia.org/wiki/Special:FilePath/";

export const PRODUCTS: Product[] = [
  { id: 1,  db_id: 11, name: "Vine Tomatoes",   farmer: "Ravi Kumar",    location: "Salem",      pric
e: 28, category: "Vegetables",   icon: "tomato", stock: 40, description: "Sun-ripened tomatoes picked
this morning, ideal for curries and chutneys. No cold storage — sold within a day of harvest.", imag
e: `${WIKI}Tomato_je.jpg` },
  { id: 2,  db_id: 13, name: "Local Carrots",    farmer: "Meena Devi",   location: "Ooty",       pric
e: 32, category: "Vegetables",   icon: "carrot", stock: 25, description: "Sweet, crisp hill-grown car
rots. Great raw, roasted, or in a home-style sambar.", image: `${WIKI}Carrots.JPG` },
  { id: 3,  db_id: 12, name: "Fresh Spinach",    farmer: "Suresh P.",    location: "Erode",      pric
e: 18, category: "Leafy Greens", icon: "leafy",  stock: 15, description: "Tender spinach leaves, harv
ested at dawn. Rich in iron, best used within two days.", image: `${WIKI}Spinach_leaves.jpg` },
  { id: 4,  db_id: 14, name: "Green Chili",      farmer: "Lakshmi N.",   location: "Coimbatore", pric
e: 45, category: "Vegetables",   icon: "chili",  stock: 12, description: "Medium-heat green chilies,
hand-picked and sorted for freshness.", image: `${WIKI}Aesthetic_green_Chillies.JPG` },
  { id: 5,  db_id: 11, name: "Country Tomatoes", farmer: "Muthu Selvam", location: "Dindigul",   pric
e: 24, category: "Vegetables",   icon: "tomato", stock: 60, description: "Slightly tangier native var
iety, well suited for rasam and pickles.", image: `${WIKI}Tomatoes_ARS.jpg` },
  { id: 6,  db_id: 13, name: "Baby Carrots",     farmer: "Meena Devi",   location: "Ooty",       pric
e: 38, category: "Vegetables",   icon: "carrot", stock: 18, description: "Small, tender carrots — a f
avourite for salads and steaming whole.", image: `${WIKI}Carrots_with_stems.jpg` },
  { id: 7,  db_id: 12, name: "Amaranth Greens",  farmer: "Suresh P.",    location: "Erode",      pric
e: 16, category: "Leafy Greens", icon: "leafy",  stock: 20, description: "Local mulaikeerai, a staple
in South Indian home cooking.", image: `${WIKI}Basella_alba_leaves_27052014.jpg` },
  { id: 8,  db_id: 14, name: "Byadgi Chili",     farmer: "Lakshmi N.",   location: "Coimbatore", pric
e: 52, category: "Vegetables",   icon: "chili",  stock:  8, description: "Mild, deep-red chilies priz
ed for colour more than heat.", image: `${WIKI}Chili_pepper.jpg` },
];

export const CATEGORIES = Array.from(new Set(PRODUCTS.map((p) => p.category)));

export function getProduct(id: number): Product | undefined {
  return PRODUCTS.find((p) => p.id === id);
}
```

app/components/AuthShell.tsx (Auth Page Shell Wrapper)

```
"use client";

import Link from "next/link";

export default function AuthShell({
  icon,
  title,
  subtitle,
  children,
  footer,
}: {
  icon: string;
  title: string;
  subtitle: string;
  children: React.ReactNode;
  footer?: React.ReactNode;
}) {
  return (
    <div className="auth-page">
      <div className="scene" aria-hidden="true">
        <div className="cube c1"><span /><span /><span /><span /><span /><span /></div>
        <div className="cube c2"><span /><span /><span /><span /><span /><span /></div>
        <div className="cube c3"><span /><span /><span /><span /><span /><span /></div>
        <div className="orb o1" />
        <div className="orb o2" />
        <div className="orb o3" />
      </div>

      <Link href="/" className="auth-logo">Farm<span>Link</span></Link>
      <div className="auth-card">
        <p className="auth-icon">{icon}</p>
        <h1>{title}</h1>
        <p className="subtitle">{subtitle}</p>
        {children}
        {footer && <div className="auth-footer">{footer}</div>}
      </div>

      <style jsx>{`
        .auth-page {
          position: relative;
          min-height: 100vh;
          overflow: hidden;
          background: radial-gradient(120% 100% at 80% 0%, #223326 0%, var(--soil) 55%, var(--soil-de
ep) 100%);
          display: flex;
          flex-direction: column;
          align-items: center;
          justify-content: center;
          padding: 40px 20px;
        }

        /* ---------------- 3D floating scene ---------------- */
        .scene {
          position: absolute;
          inset: 0;
          perspective: 1200px;
          overflow: hidden;
          z-index: 0;
        }
```

```
        .cube {
          position: absolute;
          transform-style: preserve-3d;
          animation: float-rotate 16s ease-in-out infinite;
          opacity: 0.5;
        }
        .cube span {
          position: absolute;
          inset: 0;
          background: linear-gradient(135deg, rgba(233, 178, 74, 0.35), rgba(122, 61, 26, 0.25));
          border: 1px solid rgba(239, 232, 216, 0.25);
          backdrop-filter: blur(1px);
        }
        .c1 { width: 90px; height: 90px; top: 10%; left: 8%; animation-duration: 18s; }
        .c1 span:nth-child(1) { transform: translateZ(45px); }
        .c1 span:nth-child(2) { transform: rotateY(180deg) translateZ(45px); }
        .c1 span:nth-child(3) { transform: rotateY(90deg) translateZ(45px); }
        .c1 span:nth-child(4) { transform: rotateY(-90deg) translateZ(45px); }
        .c1 span:nth-child(5) { transform: rotateX(90deg) translateZ(45px); }
        .c1 span:nth-child(6) { transform: rotateX(-90deg) translateZ(45px); }

        .c2 { width: 60px; height: 60px; top: 65%; left: 14%; animation-duration: 13s; animation-dela
y: -3s; }
        .c2 span:nth-child(1) { transform: translateZ(30px); }
        .c2 span:nth-child(2) { transform: rotateY(180deg) translateZ(30px); }
        .c2 span:nth-child(3) { transform: rotateY(90deg) translateZ(30px); }
        .c2 span:nth-child(4) { transform: rotateY(-90deg) translateZ(30px); }
        .c2 span:nth-child(5) { transform: rotateX(90deg) translateZ(30px); }
        .c2 span:nth-child(6) { transform: rotateX(-90deg) translateZ(30px); }

        .c3 { width: 110px; height: 110px; top: 20%; right: 6%; animation-duration: 22s; animation-de
lay: -8s; }
        .c3 span:nth-child(1) { transform: translateZ(55px); }
        .c3 span:nth-child(2) { transform: rotateY(180deg) translateZ(55px); }
        .c3 span:nth-child(3) { transform: rotateY(90deg) translateZ(55px); }
        .c3 span:nth-child(4) { transform: rotateY(-90deg) translateZ(55px); }
        .c3 span:nth-child(5) { transform: rotateX(90deg) translateZ(55px); }
        .c3 span:nth-child(6) { transform: rotateX(-90deg) translateZ(55px); }

        .orb {
          position: absolute;
          border-radius: 50%;
          filter: blur(1px);
          animation: drift 20s ease-in-out infinite;
        }
        .o1 {
          width: 46px; height: 46px; bottom: 14%; right: 12%;
          background: radial-gradient(circle at 35% 30%, #e9b24a, #a9631f);
          animation-duration: 14s;
        }
        .o2 {
          width: 26px; height: 26px; top: 30%; right: 22%;
          background: radial-gradient(circle at 35% 30%, #9fd18a, #4c7a3a);
          animation-duration: 11s; animation-delay: -4s;
        }
        .o3 {
          width: 34px; height: 34px; bottom: 30%; left: 6%;
          background: radial-gradient(circle at 35% 30%, #f2d3a3, #b5762f);
          animation-duration: 17s; animation-delay: -6s;
        }

        @keyframes float-rotate {
```

```
          0%   { transform: translateY(0px) rotateX(0deg) rotateY(0deg); }
          50%  { transform: translateY(-30px) rotateX(180deg) rotateY(180deg); }
          100% { transform: translateY(0px) rotateX(360deg) rotateY(360deg); }
        }
        @keyframes drift {
          0%   { transform: translate(0, 0) scale(1); }
          50%  { transform: translate(18px, -24px) scale(1.15); }
          100% { transform: translate(0, 0) scale(1); }
        }

        @media (prefers-reduced-motion: reduce) {
          .cube, .orb { animation: none; }
        }

        /* ---------------- card content (kept above the scene) ---------------- */
        .auth-logo {
          position: relative;
          z-index: 1;
          font-family: var(--font-display);
          font-weight: 900;
          font-size: 1.5rem;
          color: #efe8d8;
          margin-bottom: 26px;
        }
        .auth-logo span { color: var(--turmeric); }
        .auth-card {
          position: relative;
          z-index: 1;
          width: 100%;
          max-width: 420px;
          background: linear-gradient(160deg, #f7f2e6, #ede4cd);
          border-radius: 18px;
          padding: 36px 32px;
          box-shadow: 0 30px 60px rgba(14, 23, 18, 0.5);
        }
        .auth-icon { font-size: 1.8rem; margin: 0 0 6px; text-align: center; }
        h1 {
          font-family: var(--font-display);
          font-weight: 600;
          font-size: 1.5rem;
          text-align: center;
          margin: 0 0 6px;
          color: var(--ink);
        }
        .subtitle {
          text-align: center;
          color: #6b6555;
          font-size: 0.88rem;
          margin: 0 0 26px;
        }
        .auth-footer {
          text-align: center;
          margin-top: 20px;
          font-size: 0.85rem;
          color: #6b6555;
        }
        .auth-footer :global(a) { color: var(--rust); font-weight: 600; }
      `}</style>
    </div>
  );
}
```

app/components/Journey.tsx (Farm Journey Infographic Component)

```
"use client";

const STEPS = [
  {
    n: "01",
    title: "Picked at the farm",
    body: "Produce is listed the same morning it's harvested — no cold storage, no distributor marku
p, no guessing how old it is.",
  },
  {
    n: "02",
    title: "Farmer verified",
    body: "Every seller is Aadhaar-checked and approved by our team before their first listing goes l
ive.",
  },
  {
    n: "03",
    title: "Straight to you",
    body: "Order in kilos or in bulk — the price you see is the price the farmer set, split with no m
iddlemen in between.",
  },
];

export default function Journey() {
  return (
    <section className="journey container">
      <p className="eyebrow">How it moves</p>
      <h2 className="journey-title">Field to doorstep, in three honest steps</h2>

      <div className="rows">
        {STEPS.map((s) => (
          <div className="row" key={s.n}>
            <span className="row-n">{s.n}</span>
            <div className="row-body">
              <h3>{s.title}</h3>
              <p>{s.body}</p>
            </div>
          </div>
        ))}
      </div>

      <style jsx>{`
        .journey { padding: 96px 24px 40px; }
        .eyebrow {
          font-family: var(--font-mono);
          text-transform: uppercase;
          letter-spacing: 0.14em;
          font-size: 0.78rem;
          color: var(--rust);
          margin: 0 0 10px;
        }
        .journey-title {
          font-family: var(--font-display);
          font-weight: 600;
          font-size: clamp(1.8rem, 3.4vw, 2.6rem);
          max-width: 640px;
          margin: 0 0 52px;
          line-height: 1.15;
        }
```

```
        .rows { display: flex; flex-direction: column; }
        .row {
          display: grid;
          grid-template-columns: 90px 1fr;
          gap: 28px;
          padding: 30px 0;
          border-top: 1px solid rgba(26, 26, 21, 0.12);
          align-items: baseline;
        }
        .row:last-child { border-bottom: 1px solid rgba(26, 26, 21, 0.12); }
        .row-n {
          font-family: var(--font-display);
          font-weight: 900;
          font-size: 2.2rem;
          color: var(--paper-dim);
          -webkit-text-stroke: 1px var(--rust);
        }
        .row-body h3 {
          font-family: var(--font-display);
          font-weight: 600;
          font-size: 1.25rem;
          margin: 0 0 8px;
        }
        .row-body p {
          margin: 0;
          max-width: 560px;
          color: #4a4a3f;
          line-height: 1.55;
        }
        @media (max-width: 560px) {
          .row { grid-template-columns: 50px 1fr; gap: 16px; }
          .row-n { font-size: 1.5rem; }
        }
      `}</style>
    </section>
  );
}
```

app/components/PageLoader.tsx (Full Page Loading Spinner)

```
"use client";

import { useEffect, useState } from "react";

export default function PageLoader() {
  const [active, setActive] = useState(true);
  const [fading, setFading] = useState(false);

  useEffect(() => {
    const fadeTimer = setTimeout(() => setFading(true), 800);
    const removeTimer = setTimeout(() => setActive(false), 1050);
    return () => {
      clearTimeout(fadeTimer);
      clearTimeout(removeTimer);
    };
  }, []);

  if (!active) return null;

  return (
    <div className={`pl-backdrop ${fading ? "pl-fade" : ""}`}>
      <div className="pl-box">

        {/* Spinning ring around circle logo */}
        <div className="pl-ring-outer">
          <div className="pl-spinner" />
          <div className="pl-logo-circle">
            <img
              src="/agrenza-logo-small.jpg"
              alt="Agrenza"
              width={72}
              height={72}
              style={{
                width: "72px",
                height: "72px",
                borderRadius: "50%",
                objectFit: "cover",
                display: "block",
              }}
            />
          </div>
        </div>

        {/* Brand name */}
        <div className="pl-text">
          <span className="pl-brand">Agren<span>za</span></span>
          <span className="pl-sub">Farm to Doorstep</span>
        </div>

      </div>

      <style jsx>{`
        /* ── Backdrop ── */
        .pl-backdrop {
          position: fixed;
          inset: 0;
          z-index: 999999;
          background: #0f2314;
          display: flex;
```

```
  align-items: center;
  justify-content: center;
  opacity: 1;
  visibility: visible;
  will-change: opacity;
  transform: translateZ(0);
  transition: opacity 0.25s ease, visibility 0.25s linear;
  pointer-events: none;
}
.pl-backdrop.pl-fade {
  opacity: 0;
  visibility: hidden;
}

/* ── Content box ── */
.pl-box {
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 16px;
}

/* ── Spinning ring wrapper ── */
.pl-ring-outer {
  position: relative;
  width: 96px;
  height: 96px;
  display: flex;
  align-items: center;
  justify-content: center;
}

/* Spinning coloured ring */
.pl-spinner {
  position: absolute;
  inset: 0;
  border-radius: 50%;
  border: 3px solid transparent;
  border-top-color: #22c55e;
  border-right-color: #eab308;
  animation: pl-spin 0.9s linear infinite;
  will-change: transform;
}

@keyframes pl-spin {
  to { transform: rotate(360deg); }
}

/* Circle logo inside ring */
.pl-logo-circle {
  width: 80px;
  height: 80px;
  border-radius: 50%;
  overflow: hidden;
  background: #fff;
  border: 2px solid rgba(34, 197, 94, 0.35);
  display: flex;
  align-items: center;
  justify-content: center;
  flex-shrink: 0;
}
```

```
        /* ── Brand text ── */
        .pl-text {
          display: flex;
          flex-direction: column;
          align-items: center;
          gap: 3px;
        }

        .pl-brand {
          font-family: var(--font-display, sans-serif);
          font-weight: 900;
          font-size: 1.7rem;
          color: #efe8d8;
          letter-spacing: -0.02em;
        }

        .pl-brand span {
          color: #4ade80;
        }

        .pl-sub {
          font-size: 0.73rem;
          color: #a7f3d0;
          font-weight: 500;
          opacity: 0.85;
          letter-spacing: 0.06em;
          text-transform: uppercase;
        }
      `}</style>
    </div>
  );
}
```

app/components/EmailAuthenticityBadge.tsx (Email Verification Badge)

```
"use client";

import Link from "next/link";
import { EmailAuthenticityResult } from "@/app/lib/api";

interface Props {
  email: string;
  result: EmailAuthenticityResult | null;
  checking: boolean;
}

export default function EmailAuthenticityBadge({ email, result, checking }: Props) {
  if (!email.trim()) return null;

  if (checking) {
    return (
      <div
        style={{
          display: "flex",
          alignItems: "center",
          gap: "8px",
          marginTop: "6px",
          padding: "6px 10px",
          background: "#f0f9ff",
          border: "1px solid #bae6fd",
          borderRadius: "6px",
          fontSize: "0.78rem",
          color: "#0369a1",
        }}
      >
        <span
          style={{
            display: "inline-block",
            width: "12px",
            height: "12px",
            border: "2px solid #0284c7",
            borderTopColor: "transparent",
            borderRadius: "50%",
            animation: "spin 0.8s linear infinite",
          }}
        />
        <span>Checking email authenticity & MX records...</span>
      </div>
    );
  }

  if (!result) return null;

  // 1. Original / Authentic
  if (result.is_original && !result.is_duplicate && result.ok) {
    return (
      <div
        style={{
          marginTop: "6px",
          padding: "8px 12px",
          background: "#f0fdf4",
          border: "1px solid #86efac",
          borderRadius: "7px",
          fontSize: "0.8rem",
```

```
          color: "#166534",
          display: "flex",
          alignItems: "flex-start",
          gap: "8px",
        }}
      >
        <span style={{ fontSize: "1rem", lineHeight: 1 }}>🟢</span>
        <div>
          <strong style={{ display: "block", color: "#14532d" }}>
            Authentic {result.provider || "Email"} Confirmed
          </strong>
          <span style={{ color: "#166534", fontSize: "0.76rem" }}>
            ✓ Real mailbox ready for verification code & credentials.
          </span>
        </div>
      </div>
    );
  }

  // 2. Duplicate Email (Already Registered)
  if (result.is_duplicate || result.status === "duplicate") {
    return (
      <div
        style={{
          marginTop: "6px",
          padding: "10px 12px",
          background: "#fffbeb",
          border: "1px solid #fde68a",
          borderRadius: "7px",
          fontSize: "0.8rem",
          color: "#92400e",
        }}
      >
        <div style={{ display: "flex", alignItems: "center", gap: "6px", fontWeight: 700, color: "#b4
5309" }}>
          <span>⚠️</span>
          <span>This email is already registered!</span>
        </div>
        <p style={{ margin: "4px 0 8px", fontSize: "0.76rem", color: "#78350f" }}>
          An account with this email already exists on Agrenza for this role.
        </p>
        <Link
          href="/auth/login"
          style={{
            display: "inline-block",
            background: "#d97706",
            color: "#ffffff",
            padding: "5px 12px",
            borderRadius: "6px",
            fontSize: "0.78rem",
            fontWeight: 600,
            textDecoration: "none",
          }}
        >
          Go to Login Instead →
        </Link>
      </div>
    );
  }

  // 3. Fake / Disposable temporary email
  if (result.status === "disposable") {
```

```
    return (
      <div
        style={{
          marginTop: "6px",
          padding: "10px 12px",
          background: "#fef2f2",
          border: "1px solid #fca5a5",
          borderRadius: "7px",
          fontSize: "0.8rem",
          color: "#991b1b",
        }}
      >
        <div style={{ display: "flex", alignItems: "center", gap: "6px", fontWeight: 700, color: "#b9
1c1c" }}>
          <span>🚫</span>
          <span>Fake / Disposable Email Detected</span>
        </div>
        <p style={{ margin: "4px 0 0", fontSize: "0.76rem", color: "#7f1d1d" }}>
          Temporary or throwaway fake email domains are blocked. Please provide your real, authentic
personal Gmail ID.
        </p>
      </div>
    );
  }

  // 4. Invalid Gmail rules (length or consecutive dots)
  if (result.status === "invalid_gmail") {
    return (
      <div
        style={{
          marginTop: "6px",
          padding: "8px 12px",
          background: "#fef2f2",
          border: "1px solid #fca5a5",
          borderRadius: "7px",
          fontSize: "0.8rem",
          color: "#991b1b",
        }}
      >
        <div style={{ fontWeight: 700, color: "#b91c1c" }}>
          ✕ Invalid Gmail Username
        </div>
        <span style={{ fontSize: "0.76rem", color: "#7f1d1d" }}>
          {result.error || "Gmail usernames must be 6 to 30 characters without consecutive dots."}
        </span>
      </div>
    );
  }

  // 5. Non-existent domain / no MX records
  if (result.status === "invalid_domain") {
    return (
      <div
        style={{
          marginTop: "6px",
          padding: "8px 12px",
          background: "#fef2f2",
          border: "1px solid #fca5a5",
          borderRadius: "7px",
          fontSize: "0.8rem",
          color: "#991b1b",
        }}
```

```
      >
        <div style={{ fontWeight: 700, color: "#b91c1c" }}>
          ✕ Mail Domain Does Not Exist
        </div>
        <span style={{ fontSize: "0.76rem", color: "#7f1d1d" }}>
          {result.error || "This domain cannot receive emails. Please check for spelling mistakes."}
        </span>
      </div>
    );
  }

  // 6. Generic invalid format (if email has @ but not finished)
  if (result.status === "invalid_format" && email.includes("@")) {
    return (
      <div
        style={{
          marginTop: "4px",
          fontSize: "0.75rem",
          color: "#dc2626",
        }}
      >
        {result.error || "Please enter a valid email address (e.g. name@gmail.com)"}
      </div>
    );
  }

  return null;
}
```

app/auth/email-verify/page.tsx (Email Verification Page)

```
"use client";

import { Suspense, useEffect, useRef, useState } from "react";
import { useRouter, useSearchParams } from "next/navigation";
import Link from "next/link";
import AuthShell from "@/app/components/AuthShell";
import "@/app/components/forms.css";
import { completeEmailVerification } from "@/app/lib/otp";
import { verifyEmailWithBackend, ApiError } from "@/app/lib/api";

type Status = "checking" | "enter_code" | "success" | "pending_approval" | "error";

function EmailVerifyContent() {
  const router = useRouter();
  const searchParams = useSearchParams();

  const [status, setStatus] = useState<Status>("checking");
  const [message, setMessage] = useState("Checking verification details…");
  const [emailInput, setEmailInput] = useState("");
  const [otpInput, setOtpInput] = useState("");
  const [loading, setLoading] = useState(false);
  const [verifiedRole, setVerifiedRole] = useState<string>("");

  const attempted = useRef(false);

  useEffect(() => {
    if (attempted.current) return;
    attempted.current = true;

    const urlToken = searchParams.get("token") || "";
    const urlEmail = searchParams.get("email") || "";
    const urlOtp = searchParams.get("otp") || searchParams.get("code") || "";

    if (urlEmail) setEmailInput(urlEmail);
    if (urlOtp) setOtpInput(urlOtp);

    // Case 1: One-click link with token from email
    if (urlToken && urlEmail) {
      handleTokenVerification(urlToken, urlEmail);
      return;
    }

    // Case 2: One-click link with OTP code from email
    if (urlOtp && urlEmail) {
      handleOtpVerification(urlOtp, urlEmail);
      return;
    }

    // Case 3: Firebase passwordless email link
    if (typeof window !== "undefined" && window.location.href.includes("apiKey=")) {
      handleFirebaseVerification(urlEmail);
      return;
    }

    // Otherwise: Show the code entry form
    setStatus("enter_code");
    setMessage("Enter the 6-digit verification code sent to your Gmail inbox:");
  }, [searchParams]);
```

```
async function handleTokenVerification(token: string, email: string) {
  setLoading(true);
  setMessage("Verifying your email token…");
  try {
    const res = await verifyEmailWithBackend({ token, email });
    setVerifiedRole(res.role || "");
    if (res.requires_approval) {
      setStatus("pending_approval");
      setMessage(`${email} is verified! Your farmer account is now awaiting admin review.`);
    } else {
      setStatus("success");
      setMessage(`${email} is verified successfully! Welcome to Agrenza.`);
    }
  } catch (err: any) {
    handleError(err);
  } finally {
    setLoading(false);
  }
}

async function handleOtpVerification(otp: string, email: string) {
  setLoading(true);
  setMessage("Confirming your verification code…");
  try {
    const res = await verifyEmailWithBackend({ otp, email });
    setVerifiedRole(res.role || "");
    if (res.requires_approval) {
      setStatus("pending_approval");
      setMessage(`${email} is verified! Your farmer account is now awaiting admin review.`);
    } else {
      setStatus("success");
      setMessage(`${email} is verified successfully! Welcome to Agrenza.`);
    }
  } catch (err: any) {
    handleError(err);
  } finally {
    setLoading(false);
  }
}

async function handleFirebaseVerification(manualEmail?: string) {
  setLoading(true);
  try {
    const result = await completeEmailVerification(manualEmail);
    if (!result) {
      setStatus("enter_code");
      setMessage("Please confirm your email address and verification code:");
      return;
    }
    const res = await verifyEmailWithBackend({ idToken: result.idToken, email: result.email });
    setVerifiedRole(res.role || "");
    if (res.requires_approval) {
      setStatus("pending_approval");
      setMessage(`${result.email} is verified! Your farmer account is now awaiting admin review.`);
    } else {
      setStatus("success");
      setMessage(`${result.email} is verified! Redirecting to login…`);
    }
  } catch (err: any) {
    handleError(err);
  } finally {
    setLoading(false);
```

```
    }
  }

  function handleError(err: any) {
    setStatus("error");
    const errMsg = err instanceof ApiError ? err.message : err?.message || "";
    if (errMsg.toLowerCase().includes("already verified") || errMsg.toLowerCase().includes("already r
egistered")) {
      setMessage("This email has already been verified. You can log in directly.");
    } else {
      setMessage(errMsg || "Could not verify your email. The code or link may have expired.");
    }
  }

  async function handleManualSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (!emailInput.trim()) return alert("Enter your email address.");
    if (!otpInput.trim() || otpInput.trim().length !== 6) {
      return alert("Enter the 6-digit numeric verification code sent to your Gmail.");
    }
    await handleOtpVerification(otpInput.trim(), emailInput.trim());
  }

  return (
    <AuthShell
      icon={
        status === "success"
          ? "✅"
          : status === "pending_approval"
          ? "⌛"
          : status === "error"
          ? "⚠️"
          : "✉️"
      }
      title="Gmail Verification"
      subtitle={message}
    >
      {/* Interactive 6-digit OTP code entry form */}
      {(status === "enter_code" || status === "error") && (
        <form onSubmit={handleManualSubmit} style={{ marginTop: 16 }}>
          <div className="field">
            <label>Gmail / Email Address</label>
            <input
              type="email"
              placeholder="e.g. yourname@gmail.com"
              value={emailInput}
              onChange={(e) => setEmailInput(e.target.value)}
              required
            />
          </div>

          <div className="field">
            <label>6-Digit Verification Code</label>
            <input
              type="text"
              inputMode="numeric"
              maxLength={6}
              placeholder="123456"
              style={{ letterSpacing: "4px", fontSize: "1.2rem", fontWeight: "bold", textAlign: "cent
er" }}
              value={otpInput}
              onChange={(e) => setOtpInput(e.target.value.replace(/\D/g, "").slice(0, 6))}
```

```
              required
            />
            <small style={{ color: "#6b7280", fontSize: "0.75rem", display: "block", marginTop: "4px"
}}>
              Check your Gmail inbox for the 6-digit code.
            </small>
          </div>

          <button type="submit" className="btn-block" disabled={loading} style={{ marginTop: 10 }}>
            {loading ? "Verifying code…" : "Verify & Activate Account"}
          </button>

          <div style={{ textAlign: "center", marginTop: 16 }}>
            <Link href="/auth/login" style={{ fontSize: "0.86rem", color: "var(--leaf, #2e7d32)", fon
tWeight: 600 }}>
              Back to Login
            </Link>
          </div>
        </form>
      )}

      {/* Success or Pending Approval State */}
      {(status === "success" || status === "pending_approval") && (
        <div style={{ marginTop: 20, textAlign: "center" }}>
          {status === "success" && verifiedRole === "buyer" && (
            <div style={{ background: "#f0fdf4", border: "1px solid #bbf7d0", padding: "14px", border
Radius: "8px", margin: "14px 0", fontSize: "0.9rem", color: "#166534", textAlign: "left" }}>
              💡 <strong>Ready to Shop:</strong> Your account is activated in our database. You can
now log in using your Gmail and the auto-generated password sent to your inbox.
            </div>
          )}

          {status === "pending_approval" && (
            <div style={{ background: "#fffbeb", border: "1px solid #fde68a", padding: "14px", border
Radius: "8px", margin: "14px 0", fontSize: "0.9rem", color: "#92400e", textAlign: "left" }}>
              📋 <strong>Under Review:</strong> Your email is confirmed! Our admin team has been not
ified and is reviewing your farmer application. You will receive an approval notification email once
activated.
            </div>
          )}

          <Link href="/auth/login" className="btn-block" style={{ display: "block", textAlign: "cente
r", marginTop: 16 }}>
            Continue to Login →
          </Link>
        </div>
      )}
    </AuthShell>
  );
}

export default function EmailVerifyPage() {
  return (
    <Suspense fallback={<div style={{ padding: "80px", textAlign: "center" }}>Loading verification…</
div>}>
      <EmailVerifyContent />
    </Suspense>
  );
}
```

app/auth/verify-otp/page.tsx (OTP Verification Page)

```
"use client";

import { useEffect, useRef, useState } from "react";
import { useRouter } from "next/navigation";
import AuthShell from "@/app/components/AuthShell";
import "@/app/components/forms.css";
import { sendMobileOtp, confirmMobileOtp, resetMobileOtpState } from "@/app/lib/otp";
import { verifyMobileWithBackend, ApiError } from "@/app/lib/api";

export default function VerifyOtpPage() {
  const router = useRouter();
  const [mobile, setMobile] = useState("");
  const [digits, setDigits] = useState(["", "", "", "", "", ""]);
  const [error, setError] = useState("");
  const [info, setInfo] = useState("Sending OTP…");
  const [loading, setLoading] = useState(false);
  const [sending, setSending] = useState(true);
  const [seconds, setSeconds] = useState(30);
  const inputs = useRef<(HTMLInputElement | null)[]>([]);
  const pendingMobile = useRef<string>("");

  useEffect(() => {
    const m = sessionStorage.getItem("pending_mobile") || "";
    setMobile(m);
    pendingMobile.current = m;
    if (m) sendOtp(m);
    else {
      setSending(false);
      setInfo("");
      setError("No pending registration found. Please register again.");
    }
    return () => resetMobileOtpState();
    // eslint-disable-next-line react-hooks/exhaustive-deps
  }, []);

  useEffect(() => {
    if (seconds <= 0) return;
    const t = setTimeout(() => setSeconds((s) => s - 1), 1000);
    return () => clearTimeout(t);
  }, [seconds]);

  async function sendOtp(m: string) {
    setSending(true);
    setError("");
    try {
      // sendMobileOtp mounts an invisible reCAPTCHA against
      // #recaptcha-container and triggers Firebase's real SMS OTP.
      await sendMobileOtp(m);
      setInfo("OTP sent. Enter the 6-digit code below.");
      setSeconds(30);
    } catch (err) {
      setError(
        err instanceof Error
          ? `Could not send OTP: ${err.message}`
          : "Could not send OTP. Check your Firebase phone-auth setup."
      );
    } finally {
      setSending(false);
    }
```

```
}

function handleChange(i: number, val: string) {
  const d = val.replace(/\D/g, "").slice(-1);
  const next = [...digits];
  next[i] = d;
  setDigits(next);
  if (d && i < 5) inputs.current[i + 1]?.focus();
}

function handleKeyDown(i: number, e: React.KeyboardEvent<HTMLInputElement>) {
  if (e.key === "Backspace" && !digits[i] && i > 0) inputs.current[i - 1]?.focus();
}

async function handleVerify() {
  const code = digits.join("");
  if (code.length !== 6) {
    setError("Enter the full 6-digit code.");
    return;
  }
  setError("");
  setLoading(true);
  try {
    // 1. Confirm the code with Firebase — this proves the user really
    //    received the SMS and returns a signed ID token.
    const idToken = await confirmMobileOtp(code);
    // 2. Hand that token to the PHP backend, which verifies its
    //    signature server-side and flips is_verified = 1 for this user.
    const result = await verifyMobileWithBackend(idToken, pendingMobile.current);
    sessionStorage.removeItem("pending_mobile");
    if (result.requires_approval) {
      router.push("/auth/login?pending_approval=1");
    } else {
      router.push("/shop");
    }
  } catch (err) {
    setError(err instanceof ApiError ? err.message : "Invalid or expired code. Please try again.");
  } finally {
    setLoading(false);
  }
}

function handleResend() {
  setDigits(["", "", "", "", "", ""]);
  inputs.current[0]?.focus();
  if (pendingMobile.current) sendOtp(pendingMobile.current);
}

return (
  <AuthShell
    icon="🛡️"
    title="Mobile Verification"
    subtitle={mobile ? `Verifying +91 ${mobile}` : "Verifying your mobile number"}
  >
    {error && <div className="form-error">{error}</div>}
    {!error && info && <div className="form-info">{info}</div>}

    <div className="otp-row">
      {digits.map((d, i) => (
        <input
          key={i}
          ref={(el) => { inputs.current[i] = el; }}
```

```
            value={d}
            onChange={(e) => handleChange(i, e.target.value)}
            onKeyDown={(e) => handleKeyDown(i, e)}
            inputMode="numeric"
            maxLength={1}
            disabled={sending}
            className="otp-box"
          />
        ))}
      </div>

      <button className="btn-block" onClick={handleVerify} disabled={loading || sending}>
        {loading ? "Verifying…" : "Verify & continue"}
      </button>

      <button type="button" className="resend" onClick={handleResend} disabled={seconds > 0 || sendin
g}>
        {sending ? "Sending…" : seconds > 0 ? `Resend OTP in ${seconds}s` : "Resend OTP"}
      </button>

      {/* Required, invisible container Firebase's reCAPTCHA attaches to. */}
      <div id="recaptcha-container" />

      <style jsx>{`
        .otp-row { display: flex; gap: 10px; justify-content: center; margin-bottom: 22px; }
        .otp-box {
          width: 44px;
          height: 52px;
          text-align: center;
          font-family: var(--font-mono);
          font-size: 1.3rem;
          border: 1px solid rgba(26, 26, 21, 0.2);
          border-radius: 9px;
          background: #fff;
        }
        .otp-box:focus {
          outline: 2px solid var(--turmeric);
          outline-offset: 1px;
          border-color: var(--turmeric);
        }
        .resend {
          display: block;
          width: 100%;
          background: none;
          border: none;
          text-align: center;
          margin-top: 14px;
          font-size: 0.85rem;
          color: var(--rust);
          font-weight: 600;
        }
        .resend:disabled { color: #a39c88; font-weight: 400; }
      `}</style>
    </AuthShell>
  );
}
```

app/auth/admin-login/page.tsx (Admin Login Page)

```
"use client";

import { useState } from "react";
import { useRouter } from "next/navigation";
import AuthShell from "@/app/components/AuthShell";
import "@/app/components/forms.css";
import { loginUser, ApiError } from "@/app/lib/api";

export default function AdminLoginPage() {
  const router = useRouter();
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [error, setError] = useState("");
  const [loading, setLoading] = useState(false);

  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    setError("");
    if (!/^\S+@\S+\.\S+$/.test(email)) {
      setError("Enter a valid email address.");
      return;
    }
    setLoading(true);
    try {
      await loginUser({ role: "admin", email, password });
      router.push("/admin/dashboard");
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not reach the server. Try again.");
    } finally {
      setLoading(false);
    }
  }

  return (
    <AuthShell icon="🛡️" title="Admin Login" subtitle="Agrenza administration panel">
      <form onSubmit={handleSubmit}>
        {error && <div className="form-error">{error}</div>}

        <div className="field">
          <label>Admin email</label>
          <input
            type="email"
            placeholder="admin@farmlink.com"
            value={email}
            onChange={(e) => setEmail(e.target.value)}
            required
          />
        </div>

        <div className="field">
          <label>Password</label>
          <input
            type="password"
            value={password}
            onChange={(e) => setPassword(e.target.value)}
            required
          />
        </div>
```

```
        <button className="btn-block" type="submit" disabled={loading}>
          {loading ? "Logging in…" : "Login"}
        </button>
      </form>
    </AuthShell>
  );
}
```

app/buyer/checkout/page.tsx (Buyer Checkout Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import Link from "next/link";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import { useCart } from "@/app/lib/cart-context";
import { placeOrderApi, getMe, ApiError, AuthUser } from "@/app/lib/api";
import "@/app/components/forms.css";

export default function CheckoutPage() {
  const router = useRouter();
  const { items, total, addItem, removeItem, clearCart, placeOrder } = useCart();

  // User auth state
  const [user, setUser] = useState<AuthUser | null>(null);
  const [authLoading, setAuthLoading] = useState(true);

  // Active step (1: Login/Details, 2: Address, 3: Order Summary, 4: Payment)
  const [activeStep, setActiveStep] = useState<number>(1);

  // Address fields (Flipkart structured format)
  const [fullName, setFullName] = useState("");
  const [phone, setPhone] = useState("");
  const [pincode, setPincode] = useState("");
  const [locality, setLocality] = useState("");
  const [streetAddress, setStreetAddress] = useState("");
  const [city, setCity] = useState("");
  const [state, setState] = useState("Tamil Nadu");
  const [landmark, setLandmark] = useState("");
  const [addressType, setAddressType] = useState<"home" | "work">("home");

  // Payment mode
  const [paymentMode, setPaymentMode] = useState<"cod" | "upi" | "card" | "netbanking">("cod");
  const [upiId, setUpiId] = useState("");

  // Coupon / Buyer discount state
  const [couponInput, setCouponInput] = useState("");
  const [appliedCoupon, setAppliedCoupon] = useState<{ code: string; discount: number; label: string
} | null>(null);
  const [couponError, setCouponError] = useState("");

  const [placing, setPlacing] = useState(false);
  const [error, setError] = useState("");

  function handleApplyCoupon(codeToApply?: string) {
    const code = (codeToApply || couponInput).trim().toUpperCase();
    setCouponError("");
    if (!code) {
      setCouponError("Please enter a valid coupon code.");
      return;
    }
    if (code === "FRESHFARM10") {
      const disc = total * 0.10;
      setAppliedCoupon({ code, discount: disc, label: "10% Harvest Season Discount" });
      setCouponInput("");
    } else if (code === "ORGANIC20") {
      const disc = total * 0.20;
```

```
    setAppliedCoupon({ code, discount: disc, label: "20% Organic Produce Discount" });
    setCouponInput("");
  } else if (code === "FIRSTHARVEST") {
    const disc = Math.min(50, total);
    setAppliedCoupon({ code, discount: disc, label: "₹50 New Buyer Welcome Gift" });
    setCouponInput("");
  } else if (code === "BULKSAVER15") {
    const disc = total * 0.15;
    setAppliedCoupon({ code, discount: disc, label: "15% Wholesale Direct Discount" });
    setCouponInput("");
  } else if (code === "FREESHIP") {
    setAppliedCoupon({ code, discount: 0, label: "Free Doorstep Farm Delivery" });
    setCouponInput("");
  } else if (code === "FARMDIRECT5") {
    const disc = total * 0.05;
    setAppliedCoupon({ code, discount: disc, label: "5% Online Payment Discount" });
    setCouponInput("");
  } else {
    setCouponError("Invalid or expired coupon code. Check available codes below.");
  }
}

function handleRemoveCoupon() {
  setAppliedCoupon(null);
  setCouponError("");
}

useEffect(() => {
  (async () => {
    try {
      const me = await getMe();
      if (me?.user) {
        setUser(me.user);
        if (me.user.name) setFullName(me.user.name);
        if (me.user.mobile) setPhone(me.user.mobile);

        if (me.user.role === "buyer") {
          // Logged in buyer: automatically advance to Step 2 (Delivery Address)
          setActiveStep(2);
        } else {
          setActiveStep(1);
        }
      } else {
        setUser(null);
        setActiveStep(1);
      }
    } catch {
      setUser(null);
      setActiveStep(1);
    } finally {
      setAuthLoading(false);
    }
  })();
}, []);

// Format full combined address
const fullDeliveryAddress = [
  fullName ? `Name: ${fullName}` : "",
  phone ? `Ph: ${phone}` : "",
  streetAddress,
  locality,
  landmark ? `Near ${landmark}` : "",
```

```
    city,
    state,
    pincode ? `PIN: ${pincode}` : "",
    `(${addressType.toUpperCase()})`,
  ]
    .filter(Boolean)
    .join(", ");

  function goToStep(targetStep: number) {
    if (!user) {
      setActiveStep(1);
      setError("Please log in to your Buyer account first to proceed.");
      return;
    }
    if (user.role !== "buyer") {
      setActiveStep(1);
      setError("Only Buyer accounts can place orders. Please switch to a Buyer account.");
      return;
    }
    setError("");
    setActiveStep(targetStep);
  }

  function handleAddressSave(e: React.FormEvent) {
    e.preventDefault();
    if (!user || user.role !== "buyer") {
      setActiveStep(1);
      setError("Please log in to your Buyer account first to save address and continue.");
      return;
    }
    if (!streetAddress.trim() || !city.trim() || !pincode.trim()) {
      setError("Please fill in Street Address, City, and Pincode.");
      return;
    }
    if (pincode.trim().length !== 6) {
      setError("Please enter a valid 6-digit Pincode.");
      return;
    }
    setError("");
    setActiveStep(3);
  }

  function handleQtyChange(productId: number, newQty: number) {
    if (newQty <= 0) {
      removeItem(productId);
    } else {
      const item = items.find((i) => i.productId === productId);
      if (item) {
        const diff = newQty - item.qty;
        addItem({ id: item.productId, name: item.name, price: item.price, category: "", stock: 999, d
escription: "", farmer: "", location: "", icon: item.icon || "tomato", image: undefined }, diff);
      }
    }
  }

  async function handleFinalPlaceOrder() {
    if (!user) {
      setActiveStep(1);
      setError("Authentication required: Please log in with a Buyer account to place this order.");
      return;
    }
```

```
    if (user.role !== "buyer") {
      setActiveStep(1);
      setError("Access denied: Only Buyer accounts are permitted to place orders. Please log in as a
Buyer.");
      return;
    }

    if (!fullDeliveryAddress.trim() || !streetAddress.trim()) {
      setActiveStep(2);
      setError("Please provide your delivery address.");
      return;
    }

    setPlacing(true);
    setError("");

    try {
      const backendPayMode: "cod" | "online" = paymentMode === "cod" ? "cod" : "online";
      const orderPayload: {
        address: string;
        payment_mode: "cod" | "online";
        items: { product_id: number; qty: number }[];
      } = {
        address: fullDeliveryAddress,
        payment_mode: backendPayMode,
        items: items.map((i) => ({
          product_id: i.productId,
          qty: i.qty,
        })),
      };

      await placeOrderApi(orderPayload);

      placeOrder(backendPayMode);
      clearCart();

      router.push("/buyer/orders");
    } catch (err: any) {
      if (err instanceof ApiError) {
        setError(err.message);
      } else {
        setError("Failed to place order. Please try again.");
      }
    } finally {
      setPlacing(false);
    }
  }

  if (items.length === 0) {
    return (
      <>
        <Nav />
        <div className="container empty-wrap">
          <div className="empty-cart-card">
            <span className="empty-icon">🛒</span>
            <h2>Your Shopping Cart is Empty</h2>
            <p>Explore fresh organic harvest directly from verified local farmers!</p>
            <Link href="/buyer/shop" className="btn-shop">
              🌾 Browse Farm Shop
            </Link>
          </div>
        </div>
```

```
        <Footer />
        <style jsx>{`
          .empty-wrap { padding: 80px 24px; display: flex; justify-content: center; }
          .empty-cart-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius:
16px; padding: 48px; text-align: center; max-width: 480px; box-shadow: 0 4px 20px rgba(0,0,0,0.04); }
          .empty-icon { font-size: 3.5rem; display: block; margin-bottom: 12px; }
          h2 { font-family: var(--font-display); font-size: 1.5rem; color: #1b3821; margin: 0 0 8px;
}
          p { color: #6b6555; font-size: 0.9rem; margin-bottom: 24px; }
          .btn-shop { background: #16a34a; color: #fff; padding: 12px 26px; border-radius: 999px; tex
t-decoration: none; font-weight: 700; font-size: 0.92rem; display: inline-block; }
          .btn-shop:hover { background: #15803d; }
        `}</style>
      </>
    );
  }

  // Savings calculation (assume retail market is ~20% higher than direct farm base)
  const retailEstimatedTotal = total * 1.25;
  const savings = retailEstimatedTotal - total;

  return (
    <>
      <Nav />
      <div className="checkout-bg">
        <div className="container checkout-container">

          {error && <div className="error-banner">⚠️ {error}</div>}

          <div className="checkout-grid">

            {/* LEFT COLUMN: FLIPKART ACCORDION STEPS */}
            <div className="steps-column">

              {/* STEP 1: LOGIN / BUYER DETAILS */}
              <div className={`step-card ${user && user.role === "buyer" && activeStep > 1 ? "complet
ed" : "active"}`}>
                <div className="step-header">
                  <div className="step-num-wrap">
                    <span className="step-num">{user && user.role === "buyer" && activeStep > 1 ? "✓"
: "1"}</span>
                    <span className="step-title">
                      {user && user.role === "buyer" ? "LOGIN & BUYER PROFILE" : "LOGIN REQUIRED TO O
RDER"}
                    </span>
                  </div>
                  {user && user.role === "buyer" && activeStep > 1 && (
                    <button type="button" className="btn-step-change" onClick={() => goToStep(1)}>
                      CHANGE
                    </button>
                  )}
                </div>

                {authLoading ? (
                  <div className="step-content">
                    <p style={{ color: "#64748b", margin: 0 }}>⏳ Verifying buyer login status…</p>
                  </div>
                ) : !user ? (
                  /* GUEST / NOT LOGGED IN USER */
                  <div className="step-content">
                    <div className="auth-required-box">
                      <div className="auth-icon-wrap">🔒</div>
```

```
                    <div className="auth-info-text">
                      <h4 className="auth-title">Buyer Login Required to Place Order</h4>
                      <p className="auth-desc">
                        Only registered buyers can place orders on Agrenza. Non-logged-in visitors
cannot order produce. Please sign in to your buyer account or register below to complete your harvest
purchase.
                      </p>

                      <div className="auth-btn-row">
                        <Link href="/auth/login?redirect=/buyer/checkout" className="btn-primary-au
th">
                          🔑 Log In as Buyer
                        </Link>
                        <Link href="/auth/buyer-register" className="btn-secondary-auth">
                          🌾 Create New Buyer Account
                        </Link>
                      </div>

                      <div className="auth-perks">
                        <span>✓ Direct farm fresh price</span>
                        <span>✓ Live delivery travel tracking</span>
                        <span>✓ SMS &amp;amp; Email confirmations</span>
                      </div>
                    </div>
                  </div>
                </div>
              ) : user.role !== "buyer" ? (
                /* LOGGED IN AS FARMER OR ADMIN */
                <div className="step-content">
                  <div className="role-alert-box">
                    <span className="alert-icon">⚠️</span>
                    <div>
                      <strong>Logged in as {user.role.toUpperCase()} ({user.name})</strong>
                      <p style={{ margin: "4px 0 12px", fontSize: "0.85rem", color: "#64748b" }}>
                        Only accounts registered with the <strong>Buyer</strong> role can purchase
products and place orders.
                      </p>
                      <div className="auth-btn-row">
                        <Link href="/auth/login?redirect=/buyer/checkout" className="btn-primary-au
th">
                          Switch to a Buyer Account
                        </Link>
                        <Link href="/auth/buyer-register" className="btn-secondary-auth">
                          Register as Buyer
                        </Link>
                      </div>
                    </div>
                  </div>
                </div>
              ) : activeStep === 1 ? (
                /* LOGGED IN BUYER EDITING STEP 1 */
                <div className="step-content">
                  <div className="user-profile-row">
                    <div className="user-meta-wrap">
                      <div className="verified-shield">✓</div>
                      <div>
                        <strong>{user.name} <span className="badge-buyer">Verified Buyer</span></st
rong>
                        <div className="user-sub">{user.mobile || "Mobile verified"} &bull; {user.e
mail || "Active account"}</div>
                      </div>
                    </div>
```

```
                    <div className="user-action-wrap">
                      <Link href="/auth/login?redirect=/buyer/checkout" className="btn-switch-lin
k">
                        Switch Account
                      </Link>
                      <button type="button" className="btn-flipkart-action" onClick={() => goToStep
(2)}>
                        CONTINUE TO DELIVERY ADDRESS
                      </button>
                    </div>
                  </div>
                </div>
              ) : (
                /* LOGGED IN BUYER COMPLETED STEP 1 SUMMARY */
                <div className="step-summary-text">
                  <strong>{user.name}</strong> • {user.mobile || user.email || "Verified Buyer"}
                </div>
              )}
            </div>

            {/* STEP 2: DELIVERY ADDRESS */}
            <div className={`step-card ${activeStep === 2 ? "active" : activeStep > 2 ? "completed"
: "pending"}`}>
              <div className="step-header">
                <div className="step-num-wrap">
                  <span className="step-num">{activeStep > 2 ? "✓" : "2"}</span>
                  <span className="step-title">DELIVERY ADDRESS</span>
                </div>
                {activeStep > 2 && (
                  <button type="button" className="btn-step-change" onClick={() => goToStep(2)}>
                    CHANGE
                  </button>
                )}
              </div>

              {activeStep === 2 ? (
                <form className="step-content" onSubmit={handleAddressSave}>
                  <div className="form-grid-2">
                    <div className="form-field">
                      <label>Full Name *</label>
                      <input
                        type="text"
                        required
                        placeholder="Receiver Name"
                        value={fullName}
                        onChange={(e) => setFullName(e.target.value)}
                      />
                    </div>
                    <div className="form-field">
                      <label>10-digit Mobile Number *</label>
                      <input
                        type="tel"
                        required
                        maxLength={10}
                        placeholder="Mobile Number"
                        value={phone}
                        onChange={(e) => setPhone(e.target.value)}
                      />
                    </div>
                  </div>

                  <div className="form-grid-2">
```

```
            <div className="form-field">
              <label>Pincode (6 digits) *</label>
              <input
                type="text"
                required
                maxLength={6}
                placeholder="e.g. 641001"
                value={pincode}
                onChange={(e) => setPincode(e.target.value)}
              />
            </div>
            <div className="form-field">
              <label>Locality / Area *</label>
              <input
                type="text"
                required
                placeholder="e.g. Gandhipuram, RS Puram"
                value={locality}
                onChange={(e) => setLocality(e.target.value)}
              />
            </div>
          </div>

          <div className="form-field full">
            <label>Address (House No, Building, Street, Area) *</label>
            <textarea
              rows={2}
              required
              placeholder="Flat / Door No, Apartment name, Street name"
              value={streetAddress}
              onChange={(e) => setStreetAddress(e.target.value)}
            />
          </div>

          <div className="form-grid-3">
            <div className="form-field">
              <label>City / District *</label>
              <input
                type="text"
                required
                placeholder="e.g. Coimbatore"
                value={city}
                onChange={(e) => setCity(e.target.value)}
              />
            </div>
            <div className="form-field">
              <label>State *</label>
              <select value={state} onChange={(e) => setState(e.target.value)}>
                <option value="Tamil Nadu">Tamil Nadu</option>
                <option value="Kerala">Kerala</option>
                <option value="Karnataka">Karnataka</option>
                <option value="Andhra Pradesh">Andhra Pradesh</option>
              </select>
            </div>
            <div className="form-field">
              <label>Landmark (Optional)</label>
              <input
                type="text"
                placeholder="e.g. Near Bus Stand"
                value={landmark}
                onChange={(e) => setLandmark(e.target.value)}
              />
```

```
                    </div>
                  </div>

                  {/* Address Type */}
                  <div className="address-type-wrap">
                    <span className="type-label">Address Type:</span>
                    <div className="radio-group">
                      <label className={`type-chip ${addressType === "home" ? "selected" : ""}`}>
                        <input
                          type="radio"
                          name="addrType"
                          checked={addressType === "home"}
                          onChange={() => setAddressType("home")}
                        />
                        <span>🏠 Home (All day delivery)</span>
                      </label>
                      <label className={`type-chip ${addressType === "work" ? "selected" : ""}`}>
                        <input
                          type="radio"
                          name="addrType"
                          checked={addressType === "work"}
                          onChange={() => setAddressType("work")}
                        />
                        <span>🏢 Work (Delivery 10 AM - 5 PM)</span>
                      </label>
                    </div>
                  </div>

                  <div className="btn-row">
                    <button type="submit" className="btn-flipkart-save">
                      DELIVER HERE
                    </button>
                  </div>
                </form>
              ) : (
                activeStep > 2 && (
                  <div className="step-summary-text">
                    <div className="addr-badge">{addressType.toUpperCase()}</div>
                    <strong>{fullName || user?.name}</strong> • {phone || user?.mobile}
                    <p className="addr-line">{fullDeliveryAddress}</p>
                  </div>
                )
              )}
            </div>

            {/* STEP 3: ORDER SUMMARY */}
            <div className={`step-card ${activeStep === 3 ? "active" : activeStep > 3 ? "completed"
: "pending"}`}>
              <div className="step-header">
                <div className="step-num-wrap">
                  <span className="step-num">{activeStep > 3 ? "✓" : "3"}</span>
                  <span className="step-title">ORDER SUMMARY ({items.length} PRODUCE ITEMS)</span>
                </div>
                {activeStep > 3 && (
                  <button type="button" className="btn-step-change" onClick={() => goToStep(3)}>
                    CHANGE
                  </button>
                )}
              </div>

              {activeStep === 3 ? (
                <div className="step-content">
```

```
                <div className="produce-items-list">
                  {items.map((item) => (
                    <div className="produce-item-card" key={item.productId}>
                      <div className="prod-left">
                        <div className="item-icon-box">🌾</div>
                        <div className="prod-details">
                          <strong className="prod-name">{item.name}</strong>
                          <span className="prod-rate">₹{item.price.toFixed(2)} / kg</span>
                          <div className="delivery-badge">
                            🚚 Farm-Fresh Delivery within 24–48 Hours
                          </div>
                        </div>
                      </div>

                      <div className="prod-right">
                        <div className="qty-counter">
                          <button
                            type="button"
                            className="qty-btn"
                            onClick={() => handleQtyChange(item.productId, item.qty - 1)}
                          >
                            -
                          </button>
                          <span className="qty-display">{item.qty} kg</span>
                          <button
                            type="button"
                            className="qty-btn"
                            onClick={() => handleQtyChange(item.productId, item.qty + 1)}
                          >
                            +
                          </button>
                        </div>
                        <strong className="item-subtotal">₹{(item.price * item.qty).toFixed(2)}</st
rong>

                        <button
                          type="button"
                          className="btn-remove-item"
                          onClick={() => removeItem(item.productId)}
                        >
                          REMOVE
                        </button>
                      </div>
                    </div>
                  ))}
                </div>

                <div className="order-summary-footer">
                  <span className="email-notif">
                    🔵 Order confirmation email will be sent to your Gmail inbox.
                  </span>
                  <button type="button" className="btn-flipkart-save" onClick={() => goToStep(4)}>
                    CONTINUE TO PAYMENT
                  </button>
                </div>
              </div>
            ) : (
              activeStep > 3 && (
                <div className="step-summary-text">
                  {items.length} Farm Item(s) — Total: <strong>₹{total.toFixed(2)}</strong>
                </div>
              )
            )}
```

```
            </div>

            {/* STEP 4: PAYMENT OPTIONS */}
            <div className={`step-card ${activeStep === 4 ? "active" : "pending"}`}>
              <div className="step-header">
                <div className="step-num-wrap">
                  <span className="step-num">4</span>
                  <span className="step-title">PAYMENT OPTIONS</span>
                </div>
              </div>

              {activeStep === 4 && (
                <div className="step-content">
                  {(!user || user.role !== "buyer") ? (
                    <div className="auth-required-box">
                      <div className="auth-icon-wrap">🔒</div>
                      <div className="auth-info-text">
                        <h4 className="auth-title">Log in as a Buyer to Select Payment</h4>
                        <p className="auth-desc">
                          You must be logged into an active Buyer account before completing your paym
ent.
                        </p>
                        <div className="auth-btn-row">
                          <Link href="/auth/login?redirect=/buyer/checkout" className="btn-primary-au
th">
                            🔑 Log In as Buyer
                          </Link>
                        </div>
                      </div>
                    </div>
                  ) : (
                    <div className="payment-options-grid">

                      {/* Cash on Delivery (COD) */}
                      <label className={`pay-option-card ${paymentMode === "cod" ? "selected" : ""}`}
>
                        <div className="pay-radio-line">
                          <input
                            type="radio"
                            name="paymode"
                            checked={paymentMode === "cod"}
                            onChange={() => setPaymentMode("cod")}
                          />
                          <span className="pay-name">💵 Cash on Delivery (COD)</span>
                        </div>
                        {paymentMode === "cod" && (
                          <div className="pay-sub-box">
                            <p>Pay cash or scan UPI QR code at your doorstep when your fresh produce
arrives.</p>
                            <button
                              type="button"
                              className="btn-place-order-flipkart"
                              onClick={handleFinalPlaceOrder}
                              disabled={placing}
                            >
                              {placing ? "PLACING ORDER…" : "CONFIRM & PLACE ORDER (COD)"}
                            </button>
                          </div>
                        )}
                      </label>

                      {/* UPI / QR Code */}
```

```
                    <label className={`pay-option-card ${paymentMode === "upi" ? "selected" : ""}`}
>
                      <div className="pay-radio-line">
                        <input
                          type="radio"
                          name="paymode"
                          checked={paymentMode === "upi"}
                          onChange={() => setPaymentMode("upi")}
                        />
                        <span className="pay-name">📱 UPI / QR Code (Google Pay, PhonePe, Paytm)</
span>
                      </div>
                      {paymentMode === "upi" && (
                        <div className="pay-sub-box">
                          <label className="sub-label">Enter UPI ID (VPA):</label>
                          <input
                            type="text"
                            placeholder="e.g. mobile@upi or username@okhdfcbank"
                            value={upiId}
                            onChange={(e) => setUpiId(e.target.value)}
                            className="upi-input"
                          />
                          <button
                            type="button"
                            className="btn-place-order-flipkart"
                            onClick={handleFinalPlaceOrder}
                            disabled={placing}
                          >
                            {placing ? "PROCESSING…" : `PAY ₹${total.toFixed(2)} VIA UPI`}
                          </button>
                        </div>
                      )}
                    </label>

                    {/* Credit / Debit Card */}
                    <label className={`pay-option-card ${paymentMode === "card" ? "selected" : ""}
`}>
                      <div className="pay-radio-line">
                        <input
                          type="radio"
                          name="paymode"
                          checked={paymentMode === "card"}
                          onChange={() => setPaymentMode("card")}
                        />
                        <span className="pay-name">💳 Credit / Debit / ATM Card (Visa, RuPay, Mast
erCard)</span>
                      </div>
                      {paymentMode === "card" && (
                        <div className="pay-sub-box">
                          <p className="hint">Instant secure 256-bit encrypted card transaction.</p
>
                          <button
                            type="button"
                            className="btn-place-order-flipkart"
                            onClick={handleFinalPlaceOrder}
                            disabled={placing}
                          >
                            {placing ? "PROCESSING…" : `PAY ₹${total.toFixed(2)} VIA CARD`}
                          </button>
                        </div>
                      )}
                    </label>
```

```
                    {/* Net Banking */}
                    <label className={`pay-option-card ${paymentMode === "netbanking" ? "selected"
: ""}`}>
                      <div className="pay-radio-line">
                        <input
                          type="radio"
                          name="paymode"
                          checked={paymentMode === "netbanking"}
                          onChange={() => setPaymentMode("netbanking")}
                        />
                        <span className="pay-name">🏦 Net Banking (SBI, HDFC, ICICI, Axis &amp;amp; Al
l Major Banks)</span>
                      </div>
                      {paymentMode === "netbanking" && (
                        <div className="pay-sub-box">
                          <button
                            type="button"
                            className="btn-place-order-flipkart"
                            onClick={handleFinalPlaceOrder}
                            disabled={placing}
                          >
                            {placing ? "CONNECTING BANK…" : `PAY ₹${total.toFixed(2)} VIA NET BANKI
NG`}
                          </button>
                        </div>
                      )}
                    </label>

                  </div>
                )}
              </div>
            )}
          </div>

          </div>

          {/* RIGHT COLUMN: FLIPKART STICKY PRICE DETAILS */}
          <div className="sidebar-column">
            <div className="price-details-card">
              <h3 className="sidebar-title">PRICE DETAILS</h3>

              <div className="price-row">
                <span>Price ({items.length} items)</span>
                <span>₹{retailEstimatedTotal.toFixed(2)}</span>
              </div>

              <div className="price-row">
                <span>Direct Farm Discount</span>
                <span className="green-text">-₹{savings.toFixed(2)}</span>
              </div>

              <div className="price-row">
                <span>Delivery Charges</span>
                <span className="green-text">FREE</span>
              </div>

              <div className="price-row">
                <span>Packaging &amp;amp; Handling</span>
                <span className="green-text">FREE</span>
              </div>
```

```
              <div className="price-divider" />

              <div className="price-row total-row">
                <span>Total Payable Amount</span>
                <strong className="total-amount">₹{total.toFixed(2)}</strong>
              </div>

              <div className="savings-highlight-box">
                🌱 You will save <strong>₹{savings.toFixed(2)}</strong> on this order by buying dire
ctly from verified local farmers!
              </div>

              {(!user || user.role !== "buyer") && (
                <div className="sidebar-login-prompt">
                  <span className="lock-icon">🔒</span>
                  <div className="prompt-text">
                    <strong>Buyer Login Required</strong>
                    <p>Log in as a Buyer to place your harvest order.</p>
                  </div>
                  <Link href="/auth/login?redirect=/buyer/checkout" className="btn-sidebar-login">
                    Log In as Buyer →
                  </Link>
                </div>
              )}

              <div className="security-guarantee">
                <span className="shield-icon">🛡️</span>
                <span>100% Safe &amp;amp; Direct Farm Fresh Produce Guarantee</span>
              </div>
            </div>
          </div>

        </div>
      </div>
    </div>
    <Footer />

    <style jsx>{`
      .checkout-bg { background: #f1f3f6; min-height: 85vh; padding: 28px 0 80px; }
      .checkout-container { max-width: 1200px; margin: 0 auto; }

      .error-banner { background: #fef2f2; border: 1.5px solid #fecaca; color: #991b1b; padding: 12px
18px; border-radius: 8px; margin-bottom: 20px; font-weight: 600; font-size: 0.9rem; }

      .checkout-grid { display: grid; grid-template-columns: 1fr 360px; gap: 20px; align-items: star
t; }
      @media (max-width: 960px) {
        .checkout-grid { grid-template-columns: 1fr; }
      }

      /* Step Card Accordion */
      .steps-column { display: flex; flex-direction: column; gap: 14px; }

      .step-card { background: #ffffff; border-radius: 4px; box-shadow: 0 1px 4px rgba(0,0,0,0.08); o
verflow: hidden; border: 1px solid #e0e0e0; }
      .step-card.active .step-header { background: #16a34a; color: #ffffff; }
      .step-card.active .step-num { background: #ffffff; color: #16a34a; }
      .step-card.completed .step-header { background: #f8fafc; color: #64748b; }
      .step-card.completed .step-num { background: #e2e8f0; color: #16a34a; font-weight: 800; }
      .step-card.pending .step-header { background: #ffffff; color: #94a3b8; }

      .step-header { display: flex; justify-content: space-between; align-items: center; padding: 14p
```

```
x 20px; }
      .step-num-wrap { display: flex; align-items: center; gap: 12px; }
      .step-num { width: 24px; height: 24px; border-radius: 4px; background: #e2e8f0; color: #475569;
display: flex; align-items: center; justify-content: center; font-size: 0.8rem; font-weight: 700; }
      .step-title { font-size: 0.88rem; font-weight: 700; letter-spacing: 0.03em; }
      .btn-step-change { border: 1px solid #cbd5e1; background: #ffffff; color: #16a34a; font-size:
0.76rem; font-weight: 700; padding: 5px 14px; border-radius: 3px; cursor: pointer; }
      .btn-step-change:hover { background: #f0fdf4; }

      .step-content { padding: 22px 20px; background: #ffffff; }
      .step-summary-text { padding: 14px 20px; font-size: 0.86rem; color: #334155; background: #fffff
f; }
      .addr-badge { display: inline-block; font-size: 0.7rem; font-weight: 700; background: #f1f5f9;
color: #475569; padding: 2px 6px; border-radius: 3px; margin-bottom: 4px; }
      .addr-line { color: #64748b; font-size: 0.82rem; margin: 4px 0 0; }

      /* Auth Required Box for Non-Logged-In Users */
      .auth-required-box { display: flex; gap: 16px; align-items: flex-start; background: #f8fafc; bo
rder: 1.5px solid #e2e8f0; border-radius: 8px; padding: 20px; }
      .auth-icon-wrap { font-size: 2rem; background: #e0f2fe; width: 50px; height: 50px; border-radiu
s: 50%; display: flex; align-items: center; justify-content: center; flex-shrink: 0; }
      .auth-info-text { flex: 1; }
      .auth-title { margin: 0 0 6px; font-size: 1.1rem; color: #0f172a; font-weight: 700; }
      .auth-desc { margin: 0 0 16px; font-size: 0.86rem; color: #475569; line-height: 1.5; }
      .auth-btn-row { display: flex; gap: 12px; flex-wrap: wrap; margin-bottom: 14px; }
      .btn-primary-auth { background: #16a34a; color: #ffffff; padding: 10px 20px; border-radius: 6p
x; font-weight: 700; font-size: 0.88rem; text-decoration: none; display: inline-flex; align-items: ce
nter; gap: 6px; transition: background 0.15s; }
      .btn-primary-auth:hover { background: #15803d; }
      .btn-secondary-auth { background: #ffffff; color: #1e293b; border: 1px solid #cbd5e1; padding:
10px 18px; border-radius: 6px; font-weight: 600; font-size: 0.88rem; text-decoration: none; display:
inline-flex; align-items: center; gap: 6px; }
      .btn-secondary-auth:hover { background: #f1f5f9; }
      .auth-perks { display: flex; gap: 14px; font-size: 0.76rem; color: #64748b; font-weight: 600; f
lex-wrap: wrap; border-top: 1px dashed #e2e8f0; padding-top: 10px; }

      /* Role Alert Box */
      .role-alert-box { display: flex; gap: 14px; align-items: flex-start; background: #fffbeb; borde
r: 1px solid #fde68a; border-radius: 8px; padding: 16px; color: #92400e; }
      .alert-icon { font-size: 1.8rem; }

      /* User Profile Step 1 Row */
      .user-profile-row { display: flex; justify-content: space-between; align-items: center; flex-wr
ap: wrap; gap: 14px; }
      .user-meta-wrap { display: flex; align-items: center; gap: 12px; }
      .verified-shield { width: 34px; height: 34px; border-radius: 50%; background: #dcfce7; color: #
15803d; display: flex; align-items: center; justify-content: center; font-weight: 800; font-size: 1re
m; }
      .badge-buyer { display: inline-block; font-size: 0.72rem; font-weight: 700; background: #dcfce
7; color: #166534; padding: 2px 8px; border-radius: 10px; margin-left: 6px; vertical-align: middle; }
      .user-sub { font-size: 0.82rem; color: #64748b; margin-top: 2px; }
      .user-action-wrap { display: flex; align-items: center; gap: 12px; }
      .btn-switch-link { font-size: 0.82rem; color: #64748b; text-decoration: underline; }
      .btn-switch-link:hover { color: #0f172a; }
      .btn-flipkart-action { background: #fb641b; color: #fff; border: none; padding: 12px 22px; font
-size: 0.85rem; font-weight: 700; border-radius: 3px; cursor: pointer; box-shadow: 0 1px 3px rgba(0,
0,0,0.12); }
      .btn-flipkart-action:hover { background: #e65100; }

      /* Address Form */
      .form-grid-2 { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; margin-bottom: 14px; }
      .form-grid-3 { display: grid; grid-template-columns: 1fr 1fr 1fr; gap: 14px; margin-bottom: 14p
```

```
x; }
      @media (max-width: 640px) {
        .form-grid-2, .form-grid-3 { grid-template-columns: 1fr; }
      }

      .form-field { display: flex; flex-direction: column; gap: 4px; }
      .form-field.full { margin-bottom: 14px; }
      .form-field label { font-size: 0.78rem; font-weight: 600; color: #475569; }
      .form-field input, .form-field select, .form-field textarea {
        border: 1px solid #cbd5e1; border-radius: 4px; padding: 9px 12px; font-size: 0.88rem; outlin
e: none; transition: border 0.15s; font-family: inherit;
      }
      .form-field input:focus, .form-field select:focus, .form-field textarea:focus { border-color: #
16a34a; box-shadow: 0 0 0 2px rgba(22,163,74,0.15); }

      .address-type-wrap { margin: 12px 0 18px; }
      .type-label { font-size: 0.78rem; font-weight: 600; color: #475569; display: block; margin-bott
om: 8px; }
      .radio-group { display: flex; gap: 14px; flex-wrap: wrap; }
      .type-chip { display: flex; align-items: center; gap: 8px; border: 1px solid #cbd5e1; padding:
8px 14px; border-radius: 4px; cursor: pointer; font-size: 0.82rem; font-weight: 600; color: #334155;
}
      .type-chip.selected { border-color: #16a34a; background: #f0fdf4; color: #15803d; }

      .btn-row { margin-top: 14px; }
      .btn-flipkart-save { background: #fb641b; color: #ffffff; border: none; padding: 14px 34px; fon
t-size: 0.92rem; font-weight: 800; border-radius: 3px; cursor: pointer; box-shadow: 0 1px 4px rgba(0,
0,0,0.15); letter-spacing: 0.02em; }
      .btn-flipkart-save:hover { background: #e65100; }

      /* Order Summary Produce Items */
      .produce-items-list { display: flex; flex-direction: column; gap: 14px; }
      .produce-item-card { display: flex; justify-content: space-between; align-items: center; paddin
g: 14px; border: 1px solid #f1f5f9; border-radius: 6px; background: #fafafa; }
      @media (max-width: 600px) {
        .produce-item-card { flex-direction: column; align-items: flex-start; gap: 12px; }
      }
      .prod-left { display: flex; align-items: center; gap: 14px; }
      .item-icon-box { font-size: 1.8rem; background: #ede4cd; width: 50px; height: 50px; border-radi
us: 6px; display: flex; align-items: center; justify-content: center; }
      .prod-details { display: flex; flex-direction: column; gap: 2px; }
      .prod-name { font-size: 0.95rem; color: #1b3821; }
      .prod-rate { font-size: 0.82rem; color: #64748b; }
      .delivery-badge { font-size: 0.74rem; color: #16a34a; font-weight: 600; margin-top: 4px; }

      .prod-right { display: flex; align-items: center; gap: 16px; }
      .qty-counter { display: flex; align-items: center; border: 1px solid #cbd5e1; border-radius: 4p
x; background: #fff; }
      .qty-btn { width: 28px; height: 28px; border: none; background: #f8fafc; font-weight: 700; curs
or: pointer; color: #334155; }
      .qty-btn:hover { background: #e2e8f0; }
      .qty-display { padding: 0 10px; font-size: 0.85rem; font-weight: 700; color: #1e293b; }
      .item-subtotal { font-size: 1.05rem; color: #16a34a; font-weight: 800; }
      .btn-remove-item { border: none; background: none; color: #dc2626; font-size: 0.78rem; font-wei
ght: 700; cursor: pointer; }

      .order-summary-footer { display: flex; justify-content: space-between; align-items: center; mar
gin-top: 20px; padding-top: 14px; border-top: 1px solid #e2e8f0; flex-wrap: wrap; gap: 12px; }
      .email-notif { font-size: 0.78rem; color: #64748b; }

      /* Payment Options */
      .payment-options-grid { display: flex; flex-direction: column; gap: 12px; }
```

```
      .pay-option-card { display: block; border: 1.5px solid #e2e8f0; border-radius: 6px; padding: 14
px 16px; cursor: pointer; transition: all 0.15s; }
      .pay-option-card.selected { border-color: #16a34a; background: #fafdfa; }
      .pay-radio-line { display: flex; align-items: center; gap: 12px; }
      .pay-name { font-size: 0.9rem; font-weight: 700; color: #1e293b; }

      .pay-sub-box { margin-top: 12px; padding: 12px 14px; background: #f8fafc; border-radius: 6px; b
order: 1px solid #e2e8f0; font-size: 0.84rem; color: #475569; }
      .sub-label { display: block; font-weight: 600; margin-bottom: 6px; }
      .upi-input { width: 100%; max-width: 320px; padding: 8px 12px; border: 1px solid #cbd5e1; borde
r-radius: 4px; font-size: 0.88rem; margin-bottom: 10px; }
      .hint { font-size: 0.78rem; color: #64748b; margin-bottom: 10px; }

      .btn-place-order-flipkart { width: 100%; max-width: 340px; padding: 14px 24px; background: #fb6
41b; color: #fff; border: none; border-radius: 3px; font-size: 0.95rem; font-weight: 800; cursor: poi
nter; box-shadow: 0 2px 6px rgba(0,0,0,0.15); letter-spacing: 0.03em; margin-top: 6px; }
      .btn-place-order-flipkart:hover { background: #e65100; }
      .btn-place-order-flipkart:disabled { opacity: 0.6; cursor: not-allowed; }

      /* RIGHT COLUMN: FLIPKART PRICE DETAILS CARD */
      .sidebar-column { position: sticky; top: 20px; }
      .price-details-card { background: #ffffff; border-radius: 4px; border: 1px solid #e0e0e0; box-s
hadow: 0 1px 4px rgba(0,0,0,0.08); padding: 18px 22px; }
      .sidebar-title { font-size: 0.85rem; font-weight: 800; color: #878787; border-bottom: 1px solid
#f0f0f0; padding-bottom: 12px; margin: 0 0 16px; letter-spacing: 0.04em; }

      .price-row { display: flex; justify-content: space-between; font-size: 0.9rem; color: #212121;
margin-bottom: 14px; }
      .green-text { color: #388e3c; font-weight: 700; }
      .price-divider { height: 1px; border-top: 1px dashed #e0e0e0; margin: 16px 0; }
      .total-row { font-size: 1.05rem; font-weight: 800; color: #212121; margin-bottom: 16px; }
      .total-amount { font-size: 1.25rem; color: #16a34a; }

      .savings-highlight-box { background: #f0fdf4; border: 1px solid #bbf7d0; border-radius: 4px; pa
dding: 10px 12px; font-size: 0.82rem; color: #166534; line-height: 1.4; margin-bottom: 16px; }

      .sidebar-login-prompt { background: #fefce8; border: 1.5px solid #fde047; border-radius: 6px; p
adding: 12px; margin-bottom: 16px; display: flex; flex-direction: column; gap: 8px; }
      .sidebar-login-prompt .lock-icon { font-size: 1.2rem; }
      .sidebar-login-prompt .prompt-text strong { font-size: 0.85rem; color: #854d0e; display: block;
}
      .sidebar-login-prompt .prompt-text p { font-size: 0.78rem; color: #a16207; margin: 2px 0 0; }
      .btn-sidebar-login { background: #16a34a; color: #ffffff; text-align: center; padding: 8px 12p
x; border-radius: 4px; font-weight: 700; font-size: 0.82rem; text-decoration: none; }
      .btn-sidebar-login:hover { background: #15803d; }

      /* COUPONS & OFFERS CARD */
      .coupon-card { background: #ffffff; border-radius: 4px; border: 1px solid #e0e0e0; box-shadow:
0 1px 4px rgba(0,0,0,0.08); padding: 18px; margin-bottom: 14px; }
      .coupon-head { display: flex; align-items: center; gap: 8px; font-size: 0.88rem; color: #0f172
a; margin-bottom: 12px; }
      .coupon-icon { font-size: 1.1rem; }

      .coupon-input-row { display: flex; gap: 8px; margin-bottom: 8px; }
      .coupon-input { flex: 1; border: 1px solid #cbd5e1; border-radius: 4px; padding: 8px 12px; font
-size: 0.85rem; text-transform: uppercase; font-family: var(--font-mono); font-weight: 700; outline:
none; }
      .coupon-input:focus { border-color: #16a34a; }
      .btn-apply-coupon { background: #16a34a; color: #ffffff; border: none; padding: 8px 16px; borde
r-radius: 4px; font-weight: 800; font-size: 0.8rem; cursor: pointer; }
      .btn-apply-coupon:hover { background: #15803d; }
```

```
      .coupon-error-text { display: block; font-size: 0.75rem; color: #dc2626; margin-bottom: 8px; }

      .quick-coupon-chips { display: flex; flex-direction: column; gap: 6px; margin: 8px 0 12px; }
      .quick-label { font-size: 0.72rem; color: #64748b; font-weight: 600; text-transform: uppercase;
}
      .coupon-chip { background: #f8fafc; border: 1px dashed #cbd5e1; border-radius: 4px; padding: 6p
x 10px; font-size: 0.76rem; color: #166534; font-weight: 700; cursor: pointer; text-align: left; tran
sition: all 0.15s; }
      .coupon-chip:hover { background: #f0fdf4; border-color: #16a34a; }

      .applied-coupon-box { display: flex; justify-content: space-between; align-items: center; backg
round: #f0fdf4; border: 1.5px solid #86efac; border-radius: 6px; padding: 10px 12px; margin-bottom: 1
0px; }
      .applied-info { display: flex; flex-direction: column; }
      .applied-code { font-weight: 800; color: #166534; font-size: 0.85rem; font-family: var(--font-m
ono); }
      .applied-label { font-size: 0.75rem; color: #15803d; }
      .applied-amount { font-size: 0.78rem; color: #16a34a; font-weight: 700; margin-top: 2px; }
      .btn-remove-coupon { background: none; border: none; color: #dc2626; font-size: 0.76rem; font-w
eight: 700; cursor: pointer; }

      .view-all-offers-link { display: block; font-size: 0.78rem; font-weight: 700; color: #16a34a; t
ext-align: center; text-decoration: none; padding-top: 8px; border-top: 1px dashed #f0f0f0; }
      .view-all-offers-link:hover { text-decoration: underline; }

      .coupon-discount-row { color: #16a34a !important; font-weight: 700; }

      .security-guarantee { display: flex; align-items: center; gap: 8px; font-size: 0.74rem; color:
#878787; font-weight: 600; padding-top: 12px; border-top: 1px solid #f0f0f0; }
      .shield-icon { font-size: 1.1rem; }
    `}</style>
  </>
);
}
```

app/buyer/orders/page.tsx (Buyer Orders History Page)

```
"use client";

import { useEffect, useState } from "react";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import { listMyOrders, BuyerOrder } from "@/app/lib/api";

const ORDER_STAGES = [
  { key: "pending", label: "Order Placed", icon: "📝" },
  { key: "packing", label: "Packing & Quality Check", icon: "📦" },
  { key: "shipped", label: "Shipped & In Transit (Travel)", icon: "🚚" },
  { key: "out_for_delivery", label: "Out for Delivery", icon: "🛵" },
  { key: "delivered", label: "Delivered", icon: "✅" },
];

function getStageIndex(status: string): number {
  if (status === "pending") return 0;
  if (status === "packing" || status === "confirmed") return 1;
  if (status === "shipped") return 2;
  if (status === "out_for_delivery") return 3;
  if (status === "delivered") return 4;
  return 0;
}

const STATUS_BADGES: Record<string, { label: string; color: string; bg: string }> = {
  pending: { label: "⏳ Pending Verification", color: "#b45309", bg: "#fffbeb" },
  confirmed: { label: "📦 Confirmed & Packing", color: "#4338ca", bg: "#eef2ff" },
  packing: { label: "📦 Packing & Prep", color: "#4338ca", bg: "#eef2ff" },
  shipped: { label: "🚚 Shipped / In Transit", color: "#0284c7", bg: "#f0f9ff" },
  out_for_delivery: { label: "🛵 Out for Delivery", color: "#d97706", bg: "#fef3c7" },
  delivered: { label: "✅ Delivered", color: "#15803d", bg: "#f0fdf4" },
  cancelled: { label: "❌ Cancelled", color: "#b91c1c", bg: "#fef2f2" },
};

export default function BuyerOrdersPage() {
  const [orders, setOrders] = useState<BuyerOrder[]>([]);
  const [loading, setLoading] = useState(true);
  const [tab, setTab] = useState<"orders" | "bulk">("orders");

  async function load() {
    setLoading(true);
    try {
      const res = await listMyOrders();
      setOrders(res.orders || []);
    } catch {
      // Fallback
    } finally {
      setLoading(false);
    }
  }

  useEffect(() => {
    load();
  }, []);

  return (
    <>
      <Nav />
      <div className="container orders-wrap">
```

```
        <div className="header-box">
          <div>
            <h1>My Orders &amp;amp; Live Tracking</h1>
            <p className="subhead">Track your fresh harvest orders, packing status, shipping travel,
and delivery updates.</p>
          </div>
          <button type="button" onClick={load} className="btn-refresh">
            🔄 Refresh Status
          </button>
        </div>

        <div className="tabs">
          <button className={tab === "orders" ? "active" : ""} onClick={() => setTab("orders")}>
            🛒 My Orders ({orders.length})
          </button>
          <button className={tab === "bulk" ? "active" : ""} onClick={() => setTab("bulk")}>
            🌾 My Bulk Requests
          </button>
        </div>

        {tab === "orders" && (
          loading ? (
            <div className="empty-card">Loading your orders…</div>
          ) : orders.length === 0 ? (
            <div className="empty-card">
              <span style={{ fontSize: "2rem", display: "block", marginBottom: "8px" }}>🛒</span>
              <strong>No orders placed yet.</strong>
              <p style={{ margin: "6px 0 0", fontSize: "0.85rem", color: "#888" }}>
                Visit our fresh produce Shop to place your first farm order!
              </p>
            </div>
          ) : (
            orders.map((order) => {
              const currentStep = getStageIndex(order.status);
              const isCancelled = order.status === "cancelled";
              const badge = STATUS_BADGES[order.status] || STATUS_BADGES.pending;

              return (
                <div className="order-card" key={order.id}>
                  <div className="order-head">
                    <div>
                      <span className="order-id">Order #{order.id}</span>
                      <span className="order-date">
                        &bull; {new Date(order.date).toLocaleDateString("en-IN", { month: "short", da
y: "numeric", year: "numeric" })}
                      </span>
                    </div>
                    <span className="status-badge" style={{ color: badge.color, background: badge.bg
}}>
                      {badge.label}
                    </span>
                  </div>

                  {/* Live Order Tracking Stepper */}
                  {!isCancelled && (
                    <div className="stepper-wrap">
                      <div className="stepper">
                        {ORDER_STAGES.map((stage, idx) => {
                          const isDone = idx < currentStep;
                          const isCurrent = idx === currentStep;
                          return (
                            <div key={stage.key} className={`step-item ${isDone ? "done" : ""} ${isCu
```

```
rrent ? "current" : ""}`}>
                            <div className="step-icon-wrap">
                              <span className="step-icon">{stage.icon}</span>
                            </div>
                            <span className="step-label">{stage.label}</span>
                          </div>
                        );
                      })}
                    </div>
                    {/* Stepper Progress Bar */}
                    <div className="stepper-line-bg">
                      <div
                        className="stepper-line-fill"
                        style={{ width: `${(currentStep / (ORDER_STAGES.length - 1)) * 100}%` }}
                      />
                    </div>
                  </div>
                )}

                {/* Tracking Note Alert */}
                {order.tracking_note && (
                  <div className="tracking-note-banner">
                    <span className="note-icon">📢</span>
                    <div>
                      <strong>Latest Logistics Update from Admin:</strong>
                      <p>{order.tracking_note}</p>
                    </div>
                  </div>
                )}

                {/* Order Line Items */}
                <ul className="items-list">
                  {order.items?.map((item, i) => (
                    <li key={i} className="item-row">
                      <div className="item-name-box">
                        <span className="item-icon">🌾</span>
                        <div>
                          <strong>{item.name}</strong>
                          <span className="item-qty-tag">{item.qty} kg</span>
                        </div>
                      </div>
                      <span className="item-price">₹{item.subtotal.toFixed(2)}</span>
                    </li>
                  ))}
                </ul>

                <div className="order-foot">
                  <div className="foot-left">
                    <span className="pay-mode">Payment: <strong>{order.payment_mode?.toUpperCase()
|| "COD"}</strong></span>
                    <span className="delivery-addr" title={order.address}>📍 Delivery to: {order.a
ddress}</span>
                  </div>
                  <div className="foot-right">
                    <span className="total-label">Total Amount:</span>
                    <strong className="total-amount">₹{order.total.toFixed(2)}</strong>
                  </div>
                </div>
              </div>
            );
          })
        )
```

```
        )}

        {tab === "bulk" && (
          <div className="empty-card">
            No bulk requests submitted yet. Request one from any product page for wholesale orders ab
ove 10 kg.
          </div>
        )}
      </div>
      <Footer />

      <style jsx>{`
        .orders-wrap { padding: 44px 24px 90px; }
        .header-box { display: flex; justify-content: space-between; align-items: flex-start; margin-
bottom: 22px; gap: 14px; flex-wrap: wrap; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.8rem; margin: 0 0 4px;
color: #1b3821; }
        .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }
        .btn-refresh { background: #f0fdf4; border: 1px solid #16a34a; color: #15803d; padding: 8px 1
6px; border-radius: 999px; font-size: 0.82rem; font-weight: 700; cursor: pointer; transition: all 0.1
5s; }
        .btn-refresh:hover { background: #16a34a; color: #fff; }

        .tabs { display: flex; gap: 8px; margin-bottom: 24px; }
        .tabs button {
          border: 1px solid rgba(26,26,21,0.14); background: #fff;
          padding: 9px 20px; border-radius: 999px; font-size: 0.87rem; font-weight: 600; cursor: poin
ter;
        }
        .tabs button.active { background: #1b3821; color: #ffffff; border-color: #1b3821; }

        .empty-card {
          background: #fff; border: 1px solid rgba(26,26,21,0.08);
          border-radius: 14px; padding: 50px; text-align: center; color: #6b6555; box-shadow: 0 2px 8
px rgba(0,0,0,0.04);
        }

        .order-card {
          background: #fff; border: 1px solid rgba(26,26,21,0.08);
          border-radius: 16px; margin-bottom: 20px; overflow: hidden; box-shadow: 0 2px 10px rgba(0,
0,0,0.04);
        }
        .order-head {
          display: flex; justify-content: space-between; align-items: center;
          padding: 16px 22px; border-bottom: 1px solid rgba(26,26,21,0.06);
          font-size: 0.88rem; background: #fafafa;
        }
        .order-id { font-family: monospace; font-weight: 700; color: #1b3821; font-size: 0.95rem; }
        .order-date { color: #888273; font-size: 0.82rem; }

        .status-badge { font-size: 0.78rem; font-weight: 700; padding: 5px 12px; border-radius: 999p
x; }

        /* Stepper */
        .stepper-wrap { position: relative; padding: 22px 24px 18px; background: #faf8f5; border-bott
om: 1px solid rgba(26,26,21,0.06); }
        .stepper { display: flex; justify-content: space-between; position: relative; z-index: 2; }
        .step-item { display: flex; flex-direction: column; align-items: center; text-align: center;
flex: 1; opacity: 0.45; transition: all 0.2s; }
        .step-item.done, .step-item.current { opacity: 1; }
        .step-icon-wrap { width: 36px; height: 36px; border-radius: 50%; background: #ffffff; border:
2px solid #d1d5db; display: flex; align-items: center; justify-content: center; font-size: 1.1rem; ma
```

```
rgin-bottom: 6px; box-shadow: 0 2px 4px rgba(0,0,0,0.06); }
        .step-item.done .step-icon-wrap { border-color: #16a34a; background: #f0fdf4; }
        .step-item.current .step-icon-wrap { border-color: #16a34a; background: #16a34a; transform: s
cale(1.15); }
        .step-label { font-size: 0.74rem; font-weight: 600; color: #4b5563; max-width: 90px; line-hei
ght: 1.25; }
        .step-item.current .step-label { color: #15803d; font-weight: 700; }

        .stepper-line-bg { position: absolute; top: 38px; left: 8%; right: 8%; height: 4px; backgroun
d: #e5e7eb; z-index: 1; border-radius: 999px; }
        .stepper-line-fill { height: 100%; background: #16a34a; transition: width 0.3s ease; border-r
adius: 999px; }

        /* Tracking Note */
        .tracking-note-banner { display: flex; align-items: flex-start; gap: 10px; padding: 12px 20p
x; background: #eff6ff; border-bottom: 1px solid #dbeafe; font-size: 0.82rem; color: #1e40af; }
        .note-icon { font-size: 1.2rem; }
        .tracking-note-banner p { margin: 2px 0 0; color: #1d4ed8; }

        /* Items List */
        .items-list { list-style: none; margin: 0; padding: 8px 22px; }
        .item-row { display: flex; justify-content: space-between; align-items: center; padding: 12px
0; font-size: 0.9rem; border-bottom: 1px solid rgba(26,26,21,0.05); }
        .item-row:last-child { border-bottom: none; }
        .item-name-box { display: flex; align-items: center; gap: 10px; }
        .item-icon { font-size: 1.3rem; }
        .item-qty-tag { display: inline-block; margin-left: 8px; font-size: 0.76rem; background: #f3f
4f6; color: #4b5563; padding: 2px 7px; border-radius: 4px; font-weight: 600; }
        .item-price { font-weight: 700; color: #1b3821; font-size: 0.95rem; }

        .order-foot {
          display: flex; justify-content: space-between; align-items: center;
          padding: 14px 22px; background: #faf7ee; font-size: 0.85rem; border-top: 1px solid rgba(26,
26,21,0.06);
          flex-wrap: wrap; gap: 12px;
        }
        .foot-left { display: flex; flex-direction: column; gap: 2px; }
        .pay-mode { color: #6b6555; }
        .delivery-addr { font-size: 0.78rem; color: #888273; max-width: 320px; }
        .foot-right { display: flex; align-items: center; gap: 8px; }
        .total-label { color: #6b6555; }
        .total-amount { font-size: 1.2rem; color: #16a34a; font-weight: 800; }
      `}</style>
    </>
  );
}
```

app/buyer/bulk-request/page.tsx (Buyer Wholesale Bulk Request Page)

```
"use client";

import { Suspense, useState } from "react";
import { useRouter, useSearchParams } from "next/navigation";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import { getProduct } from "@/app/lib/products-data";
import { submitBulkRequest } from "@/app/lib/api";
import "@/app/components/forms.css";

export default function BulkRequestPage() {
  return (
    <Suspense fallback={null}>
      <BulkRequestForm />
    </Suspense>
  );
}

function BulkRequestForm() {
  const router = useRouter();
  const searchParams = useSearchParams();
  const productId = Number(searchParams.get("product_id") || 0);
  const product = getProduct(productId);

  const [qty, setQty] = useState(11);
  const [reason, setReason] = useState("");
  const [submitted, setSubmitted] = useState(false);
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState("");

  if (!product) {
    return (
      <>
        <Nav />
        <div className="container" style={{ padding: "80px 24px", textAlign: "center" }}>
          <p>Product not found.</p>
        </div>
        <Footer />
      </>
    );
  }

  async function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    setLoading(true);
    setError("");
    try {
      // Use db_id (real MySQL product ID) to avoid FK constraint errors
      const dbProductId = product?.db_id ?? productId;
      await submitBulkRequest(dbProductId, qty, reason);
      setSubmitted(true);
    } catch (err: any) {
      setError(err?.message ?? "Submission failed. Please try again.");
    } finally {
      setLoading(false);
    }
  }

  return (
```

```
    <>
      <Nav />
      <div className="container bulk-wrap">
        <div className="card">
          <p className="icon">📦</p>
          <h1>Bulk purchase request</h1>
          <p className="subtitle">
            Requesting <strong>{product.name}</strong> in bulk (above 10 kg) requires
            admin approval before you can buy it.
          </p>

          {submitted ? (
            <div className="form-info">
              Request submitted. You&rsquo;ll see it under &ldquo;My bulk requests&rdquo;
              once an admin reviews it. <a onClick={() => router.push("/buyer/orders")}>View requests
→</a>
            </div>
          ) : (
            <form onSubmit={handleSubmit}>
              <div className="field">
                <label>Required quantity (kg)</label>
                <input
                  type="number" step={1} min={11} max={product.stock}
                  value={qty} onChange={(e) => setQty(parseInt(e.target.value) || 11)}
                  required
                />
                <p className="hint">Available stock: {product.stock} kg</p>
              </div>
              <div className="field">
                <label>Reason for bulk purchase</label>
                <textarea
                  rows={4}
                  placeholder="E.g. for restaurant supply, wholesale reselling, event catering, etc."
                  value={reason}
                  onChange={(e) => setReason(e.target.value)}
                  required
                />
              </div>
              {error && (
                <p style={{ color: "#dc2626", fontSize: "0.85rem", margin: "0 0 12px", padding: "10px
12px", background: "#fef2f2", border: "1px solid #fecaca", borderRadius: "8px" }}>
                  ⚠️ {error}
                </p>
              )}
              <button className="btn-block" type="submit" disabled={loading}>
                {loading ? "Submitting…" : "Submit request to admin"}
              </button>
            </form>
          )}
        </div>
      </div>
      <Footer />

      <style jsx>{`
        .bulk-wrap { padding: 44px 24px 90px; display: flex; justify-content: center; }
        .card {
          width: 100%; max-width: 480px; background: #fff;
          border: 1px solid rgba(26,26,21,0.08); border-radius: 16px; padding: 32px;
        }
        .icon { font-size: 1.6rem; margin: 0 0 6px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.4rem; margin: 0 0 8px;
}
```

```
        .subtitle { color: #6b6555; font-size: 0.9rem; line-height: 1.5; margin: 0 0 24px; }
        .hint { font-size: 0.78rem; color: #6b6555; margin: 6px 0 0; }
        .form-info a { cursor: pointer; font-weight: 600; }
      `}</style>
    </>
  );
}
```

app/buyer/dashboard/page.tsx (Buyer Dashboard Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import Link from "next/link";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";
import { getMe, logoutUser, listMyOrders, BuyerOrder, AuthUser } from "@/app/lib/api";

const OFFERS = [
  {
    code: "FRESHFARM10",
    title: "10% Harvest Season Discount",
    desc: "Get 10% instant discount on all fresh vegetables and fruits directly from verified local f
arms.",
    minOrder: "₹299",
    tag: "POPULAR",
    color: "#16a34a",
    bg: "#f0fdf4",
    border: "#bbf7d0",
  },
  {
    code: "ORGANIC20",
    title: "20% OFF Organic Greens & Herbs",
    desc: "Special discount on all certified chemical-free leafy vegetables, spinach, coriander, and
mint.",
    minOrder: "₹199",
    tag: "VEGGIE SPECIAL",
    color: "#0284c7",
    bg: "#f0f9ff",
    border: "#bae6fd",
  },
  {
    code: "FREESHIP",
    title: "100% Free Doorstep Delivery",
    desc: "Enjoy zero delivery fee on all direct farm produce orders delivered fresh within 24-48 hou
rs.",
    minOrder: "₹499",
    tag: "ZERO SHIPPING",
    color: "#7c3aed",
    bg: "#f5f3ff",
    border: "#ddd6fe",
  },
  {
    code: "FIRSTHARVEST",
    title: "Flat ₹50 Welcome Discount",
    desc: "Exclusive welcome gift for your first harvest order on the FarmLink marketplace.",
    minOrder: "₹250",
    tag: "NEW BUYER",
    color: "#ea580c",
    bg: "#fff7ed",
    border: "#fed7aa",
  },
  {
    code: "BULKSAVER15",
    title: "15% Direct Wholesale Discount",
    desc: "Save 15% when purchasing 5kg or more for restaurants, hotels, hostels, or large familie
s.",
    minOrder: "5 kg",
```

```
    tag: "BULK ORDER",
    color: "#b45309",
    bg: "#fffbeb",
    border: "#fde68a",
  },
];

export default function BuyerDashboardPage() {
  const router = useRouter();
  const [user, setUser] = useState<AuthUser | null>(null);
  const [loading, setLoading] = useState(true);
  const [orders, setOrders] = useState<BuyerOrder[]>([]);
  const [copiedCode, setCopiedCode] = useState<string | null>(null);

  useEffect(() => {
    (async () => {
      try {
        const me = await getMe();
        if (me?.user && me.user.role === "buyer") {
          setUser(me.user);
          const ordersRes = await listMyOrders().catch(() => ({ orders: [] }));
          setOrders(ordersRes.orders || []);
        } else if (me?.user && me.user.role !== "buyer") {
          if (me.user.role === "farmer") router.push("/farmer/dashboard");
          else if (me.user.role === "admin") router.push("/admin/dashboard");
        } else {
          router.push("/auth/login?redirect=/buyer/dashboard");
        }
      } catch {
        router.push("/auth/login?redirect=/buyer/dashboard");
      } finally {
        setLoading(false);
      }
    })();
  }, [router]);

  async function handleLogout() {
    try {
      await logoutUser();
    } catch {
      // ignore
    }
    router.push("/auth/login");
  }

  function handleCopy(code: string) {
    navigator.clipboard.writeText(code);
    setCopiedCode(code);
    setTimeout(() => setCopiedCode(null), 2200);
  }

  if (loading) {
    return (
      <>
        <Nav />
        <div className="container" style={{ padding: "80px 24px", textAlign: "center" }}>
          <p style={{ color: "#64748b", fontSize: "1.1rem" }}>⏳ Loading your Buyer Dashboard…</p>
        </div>
        <Footer />
      </>
    );
  }
```

```
  if (!user) return null;

  const activeOrdersCount = orders.filter((o) => o.status !== "delivered" && o.status !== "cancelle
d").length;
  const totalSpent = orders.reduce((sum, o) => sum + (o.total || 0), 0);
  const estimatedSavings = totalSpent * 0.25;

  return (
    <>
      <Nav />
      <div className="dashboard-bg">
        <div className="container dashboard-container">

          {/* TOP PROFILE HEADER CARD */}
          <div className="profile-banner-card">
            <div className="profile-left">
              <div className="avatar-circle">
                {user.name.charAt(0).toUpperCase()}
              </div>
              <div className="profile-details">
                <div className="profile-name-row">
                  <h1>{user.name}</h1>
                  <span className="badge-role">🌾 Verified Buyer</span>
                </div>
                <div className="profile-meta-grid">
                  <div className="meta-item">
                    <span className="meta-icon">📧</span>
                    <span className="meta-val">{user.email || "Email Account"}</span>
                    <span className="badge-verified">✓ Verified</span>
                  </div>
                  {user.mobile && (
                    <div className="meta-item">
                      <span className="meta-icon">📱</span>
                      <span className="meta-val">{user.mobile}</span>
                    </div>
                  )}
                  {user.created_at && (
                    <div className="meta-item">
                      <span className="meta-icon">📅</span>
                      <span className="meta-val">
                        Member since {new Date(user.created_at).toLocaleDateString("en-IN", { month:
"short", year: "numeric" })}
                      </span>
                    </div>
                  )}
                </div>
              </div>
            </div>

            <div className="profile-right-actions">
              <Link href="/shop" className="btn-action-primary">
                🌾 Browse Fresh Shop
              </Link>
              <button type="button" onClick={handleLogout} className="btn-action-logout">
                🚪 Log Out
              </button>
            </div>
          </div>

          {/* QUICK STATS ROW */}
          <div className="stats-grid">
```

```
            <div className="stat-card">
              <div className="stat-icon">🛒</div>
              <div className="stat-info">
                <span className="stat-label">Total Orders</span>
                <strong className="stat-number">{orders.length}</strong>
              </div>
            </div>

            <div className="stat-card highlight">
              <div className="stat-icon">🚚</div>
              <div className="stat-info">
                <span className="stat-label">In-Transit / Active</span>
                <strong className="stat-number">{activeOrdersCount}</strong>
              </div>
            </div>

            <div className="stat-card green">
              <div className="stat-icon">💰</div>
              <div className="stat-info">
                <span className="stat-label">Direct Farm Savings</span>
                <strong className="stat-number">₹{estimatedSavings.toFixed(0)}</strong>
              </div>
            </div>

            <div className="stat-card">
              <div className="stat-icon">🏷️</div>
              <div className="stat-info">
                <span className="stat-label">Active Coupons</span>
                <strong className="stat-number">{OFFERS.length} Offers</strong>
              </div>
            </div>
          </div>

          {/* BUYER DISCOUNTS & SPECIAL OFFERS SECTION */}
          <section className="dashboard-section" id="offers">
            <div className="section-head">
              <div>
                <h2>🎁 Buyer Discounts &amp;amp; Exclusive Farm Offers</h2>
                <p className="section-subhead">
                  Use these promotional coupon codes during checkout to unlock instant discounts and
free delivery directly from local farms.
                </p>
              </div>
              <Link href="/shop" className="btn-shop-deals">
                Shop Deals Now →
              </Link>
            </div>

            <div className="offers-grid">
              {OFFERS.map((offer) => (
                <div
                  className="offer-card"
                  key={offer.code}
                  style={{ background: offer.bg, borderColor: offer.border }}
                >
                  <div className="offer-header">
                    <span className="offer-tag" style={{ background: offer.color }}>
                      {offer.tag}
                    </span>
                    <span className="min-order">Min Order: {offer.minOrder}</span>
                  </div>
```

```
                  <h3 className="offer-title">{offer.title}</h3>
                  <p className="offer-desc">{offer.desc}</p>

                  <div className="offer-bottom">
                    <div className="coupon-code-pill">
                      <span className="code-text">{offer.code}</span>
                      <button
                        type="button"
                        className="btn-copy-code"
                        onClick={() => handleCopy(offer.code)}
                        title="Click to copy coupon code"
                      >
                        {copiedCode === offer.code ? "✓ Copied!" : "📋 Copy"}
                      </button>
                    </div>

                    <Link href="/shop" className="btn-use-coupon">
                      Apply in Shop →
                    </Link>
                  </div>
                </div>
              ))}
            </div>
          </section>

          {/* RECENT ORDERS & TRACKING SECTION */}
          <section className="dashboard-section">
            <div className="section-head">
              <div>
                <h2>📦 Recent Orders &amp;amp; Live Shipment Tracking</h2>
                <p className="section-subhead">Track your packing and delivery journey from the farm
to your table.</p>
              </div>
              <Link href="/buyer/orders" className="btn-outline-link">
                View All Orders ({orders.length}) →
              </Link>
            </div>

            {orders.length === 0 ? (
              <div className="empty-orders-box">
                <span className="empty-icon">🌱</span>
                <h3>No Orders Placed Yet</h3>
                <p>Support your local farmers by purchasing fresh organic harvest today.</p>
                <Link href="/shop" className="btn-action-primary" style={{ display: "inline-block", m
arginTop: "12px" }}>
                  Start Shopping
                </Link>
              </div>
            ) : (
              <div className="recent-orders-list">
                {orders.slice(0, 3).map((order) => (
                  <div className="order-item-row" key={order.id}>
                    <div className="order-main-info">
                      <span className="order-badge-id">Order #{order.id}</span>
                      <span className="order-date-text">
                        {new Date(order.date).toLocaleDateString("en-IN", { month: "short", day: "num
eric", year: "numeric" })}
                      </span>
                      <span className="order-items-count">
                        &bull; {order.items?.length || 0} produce item(s)
                      </span>
                    </div>
```

```
                  <div className="order-status-and-action">
                    <strong className="order-amount-text">₹{order.total.toFixed(2)}</strong>
                    <span className={`status-pill status-${order.status}`}>
                      {order.status.toUpperCase()}
                    </span>
                    <Link href="/buyer/orders" className="btn-track-order">
                      🚚 Track Shipment
                    </Link>
                  </div>
                </div>
              ))}
            </div>
          )}
        </section>

        {/* QUICK LINKS GRID */}
        <div className="quick-links-grid">
          <Link href="/shop" className="quick-link-card">
            <span className="quick-icon">🌾</span>
            <h4>Browse Farm Produce</h4>
            <p>Explore today’s freshest fruits and vegetables.</p>
          </Link>

          <Link href="/buyer/orders" className="quick-link-card">
            <span className="quick-icon">📦</span>
            <h4>Order History &amp;amp; Tracking</h4>
            <p>Check dispatch, travel notes, and delivery updates.</p>
          </Link>

          <Link href="/buyer/bulk-request" className="quick-link-card">
            <span className="quick-icon">🧺</span>
            <h4>Bulk Wholesale Orders</h4>
            <p>Direct wholesale quotes for institutions &amp;amp; events.</p>
          </Link>

          <Link href="/buyer/cart" className="quick-link-card">
            <span className="quick-icon">🛒</span>
            <h4>My Shopping Cart</h4>
            <p>Review selected produce items and checkout.</p>
          </Link>
        </div>

      </div>
    </div>
    <Footer />

    <style jsx>{`
      .dashboard-bg { background: #f8fafc; min-height: 85vh; padding: 32px 0 80px; }
      .dashboard-container { max-width: 1140px; margin: 0 auto; display: flex; flex-direction: colu
mn; gap: 28px; }

      /* Profile Header Banner */
      .profile-banner-card {
        background: linear-gradient(135deg, #1b3821 0%, #0d2313 100%);
        color: #ffffff;
        border-radius: 16px;
        padding: 32px;
        display: flex;
        justify-content: space-between;
        align-items: center;
        gap: 24px;
```

```
          box-shadow: 0 4px 20px rgba(0,0,0,0.12);
        }
        @media (max-width: 860px) {
          .profile-banner-card { flex-direction: column; align-items: flex-start; }
        }

        .profile-left { display: flex; align-items: center; gap: 20px; flex-wrap: wrap; }
        .avatar-circle {
          width: 72px; height: 72px; border-radius: 50%;
          background: linear-gradient(135deg, #16a34a, #22c55e);
          color: #ffffff; font-size: 2rem; font-weight: 800;
          display: flex; align-items: center; justify-content: center;
          box-shadow: 0 4px 12px rgba(0,0,0,0.2);
          border: 3px solid rgba(255,255,255,0.2);
        }

        .profile-name-row { display: flex; align-items: center; gap: 12px; flex-wrap: wrap; margin-bo
ttom: 8px; }
        .profile-name-row h1 { margin: 0; font-family: var(--font-display); font-size: 1.6rem; color:
#ffffff; }
        .badge-role { background: #22c55e; color: #052e16; font-size: 0.76rem; font-weight: 800; padd
ing: 3px 10px; border-radius: 999px; }

        .profile-meta-grid { display: flex; gap: 18px; flex-wrap: wrap; font-size: 0.88rem; color: #c
bd5e1; }
        .meta-item { display: flex; align-items: center; gap: 6px; }
        .meta-icon { font-size: 1rem; }
        .meta-val { color: #f1f5f9; font-weight: 600; }
        .badge-verified { background: rgba(34, 197, 94, 0.2); color: #4ade80; font-size: 0.72rem; fon
t-weight: 700; padding: 2px 6px; border-radius: 4px; border: 1px solid rgba(34, 197, 94, 0.4); }

        .profile-right-actions { display: flex; align-items: center; gap: 12px; flex-wrap: wrap; }
        .btn-action-primary { background: #16a34a; color: #ffffff; padding: 10px 20px; border-radius:
8px; font-weight: 700; font-size: 0.9rem; text-decoration: none; transition: background 0.15s; }
        .btn-action-primary:hover { background: #15803d; }
        .btn-action-logout { background: rgba(255,255,255,0.1); border: 1px solid rgba(255,255,255,0.
25); color: #f87171; padding: 10px 18px; border-radius: 8px; font-weight: 700; font-size: 0.9rem; cur
sor: pointer; transition: all 0.15s; }
        .btn-action-logout:hover { background: #ef4444; color: #ffffff; border-color: #ef4444; }

        /* Stats Grid */
        .stats-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: 18px; }
        @media (max-width: 900px) {
          .stats-grid { grid-template-columns: repeat(2, 1fr); }
        }
        @media (max-width: 480px) {
          .stats-grid { grid-template-columns: 1fr; }
        }

        .stat-card { background: #ffffff; border: 1px solid #e2e8f0; border-radius: 12px; padding: 20
px; display: flex; align-items: center; gap: 16px; box-shadow: 0 2px 6px rgba(0,0,0,0.03); }
        .stat-card.highlight { border-color: #bae6fd; background: #f0f9ff; }
        .stat-card.green { border-color: #bbf7d0; background: #f0fdf4; }
        .stat-icon { width: 48px; height: 48px; border-radius: 10px; background: #f1f5f9; display: fl
ex; align-items: center; justify-content: center; font-size: 1.5rem; flex-shrink: 0; }
        .stat-card.highlight .stat-icon { background: #e0f2fe; }
        .stat-card.green .stat-icon { background: #dcfce7; }
        .stat-info { display: flex; flex-direction: column; }
        .stat-label { font-size: 0.78rem; font-weight: 600; color: #64748b; text-transform: uppercas
e; letter-spacing: 0.03em; }
        .stat-number { font-size: 1.35rem; font-weight: 800; color: #0f172a; margin-top: 2px; }
```

```
        /* Sections */
        .dashboard-section { background: #ffffff; border: 1px solid #e2e8f0; border-radius: 14px; pad
ding: 28px; box-shadow: 0 2px 8px rgba(0,0,0,0.03); }
        .section-head { display: flex; justify-content: space-between; align-items: center; margin-bo
ttom: 22px; flex-wrap: wrap; gap: 12px; }
        .section-head h2 { font-family: var(--font-display); font-size: 1.35rem; color: #0f172a; marg
in: 0 0 4px; }
        .section-subhead { font-size: 0.86rem; color: #64748b; margin: 0; }
        .btn-shop-deals { color: #16a34a; font-weight: 700; font-size: 0.88rem; text-decoration: non
e; }
        .btn-outline-link { font-size: 0.86rem; font-weight: 700; color: #0284c7; text-decoration: no
ne; }

        /* Offers Grid */
        .offers-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(320px, 1fr)); g
ap: 18px; }
        .offer-card { border-radius: 12px; border: 1.5px solid; padding: 20px; display: flex; flex-di
rection: column; justify-content: space-between; }
        .offer-header { display: flex; justify-content: space-between; align-items: center; margin-bo
ttom: 12px; }
        .offer-tag { color: #ffffff; font-size: 0.7rem; font-weight: 800; padding: 3px 8px; border-ra
dius: 4px; letter-spacing: 0.04em; }
        .min-order { font-size: 0.76rem; color: #64748b; font-weight: 600; }
        .offer-title { font-size: 1.05rem; font-weight: 700; color: #0f172a; margin: 0 0 6px; }
        .offer-desc { font-size: 0.82rem; color: #475569; line-height: 1.45; margin: 0 0 16px; flex:
1; }

        .offer-bottom { display: flex; justify-content: space-between; align-items: center; gap: 10p
x; border-top: 1px dashed rgba(0,0,0,0.12); padding-top: 14px; flex-wrap: wrap; }
        .coupon-code-pill { display: inline-flex; align-items: center; background: #ffffff; border:
1.5px dashed #94a3b8; border-radius: 6px; padding: 4px 10px; gap: 8px; }
        .code-text { font-family: var(--font-mono); font-weight: 800; color: #0f172a; font-size: 0.88
rem; letter-spacing: 0.05em; }
        .btn-copy-code { background: none; border: none; font-size: 0.76rem; font-weight: 700; color:
#16a34a; cursor: pointer; padding: 2px 4px; }
        .btn-use-coupon { font-size: 0.8rem; font-weight: 700; color: #16a34a; text-decoration: none;
}
        .btn-use-coupon:hover { text-decoration: underline; }

        /* Recent Orders */
        .empty-orders-box { text-align: center; padding: 36px 20px; color: #64748b; }
        .empty-icon { font-size: 2.5rem; display: block; margin-bottom: 8px; }
        .recent-orders-list { display: flex; flex-direction: column; gap: 12px; }
        .order-item-row { display: flex; justify-content: space-between; align-items: center; paddin
g: 14px 18px; background: #f8fafc; border: 1px solid #e2e8f0; border-radius: 8px; flex-wrap: wrap; ga
p: 12px; }
        .order-main-info { display: flex; align-items: center; gap: 10px; font-size: 0.88rem; }
        .order-badge-id { font-weight: 800; color: #0f172a; }
        .order-date-text { color: #64748b; font-size: 0.82rem; }
        .order-items-count { color: #64748b; font-size: 0.82rem; }

        .order-status-and-action { display: flex; align-items: center; gap: 14px; }
        .order-amount-text { font-size: 0.95rem; color: #0f172a; font-weight: 800; }
        .status-pill { font-size: 0.7rem; font-weight: 800; padding: 3px 8px; border-radius: 4px; }
        .status-pending { background: #fef3c7; color: #b45309; }
        .status-packing { background: #e0e7ff; color: #4338ca; }
        .status-shipped { background: #e0f2fe; color: #0369a1; }
        .status-out_for_delivery { background: #fef3c7; color: #d97706; }
        .status-delivered { background: #dcfce7; color: #15803d; }
        .status-cancelled { background: #fee2e2; color: #b91c1c; }

        .btn-track-order { background: #ffffff; border: 1px solid #cbd5e1; color: #0f172a; padding: 5
```

```
px 12px; border-radius: 6px; font-size: 0.78rem; font-weight: 700; text-decoration: none; }
        .btn-track-order:hover { background: #f1f5f9; }

        /* Quick Links */
        .quick-links-grid { display: grid; grid-template-columns: repeat(4, 1fr); gap: 16px; }
        @media (max-width: 800px) {
          .quick-links-grid { grid-template-columns: repeat(2, 1fr); }
        }
        @media (max-width: 480px) {
          .quick-links-grid { grid-template-columns: 1fr; }
        }

        .quick-link-card { background: #ffffff; border: 1px solid #e2e8f0; border-radius: 12px; paddi
ng: 20px; text-decoration: none; color: inherit; transition: all 0.15s; }
        .quick-link-card:hover { transform: translateY(-2px); border-color: #16a34a; box-shadow: 0 4p
x 12px rgba(22,163,74,0.08); }
        .quick-icon { font-size: 1.8rem; display: block; margin-bottom: 8px; }
        .quick-link-card h4 { margin: 0 0 4px; font-size: 0.95rem; font-weight: 700; color: #0f172a;
}
        .quick-link-card p { margin: 0; font-size: 0.78rem; color: #64748b; line-height: 1.4; }
      `}</style>
    </>
  );
}
```

app/farmer/products/new/page.tsx (Farmer New Product Listing Form)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import FarmerNav from "@/app/components/FarmerNav";
import { listCategories, createProduct, ApiError } from "@/app/lib/api";
import "@/app/components/forms.css";

const GST_RATES = [
  { rate: 0, label: "0% (Exempted / Fresh Farm Produce)" },
  { rate: 5, label: "5% (Standard Agricultural Products)" },
  { rate: 12, label: "12% (Processed / Packaged Produce)" },
  { rate: 18, label: "18% (Specialty Items)" },
];

export default function AddProductPage() {
  const router = useRouter();
  const [categories, setCategories] = useState<{ id: number; name: string }[]>([]);
  const [name, setName] = useState("");
  const [categoryId, setCategoryId] = useState<number>(1);
  const [basePrice, setBasePrice] = useState("");
  const [gstPercent, setGstPercent] = useState(5);
  const [commissionPercent] = useState(5); // 5% website platform commission
  const [stock, setStock] = useState("");
  const [description, setDescription] = useState("");
  const [file, setFile] = useState<File | null>(null);
  const [error, setError] = useState("");
  const [loading, setLoading] = useState(false);

  useEffect(() => {
    (async () => {
      try {
        const res = await listCategories();
        if (res.categories && res.categories.length > 0) {
          setCategories(res.categories);
          setCategoryId(res.categories[0].id);
        }
      } catch {
        // Fallback default categories if API fails
        setCategories([
          { id: 1, name: "Vegetables" },
          { id: 2, name: "Fruits" },
          { id: 3, name: "Grains & Cereals" },
          { id: 4, name: "Pulses" },
          { id: 5, name: "Spices" },
          { id: 6, name: "Dairy" },
          { id: 17, name: "Herbs & Greens" },
        ]);
      }
    })();
  }, []);

  // Live price calculations
  const numBase = parseFloat(basePrice) || 0;
  const numGstAmount = Math.round(numBase * (gstPercent / 100) * 100) / 100;
  const numCommissionAmount = Math.round(numBase * (commissionPercent / 100) * 100) / 100;
  const numFinalPrice = Math.round((numBase + numGstAmount + numCommissionAmount) * 100) / 100;

  async function handleSubmit(e: React.FormEvent) {
```

```
    e.preventDefault();
    if (!name.trim()) return setError("Product name is required.");
    if (numBase <= 0) return setError("Enter a valid base price per kg.");
    if (parseFloat(stock) <= 0) return setError("Enter a valid stock quantity.");
    if (!file) return setError("Please select a product image.");

    setError("");
    setLoading(true);

    try {
      const data = new FormData();
      data.append("name", name.trim());
      data.append("category_id", String(categoryId));
      data.append("base_price", String(numBase));
      data.append("gst_percent", String(gstPercent));
      data.append("commission_percent", String(commissionPercent));
      data.append("price_per_kg", String(numFinalPrice));
      data.append("stock_kg", stock);
      data.append("description", description.trim());
      data.append("product_image", file);

      await createProduct(data);
      router.push("/farmer/products");
    } catch (err: any) {
      setError(err instanceof ApiError ? err.message : "Failed to add product. Please try again.");
    } finally {
      setLoading(false);
    }
  }

  return (
    <>
      <FarmerNav />
      <div className="container form-wrap">
        <div className="form-card">
          <div className="header-info">
            <h1>Add New Farm Product</h1>
            <p className="subtitle">
              Enter your base price — GST and website commission are automatically calculated and add
ed to the buyer price.
            </p>
          </div>

          <form onSubmit={handleSubmit}>
            {error && <div className="form-error">{error}</div>}

            <div className="row">
              <div className="field">
                <label>Product Name</label>
                <input
                  placeholder="e.g. Organic Tomatoes, Fresh Wheat"
                  value={name}
                  onChange={(e) => setName(e.target.value)}
                  required
                />
              </div>
              <div className="field">
                <label>Category</label>
                <select value={categoryId} onChange={(e) => setCategoryId(Number(e.target.value))}>
                  {categories.map((c) => (
                    <option key={c.id} value={c.id}>{c.name}</option>
                  ))}
```

```
                </select>
              </div>
            </div>

            {/* Pricing Section with GST and Commission */}
            <div className="pricing-box">
              <h3>Pricing, Commission &amp;amp; GST Details</h3>

              <div className="row">
                <div className="field">
                  <label>Farmer Base Price (₹ / kg) <span className="req">*</span></label>
                  <input
                    type="number"
                    step="0.01"
                    min="1"
                    placeholder="e.g. 100"
                    value={basePrice}
                    onChange={(e) => setBasePrice(e.target.value)}
                    required
                  />
                  <small className="field-hint">The net amount you will receive per kg.</small>
                </div>

                <div className="field">
                  <label>GST Rate</label>
                  <select value={gstPercent} onChange={(e) => setGstPercent(Number(e.target.value))}>
                    {GST_RATES.map((g) => (
                      <option key={g.rate} value={g.rate}>{g.label}</option>
                    ))}
                  </select>
                  <small className="field-hint">Goods and Services Tax on produce.</small>
                </div>
              </div>

              {/* Real-time Transparent Price Breakdown Card */}
              {numBase > 0 && (
                <div className="price-breakdown-card">
                  <div className="breakdown-title">Automated Price Breakdown:</div>
                  <div className="breakdown-row">
                    <span>👨‍🌾 Farmer Net Earnings (Base Price):</span>
                    <strong>₹{numBase.toFixed(2)} / kg</strong>
                  </div>
                  <div className="breakdown-row">
                    <span>🏢 Website Platform Commission ({commissionPercent}%):</span>
                    <span className="add-amount">+ ₹{numCommissionAmount.toFixed(2)} / kg</span>
                  </div>
                  <div className="breakdown-row">
                    <span>🏛️ Applicable GST ({gstPercent}%):</span>
                    <span className="add-amount">+ ₹{numGstAmount.toFixed(2)} / kg</span>
                  </div>
                  <div className="breakdown-divider"></div>
                  <div className="breakdown-row total">
                    <span>🛒 Final Selling Price on Website:</span>
                    <strong className="final-price">₹{numFinalPrice.toFixed(2)} / kg</strong>
                  </div>
                  <small className="breakdown-note">
                    Buyers in the online shop will purchase this item at ₹{numFinalPrice.toFixed(2)}/
kg.
                  </small>
                </div>
              )}
            </div>
```

```
            <div className="row">
              <div className="field">
                <label>Available Stock (kg) <span className="req">*</span></label>
                <input
                  type="number"
                  step="0.01"
                  min="1"
                  placeholder="e.g. 500"
                  value={stock}
                  onChange={(e) => setStock(e.target.value)}
                  required
                />
              </div>

              <div className="field">
                <label>Product Image <span className="req">*</span></label>
                <input
                  type="file"
                  accept="image/*"
                  onChange={(e) => setFile(e.target.files?.[0] ?? null)}
                  required
                />
              </div>
            </div>

            <div className="field">
              <label>Description</label>
              <textarea
                rows={3}
                placeholder="Describe harvest date, pesticide-free, organic status, quality detail
s..."
                value={description}
                onChange={(e) => setDescription(e.target.value)}
              />
            </div>

            <button className="btn-block" type="submit" disabled={loading}>
              {loading ? "Adding Product…" : "Submit Product for Approval"}
            </button>
          </form>
        </div>
      </div>

      <style jsx>{`
        .form-wrap { padding: 40px 24px 90px; }
        .form-card {
          max-width: 680px; margin: 0 auto; background: #fff;
          border: 1px solid rgba(26,26,21,0.08); border-radius: 16px; padding: 32px;
          box-shadow: 0 4px 20px rgba(0,0,0,0.03);
        }
        .header-info { margin-bottom: 24px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.5rem; margin: 0 0 6px;
}
        .subtitle { color: #6b6555; font-size: 0.88rem; margin: 0; line-height: 1.4; }
        .req { color: var(--rust); }
        .field-hint { font-size: 0.75rem; color: #888273; margin-top: 4px; display: block; }
        .row { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; }

        .pricing-box {
          background: #faf7ee; border: 1px solid rgba(232,163,61,0.25); border-radius: 12px;
          padding: 20px; margin: 18px 0;
```

```
        }
        .pricing-box h3 {
          margin: 0 0 14px; font-size: 0.95rem; font-family: var(--font-display);
          color: var(--soil); font-weight: 700;
        }

        .price-breakdown-card {
          background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 10px;
          padding: 16px 18px; margin-top: 14px;
        }
        .breakdown-title {
          font-size: 0.78rem; text-transform: uppercase; color: #888273; font-weight: 700;
          letter-spacing: 0.03em; margin-bottom: 10px;
        }
        .breakdown-row {
          display: flex; justify-content: space-between; align-items: center;
          font-size: 0.88rem; margin-bottom: 6px; color: var(--soil);
        }
        .add-amount { color: var(--rust); font-weight: 600; }
        .breakdown-divider {
          height: 1px; background: rgba(26,26,21,0.1); margin: 10px 0;
        }
        .breakdown-row.total {
          font-size: 1.05rem; font-weight: 700; color: var(--soil);
        }
        .final-price { color: var(--leaf); font-size: 1.25rem; font-family: var(--font-display); }
        .breakdown-note { display: block; font-size: 0.76rem; color: #777; margin-top: 6px; }

        @media (max-width: 600px) {
          .row { grid-template-columns: 1fr; }
        }
      `}</style>
    </>
  );
}
```

app/farmer/orders/page.tsx (Farmer Incoming Orders Management Page)

```
"use client";

import { useEffect, useState } from "react";
import Link from "next/link";
import FarmerNav from "@/app/components/FarmerNav";
import { listFarmerOrderItems, FarmerOrderItem, refillProduct } from "@/app/lib/api";

const STATUS_BADGES: Record<string, { label: string; color: string; bg: string }> = {
  pending: { label: "⏳ Pending", color: "#b45309", bg: "#fffbeb" },
  confirmed: { label: "📦 Confirmed & Packing", color: "#4338ca", bg: "#eef2ff" },
  packing: { label: "📦 Packing & Prep", color: "#4338ca", bg: "#eef2ff" },
  shipped: { label: "🚚 Shipped / In Transit", color: "#0284c7", bg: "#f0f9ff" },
  out_for_delivery: { label: "🛵 Out for Delivery", color: "#d97706", bg: "#fef3c7" },
  delivered: { label: "✅ Delivered", color: "#15803d", bg: "#f0fdf4" },
  cancelled: { label: "❌ Cancelled", color: "#b91c1c", bg: "#fef2f2" },
};

export default function FarmerOrdersPage() {
  const [items, setItems] = useState<FarmerOrderItem[]>([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");
  const [msg, setMsg] = useState("");

  // Quick Refill Modal
  const [refillModalOpen, setRefillModalOpen] = useState(false);
  const [refillTarget, setRefillTarget] = useState<{ id: number; name: string; stock: number } | null
>(null);
  const [refillKg, setRefillKg] = useState<number | string>(50);
  const [refillLoading, setRefillLoading] = useState(false);

  async function load() {
    setLoading(true);
    setError("");
    try {
      const res = await listFarmerOrderItems();
      setItems(res.items || []);
    } catch {
      setError("Could not load received orders.");
    } finally {
      setLoading(false);
    }
  }

  useEffect(() => {
    load();
  }, []);

  function openRefill(prodId: number, prodName: string, stock: number) {
    setRefillTarget({ id: prodId, name: prodName, stock });
    setRefillKg(50);
    setRefillModalOpen(true);
  }

  async function handleRefillSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (!refillTarget) return;
    const amount = Number(refillKg);
    if (isNaN(amount) || amount <= 0) return;
```

```
    setRefillLoading(true);
    try {
      const res = await refillProduct(refillTarget.id, amount, "add");
      setItems((prev) =>
        prev.map((i) => (i.product_id === refillTarget.id ? { ...i, current_stock: res.new_stock } :
i))
      );
      setMsg(`✅ Refilled ${refillTarget.name}! New stock is ${res.new_stock} kg.`);
      setTimeout(() => setMsg(""), 4000);
      setRefillModalOpen(false);
      setRefillTarget(null);
    } catch (err: any) {
      alert(err?.message || "Failed to refill stock.");
    } finally {
      setRefillLoading(false);
    }
  }

  return (
    <>
      <FarmerNav />
      <div className="container orders-wrap">
        <div className="head">
          <div>
            <h1>Orders Received &amp;amp; Stock Tracking</h1>
            <p className="subhead">
              Track customer orders and live stock. Order logistics (packing, travel, delivery) are m
anaged directly by Admin.
            </p>
          </div>
          <div className="head-actions">
            <button type="button" onClick={load} className="btn-refresh">
              🔄 Refresh Orders
            </button>
            <Link href="/farmer/products" className="btn-manage-products">
              📦 My Products &amp;amp; Stock
            </Link>
          </div>
        </div>

        {error && <div className="form-error">{error}</div>}
        {msg && <div className="alert-success">{msg}</div>}

        <div className="table-card">
          <table>
            <thead>
              <tr>
                <th>Date &amp;amp; Order ID</th>
                <th>Produce Item</th>
                <th>Ordered Qty</th>
                <th>Remaining Farm Stock</th>
                <th>Buyer Contact</th>
                <th>Earnings Subtotal</th>
                <th>Delivery Stage (Admin Controlled)</th>
              </tr>
            </thead>
            <tbody>
              {loading && (
                <tr><td colSpan={7} className="empty">Loading received orders…</td></tr>
              )}
              {!loading && items.length === 0 && (
                <tr><td colSpan={7} className="empty">No customer orders received yet.</td></tr>
```

```
              )}
              {items.map((item) => {
                const isOutOfStock = item.current_stock <= 0;
                const isLowStock = item.current_stock > 0 && item.current_stock <= 15;
                const badge = STATUS_BADGES[item.status] || STATUS_BADGES.pending;

                return (
                  <tr key={item.item_id}>
                    <td>
                      <div className="date-box">
                        <strong className="order-id">#{item.order_id}</strong>
                        <span>{new Date(item.order_date).toLocaleDateString("en-IN", { month: "shor
t", day: "numeric", year: "numeric" })}</span>
                      </div>
                    </td>
                    <td>
                      <div className="prod-cell">
                        {item.product_image ? (
                          // eslint-disable-next-line @next/next/no-img-element
                          <img src={item.product_image} alt={item.product_name} className="prod-thum
b" />
                        ) : (
                          <div className="prod-thumb placeholder">🌾</div>
                        )}
                        <div>
                          <strong>{item.product_name}</strong>
                          <div className="rate-hint">₹{item.price_per_kg.toFixed(2)} / kg</div>
                        </div>
                      </div>
                    </td>
                    <td>
                      <div className="ordered-qty-box">
                        <strong className="qty-number">📦 {item.qty} kg</strong>
                        <span className="qty-sub">deducted from stock</span>
                      </div>
                    </td>
                    <td>
                      <div className="stock-col">
                        {isOutOfStock ? (
                          <span className="badge-stock out">🔴 0 kg Left (Out of Stock)</span>
                        ) : isLowStock ? (
                          <span className="badge-stock low">⚠️ {item.current_stock} kg Left (Low)</s
pan>
                        ) : (
                          <span className="badge-stock good">🟢 {item.current_stock} kg Remaining</s
pan>
                        )}
                        <button
                          type="button"
                          className="btn-quick-refill"
                          onClick={() => openRefill(item.product_id, item.product_name, item.current_
stock)}
                        >
                          ➕ Refill
                        </button>
                      </div>
                    </td>
                    <td>
                      <div className="buyer-info">
                        <strong>👤 {item.buyer_name}</strong>
                        <span className="muted">📞 {item.buyer_mobile}</span>
                        <span className="address-text">📍 {item.address}</span>
```

```
                    </div>
                  </td>
                  <td>
                    <strong className="subtotal-val">₹{item.subtotal.toFixed(2)}</strong>
                  </td>
                  <td>
                    <span className="status-badge" style={{ color: badge.color, background: badge.b
g }}>
                      {badge.label}
                    </span>
                  </td>
                </tr>
              );
            })}
          </tbody>
        </table>
      </div>
    </div>

    {/* Quick Refill Modal */}
    {refillModalOpen && refillTarget && (
      <div className="modal-backdrop" onClick={() => setRefillModalOpen(false)}>
        <div className="modal-card" onClick={(e) => e.stopPropagation()}>
          <div className="modal-header">
            <h3>🌱 Quick Refill Stock: {refillTarget.name}</h3>
            <button type="button" className="btn-close" onClick={() => setRefillModalOpen(false)}>
✕</button>
          </div>
          <form onSubmit={handleRefillSubmit}>
            <div className="modal-body">
              <p style={{ fontSize: "0.88rem", color: "#4b5563", marginBottom: "14px" }}>
                Current remaining stock: <strong>{refillTarget.stock} kg</strong>
              </p>
              <label style={{ display: "block", fontSize: "0.82rem", fontWeight: "600", marginBotto
m: "6px" }}>
                Quantity to Add (kg):
              </label>
              <div style={{ display: "flex", gap: "8px", alignItems: "center" }}>
                <input
                  type="number"
                  step="0.5"
                  min="0.1"
                  required
                  value={refillKg}
                  onChange={(e) => setRefillKg(e.target.value)}
                  style={{ flex: 1, padding: "8px 12px", border: "1.5px solid #d1d5db", borderRadiu
s: "8px", fontSize: "1rem" }}
                />
                <span style={{ fontWeight: "600", color: "#6b7280" }}>kg</span>
              </div>
              <div style={{ display: "flex", gap: "6px", marginTop: "10px" }}>
                {[10, 25, 50, 100, 200].map((v) => (
                  <button
                    key={v}
                    type="button"
                    onClick={() => setRefillKg(v)}
                    style={{ padding: "4px 8px", border: "1px solid #d1d5db", borderRadius: "6px",
background: "#fff", cursor: "pointer", fontSize: "0.78rem" }}
                  >
                    +{v}kg
                  </button>
                ))}
```

```
              </div>
            </div>
            <div className="modal-footer">
              <button type="button" onClick={() => setRefillModalOpen(false)} className="btn-cance
l">Cancel</button>
              <button type="submit" disabled={refillLoading} className="btn-confirm">
                {refillLoading ? "Refilling…" : "Confirm Refill"}
              </button>
            </div>
          </form>
        </div>
      </div>
    )}

    <style jsx>{`
      .orders-wrap { padding: 40px 24px 90px; }
      .head { display: flex; justify-content: space-between; align-items: flex-start; margin-botto
m: 24px; gap: 16px; flex-wrap: wrap; }
      h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 4px;
color: #1b3821; }
      .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }

      .head-actions { display: flex; gap: 10px; align-items: center; }
      .btn-refresh { background: #fff; border: 1px solid #d1d5db; color: #374151; padding: 8px 16p
x; border-radius: 999px; font-size: 0.82rem; font-weight: 600; cursor: pointer; }
      .btn-refresh:hover { background: #f3f4f6; }
      .btn-manage-products { background: #f0fdf4; border: 1px solid #16a34a; color: #15803d; paddin
g: 8px 16px; border-radius: 999px; font-size: 0.82rem; font-weight: 600; text-decoration: none; }
      .btn-manage-products:hover { background: #16a34a; color: #fff; }

      .form-error { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; padding: 12px 1
6px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; }
      .alert-success { background: #f0fdf4; border: 1px solid #bbf7d0; color: #166534; padding: 12p
x 16px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; font-weight: 600; }

      .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow-x: auto; box-shadow: 0 2px 8px rgba(0,0,0,0.04); }
      table { width: 100%; border-collapse: collapse; font-size: 0.86rem; min-width: 900px; }
      th { text-align: left; padding: 14px 16px; background: #faf7ee; font-size: 0.76rem; text-tran
sform: uppercase; letter-spacing: 0.04em; color: #6b6555; white-space: nowrap; }
      td { padding: 14px 16px; border-top: 1px solid rgba(26,26,21,0.06); vertical-align: middle; }

      .date-box { display: flex; flex-direction: column; gap: 2px; }
      .order-id { font-family: monospace; font-size: 0.9rem; color: #1b3821; }
      .date-box span { font-size: 0.76rem; color: #888273; }

      .prod-cell { display: flex; align-items: center; gap: 10px; }
      .prod-thumb { width: 42px; height: 42px; border-radius: 8px; object-fit: cover; }
      .prod-thumb.placeholder { background: #ede4cd; display: flex; align-items: center; justify-co
ntent: center; font-size: 1.2rem; }
      .rate-hint { font-size: 0.74rem; color: #888273; }

      .ordered-qty-box { display: flex; flex-direction: column; gap: 2px; }
      .qty-number { font-size: 0.95rem; color: #1b3821; }
      .qty-sub { font-size: 0.72rem; color: #888273; }

      .stock-col { display: flex; flex-direction: column; gap: 4px; }
      .badge-stock { font-size: 0.74rem; padding: 3px 8px; border-radius: 6px; font-weight: 600; wh
ite-space: nowrap; width: fit-content; }
      .badge-stock.good { background: #f0fdf4; color: #166534; border: 1px solid #bbf7d0; }
      .badge-stock.low { background: #fffbeb; color: #92400e; border: 1px solid #fde68a; }
      .badge-stock.out { background: #fef2f2; color: #991b1b; border: 1px solid #fecaca; }
```

```
        .btn-quick-refill { background: #f0fdf4; border: 1px solid #16a34a; color: #15803d; padding:
2px 7px; border-radius: 4px; font-size: 0.72rem; font-weight: 700; cursor: pointer; width: fit-conten
t; }
        .btn-quick-refill:hover { background: #16a34a; color: #fff; }

        .buyer-info { display: flex; flex-direction: column; gap: 2px; font-size: 0.82rem; }
        .muted { color: #6b6555; font-size: 0.76rem; }
        .address-text { color: #888273; font-size: 0.74rem; max-width: 180px; white-space: nowrap; ov
erflow: hidden; text-overflow: ellipsis; }

        .subtotal-val { color: #16a34a; font-size: 0.95rem; }

        .status-badge { font-size: 0.78rem; font-weight: 700; padding: 5px 12px; border-radius: 999p
x; display: inline-block; white-space: nowrap; }
        .empty { text-align: center; color: #6b6555; padding: 40px; }

        /* Modal */
        .modal-backdrop { position: fixed; inset: 0; background: rgba(0,0,0,0.4); backdrop-filter: bl
ur(2px); z-index: 9999; display: flex; align-items: center; justify-content: center; padding: 16px; }
        .modal-card { background: #fff; border-radius: 14px; max-width: 420px; width: 100%; box-shado
w: 0 10px 30px rgba(0,0,0,0.15); overflow: hidden; }
        .modal-header { display: flex; justify-content: space-between; align-items: center; padding:
16px 20px; background: #1b3821; color: #fff; }
        .modal-header h3 { margin: 0; font-size: 1.05rem; font-weight: 700; color: #fff; }
        .btn-close { background: none; border: none; color: #fff; font-size: 1.2rem; cursor: pointer;
}
        .modal-body { padding: 18px 20px; }
        .modal-footer { display: flex; justify-content: flex-end; gap: 8px; padding: 14px 20px; backg
round: #fafafa; border-top: 1px solid #e5e7eb; }
        .btn-cancel { padding: 7px 14px; border: 1px solid #d1d5db; background: #fff; border-radius:
999px; font-size: 0.8rem; cursor: pointer; }
        .btn-confirm { padding: 7px 16px; border: none; background: #16a34a; color: #fff; border-radi
us: 999px; font-size: 0.8rem; font-weight: 700; cursor: pointer; }
        .btn-confirm:hover { background: #15803d; }
      `}</style>
    </>
  );
}
```

app/admin/farmers/page.tsx (Admin Farmer Account Management Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminListFarmers, adminFarmerAction, ApiError, AdminFarmer, FarmerStatus } from "@/ap
p/lib/api";

const TABS: { key: FarmerStatus; label: string }[] = [
  { key: "pending", label: "Pending Approval" },
  { key: "approved", label: "Approved" },
  { key: "rejected", label: "Rejected" },
  { key: "all", label: "All Farmers" },
];

export default function AdminFarmersPage() {
  const router = useRouter();
  const [tab, setTab] = useState<FarmerStatus>("pending");
  const [farmers, setFarmers] = useState<AdminFarmer[]>([]);
  const [error, setError] = useState("");
  const [checking, setChecking] = useState(true);
  const [busyId, setBusyId] = useState<number | null>(null);
  const [remarkFor, setRemarkFor] = useState<number | null>(null);
  const [remark, setRemark] = useState("");
  const [selectedFarmer, setSelectedFarmer] = useState<AdminFarmer | null>(null);

  useEffect(() => {
    (async () => {
      const me = await getMe().catch(() => ({ user: null }));
      if (!me.user || me.user.role !== "admin") {
        router.replace("/auth/admin-login");
        return;
      }
      setChecking(false);
    })();
  }, [router]);

  useEffect(() => {
    if (checking) return;
    load();
  }, [tab, checking]);

  async function load() {
    setError("");
    try {
      const res = await adminListFarmers(tab);
      setFarmers(res.farmers);
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load farmers.");
    }
  }

  async function act(id: number, action: "approve" | "reject", withRemark?: string) {
    setBusyId(id);
    setError("");
    try {
      await adminFarmerAction(id, action, withRemark);
      setRemarkFor(null);
      setRemark("");
```

```
      if (selectedFarmer && selectedFarmer.id === id) {
        setSelectedFarmer({
          ...selectedFarmer,
          status: action === "approve" ? "approved" : "rejected",
          remark: withRemark || selectedFarmer.remark,
        });
      }
      await load();
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Action failed.");
    } finally {
      setBusyId(null);
    }
  }

  if (checking) return null;

  return (
    <>
      <AdminNav />
      <div className="container wrap">
        <div className="page-header">
          <div>
            <h1>Farmer Management & Verification</h1>
            <p className="subhead">Review farmer registrations, Aadhaar details, and approve or rejec
t accounts.</p>
          </div>
        </div>

        {error && <div className="form-error">{error}</div>}

        <div className="tabs">
          {TABS.map((t) => (
            <button key={t.key} className={tab === t.key ? "active" : ""} onClick={() => setTab(t.ke
y)}>
              {t.label}
            </button>
          ))}
        </div>

        <div className="table-card">
          <table>
            <thead>
              <tr>
                <th>Farmer</th>
                <th>Contact</th>
                <th>Aadhaar</th>
                <th>Products</th>
                <th>Total Sales</th>
                <th>Status</th>
                <th>Joined</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {farmers.length === 0 && (
                <tr>
                  <td colSpan={8} className="empty">No farmers found in this tab.</td>
                </tr>
              )}
              {farmers.map((f) => (
                <tr key={f.id}>
```

```
                  <td>
                    <div className="farmer-name-cell">
                      <strong>{f.name}</strong>
                      <span className="farmer-id">ID: #{f.id}</span>
                    </div>
                  </td>
                  <td>
                    <div>  {f.mobile}</div>
                    {f.email && <div className="sub">  {f.email}</div>}
                  </td>
                  <td>
                    {f.aadhaar ? <span className="aadhaar-badge">  {f.aadhaar}</span> : <span class
Name="sub">—</span>}
                  </td>
                  <td>
                    <span className="count-badge">{f.total_products ?? 0} items</span>
                  </td>
                  <td>
                    <strong>₹{(f.total_sales ?? 0).toLocaleString("en-IN", { minimumFractionDigits: 2
})}</strong>
                  </td>
                  <td>
                    <span className={`badge ${f.status}`}>{f.status}</span>
                  </td>
                  <td>{new Date(f.joined).toLocaleDateString("en-IN")}</td>
                  <td className="row-actions">
                    <button className="btn-detail" onClick={() => setSelectedFarmer(f)}>
                      Details
                    </button>

                    {/* Pending Approval: Approve or Reject */}
                    {f.status === "pending" && (
                      <>
                        <button className="ok" disabled={busyId === f.id} onClick={() => act(f.id, "a
pprove")}>

                          Approve
                        </button>
                        {remarkFor === f.id ? (
                          <span className="remark-inline">
                            <input
                              placeholder="Reason (optional)"
                              value={remark}
                              onChange={(e) => setRemark(e.target.value)}
                            />
                            <button className="danger" disabled={busyId === f.id} onClick={() => act
(f.id, "reject", remark)}>

                              Confirm
                            </button>
                          </span>
                        ) : (
                          <button className="danger" disabled={busyId === f.id} onClick={() => setRem
arkFor(f.id)}>

                            Reject
                          </button>
                        )}
                      </>
                    )}

                    {/* Already Approved: Option to Reject/Suspend */}
                    {f.status === "approved" && (
                      <>
                        {remarkFor === f.id ? (
```

```
                        <span className="remark-inline">
                          <input
                            placeholder="Reason (optional)"
                            value={remark}
                            onChange={(e) => setRemark(e.target.value)}
                          />
                          <button className="danger" disabled={busyId === f.id} onClick={() => act
(f.id, "reject", remark)}>
                            Confirm Suspend
                          </button>
                        </span>
                      ) : (
                        <button className="danger-outline" disabled={busyId === f.id} onClick={() =
> setRemarkFor(f.id)}>
                          Reject
                        </button>
                      )}
                    </>
                  )}

                  {/* Already Rejected: Option to Re-approve */}
                  {f.status === "rejected" && (
                    <button className="ok-outline" disabled={busyId === f.id} onClick={() => act(f.
id, "approve")}>
                      Re-Approve
                    </button>
                  )}
                </td>
              </tr>
            ))}
          </tbody>
        </table>
      </div>

      {/* Farmer Details Modal */}
      {selectedFarmer && (
        <div className="modal-backdrop" onClick={() => setSelectedFarmer(null)}>
          <div className="modal-content" onClick={(e) => e.stopPropagation()}>
            <div className="modal-header">
              <h2>Farmer Details — {selectedFarmer.name}</h2>
              <button className="close-btn" onClick={() => setSelectedFarmer(null)}>✕</button>
            </div>

            <div className="modal-body">
              <div className="detail-grid">
                <div className="detail-item">
                  <label>Full Name</label>
                  <div>{selectedFarmer.name}</div>
                </div>
                <div className="detail-item">
                  <label>Mobile Number</label>
                  <div><a href={`tel:${selectedFarmer.mobile}`}>📞 {selectedFarmer.mobile}</a></di
v>
                </div>
                <div className="detail-item">
                  <label>Email Address</label>
                  <div>{selectedFarmer.email ? <a href={`mailto:${selectedFarmer.email}`}>✉ {sele
ctedFarmer.email}</a> : "—"}</div>
                </div>
                <div className="detail-item">
                  <label>Aadhaar Card</label>
                  <div>{selectedFarmer.aadhaar || "Not provided"}</div>
```

```
                </div>
                <div className="detail-item">
                  <label>Current Status</label>
                  <div><span className={`badge ${selectedFarmer.status}`}>{selectedFarmer.status}</
span></div>
                </div>
                <div className="detail-item">
                  <label>Registration Date</label>
                  <div>{new Date(selectedFarmer.joined).toLocaleString("en-IN")}</div>
                </div>
                <div className="detail-item">
                  <label>Listed Products</label>
                  <div>{selectedFarmer.total_products ?? 0} items</div>
                </div>
                <div className="detail-item">
                  <label>Total Sales Volume</label>
                  <div>₹{(selectedFarmer.total_sales ?? 0).toLocaleString("en-IN", { minimumFractio
nDigits: 2 })}</div>
                </div>
                {selectedFarmer.remark && (
                  <div className="detail-item full-width">
                    <label>Admin Note / Remark</label>
                    <div className="remark-box">{selectedFarmer.remark}</div>
                  </div>
                )}
              </div>

              <div className="modal-actions">
                {selectedFarmer.status !== "approved" && (
                  <button className="ok-large" disabled={busyId === selectedFarmer.id} onClick={()
=> act(selectedFarmer.id, "approve")}>
                    ✓ Approve Farmer
                  </button>
                )}
                {selectedFarmer.status !== "rejected" && (
                  <button className="danger-large" disabled={busyId === selectedFarmer.id} onClick=
{() => act(selectedFarmer.id, "reject", remark)}>
                    ✕ Reject Farmer
                  </button>
                )}
              </div>
            </div>
          </div>
        </div>
      )}
    </div>

    <style jsx>{`
      .wrap { padding: 40px 24px 90px; }
      .page-header { margin-bottom: 22px; }
      h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 4px;
}
      .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }
      .tabs { display: flex; gap: 8px; margin-bottom: 18px; flex-wrap: wrap; }
      .tabs button { border: 1px solid rgba(26,26,21,0.12); background: #fff; padding: 7px 16px; bo
rder-radius: 999px; font-size: 0.82rem; cursor: pointer; }
      .tabs button.active { background: var(--soil); color: #efe8d8; border-color: var(--soil); }
      .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow: hidden; overflow-x: auto; }
      table { width: 100%; border-collapse: collapse; font-size: 0.86rem; }
      th { text-align: left; padding: 12px 16px; background: #faf7ee; font-size: 0.74rem; text-tran
sform: uppercase; color: #6b6555; white-space: nowrap; }
```

```
        td { padding: 12px 16px; border-top: 1px solid rgba(26,26,21,0.06); vertical-align: middle; w
hite-space: nowrap; }
        .farmer-name-cell { display: flex; flex-direction: column; }
        .farmer-id { font-size: 0.74rem; color: #888273; }
        .aadhaar-badge { font-family: var(--font-mono); font-size: 0.8rem; background: #f4f1ea; paddi
ng: 3px 8px; border-radius: 6px; }
        .count-badge { background: #f2f7ef; color: var(--leaf); padding: 3px 8px; border-radius: 6px;
font-weight: 600; font-size: 0.8rem; }
        .sub { font-size: 0.78rem; color: #6b6555; }
        .badge { font-size: 0.72rem; padding: 3px 10px; border-radius: 999px; text-transform: capital
ize; font-weight: 600; }
        .badge.pending { background: rgba(232,163,61,0.18); color: var(--rust); }
        .badge.approved { background: rgba(76,122,61,0.15); color: var(--leaf); }
        .badge.rejected { background: rgba(200,60,40,0.12); color: #b8391f; }
        .row-actions { display: flex; gap: 8px; align-items: center; white-space: normal; }
        .row-actions button { border: none; padding: 6px 12px; border-radius: 999px; font-size: 0.78r
em; font-weight: 600; cursor: pointer; }
        .btn-detail { background: #f4f1ea; color: var(--soil); border: 1px solid rgba(26,26,21,0.1) !
important; }
        .btn-detail:hover { background: #e8e2d4; }
        .row-actions button.ok { background: var(--leaf); color: #fff; }
        .row-actions button.ok-outline { background: transparent; border: 1px solid var(--leaf) !impo
rtant; color: var(--leaf); }
        .row-actions button.danger { background: #f4dcd6; color: #b8391f; }
        .row-actions button.danger-outline { background: transparent; border: 1px solid #b8391f !impo
rtant; color: #b8391f; }
        .row-actions button:disabled { opacity: 0.6; cursor: default; }
        .remark-inline { display: flex; gap: 6px; align-items: center; }
        .remark-inline input { border: 1px solid rgba(26,26,21,0.15); border-radius: 8px; padding: 5p
x 8px; font-size: 0.78rem; width: 140px; }
        .empty { text-align: center; color: #6b6555; padding: 40px; }

        /* Modal */
        .modal-backdrop {
          position: fixed; inset: 0; background: rgba(0,0,0,0.5);
          display: flex; align-items: center; justify-content: center;
          z-index: 1000; padding: 20px;
        }
        .modal-content {
          background: #fff; border-radius: 16px; width: 100%; max-width: 580px;
          box-shadow: 0 20px 40px rgba(0,0,0,0.2); overflow: hidden;
        }
        .modal-header {
          display: flex; justify-content: space-between; align-items: center;
          padding: 18px 24px; border-bottom: 1px solid rgba(26,26,21,0.08); background: #faf7ee;
        }
        .modal-header h2 { font-family: var(--font-display); font-size: 1.2rem; margin: 0; }
        .close-btn { background: none; border: none; font-size: 1.2rem; cursor: pointer; color: #6b65
55; }
        .modal-body { padding: 24px; }
        .detail-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; margin-bottom: 24px;
}
        .detail-item { display: flex; flex-direction: column; gap: 4px; }
        .detail-item.full-width { grid-column: span 2; }
        .detail-item label { font-size: 0.75rem; text-transform: uppercase; color: #888273; font-weig
ht: 600; }
        .detail-item div { font-size: 0.95rem; color: var(--soil); }
        .detail-item a { color: var(--leaf); text-decoration: none; font-weight: 600; }
        .remark-box { background: #faf7ee; padding: 10px 14px; border-radius: 8px; font-style: itali
c; border-left: 3px solid var(--rust); }
        .modal-actions { display: flex; gap: 12px; justify-content: flex-end; }
        .ok-large { background: var(--leaf); color: #fff; border: none; padding: 10px 20px; border-ra
```

```
dius: 8px; font-weight: 600; cursor: pointer; }
        .danger-large { background: #f4dcd6; color: #b8391f; border: none; padding: 10px 20px; border
-radius: 8px; font-weight: 600; cursor: pointer; }
      `}</style>
    </>
  );
}
```

app/admin/orders/page.tsx (Admin All Orders Management Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminListOrders, adminUpdateOrderStatus, ApiError, AdminOrder } from "@/app/lib/api";

function round(val: number, decimals: number) {
  return Number(Math.round(Number(val + "e" + decimals)) + "e-" + decimals);
}

const STATUS_STAGES = [
  { value: "pending", label: "⏳ Pending (Order Placed)", color: "#b45309", bg: "#fffbeb" },
  { value: "packing", label: "📦 Packing & Prep", color: "#4338ca", bg: "#eef2ff" },
  { value: "shipped", label: "🚚 Shipped / Travel (In Transit)", color: "#0284c7", bg: "#f0f9ff" },
  { value: "out_for_delivery", label: "🛵 Out for Delivery", color: "#d97706", bg: "#fef3c7" },
  { value: "delivered", label: "✅ Delivered", color: "#15803d", bg: "#f0fdf4" },
  { value: "cancelled", label: "❌ Cancelled", color: "#b91c1c", bg: "#fef2f2" },
];

export default function AdminOrdersPage() {
  const router = useRouter();
  const [orders, setOrders] = useState<AdminOrder[]>([]);
  const [totalRevenue, setTotalRevenue] = useState(0);
  const [error, setError] = useState("");
  const [successMsg, setSuccessMsg] = useState("");
  const [checking, setChecking] = useState(true);
  const [selectedOrder, setSelectedOrder] = useState<AdminOrder | null>(null);
  const [modalTrackingNote, setModalTrackingNote] = useState("");
  const [modalStatus, setModalStatus] = useState("pending");
  const [updatingId, setUpdatingId] = useState<number | null>(null);

  useEffect(() => {
    (async () => {
      const me = await getMe().catch(() => ({ user: null }));
      if (!me.user || me.user.role !== "admin") {
        router.replace("/auth/admin-login");
        return;
      }
      setChecking(false);
      load();
    })();
  }, [router]);

  async function load() {
    setError("");
    try {
      const res = await adminListOrders();
      setOrders(res.orders);
      setTotalRevenue(res.total_revenue);
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load orders.");
    }
  }

  async function handleStatusChange(orderId: number, newStatus: string, note?: string) {
    setUpdatingId(orderId);
    setError("");
    try {
```

```
      const res = await adminUpdateOrderStatus(orderId, newStatus, note);
      setOrders((prev) =>
        prev.map((o) => (o.id === orderId ? { ...o, status: newStatus, tracking_note: note ?? o.track
ing_note } : o))
      );
      if (selectedOrder && selectedOrder.id === orderId) {
        setSelectedOrder((prev) => prev ? { ...prev, status: newStatus, tracking_note: note ?? prev.t
racking_note } : null);
      }
      setSuccessMsg(`✅ ${res.message || `Order #${orderId} updated to ${newStatus}`}`);
      setTimeout(() => setSuccessMsg(""), 4000);
    } catch (err: any) {
      setError(err?.message || "Failed to update order status.");
    } finally {
      setUpdatingId(null);
    }
  }

  function openOrderModal(o: AdminOrder) {
    setSelectedOrder(o);
    setModalStatus(o.status);
    setModalTrackingNote(o.tracking_note || "");
  }

  if (checking) return null;

  const commission = round(totalRevenue * 0.05, 2);

  return (
    <>
      <AdminNav />
      <div className="container wrap">
        <div className="page-header">
          <div>
            <h1>Admin Logistics &amp;amp; Order Status Control</h1>
            <p className="subhead">
              Master control panel for shipping, packing, transit/travel, and delivery status update
s.
            </p>
          </div>
        </div>

        {error && <div className="form-error">{error}</div>}
        {successMsg && <div className="alert-success">{successMsg}</div>}

        {/* Income Metrics Banner */}
        <div className="income-banner">
          <div className="metric">
            <span className="label">Total Platform Revenue</span>
            <strong className="val green">₹{totalRevenue.toLocaleString("en-IN", { minimumFractionDig
its: 2 })}</strong>
            <span className="hint">Gross transaction volume</span>
          </div>
          <div className="metric">
            <span className="label">Platform Commission (5%)</span>
            <strong className="val gold">₹{commission.toLocaleString("en-IN", { minimumFractionDigit
s: 2 })}</strong>
            <span className="hint">Net fee earnings</span>
          </div>
          <div className="metric">
            <span className="label">Total Orders Placed</span>
            <strong className="val">{orders.length}</strong>
```

```
            <span className="hint">Buyer purchase count</span>
          </div>
        </div>

        <div className="table-card">
          <table>
            <thead>
              <tr>
                <th>Order ID</th>
                <th>Buyer</th>
                <th>Contact</th>
                <th>Ordered Items &amp;amp; Farm Stock</th>
                <th>Amount</th>
                <th>Payment Mode</th>
                <th>Logistics Stage (Admin Only)</th>
                <th>Delivery Address</th>
                <th>Date</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {orders.length === 0 && (
                <tr>
                  <td colSpan={10} className="empty">No customer orders placed yet.</td>
                </tr>
              )}
              {orders.map((o) => (
                <tr key={o.id}>
                  <td>
                    <span className="order-id-badge">#{o.id}</span>
                  </td>
                  <td><strong>{o.buyer_name}</strong></td>
                  <td>
                    <div>  {o.buyer_mobile}</div>
                    {o.buyer_email && <div className="sub">  {o.buyer_email}</div>}
                  </td>
                  <td>
                    <div className="items-summary-col">
                      <span className="items-badge">{o.item_count} produce item(s)</span>
                      {o.items && o.items.length > 0 && (
                        <div className="item-chips">
                          {o.items.map((it) => (
                            <span key={it.id} className="item-chip">
                              • <strong>{it.product_name}</strong>: {it.qty} kg (Farm Stock Left: {i
t.current_stock} kg)
                            </span>
                          ))}
                        </div>
                      )}
                    </div>
                  </td>
                  <td>
                    <strong className="order-amount">₹{o.total_amount.toLocaleString("en-IN", { minim
umFractionDigits: 2 })}</strong>
                  </td>
                  <td>
                    <span className="pay-mode">{o.payment_mode.toUpperCase()}</span>
                  </td>
                  <td>
                    <div className="status-cell">
                      <select
                        value={o.status}
```

```
                    disabled={updatingId === o.id}
                    onChange={(e) => handleStatusChange(o.id, e.target.value)}
                    className={`status-select ${o.status}`}
                  >
                    {STATUS_STAGES.map((st) => (
                      <option key={st.value} value={st.value}>
                        {st.label}
                      </option>
                    ))}
                  </select>
                  {o.tracking_note && (
                    <span className="tracking-hint" title={o.tracking_note}>
                      📝 {o.tracking_note}
                    </span>
                  )}
                </div>
              </td>
              <td className="address-cell" title={o.address}>
                {o.address}
              </td>
              <td>{new Date(o.created_at).toLocaleDateString("en-IN")}</td>
              <td>
                <button type="button" className="btn-details" onClick={() => openOrderModal(o)}>
                  View &amp;amp; Track
                </button>
              </td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  </div>

  {/* Order Items & Logistics Modal */}
  {selectedOrder && (
    <div className="modal-backdrop" onClick={() => setSelectedOrder(null)}>
      <div className="modal-card" onClick={(e) => e.stopPropagation()}>
        <div className="modal-header">
          <h3>📦 Order #{selectedOrder.id} Details &amp;amp; Logistics Control</h3>
          <button type="button" className="btn-close" onClick={() => setSelectedOrder(null)}>✕</
button>
        </div>
        <div className="modal-body">
          <div className="buyer-summary">
            <div><strong>Buyer:</strong> {selectedOrder.buyer_name} ({selectedOrder.buyer_mobil
e})</div>
            <div><strong>Delivery Address:</strong> {selectedOrder.address}</div>
            <div><strong>Total Amount:</strong> ₹{selectedOrder.total_amount.toFixed(2)} ({select
edOrder.payment_mode.toUpperCase()})</div>
            <div><strong>Order Date:</strong> {new Date(selectedOrder.created_at).toLocaleString
("en-IN")}</div>
          </div>

          {/* Status Update Form in Modal */}
          <div className="status-update-box">
            <label className="section-label">Update Logistics Stage:</label>
            <div className="modal-select-grid">
              <select
                value={modalStatus}
                onChange={(e) => setModalStatus(e.target.value)}
                className="modal-status-select"
              >
```

```
                    {STATUS_STAGES.map((st) => (
                      <option key={st.value} value={st.value}>
                        {st.label}
                      </option>
                    ))}
                  </select>
                </div>

                <label className="section-label" style={{ marginTop: "10px" }}>
                  Tracking Note / Logistics Note (Visible to Buyer):
                </label>
                <input
                  type="text"
                  value={modalTrackingNote}
                  onChange={(e) => setModalTrackingNote(e.target.value)}
                  placeholder="e.g. Packed at warehouse, dispatched via vehicle TN-38-1234"
                  className="modal-note-input"
                />

                <button
                  type="button"
                  onClick={() => handleStatusChange(selectedOrder.id, modalStatus, modalTrackingNot
e)}
                  className="btn-modal-update"
                  disabled={updatingId === selectedOrder.id}
                >
                  {updatingId === selectedOrder.id ? "Updating…" : "💾 Save Logistics Status & Not
e"}
                </button>
              </div>

              <h4 style={{ margin: "16px 0 8px", fontSize: "0.95rem", color: "#1b3821" }}>
                Ordered Produce Items &amp;amp; Farm Inventory:
              </h4>
              <div className="item-table-wrap">
                <table className="modal-table">
                  <thead>
                    <tr>
                      <th>Product</th>
                      <th>Farmer</th>
                      <th>Ordered Qty</th>
                      <th>Remaining Farm Stock</th>
                      <th>Rate</th>
                      <th>Subtotal</th>
                    </tr>
                  </thead>
                  <tbody>
                    {selectedOrder.items?.map((it) => (
                      <tr key={it.id}>
                        <td><strong>{it.product_name}</strong></td>
                        <td>{it.farmer_name}<br /><span style={{ fontSize: "0.74rem", color: "#666"
}}>{it.farmer_mobile}</span></td>
                        <td><span style={{ color: "#16a34a", fontWeight: "700" }}>📦 {it.qty} kg</sp
an></td>
                        <td>
                          {it.current_stock <= 0 ? (
                            <span style={{ color: "#dc2626", fontWeight: "700" }}>🔴 0 kg (Out of St
ock)</span>
                          ) : it.current_stock <= 15 ? (
                            <span style={{ color: "#b45309", fontWeight: "700" }}>⚠️ {it.current_sto
ck} kg (Low)</span>
                          ) : (
```

```
                          <span style={{ color: "#166534", fontWeight: "700" }}>🟢 {it.current_sto
ck} kg left</span>
                        )}
                      </td>
                      <td>₹{it.price_per_kg.toFixed(2)}/kg</td>
                      <td><strong>₹{it.subtotal.toFixed(2)}</strong></td>
                    </tr>
                  ))}
                </tbody>
              </table>
            </div>
          </div>
          <div className="modal-footer">
            <button type="button" className="btn-close-modal" onClick={() => setSelectedOrder(nul
l)}>
              Close
            </button>
          </div>
        </div>
      </div>
    )}

    <style jsx>{`
      .wrap { padding: 40px 24px 90px; }
      .page-header { margin-bottom: 22px; }
      h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 4px;
color: #1b3821; }
      .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }

      .form-error { background: #fef2f2; border: 1px solid #fecaca; color: #991b1b; padding: 12px 1
6px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; }
      .alert-success { background: #f0fdf4; border: 1px solid #bbf7d0; color: #166534; padding: 12p
x 16px; border-radius: 8px; margin-bottom: 20px; font-size: 0.88rem; font-weight: 600; }

      .income-banner {
        display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
        gap: 16px; margin-bottom: 24px;
      }
      .metric {
        background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px;
        padding: 20px 22px; display: flex; flex-direction: column; gap: 4px;
      }
      .metric .label { font-size: 0.78rem; text-transform: uppercase; color: #888273; font-weight:
600; letter-spacing: 0.03em; }
      .metric .val { font-family: var(--font-display); font-size: 1.8rem; font-weight: 800; color:
#1b3821; }
      .metric .val.green { color: #16a34a; }
      .metric .val.gold { color: #b45309; }
      .metric .hint { font-size: 0.75rem; color: #6b6555; }

      .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow: hidden; overflow-x: auto; box-shadow: 0 2px 8px rgba(0,0,0,0.04); }
      table { width: 100%; border-collapse: collapse; font-size: 0.86rem; min-width: 1050px; }
      th { text-align: left; padding: 12px 16px; background: #faf7ee; font-size: 0.74rem; text-tran
sform: uppercase; color: #6b6555; white-space: nowrap; }
      td { padding: 12px 16px; border-top: 1px solid rgba(26,26,21,0.06); vertical-align: middle; }
      .order-id-badge { font-family: monospace; font-weight: 700; color: #1b3821; background: #faf7
ee; padding: 3px 8px; border-radius: 6px; }
      .order-amount { color: #16a34a; font-size: 0.95rem; }

      .items-summary-col { display: flex; flex-direction: column; gap: 4px; }
      .items-badge { background: #f0fdf4; color: #166534; padding: 2px 7px; border-radius: 4px; fon
```

```
t-weight: 600; font-size: 0.76rem; border: 1px solid #bbf7d0; width: fit-content; }
        .item-chips { display: flex; flex-direction: column; gap: 2px; }
        .item-chip { font-size: 0.72rem; color: #4b5563; }

        .pay-mode { font-family: monospace; font-size: 0.75rem; padding: 3px 8px; background: #faf7e
e; border-radius: 4px; font-weight: 600; }
        .sub { font-size: 0.76rem; color: #6b6555; }
        .address-cell { max-width: 180px; overflow: hidden; text-overflow: ellipsis; white-space: now
rap; }

        /* Logistics Status Select */
        .status-cell { display: flex; flex-direction: column; gap: 4px; }
        .status-select { padding: 6px 10px; border-radius: 8px; border: 1.5px solid #d1d5db; font-siz
e: 0.8rem; font-weight: 700; cursor: pointer; outline: none; }
        .status-select.pending { background: #fffbeb; color: #92400e; border-color: #fde68a; }
        .status-select.packing { background: #eef2ff; color: #4338ca; border-color: #c7d2fe; }
        .status-select.shipped { background: #f0f9ff; color: #0369a1; border-color: #bae6fd; }
        .status-select.out_for_delivery { background: #fef3c7; color: #b45309; border-color: #fcd34d;
}
        .status-select.delivered { background: #f0fdf4; color: #15803d; border-color: #bbf7d0; }
        .status-select.cancelled { background: #fef2f2; color: #991b1b; border-color: #fecaca; }
        .tracking-hint { font-size: 0.72rem; color: #6b7280; max-width: 180px; overflow: hidden; text
-overflow: ellipsis; white-space: nowrap; }

        .btn-details { background: #f0fdf4; border: 1px solid #16a34a; color: #15803d; padding: 5px 1
2px; border-radius: 6px; font-size: 0.78rem; font-weight: 700; cursor: pointer; white-space: nowrap;
}
        .btn-details:hover { background: #16a34a; color: #fff; }
        .empty { text-align: center; color: #6b6555; padding: 40px; }

        /* Modal */
        .modal-backdrop { position: fixed; inset: 0; background: rgba(0,0,0,0.45); backdrop-filter: b
lur(2px); z-index: 9999; display: flex; align-items: center; justify-content: center; padding: 16px;
}
        .modal-card { background: #fff; border-radius: 16px; max-width: 680px; width: 100%; box-shado
w: 0 10px 40px rgba(0,0,0,0.25); overflow: hidden; max-height: 90vh; display: flex; flex-direction: c
olumn; }
        .modal-header { display: flex; justify-content: space-between; align-items: center; padding:
16px 22px; background: #1b3821; color: #fff; }
        .modal-header h3 { margin: 0; font-size: 1.1rem; color: #fff; }
        .btn-close { background: none; border: none; color: #fff; font-size: 1.2rem; cursor: pointer;
}
        .modal-body { padding: 20px 22px; overflow-y: auto; }
        .buyer-summary { background: #faf7ee; border: 1px solid rgba(26,26,21,0.08); padding: 12px 14
px; border-radius: 8px; font-size: 0.84rem; display: flex; flex-direction: column; gap: 4px; }

        .status-update-box { background: #f8fafc; border: 1px solid #e2e8f0; border-radius: 10px; pad
ding: 14px; margin-top: 14px; }
        .section-label { display: block; font-size: 0.8rem; font-weight: 700; color: #334155; margin-
bottom: 6px; }
        .modal-status-select { width: 100%; padding: 8px 12px; border: 1.5px solid #cbd5e1; border-ra
dius: 8px; font-size: 0.9rem; font-weight: 600; }
        .modal-note-input { width: 100%; padding: 8px 12px; border: 1.5px solid #cbd5e1; border-radiu
s: 8px; font-size: 0.86rem; margin-top: 4px; }
        .btn-modal-update { margin-top: 10px; width: 100%; padding: 10px; background: #16a34a; color:
#fff; border: none; border-radius: 8px; font-size: 0.88rem; font-weight: 700; cursor: pointer; }
        .btn-modal-update:hover { background: #15803d; }
        .btn-modal-update:disabled { opacity: 0.6; }

        .item-table-wrap { border: 1px solid #e5e7eb; border-radius: 8px; overflow: hidden; margin-to
p: 8px; }
        .modal-table { width: 100%; border-collapse: collapse; font-size: 0.82rem; }
```

```
        .modal-table th { background: #f9fafb; padding: 10px 12px; font-size: 0.74rem; text-transfor
m: uppercase; color: #6b7280; text-align: left; }
        .modal-table td { padding: 10px 12px; border-top: 1px solid #e5e7eb; vertical-align: middle;
}
        .modal-footer { display: flex; justify-content: flex-end; padding: 14px 22px; background: #fa
fafa; border-top: 1px solid #e5e7eb; }
        .btn-close-modal { padding: 8px 18px; border: 1px solid #d1d5db; background: #fff; border-rad
ius: 999px; font-size: 0.82rem; font-weight: 600; cursor: pointer; }
      `}</style>
    </>
  );
}
```

app/admin/products/page.tsx (Admin Product Moderation Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminListProducts, adminProductAction, ApiError, AdminProduct, FarmerStatus } from
"@/app/lib/api";

const TABS: { key: FarmerStatus; label: string }[] = [
  { key: "pending", label: "Pending Approval" },
  { key: "approved", label: "Live on Website" },
  { key: "rejected", label: "Rejected" },
  { key: "all", label: "All Products" },
];

export default function AdminProductsPage() {
  const router = useRouter();
  const [tab, setTab] = useState<FarmerStatus>("pending");
  const [products, setProducts] = useState<AdminProduct[]>([]);
  const [error, setError] = useState("");
  const [checking, setChecking] = useState(true);
  const [busyId, setBusyId] = useState<number | null>(null);
  const [remarkFor, setRemarkFor] = useState<number | null>(null);
  const [remark, setRemark] = useState("");
  const [selectedProduct, setSelectedProduct] = useState<AdminProduct | null>(null);

  useEffect(() => {
    (async () => {
      const me = await getMe().catch(() => ({ user: null }));
      if (!me.user || me.user.role !== "admin") {
        router.replace("/auth/admin-login");
        return;
      }
      setChecking(false);
    })();
  }, [router]);

  useEffect(() => {
    if (checking) return;
    load();
  }, [tab, checking]);

  async function load() {
    setError("");
    try {
      const res = await adminListProducts(tab);
      setProducts(res.products);
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load products.");
    }
  }

  async function act(id: number, action: "approve" | "reject", withRemark?: string) {
    setBusyId(id);
    setError("");
    try {
      await adminProductAction(id, action, withRemark);
      setRemarkFor(null);
      setRemark("");
```

```
      if (selectedProduct && selectedProduct.id === id) {
        setSelectedProduct({
          ...selectedProduct,
          approval_status: action === "approve" ? "approved" : "rejected",
          remark: withRemark || selectedProduct.remark,
        });
      }
      await load();
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Action failed.");
    } finally {
      setBusyId(null);
    }
  }

  if (checking) return null;

  return (
    <>
      <AdminNav />
      <div className="container wrap">
        <div className="page-header">
          <div>
            <h1>Farmer Product Approvals</h1>
            <p className="subhead">Approve or reject farmer products before they appear in the public
online shop.</p>
          </div>
        </div>

        {error && <div className="form-error">{error}</div>}

        <div className="tabs">
          {TABS.map((t) => (
            <button key={t.key} className={tab === t.key ? "active" : ""} onClick={() => setTab(t.ke
y)}>
              {t.label}
            </button>
          ))}
        </div>

        <div className="table-card">
          <table>
            <thead>
              <tr>
                <th>Item</th>
                <th>Product Name</th>
                <th>Farmer Contact</th>
                <th>Category</th>
                <th>Farmer Base</th>
                <th>Fee &amp;amp; GST</th>
                <th>Selling Price</th>
                <th>Stock</th>
                <th>Website Status</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {products.length === 0 && (
                <tr>
                  <td colSpan={10} className="empty">No products in this category.</td>
                </tr>
              )}
```

```
              {products.map((p) => {
                const base = p.base_price ?? p.price;
                const gst = p.gst_amount ?? (base * ((p.gst_percent ?? 5) / 100));
                const comm = p.commission_amount ?? (base * ((p.commission_percent ?? 5) / 100));
                const fees = gst + comm;

                return (
                <tr key={p.id}>
                  <td>
                    {p.image ? (
                      // eslint-disable-next-line @next/next/no-img-element
                      <img src={p.image} alt={p.name} className="thumb" />
                    ) : (
                      <div className="thumb placeholder">🌾</div>
                    )}
                  </td>
                  <td>
                    <div className="product-title-cell">
                      <strong>{p.name}</strong>
                      <span className="product-id">ID: #{p.id}</span>
                    </div>
                  </td>
                  <td>
                    <div>🧑‍🌾 {p.farmer_name}</div>
                    <div className="sub">{p.farmer_mobile}</div>
                  </td>
                  <td><span className="cat-badge">{p.category}</span></td>
                  <td>₹{base.toFixed(2)}</td>
                  <td>
                    <span style={{ fontSize: "0.76rem", color: "#888" }}>+₹{fees.toFixed(2)}</span>
                  </td>
                  <td><strong style={{ color: "var(--leaf)" }}>₹{p.price.toFixed(2)}</strong></td>
                  <td>{p.stock} kg</td>
                  <td>
                    {p.approval_status === "approved" && (
                      <span className="badge approved">✓ Live on Website</span>
                    )}
                    {p.approval_status === "pending" && (
                      <span className="badge pending">⏳ Pending Review</span>
                    )}
                    {p.approval_status === "rejected" && (
                      <span className="badge rejected">✕ Hidden / Rejected</span>
                    )}
                  </td>
                  <td className="row-actions">
                    <button className="btn-detail" onClick={() => setSelectedProduct(p)}>
                      View
                    </button>

                    {/* Pending Approval: Approve or Reject */}
                    {p.approval_status === "pending" && (
                      <>
                        <button className="ok" disabled={busyId === p.id} onClick={() => act(p.id, "a
pprove")}>
                          ✓ Approve for Website
                        </button>
                        {remarkFor === p.id ? (
                          <span className="remark-inline">
                            <input
                              placeholder="Reason (optional)"
                              value={remark}
                              onChange={(e) => setRemark(e.target.value)}
```

```
                          />
                          <button className="danger" disabled={busyId === p.id} onClick={() => act
(p.id, "reject", remark)}>
                            Confirm
                          </button>
                        </span>
                      ) : (
                        <button className="danger" disabled={busyId === p.id} onClick={() => setRem
arkFor(p.id)}>
                          ✕ Reject
                        </button>
                      )}
                    </>
                  )}

                  {/* Already Approved: Option to Take Down / Reject */}
                  {p.approval_status === "approved" && (
                    <>
                      {remarkFor === p.id ? (
                        <span className="remark-inline">
                          <input
                            placeholder="Reason (optional)"
                            value={remark}
                            onChange={(e) => setRemark(e.target.value)}
                          />
                          <button className="danger" disabled={busyId === p.id} onClick={() => act
(p.id, "reject", remark)}>
                            Confirm Take Down
                          </button>
                        </span>
                      ) : (
                        <button className="danger-outline" disabled={busyId === p.id} onClick={() =
> setRemarkFor(p.id)}>
                          Take Down
                        </button>
                      )}
                    </>
                  )}

                  {/* Already Rejected: Option to Re-Approve */}
                  {p.approval_status === "rejected" && (
                    <button className="ok-outline" disabled={busyId === p.id} onClick={() => act(p.
id, "approve")}>
                      Publish to Website
                    </button>
                  )}
                </td>
              </tr>
            );
            })}
          </tbody>
        </table>
      </div>

      {/* Product Details Modal */}
      {selectedProduct && (
        <div className="modal-backdrop" onClick={() => setSelectedProduct(null)}>
          <div className="modal-content" onClick={(e) => e.stopPropagation()}>
            <div className="modal-header">
              <h2>Product Details — {selectedProduct.name}</h2>
              <button className="close-btn" onClick={() => setSelectedProduct(null)}>✕</button>
            </div>
```

```
              <div className="modal-body">
                {selectedProduct.image && (
                  <div className="modal-img-wrap">
                    {/* eslint-disable-next-line @next/next/no-img-element */}
                    <img src={selectedProduct.image} alt={selectedProduct.name} className="modal-img"
/>
                  </div>
                )}

                <div className="detail-grid">
                  <div className="detail-item">
                    <label>Product Name</label>
                    <div>{selectedProduct.name}</div>
                  </div>
                  <div className="detail-item">
                    <label>Category</label>
                    <div>{selectedProduct.category}</div>
                  </div>
                  <div className="detail-item full-width" style={{ background: "rgba(46,125,50,0.0
5)", padding: "12px", borderRadius: "8px", border: "1px solid rgba(46,125,50,0.15)" }}>
                    <label style={{ fontWeight: 600, color: "var(--leaf)" }}>Price Breakdown (per kg)
</label>
                    <div style={{ display: "grid", gridTemplateColumns: "repeat(auto-fit, minmax(140p
x, 1fr))", gap: "10px", marginTop: "6px" }}>
                      <div>
                        <span style={{ fontSize: "0.78rem", color: "#666" }}>Farmer Base Price:</span
>
                        <div style={{ fontWeight: 600 }}>₹{(selectedProduct.base_price ?? selectedPro
duct.price).toFixed(2)}</div>
                      </div>
                      <div>
                        <span style={{ fontSize: "0.78rem", color: "#666" }}>Platform Commission ({se
lectedProduct.commission_percent ?? 5}%):</span>
                        <div style={{ fontWeight: 600, color: "#e65100" }}>+₹{(selectedProduct.commis
sion_amount ?? ((selectedProduct.base_price ?? selectedProduct.price) * ((selectedProduct.commission_
percent ?? 5) / 100))).toFixed(2)}</div>
                      </div>
                      <div>
                        <span style={{ fontSize: "0.78rem", color: "#666" }}>GST ({selectedProduct.gs
t_percent ?? 5}%):</span>
                        <div style={{ fontWeight: 600, color: "#1565c0" }}>+₹{(selectedProduct.gst_am
ount ?? ((selectedProduct.base_price ?? selectedProduct.price) * ((selectedProduct.gst_percent ?? 5)
/ 100))).toFixed(2)}</div>
                      </div>
                      <div>
                        <span style={{ fontSize: "0.78rem", color: "#666" }}>Website Selling Price:</
span>
                        <div style={{ fontWeight: 700, fontSize: "1.1rem", color: "var(--leaf)" }}>₹
{selectedProduct.price.toFixed(2)} / kg</div>
                      </div>
                    </div>
                  </div>
                  <div className="detail-item">
                    <label>Current Stock</label>
                    <div>{selectedProduct.stock} kg available</div>
                  </div>
                  <div className="detail-item">
                    <label>Farmer Name</label>
                    <div>{selectedProduct.farmer_name}</div>
                  </div>
                  <div className="detail-item">
```

```
                    <label>Farmer Contact</label>
                    <div><a href={`tel:${selectedProduct.farmer_mobile}`}>📞 {selectedProduct.farmer
_mobile}</a></div>
                  </div>
                  <div className="detail-item full-width">
                    <label>Description</label>
                    <div className="desc-box">{selectedProduct.description || "No description provide
d."}</div>
                  </div>
                  <div className="detail-item">
                    <label>Website Visibility</label>
                    <div>
                      <span className={`badge ${selectedProduct.approval_status}`}>
                        {selectedProduct.approval_status === "approved" ? "Live in Shop" : selectedPr
oduct.approval_status}
                      </span>
                    </div>
                  </div>
                  <div className="detail-item">
                    <label>Added On</label>
                    <div>{new Date(selectedProduct.created_at).toLocaleString("en-IN")}</div>
                  </div>
                  {selectedProduct.remark && (
                    <div className="detail-item full-width">
                      <label>Admin Note / Rejection Reason</label>
                      <div className="remark-box">{selectedProduct.remark}</div>
                    </div>
                  )}
                </div>

                <div className="modal-actions">
                  {selectedProduct.approval_status !== "approved" && (
                    <button className="ok-large" disabled={busyId === selectedProduct.id} onClick={()
=> act(selectedProduct.id, "approve")}>
                      ✓ Approve & Publish to Shop
                    </button>
                  )}
                  {selectedProduct.approval_status !== "rejected" && (
                    <button className="danger-large" disabled={busyId === selectedProduct.id} onClick
={() => act(selectedProduct.id, "reject", remark)}>
                      ✕ Reject / Hide from Shop
                    </button>
                  )}
                </div>
              </div>
            </div>
          </div>
        )}
      </div>

      <style jsx>{`
        .wrap { padding: 40px 24px 90px; }
        .page-header { margin-bottom: 22px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 4px;
}
        .subhead { color: #6b6555; font-size: 0.88rem; margin: 0; }
        .tabs { display: flex; gap: 8px; margin-bottom: 18px; flex-wrap: wrap; }
        .tabs button { border: 1px solid rgba(26,26,21,0.12); background: #fff; padding: 7px 16px; bo
rder-radius: 999px; font-size: 0.82rem; cursor: pointer; }
        .tabs button.active { background: var(--soil); color: #efe8d8; border-color: var(--soil); }
        .table-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; o
verflow: hidden; overflow-x: auto; }
```

```
        table { width: 100%; border-collapse: collapse; font-size: 0.86rem; }
        th { text-align: left; padding: 12px 16px; background: #faf7ee; font-size: 0.74rem; text-tran
sform: uppercase; color: #6b6555; white-space: nowrap; }
        td { padding: 12px 16px; border-top: 1px solid rgba(26,26,21,0.06); vertical-align: middle; w
hite-space: nowrap; }
        .product-title-cell { display: flex; flex-direction: column; }
        .product-id { font-size: 0.74rem; color: #888273; }
        .thumb { width: 44px; height: 44px; border-radius: 8px; object-fit: cover; }
        .thumb.placeholder { background: linear-gradient(160deg,#f7f2e6,#ede4cd); display: flex; alig
n-items: center; justify-content: center; font-size: 1.2rem; }
        .cat-badge { background: #faf7ee; border: 1px solid rgba(26,26,21,0.08); padding: 3px 8px; bo
rder-radius: 6px; font-size: 0.8rem; }
        .sub { font-size: 0.76rem; color: #6b6555; }
        .badge { font-size: 0.72rem; padding: 3px 10px; border-radius: 999px; text-transform: capital
ize; font-weight: 600; }
        .badge.pending { background: rgba(232,163,61,0.18); color: var(--rust); }
        .badge.approved { background: rgba(76,122,61,0.15); color: var(--leaf); }
        .badge.rejected { background: rgba(200,60,40,0.12); color: #b8391f; }
        .row-actions { display: flex; gap: 8px; align-items: center; white-space: normal; }
        .row-actions button { border: none; padding: 6px 12px; border-radius: 999px; font-size: 0.78r
em; font-weight: 600; cursor: pointer; }
        .btn-detail { background: #f4f1ea; color: var(--soil); border: 1px solid rgba(26,26,21,0.1) !
important; }
        .btn-detail:hover { background: #e8e2d4; }
        .row-actions button.ok { background: var(--leaf); color: #fff; }
        .row-actions button.ok-outline { background: transparent; border: 1px solid var(--leaf) !impo
rtant; color: var(--leaf); }
        .row-actions button.danger { background: #f4dcd6; color: #b8391f; }
        .row-actions button.danger-outline { background: transparent; border: 1px solid #b8391f !impo
rtant; color: #b8391f; }
        .row-actions button:disabled { opacity: 0.6; cursor: default; }
        .remark-inline { display: flex; gap: 6px; align-items: center; }
        .remark-inline input { border: 1px solid rgba(26,26,21,0.15); border-radius: 8px; padding: 5p
x 8px; font-size: 0.78rem; width: 140px; }
        .empty { text-align: center; color: #6b6555; padding: 40px; }

        /* Modal */
        .modal-backdrop {
          position: fixed; inset: 0; background: rgba(0,0,0,0.5);
          display: flex; align-items: center; justify-content: center;
          z-index: 1000; padding: 20px;
        }
        .modal-content {
          background: #fff; border-radius: 16px; width: 100%; max-width: 600px; max-height: 90vh;
          box-shadow: 0 20px 40px rgba(0,0,0,0.2); overflow-y: auto;
        }
        .modal-header {
          display: flex; justify-content: space-between; align-items: center;
          padding: 18px 24px; border-bottom: 1px solid rgba(26,26,21,0.08); background: #faf7ee;
        }
        .modal-header h2 { font-family: var(--font-display); font-size: 1.2rem; margin: 0; }
        .close-btn { background: none; border: none; font-size: 1.2rem; cursor: pointer; color: #6b65
55; }
        .modal-body { padding: 24px; }
        .modal-img-wrap { margin-bottom: 20px; text-align: center; background: #faf7ee; border-radiu
s: 12px; padding: 12px; }
        .modal-img { max-height: 220px; max-width: 100%; object-fit: contain; border-radius: 8px; }
        .detail-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; margin-bottom: 24px;
}
        .detail-item { display: flex; flex-direction: column; gap: 4px; }
        .detail-item.full-width { grid-column: span 2; }
        .detail-item label { font-size: 0.75rem; text-transform: uppercase; color: #888273; font-weig
```

```
ht: 600; }
        .detail-item div { font-size: 0.95rem; color: var(--soil); }
        .detail-item a { color: var(--leaf); text-decoration: none; font-weight: 600; }
        .desc-box { background: #faf8f5; padding: 10px 14px; border-radius: 8px; font-size: 0.88rem !
important; line-height: 1.5; color: #444 !important; }
        .remark-box { background: #faf7ee; padding: 10px 14px; border-radius: 8px; font-style: itali
c; border-left: 3px solid var(--rust); }
        .modal-actions { display: flex; gap: 12px; justify-content: flex-end; }
        .ok-large { background: var(--leaf); color: #fff; border: none; padding: 10px 20px; border-ra
dius: 8px; font-weight: 600; cursor: pointer; }
        .danger-large { background: #f4dcd6; color: #b8391f; border: none; padding: 10px 20px; border
-radius: 8px; font-weight: 600; cursor: pointer; }
      `}</style>
    </>
  );
}
```

app/admin/bulk-requests/page.tsx (Admin Bulk Requests Management Page)

```
"use client";

import { useEffect, useState } from "react";
import { useRouter } from "next/navigation";
import AdminNav from "@/app/components/AdminNav";
import { getMe, adminListBulkRequests, adminBulkRequestAction, ApiError, AdminBulkRequest, FarmerStat
us } from "@/app/lib/api";

const TABS: { key: FarmerStatus; label: string }[] = [
  { key: "pending", label: "Pending" },
  { key: "approved", label: "Approved" },
  { key: "rejected", label: "Rejected" },
  { key: "all", label: "All" },
];

export default function AdminBulkRequestsPage() {
  const router = useRouter();
  const [tab, setTab] = useState<FarmerStatus>("pending");
  const [requests, setRequests] = useState<AdminBulkRequest[]>([]);
  const [error, setError] = useState("");
  const [checking, setChecking] = useState(true);
  const [busyId, setBusyId] = useState<number | null>(null);
  const [remarkFor, setRemarkFor] = useState<number | null>(null);
  const [remark, setRemark] = useState("");

  useEffect(() => {
    (async () => {
      const me = await getMe().catch(() => ({ user: null }));
      if (!me.user || me.user.role !== "admin") {
        router.replace("/auth/admin-login");
        return;
      }
      setChecking(false);
    })();
  }, [router]);

  useEffect(() => {
    if (checking) return;
    load();
  }, [tab, checking]);

  async function load() {
    setError("");
    try {
      const res = await adminListBulkRequests(tab);
      setRequests(res.requests);
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Could not load bulk requests.");
    }
  }

  async function act(id: number, action: "approve" | "reject", withRemark?: string) {
    setBusyId(id);
    setError("");
    try {
      await adminBulkRequestAction(id, action, withRemark);
      setRemarkFor(null);
      setRemark("");
      await load();
```

```
    } catch (err) {
      setError(err instanceof ApiError ? err.message : "Action failed.");
    } finally {
      setBusyId(null);
    }
  }

  if (checking) return null;

  return (
    <>
      <AdminNav />
      <div className="container wrap">
        <h1>Bulk order requests</h1>
        {error && <div className="form-error">{error}</div>}

        <div className="tabs">
          {TABS.map((t) => (
            <button key={t.key} className={tab === t.key ? "active" : ""} onClick={() => setTab(t.ke
y)}>
              {t.label}
            </button>
          ))}
        </div>

        <div className="cards">
          {requests.length === 0 && <div className="empty-card">No bulk requests in this list.</div>}
          {requests.map((r) => (
            <div className="request-card" key={r.id}>
              <div className="head">
                <div>
                  <strong>{r.product_name}</strong> — {r.qty} kg
                </div>
                <span className={`badge ${r.status}`}>{r.status}</span>
              </div>
              <p className="reason">“{r.reason}”</p>
              <div className="meta">
                <span>{r.buyer_name}</span>
                <span>{r.buyer_email || r.buyer_mobile}</span>
                <span>{new Date(r.created_at).toLocaleDateString("en-IN")}</span>
              </div>

              {r.status === "pending" && (
                <div className="actions">
                  <button className="ok" disabled={busyId === r.id} onClick={() => act(r.id, "approv
e")}>
                    Approve
                  </button>
                  {remarkFor === r.id ? (
                    <span className="remark-inline">
                      <input
                        placeholder="Reason (optional)"
                        value={remark}
                        onChange={(e) => setRemark(e.target.value)}
                      />
                      <button className="danger" disabled={busyId === r.id} onClick={() => act(r.id,
"reject", remark)}>
                        Confirm reject
                      </button>
                    </span>
                  ) : (
                    <button className="danger" disabled={busyId === r.id} onClick={() => setRemarkFor
```

```
(r.id)}>
                      Reject
                    </button>
                  )}
                </div>
              )}
              {r.status !== "pending" && r.remark && <p className="remark">Admin note: “{r.remark}”</
p>}
            </div>
          ))}
        </div>
      </div>

      <style jsx>{`
        .wrap { padding: 40px 24px 90px; }
        h1 { font-family: var(--font-display); font-weight: 700; font-size: 1.6rem; margin: 0 0 18px;
}
        .tabs { display: flex; gap: 8px; margin-bottom: 18px; }
        .tabs button { border: 1px solid rgba(26,26,21,0.12); background: #fff; padding: 7px 16px; bo
rder-radius: 999px; font-size: 0.82rem; cursor: pointer; }
        .tabs button.active { background: var(--soil); color: #efe8d8; border-color: var(--soil); }
        .cards { display: flex; flex-direction: column; gap: 14px; }
        .empty-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px; p
adding: 40px; text-align: center; color: #6b6555; }
        .request-card { background: #fff; border: 1px solid rgba(26,26,21,0.08); border-radius: 14px;
padding: 18px 20px; }
        .head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8p
x; }
        .badge { font-size: 0.72rem; padding: 3px 10px; border-radius: 999px; text-transform: capital
ize; }
        .badge.pending { background: rgba(232,163,61,0.18); color: var(--rust); }
        .badge.approved { background: rgba(76,122,61,0.15); color: var(--leaf); }
        .badge.rejected { background: rgba(200,60,40,0.12); color: #b8391f; }
        .reason { color: #4a4638; font-style: italic; margin: 6px 0; }
        .meta { display: flex; gap: 16px; font-size: 0.78rem; color: #6b6555; margin-bottom: 10px; }
        .actions { display: flex; gap: 8px; align-items: center; }
        .actions button { border: none; padding: 6px 14px; border-radius: 999px; font-size: 0.8rem; f
ont-weight: 600; cursor: pointer; }
        .actions button.ok { background: var(--leaf); color: #fff; }
        .actions button.danger { background: #f4dcd6; color: #b8391f; }
        .actions button:disabled { opacity: 0.6; cursor: default; }
        .remark-inline { display: flex; gap: 6px; align-items: center; }
        .remark-inline input { border: 1px solid rgba(26,26,21,0.15); border-radius: 8px; padding: 5p
x 8px; font-size: 0.78rem; width: 160px; }
        .remark { color: #6b6555; font-size: 0.8rem; margin: 4px 0 0; }
      `}</style>
    </>
  );
}
```

app/privacy-policy/page.tsx (Privacy Policy Page)

```
import type { Metadata } from "next";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";

export const metadata: Metadata = {
  title: "Privacy Policy – Agrenza",
  description:
    "Read Agrenza's Privacy Policy. Learn how we collect, use, and protect your personal data when yo
u use our farm-to-doorstep marketplace.",
  robots: { index: true, follow: true },
};

export default function PrivacyPolicyPage() {
  return (
    <>
      <Nav />

      <div className="doc-hero">
        <div className="container">
          <h1>Privacy Policy</h1>
          <p className="effective">Effective Date: September 2026</p>
          <p className="intro">
            At Agrenza, your privacy is important to us. This Privacy Policy explains how we collect,
use, share,
            and protect your personal information when you visit our website or use our services.
          </p>
        </div>
      </div>

      <div className="doc-body container">

        <section className="doc-section">
          <h2>1. Information We Collect</h2>
          <p>We collect the following types of information when you register or use Agrenza:</p>
          <ul>
            <li><strong>Registration Data:</strong> Name, email address, mobile number, and role (buy
er or farmer).</li>
            <li><strong>Profile Information:</strong> Address, state, and delivery preferences.</li>
            <li><strong>Order Data:</strong> Products ordered, quantities, delivery address, and orde
r history.</li>
            <li><strong>Usage Data:</strong> Pages visited, products browsed, device type, IP addres
s, and browser information.</li>
            <li><strong>Payment Information:</strong> We do not store payment card details. Payments
are processed securely by third-party payment gateways.</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>2. How We Use Your Information</h2>
          <ul>
            <li>To process and fulfil your orders.</li>
            <li>To verify your identity and authenticate your account.</li>
            <li>To send order confirmations, shipping updates, and support communications.</li>
            <li>To improve our platform, products, and user experience.</li>
            <li>To send promotional offers and discounts (you can unsubscribe at any time).</li>
            <li>To comply with legal obligations and resolve disputes.</li>
          </ul>
        </section>
```

```
        <section className="doc-section">
          <h2>3. Data Sharing</h2>
          <p>We <strong>never sell your personal data</strong> to third parties. We share data only i
n these limited circumstances:</p>
          <ul>
            <li><strong>Delivery Partners:</strong> Your name and address are shared with delivery pa
rtners to fulfil orders.</li>
            <li><strong>Payment Processors:</strong> Payment information is handled by PCI-compliant
third-party gateways.</li>
            <li><strong>Legal Compliance:</strong> We may disclose information when required by India
n law or a court order.</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>4. Cookies</h2>
          <p>Agrenza uses cookies for the following purposes:</p>
          <ul>
            <li><strong>Session Cookies:</strong> To keep you logged in during your visit.</li>
            <li><strong>Cart Cookies:</strong> To remember your cart items between sessions.</li>
            <li><strong>Preference Cookies:</strong> To remember your category filters and settings.
</li>
            <li><strong>Analytics Cookies:</strong> To understand how users interact with our platfor
m (Google Analytics).</li>
          </ul>
          <p>You can disable cookies in your browser settings, though some features may not function
correctly.</p>
        </section>

        <section className="doc-section">
          <h2>5. Data Security</h2>
          <p>
            We take reasonable security measures to protect your information, including encrypted sto
rage of passwords
            (using bcrypt hashing), HTTPS connections, and restricted database access. However, no in
ternet transmission
            is 100% secure and we cannot guarantee absolute security.
          </p>
        </section>

        <section className="doc-section">
          <h2>6. Your Rights</h2>
          <p>Under applicable Indian data protection law, you have the right to:</p>
          <ul>
            <li><strong>Access</strong> your personal data we hold.</li>
            <li><strong>Correct</strong> inaccurate or incomplete information.</li>
            <li><strong>Delete</strong> your account and associated data (subject to legal retention
requirements).</li>
            <li><strong>Withdraw consent</strong> for marketing communications at any time.</li>
          </ul>
          <p>To exercise these rights, contact us at <a href="mailto:privacy@agrenza.com">privacy@agr
enza.com</a>.</p>
        </section>

        <section className="doc-section">
          <h2>7. Children's Privacy</h2>
          <p>
            Agrenza is not intended for children under the age of 13. We do not knowingly collect per
sonal information from
            children. If you believe a child has provided us with their data, please contact us immed
iately.
          </p>
```

```
        </section>

        <section className="doc-section">
          <h2>8. Changes to This Policy</h2>
          <p>
            We may update this Privacy Policy from time to time. When we do, we will update the effec
tive date at the top
            of this page and notify registered users by email.
          </p>
        </section>

        <section className="doc-section">
          <h2>9. Contact Us</h2>
          <p>If you have questions about this Privacy Policy, please contact:</p>
          <div className="contact-block">
            <p><strong>Agrenza Technologies Pvt Ltd</strong></p>
            <p>Email: <a href="mailto:privacy@agrenza.com">privacy@agrenza.com</a></p>
            <p>Phone: 1800-000-000 (Toll Free)</p>
            <p>Address: Chennai, Tamil Nadu, India</p>
          </div>
        </section>

      </div>

      <Footer />

      <style>{`
        .doc-hero {
          background: #f8fafc;
          border-bottom: 1px solid #e2e8f0;
          padding: 60px 0 40px;
        }
        .doc-hero h1 {
          font-family: var(--font-display, serif);
          font-size: 2.4rem;
          font-weight: 900;
          color: #16241a;
          margin: 0 0 8px;
        }
        .effective {
          font-size: 0.85rem;
          color: #2e7d32;
          font-weight: 600;
          margin: 0 0 16px;
        }
        .intro {
          font-size: 1rem;
          color: #4a5548;
          line-height: 1.65;
          max-width: 680px;
          margin: 0;
        }
        .doc-body {
          padding: 56px 0 80px;
          max-width: 800px;
        }
        .doc-section {
          margin-bottom: 40px;
          padding-bottom: 36px;
          border-bottom: 1px solid #f1f5f9;
        }
        .doc-section:last-child { border-bottom: none; }
```

```
        .doc-section h2 {
          font-family: var(--font-display, serif);
          font-size: 1.2rem;
          font-weight: 800;
          color: #2e7d32;
          margin: 0 0 14px;
        }
        .doc-section p, .doc-section li {
          font-size: 0.93rem;
          color: #334155;
          line-height: 1.7;
        }
        .doc-section ul {
          padding-left: 20px;
          margin: 8px 0 0;
        }
        .doc-section li { margin-bottom: 6px; }
        .doc-section a { color: #2e7d32; font-weight: 600; }
        .contact-block {
          background: #f0fdf4;
          border: 1px solid #bbf7d0;
          border-radius: 10px;
          padding: 20px 24px;
          margin-top: 12px;
        }
        .contact-block p { margin: 4px 0; font-size: 0.92rem; color: #334155; }
        .contact-block strong { color: #16241a; }
      `}</style>
    </>
  );
}
```

app/terms/page.tsx (Terms of Service Page)

```
import type { Metadata } from "next";
import Nav from "@/app/components/Nav";
import Footer from "@/app/components/Footer";

export const metadata: Metadata = {
  title: "Terms of Service — Agrenza",
  description:
    "Read Agrenza's Terms of Service. Understand your rights and responsibilities when using our farm
-to-doorstep marketplace platform in India.",
  robots: { index: true, follow: true },
};

export default function TermsPage() {
  return (
    <>
      <Nav />

      <div className="doc-hero">
        <div className="container">
          <h1>Terms of Service</h1>
          <p className="effective">Effective Date: September 2026</p>
          <p className="intro">
            Please read these Terms of Service carefully before using the Agrenza platform.
            By accessing or using our services, you agree to be bound by these terms.
          </p>
        </div>
      </div>

      <div className="doc-body container">

        <section className="doc-section">
          <h2>1. Acceptance of Terms</h2>
          <p>
            By creating an account or using Agrenza's website or services, you agree to these Terms o
f Service
            and our Privacy Policy. If you do not agree with any part of these terms, you must not us
e the platform.
          </p>
        </section>

        <section className="doc-section">
          <h2>2. Eligibility</h2>
          <p>
            You must be at least <strong>18 years of age</strong> to use Agrenza. By using the platfo
rm, you represent
            and warrant that you are 18 or older and have the legal capacity to enter into a binding
agreement.
          </p>
        </section>

        <section className="doc-section">
          <h2>3. Use of Platform</h2>
          <ul>
            <li>You must provide accurate, complete, and current information when registering.</li>
            <li>You are responsible for maintaining the confidentiality of your account and password.
</li>
            <li>You agree not to misuse the platform for fraudulent, illegal, or harmful purposes.</l
i>
            <li>You must not scrape, copy, or reproduce platform content without written permission.
```

```
</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>4. Farmer Responsibilities</h2>
          <p>If you register as a farmer on Agrenza, you agree to:</p>
          <ul>
            <li>Provide accurate product listings with honest descriptions, weights, and prices.</li>
            <li>Ensure all listed products are fresh, safe for consumption, and comply with FSSAI sta
ndards.</li>
            <li>Fulfil orders on time and communicate delays promptly.</li>
            <li>Not list products that are banned, hazardous, or otherwise prohibited by Indian law.
</li>
            <li>Accept that Agrenza may remove listings that violate quality standards.</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>5. Buyer Responsibilities</h2>
          <p>If you register as a buyer on Agrenza, you agree to:</p>
          <ul>
            <li>Provide an accurate delivery address at the time of checkout.</li>
            <li>Make payment in full at the time of placing an order.</li>
            <li>Not abuse the returns or refund process.</li>
            <li>Treat farmers and Agrenza support staff with respect.</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>6. Orders &amp;amp; Cancellations</h2>
          <ul>
            <li>Orders can be cancelled within <strong>2 hours</strong> of placement for a full refun
d.</li>
            <li>After 2 hours, cancellation is at the discretion of the farmer.</li>
            <li>If you receive a damaged or incorrect product, contact us within <strong>24 hours</st
rong> of delivery.</li>
            <li>Refunds are processed within 5–7 business days to the original payment method.</li>
          </ul>
        </section>

        <section className="doc-section">
          <h2>7. Pricing &amp;amp; GST</h2>
          <p>
            All prices displayed on Agrenza are <strong>GST-inclusive</strong>. The price you see is
the final price you pay.
            Farmers set their own prices. Agrenza does not guarantee any specific price for any produ
ct.
            Prices may change without notice based on market conditions.
          </p>
        </section>

        <section className="doc-section">
          <h2>8. Intellectual Property</h2>
          <p>
            All content on the Agrenza platform — including the logo, design, software, text, and ima
ges — is the property
            of Agrenza Technologies Pvt Ltd and is protected by Indian copyright law. You may not rep
roduce, distribute,
            or create derivative works without written permission.
          </p>
        </section>
```

```
        <section className="doc-section">
          <h2>9. Limitation of Liability</h2>
          <p>
            Agrenza acts as a marketplace connecting farmers and buyers. We are not responsible for:
          </p>
          <ul>
            <li>Product quality beyond what is described in the listing.</li>
            <li>Delays caused by delivery partners.</li>
            <li>Any indirect, incidental, or consequential loss arising from use of the platform.</li
>
          </ul>
          <p>Our maximum liability shall not exceed the value of the order in question.</p>
        </section>

        <section className="doc-section">
          <h2>10. Governing Law</h2>
          <p>
            These Terms of Service are governed by and construed in accordance with the laws of <stro
ng>India</strong>.
            Any disputes arising out of or in connection with these terms shall be subject to the exc
lusive jurisdiction
            of the courts of <strong>Chennai, Tamil Nadu, India</strong>.
          </p>
        </section>

        <section className="doc-section">
          <h2>11. Changes to Terms</h2>
          <p>
            We reserve the right to update these Terms of Service at any time. When changes are made,
we will update
            the effective date at the top of this page. Continued use of the platform after changes c
onstitutes acceptance
            of the new terms.
          </p>
        </section>

        <section className="doc-section">
          <h2>12. Contact Us</h2>
          <p>For legal inquiries or questions about these terms, contact:</p>
          <div className="contact-block">
            <p><strong>Agrenza Technologies Pvt Ltd</strong></p>
            <p>Email: <a href="mailto:legal@agrenza.com">legal@agrenza.com</a></p>
            <p>Phone: 1800-000-000 (Toll Free)</p>
            <p>Address: Chennai, Tamil Nadu, India</p>
          </div>
        </section>

      </div>

      <Footer />

      <style>{`
        .doc-hero {
          background: #f8fafc;
          border-bottom: 1px solid #e2e8f0;
          padding: 60px 0 40px;
        }
        .doc-hero h1 {
          font-family: var(--font-display, serif);
          font-size: 2.4rem;
          font-weight: 900;
```

```
          color: #16241a;
          margin: 0 0 8px;
        }
        .effective {
          font-size: 0.85rem;
          color: #2e7d32;
          font-weight: 600;
          margin: 0 0 16px;
        }
        .intro {
          font-size: 1rem;
          color: #4a5548;
          line-height: 1.65;
          max-width: 680px;
          margin: 0;
        }
        .doc-body {
          padding: 56px 0 80px;
          max-width: 800px;
        }
        .doc-section {
          margin-bottom: 40px;
          padding-bottom: 36px;
          border-bottom: 1px solid #f1f5f9;
        }
        .doc-section:last-child { border-bottom: none; }
        .doc-section h2 {
          font-family: var(--font-display, serif);
          font-size: 1.2rem;
          font-weight: 800;
          color: #2e7d32;
          margin: 0 0 14px;
        }
        .doc-section p, .doc-section li {
          font-size: 0.93rem;
          color: #334155;
          line-height: 1.7;
        }
        .doc-section ul {
          padding-left: 20px;
          margin: 8px 0 0;
        }
        .doc-section li { margin-bottom: 6px; }
        .doc-section a { color: #2e7d32; font-weight: 600; }
        .contact-block {
          background: #f0fdf4;
          border: 1px solid #bbf7d0;
          border-radius: 10px;
          padding: 20px 24px;
          margin-top: 12px;
        }
        .contact-block p { margin: 4px 0; font-size: 0.92rem; color: #334155; }
        .contact-block strong { color: #16241a; }
      `}</style>
    </>
  );
}
```

CHAPTER 10: APPENDICES

SECTION 3: SCREENLAYOUT & APPLICATION OUTPUTS

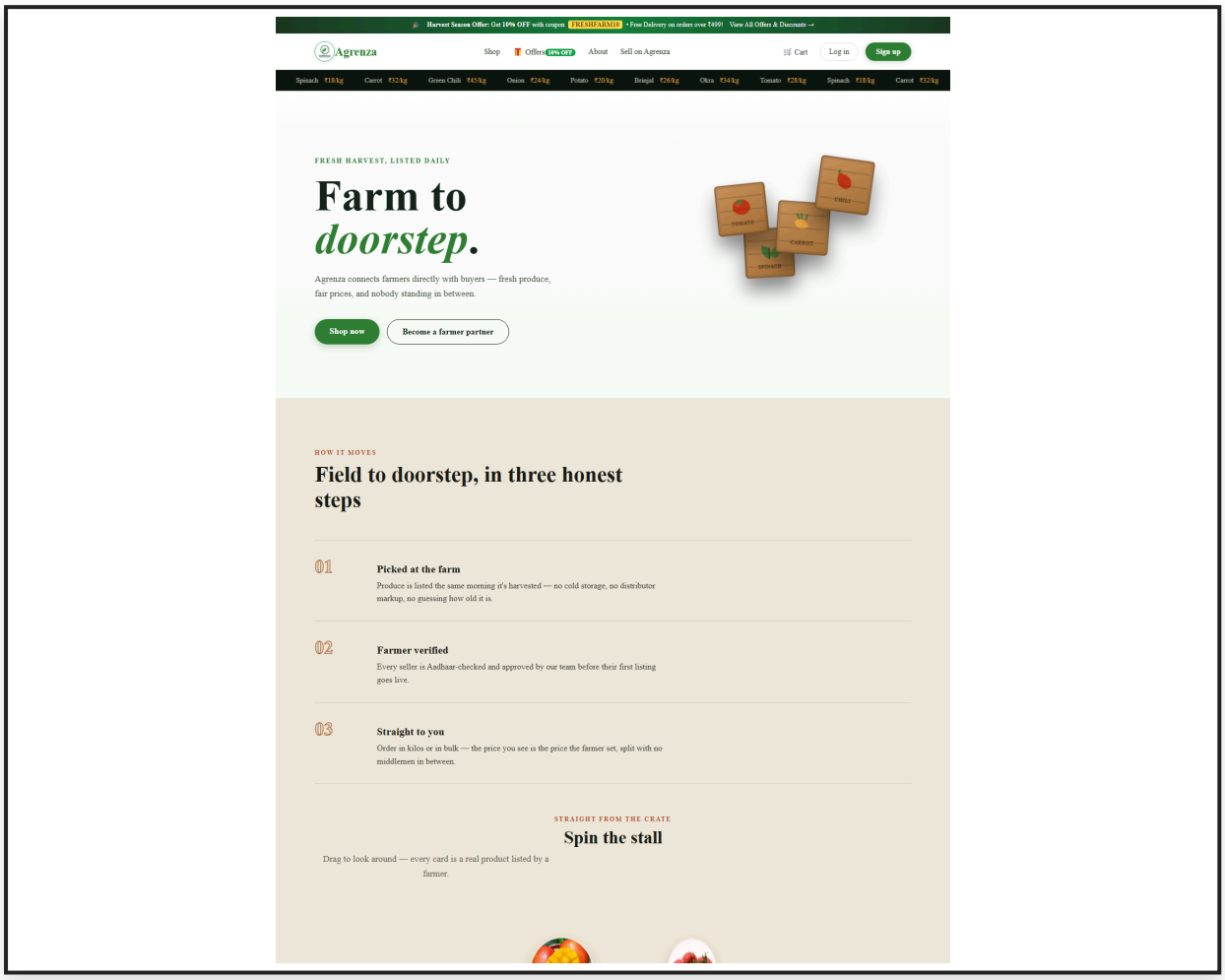


Figure 10.1: Home Page — Offer Banner, Navigation Header & Live Mandi Price Ticker

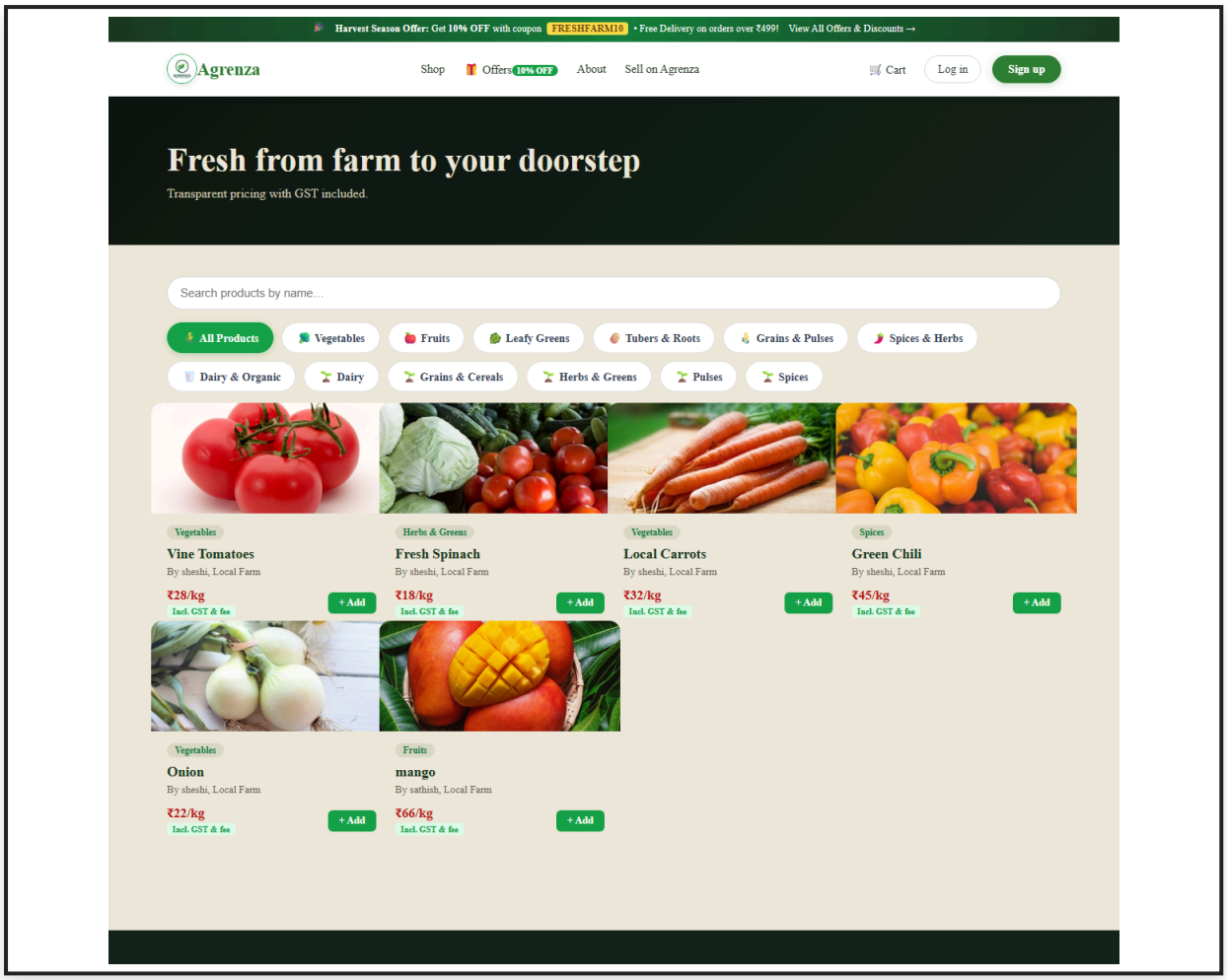


Figure 10.2: Public Marketplace — Categorized Fresh Produce Listings with Filter Tabs

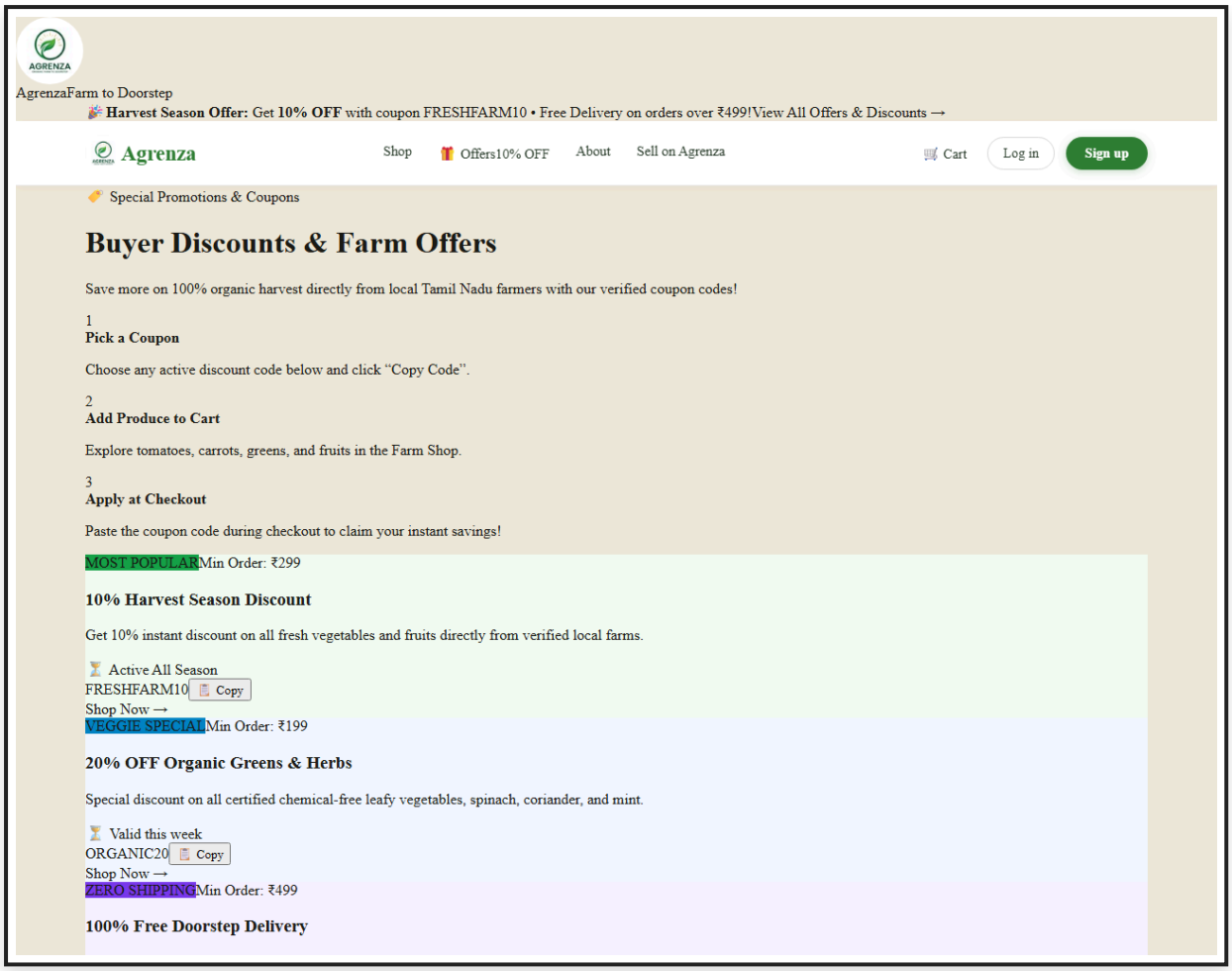


Figure 10.3: Promotional Offers & Harvest Season Discount Coupons

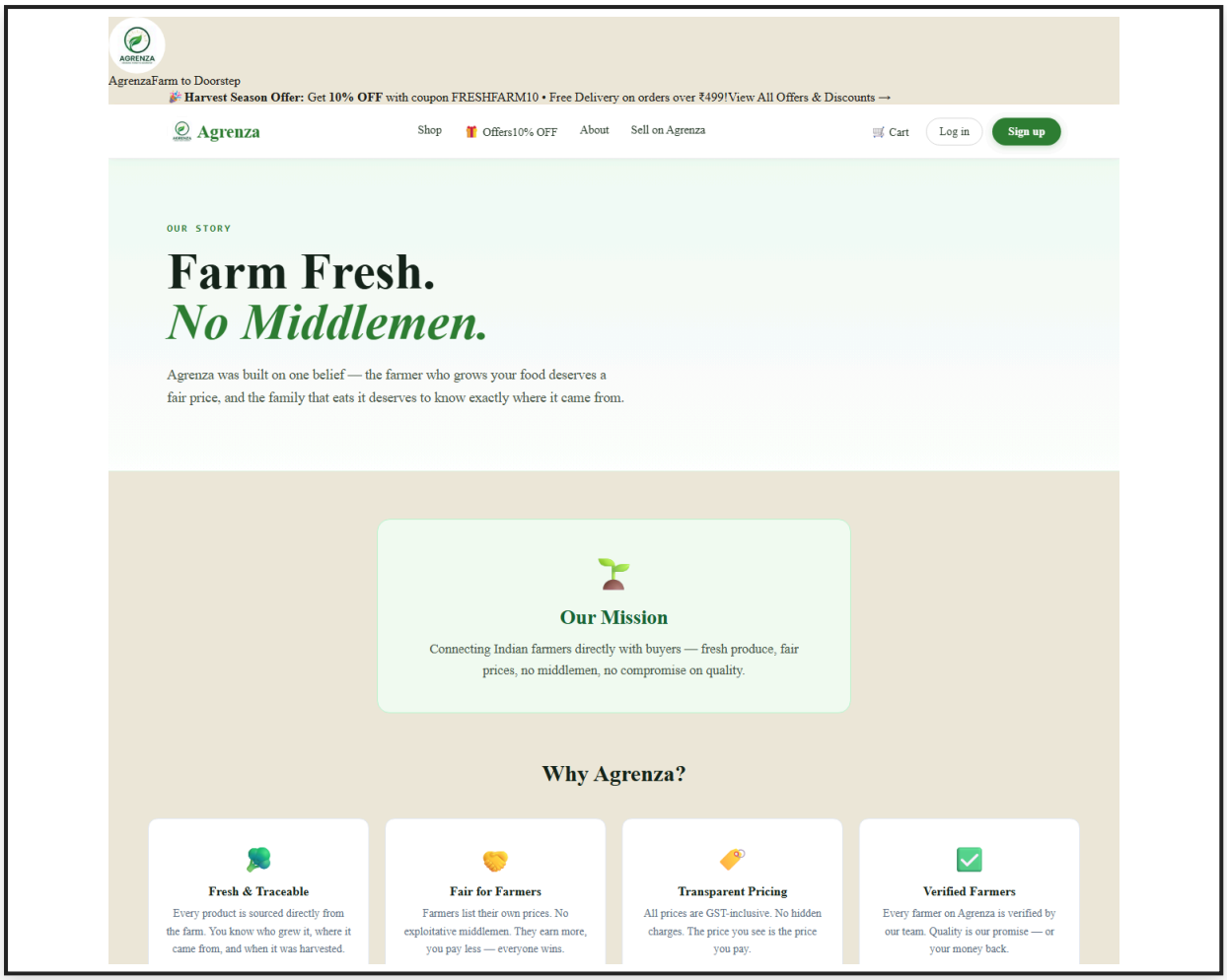


Figure 10.4: About Agrenza — Mission, Story & Trust Statistics

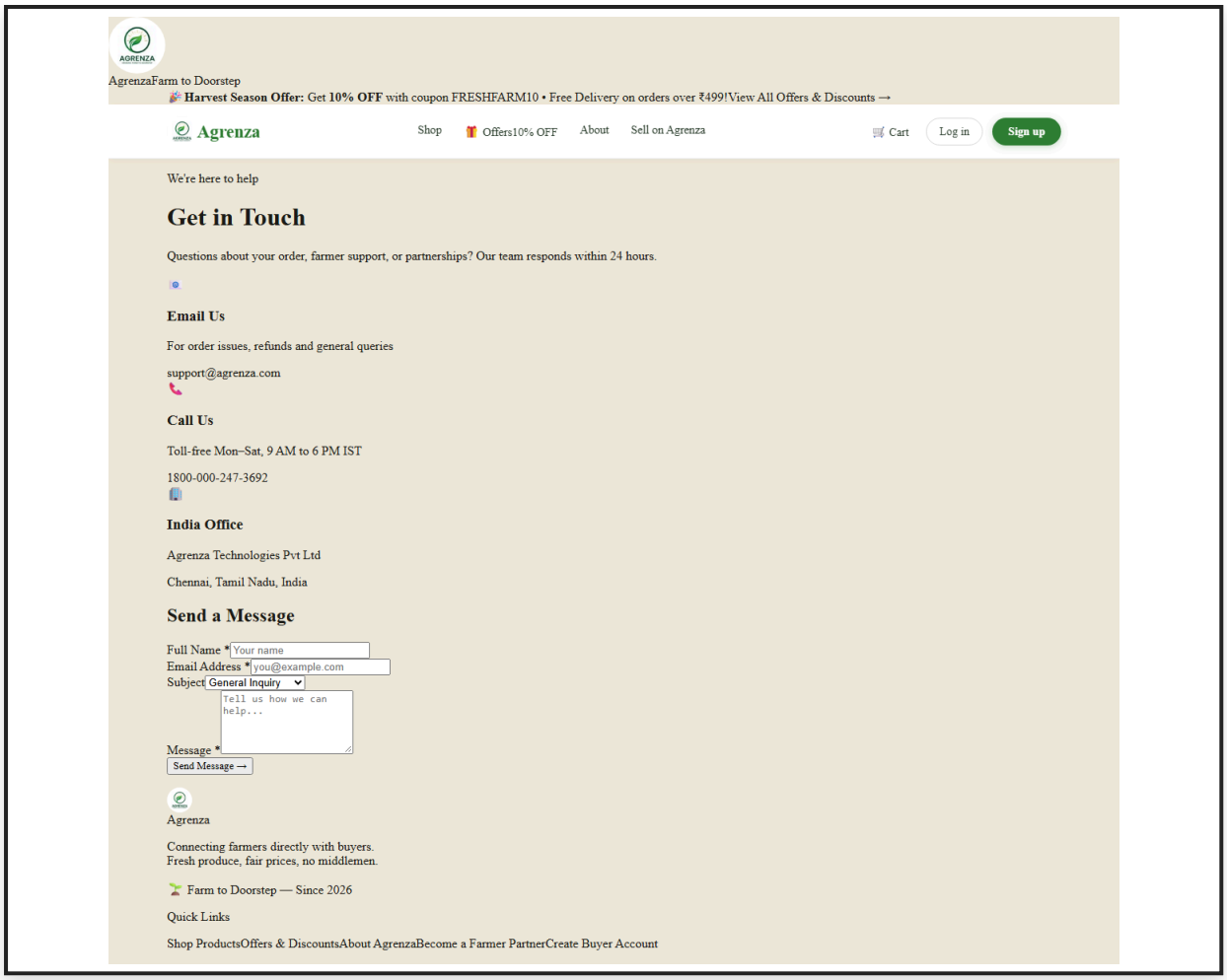


Figure 10.5: Contact & Customer Support Inquiry Portal

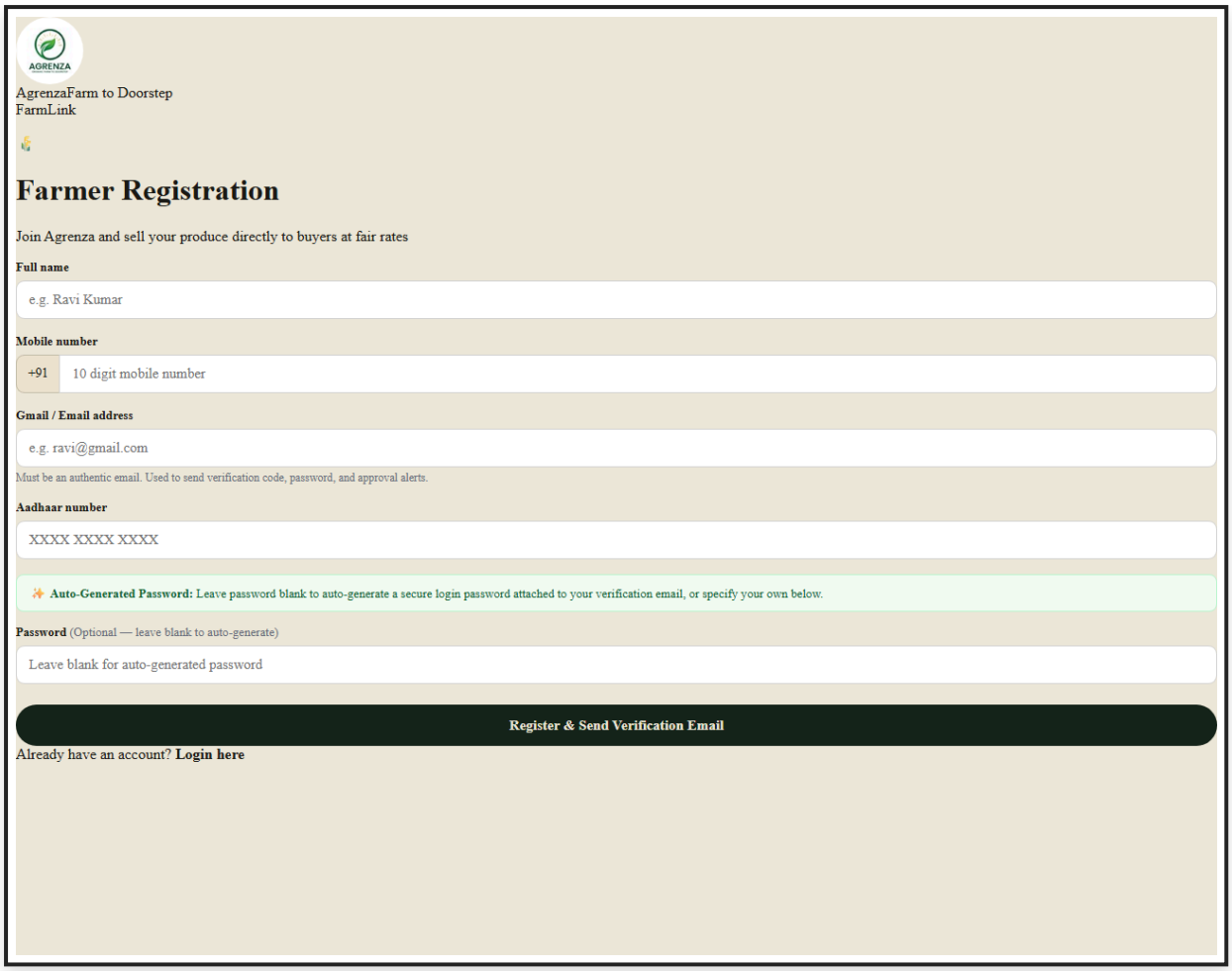


Figure 10.6: Farmer Partner Registration Portal (Dual-Role Enabled)

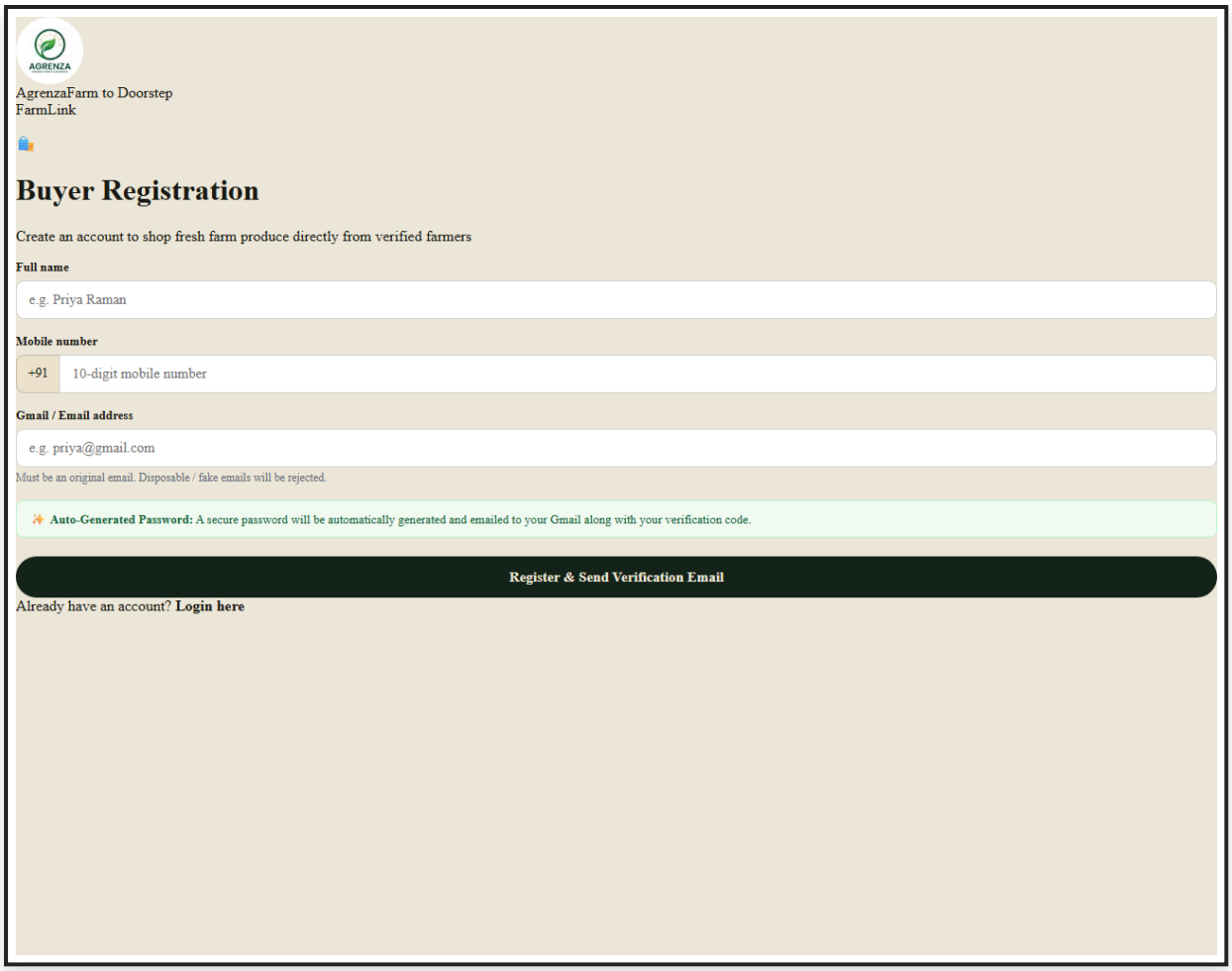


Figure 10.7: Buyer Registration Portal (Same Email & Mobile Permitted)

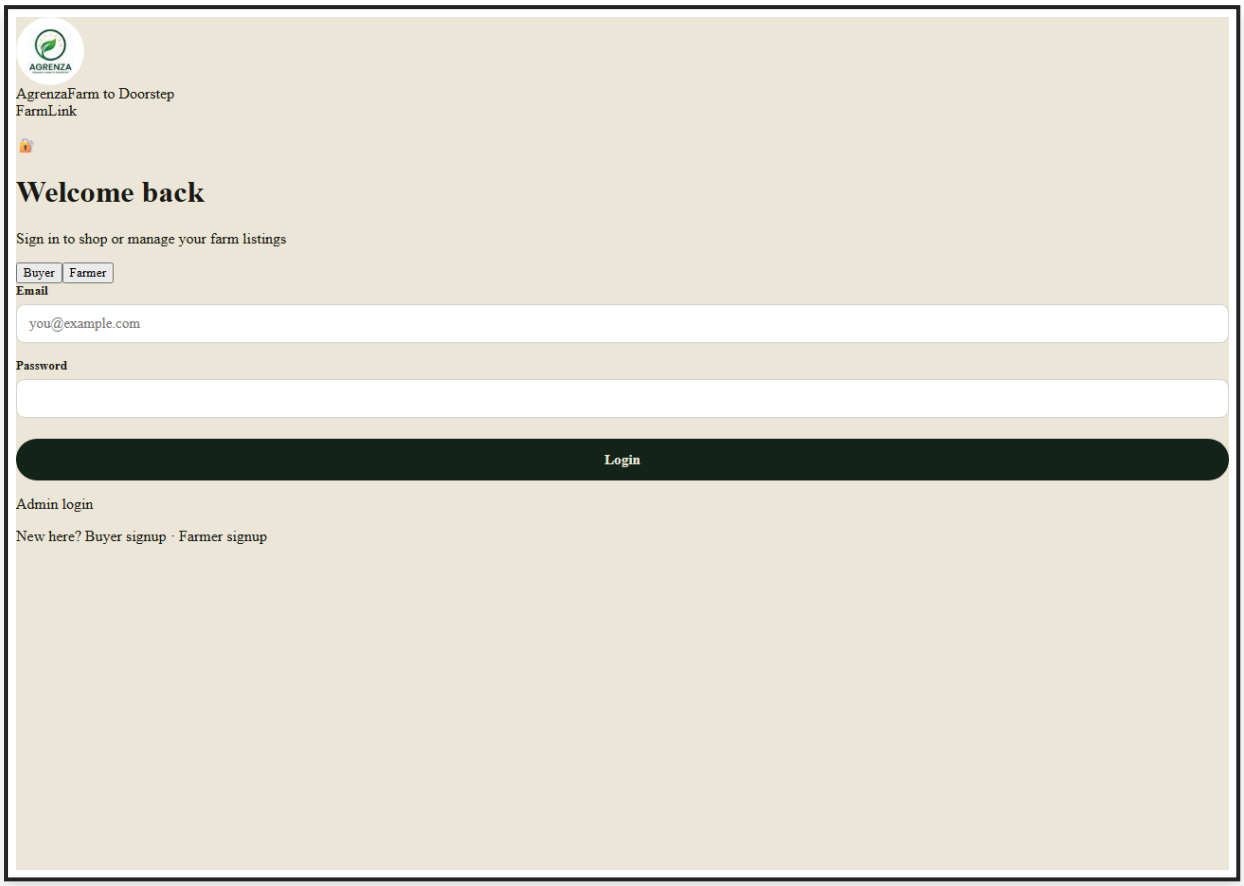


Figure 10.8: User Authentication & Role-Based Login Portal

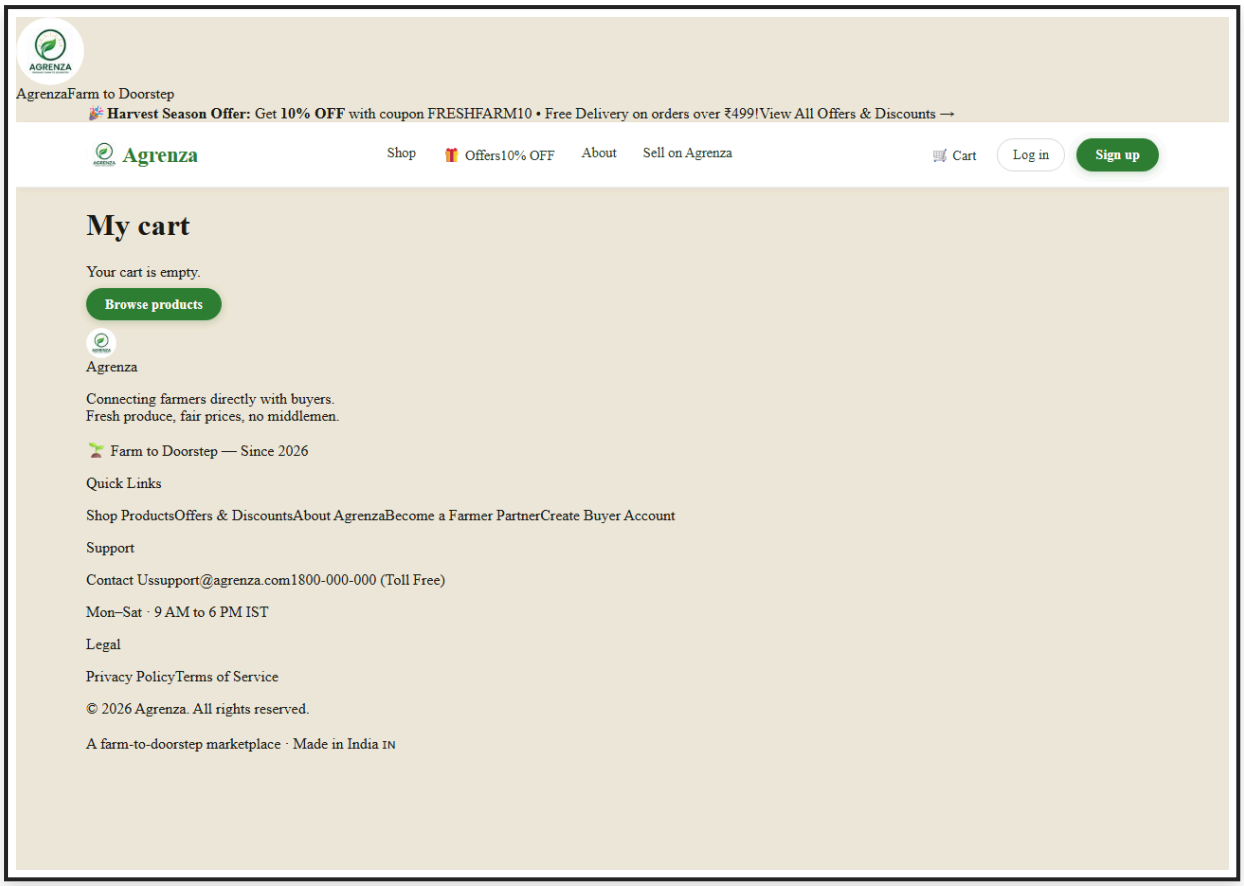


Figure 10.9: Shopping Cart & Order Checkout with Itemized GST Breakdown

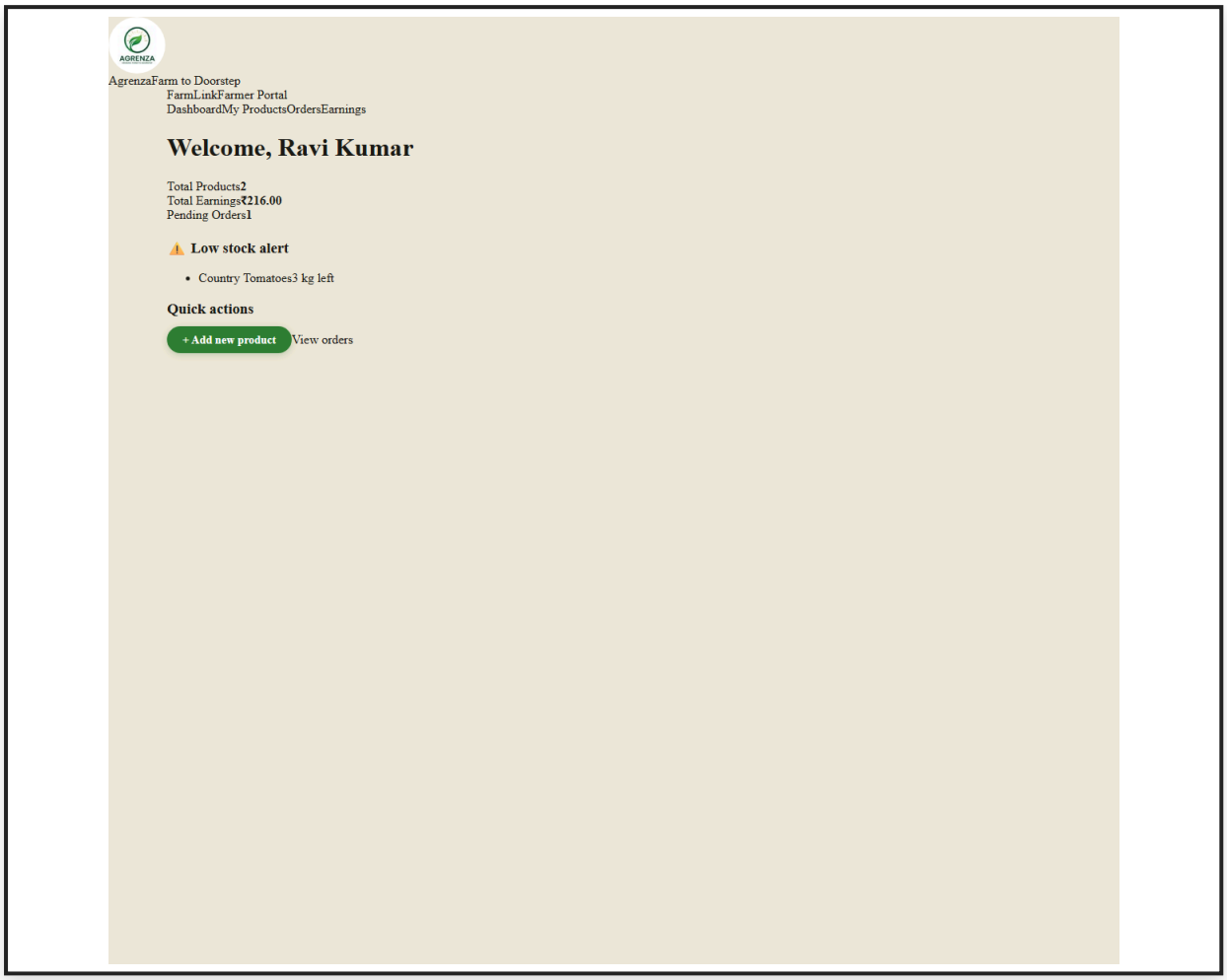


Figure 10.10: Farmer Dashboard — Revenue Analytics & Order Status Controls

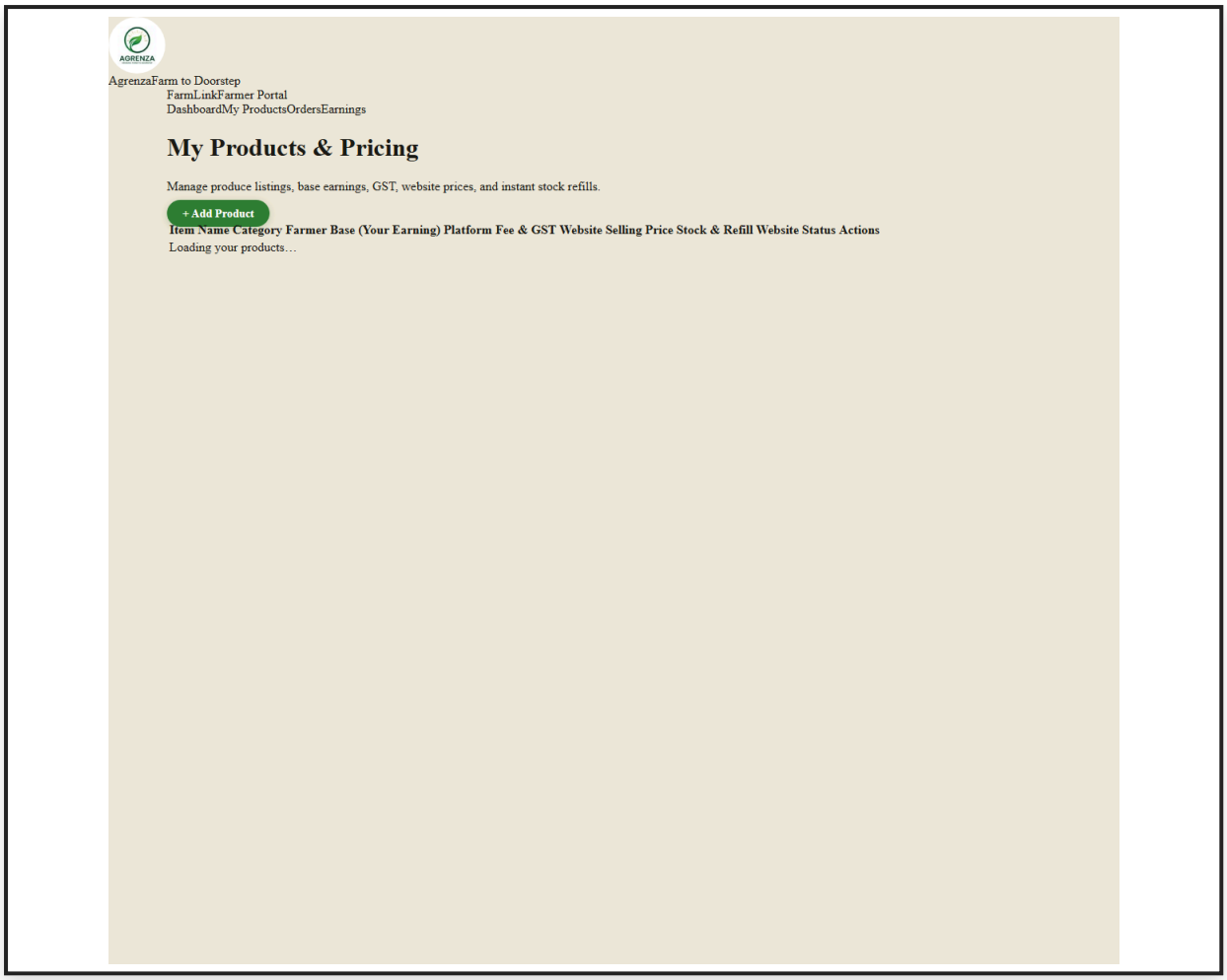


Figure 10.11: Farmer Product Inventory & Stock Refill Control Console



Figure 10.12: Admin Governance Dashboard & Platform KPI Metrics

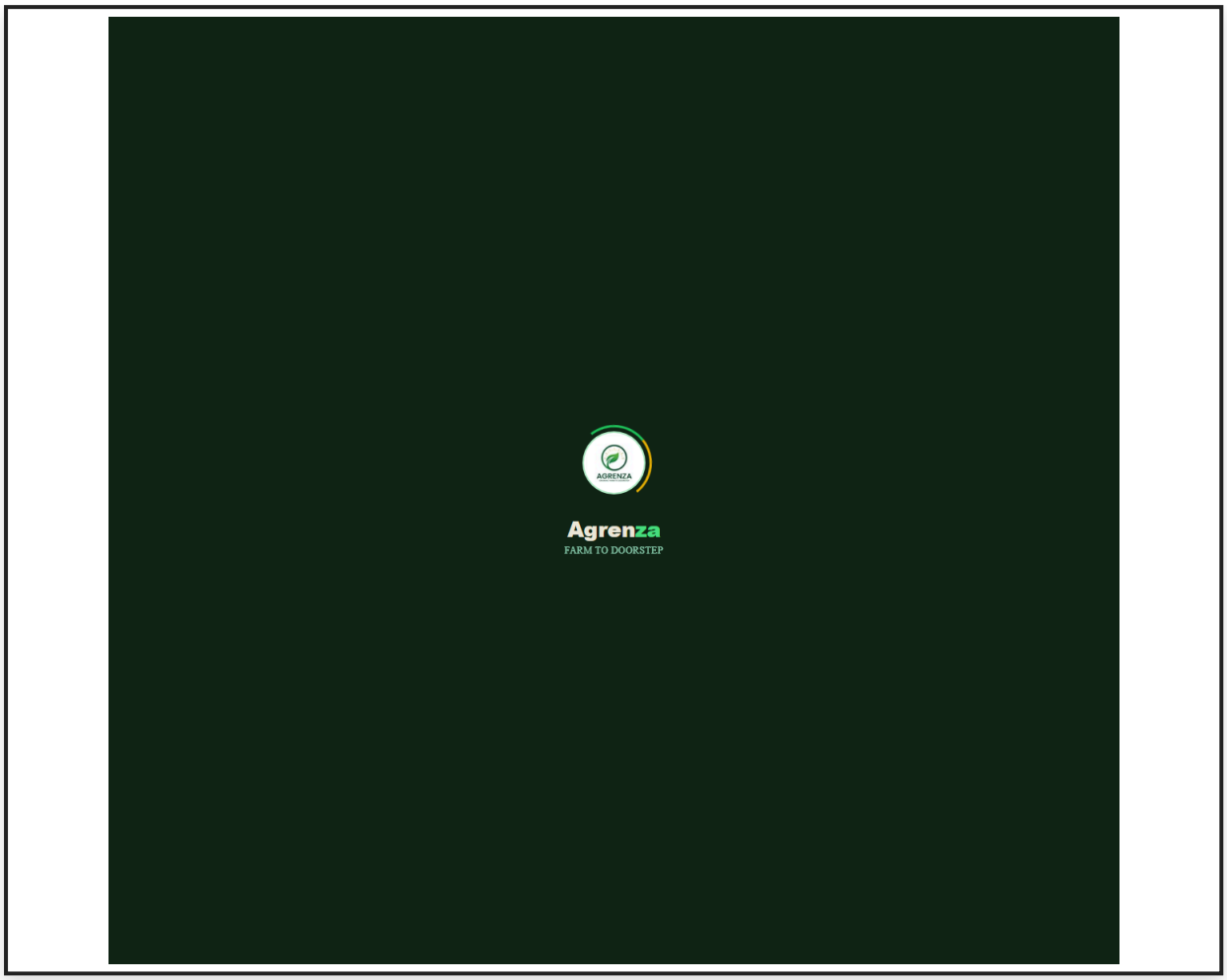


Figure 10.13: Admin Dual-Role Shared Accounts Auditor